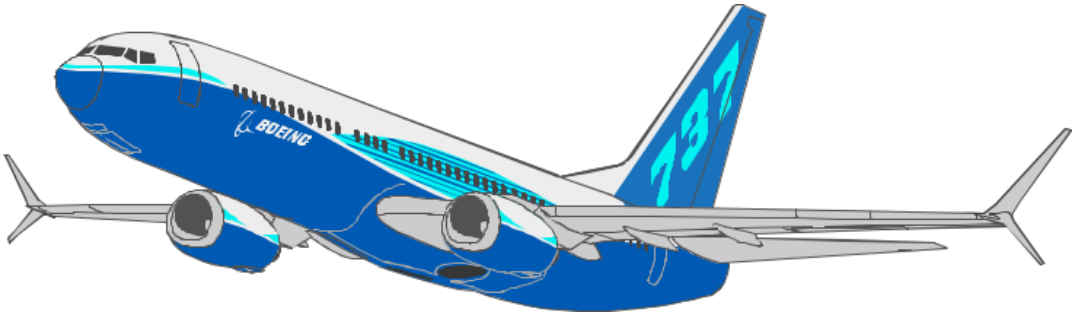
CARIBBEAN AIRLINES

BOEING 737-8



MINIMUM EQUIPMENT LIST

COPY NO.:……..

DOC. NO.: BW-737-8-MEL-0046

CARIBBEAN AIRLINES

BOEING 737-8

MINIMUM EQUIPMENT LIST

Document No.: BW-737-8-MEL-0046

CIVIL AVIATION AUTHORITY

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OUR REF: CAA/611/26/FOI
File No.: 6/1/2

12 June 2026

Capt. Daniel Galy
Chief Pilot
Caribbean Airlines Limited
Iere House
Golden Grove Road
PIARCO

Dear Captain Galy,

APPROVAL OF B737-8 MINIMUM EQUIPMENT LIST (MEL)
DOC. NO.: BW-737-8-MEL-0046 REVISION No.: 6b (CAL) DATED 05 JUNE 2026

Reference: OY: 103/2026 dated 09 June 2026

In accordance with Trinidad and Tobago Civil Aviation Regulations **No. 3:**34 and **3**:46, the Trinidad and Tobago Civil Aviation Authority (TTCAA) hereby approves **Revision No. 6b (CAL)** dated **05 June 2026**, of the B737-8 Minimum Equipment List – Document No.: BW-737-8-MEL-0046.

The B737-8 MEL [Doc. No.: BW-737-8-MEL-0046] shall be current and appropriate for the prescribed functions and the TTCAA must be advised of any amendments to the document.

Please find attached the original List of Effective Pages dated and signed by an authorized TTCAA Inspector.

Yours Respectfully,
TRINIDAD AND TOBAGO CIVIL AVIATION AUTHORITY

Captain Richard Martin
/f/ Director General Civil Aviation

xc. Mr. Cary Price - Director-General, TTCAA
Mr. Gerard A. Mark - Executive Manager Safety Regulation (Ag.), TTCAA
Mr. Ronald Bidaisee - Manager Flight Standards (Ag.), TTCAA
Capt. Raj Narwani - Flight Operations Inspector, TTCAA
Capt. René Seow - Executive Manager Flight Operations, Caribbean Airlines
Capt. Simon Laughlin - Manager Flight Standard and Training, Caribbean Airlines
Capt. Glen Shaw – Fleet Manager B737 POS (Ag.), Caribbean Airlines
Capt. Christopher Kelly - Fleet Manager B737 KIN, Caribbean Airlines
Mr. Dion Nagee - Executive Manager Quality & Air Safety, Caribbean Airlines
Mr. Dinesh Rampersad - Manager Service Engineering (Ag.), Caribbean Airlines

Board Members: Mr. Kimchand Rampaul *(Chairman)*, Mr. Yves Jacques Nicholson (*Deputy Chairman*), Mr. Trevor Benjamin, Mr. Brandon Baksh, Mr. Doolar Ramlal, Ms. Sarah Awai-King, Mr. Javan Edwards, Mr. Cary Price *(Director-General)*

FLIGHT OPERATIONS PUBLICATIONS - FILING INSTRUCTIONS

DOCUMENT TITLE : **B737-8 Minimum Equipment List**
DOCUMENT NO. : **BW-737-8-MEL-0046**
REVISION NO. : **6b (CAL)** DATE: **2026 June, 05**

HIGHLIGHTS OF CHANGE

MEL Revision 6b (CAL), dated 2026 June 05, incorporates changes made by Caribbean Airlines for the induction of 9Y-CUB.

The following changes are the Highlights of Changes for Revision 6b (CAL).

PAGE	EXPLANATION OF CHANGE
INDTRODUCTION	Editorial correction. Added item **2. Aircraft Applicability**.
RECORD OF REVISION APPROVAL	Revised to include the revision number, signature and date.
LIST OF EFFECTIVE PAGES	Revised.
MANUAL ORGANISATION & ADMINISTRATION	Revised to add Aircraft Number: 9644, Registration Number: 9Y-CUB, Serial Number: 65286, and Tabulation Number: 1B359.
DISTRIBUTION LIST / ACCESS	Revised to Assign No. 15 to Senior Engineer – Base Maintenance (Office).

SYSTEM/SEQUENCE NUMBERS	REMARKS
ATA 23 – COMMUNICATION	
-00.1 - -00.4	Updated Table of Contents.
-24-01A	Added (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO). Modified Maintenance (M) Procedures step.

…/2

Insert updated pages dated 2026 June 05, following the sequence on the Lists of Effective Pages. On the List of Effective Pages new pages are identified by '*n*', '*' for revised pages and '*d*' for pages to be deleted.

NOTE: When updating your manual, please audit your book using the current List of Effective Pages to ensure that your manual is current.

Caribbean Airlines

FLIGHT OPERATIONS PUBLICATIONS - FILING INSTRUCTIONS

DOCUMENT TITLE : **B737-8 Minimum Equipment List**
DOCUMENT NO. : **BW-737-8-MEL-0046**
REVISION NO. : **6b (CAL)** DATE: **2026 June, 05**

SYSTEM/SEQUENCE NUMBERS	REMARKS
ATA 23 – COMMUNICATION	
-24-01B	Added (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO).
-24-01-01	Added (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO). Removed NOTE. Modified Maintenance (M) Procedures steps.
-24-01-02	Added (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO).
-24-02-01A	Added new sub-item for CUB ONLY.
-24-02-01B	Added new sub-item for CUB ONLY.
-24-02-01-01	Added new sub-item for CUB ONLY.
-24-02-01-02	Added new sub-item for CUB ONLY.
-24-02-02A	Added new sub-item for CUB ONLY.
-24-02-02B	Added new sub-item for CUB ONLY.
-25-01B	Added new sub-item.
ATA 34 – NAVIGATION	
-00.1	Updated Table of Content.
-31-02	Added new sub-item for 9Y-CUB ONLY.

Record of Revisions

Temporary Revision Record

Introduction

Definitions

Preamble

Record of Revision Approval

Record of Temporary Revision Approval

RVSM Operations

FLY Away Kit

List of Effective Pages

Manual Organization & Administration

Contents - ATA Systems

Distribution List / Access

Miscellaneous

Status Messages

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Rev. No.	Issue Date	Date Inserted	Inserted By	Rev. No.	Issue Date	Date Inserted	Inserted By
3	2021-04-30						
3a	2022-01-21						
3b	2022-02-15						
3c	2022-03-01						
3d	2022-05-04						
4	2022-05-10						
5	2022-07-15						
5a (CAL)	2023-03-23						
5b (CAL)	2023-11-09						
5c (CAL)	2024-05-06						
6	2024-08-20						
6a (CAL)	2025-06-02						
6b (CAL)	2026-06-05						

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Rev. No.	Issue Date	Date Inserted	Inserted By	Rev. No.	Issue Date	Date Inserted	Inserted By
1	22/11/29						
2	23/08/16						

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POLICY FOR USE OF THE MINIMUM EQUIPMENT LIST

1. GENERAL

1.1 The Minimum Equipment List (MEL), which is authorised by the TTCAA, details which equipment, under what conditions, may be inoperative and the Aircraft as a whole still be operative. This is possible because of duplication of, or redundancy in certain systems. In addition, alternate methods or procedures may be used.

1.2 All components/systems on an aircraft fall into three classes in regard to airworthiness requirements.

- Units obviously required for Aircraft to be airworthy such as tyres, primary control systems, wings, units required by original certification, etc. such as First Aid Kits, impact lights etc. These are generally not included in the MEL.
- Units obviously not required for Aircraft to be airworthy such as pax entertainment systems, carpets, curtains, etc. are not included in the MEL.
- Units which do not clearly fall into either of the above classes or for which some deviations for the normal complement of equipment have been approved.

1.3 It is important to note therefore that ANY ITEM WHICH IS RELATED TO THE AIRWORTHINESS OF THE AIRCRAFT AND WHICH IS NOT INCLUDED IN THE MEL IS ALWAYS REQUIRED TO BE OPERATIVE BEFORE A FLIGHT IS DISPATCHED.

1.4 Multiple Deferments

- Caribbean Airlines is responsible for exercising the necessary operational controls to ensure that no aircraft is dispatched or flown with multiple MEL items inoperative without first determining that any interface or inter-relationship between inoperative systems or components will not result in a degradation in the level of safety and/or an undue increase in crew workload.

1.5 Under some conditions (i.e., poor weather conditions, heavy air traffic) it may be inadvisable to dispatch an Aircraft with several MEL deferments due to considerably increased pilot workload. Final responsibility for dispatch with any deferment rests with the Captain.

1.6 Continued Deferments - The MEL is of course not intended to provide for continued operation of the Aircraft for an indefinite period with MEL item(s) inoperative. Final responsibility for scheduling repairs or replacements rests with Maintenance, in agreement with Flight Operations.

Note: "STATION WHERE REPAIRS OR REPLACEMENTS CAN BE MADE" is defined as a Maintenance Station where mechanics are normally on duty and where the part required is normally allocated.

2. AIRCRAFT APPLICABILITY

When certain MEL ATA item numbers apply only to specific aircraft in Caribbean Airlines' fleet, the applicable aircraft will be identified by including the registration numbers in the title. The aircraft registration will be abbreviated to the last three letters (e.g., CAL, GUY, ANT, BAR).

3. RELEASING AN AIRCRAFT WITH INOPERATIVE UNIT

3.1 The exposure to additional failures during continued operation with inoperative systems or components must also be considered in determining that an acceptable level of safety is being maintained. The MEL may not deviate from requirements of the AFM Limitations section, emergency procedures or airworthiness directives unless the airworthiness directive or AFM provides otherwise.

3.2 Appropriate action must be taken to ensure that no secondary hazard can be introduced by an inoperative component. Such action may consist typically of removing the unit(s) involved, capping the openings, including lines, disconnecting and securing electrical connections, plugs, circuit breakers, etc. to ensure complete isolation from the Aircraft or associated systems as required.
APPROPRIATE ENTRIES SHALL BE MADE IN THE AIRCRAFT TECHNICAL LOG. ALL INOPERATIVE EQUIPMENT MUST BE PROPERLY PLACARDED WHERE THIS IS POSSIBLE.

3.3 When the indicating portion of an aircraft system or component is malfunctioning, appropriate troubleshooting procedure must be accomplished to determine that the fault exists in the indicating system and not in the aircraft system or component, in order to properly apply the requirements of this Section.

4. THE B737-8 MEL IS NO LESS RESTRICTIVE THAN THE FAA MMEL.

5. MEL APPLICABILITY

5.1 MEL relief is applied to an MEL item newly identified as inoperative up until the start of the takeoff (the point at which the pilot physically begins to apply power to accelerate an aircraft from a state of rest to flight). However, item failures that occur after an aircraft leaves the gate or ramp area, during pushback or taxi, and prior to takeoff should be dealt with in the following manner:

- Where the defect is identified in the MEL, is eligible for relief and no MAINTENANCE (M) PROCEDURES are required for dispatch, the flight may continue with the approval of the Captain after determining that an inoperative item will not affect the safety of the flight. Consideration should be given to circumstances such as airspace requirements, weather, performance, weight & balance, or fuel limitations.
- The flight crew will then carry out all necessary OPERATIONS (O) PROCEDURES as required.
- Where the defect requires MAINTENANCE (M) PROCEDURES, takeoff is prohibited until such procedures are carried out by the approved maintenance personnel.

5. MEL APPLICABILITY (CONT'D)

5.2 If the flight crew becomes aware of a defect after the start of the takeoff, none of the requirements of the MEL apply. However, it may be prudent to consider the requirements of the MEL as the defect may affect dispatch from the destination airport. Flight crew may consult with MCC and SOC to determine the best course of action.

5.3 All defects must be recorded in the Technical Log after landing at the destination airport.

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1. **"Airplane Flight Manual"** (AFM) is the document required for type certification and approved by the responsible Aircraft Service Certification Office. The approved AFM for the specific aircraft is listed on the applicable Type Certificate Data Sheet (TCDS). The approved flight manual is the governing document for operational limitations and performance parameters for an aircraft.

2. **"Ballast"** For the purposes of this MEL, Ballast, when required, shall consist only of; any or all of the items listed as basic items of a fly away kit on FLY AWAY KIT Page 1 of this MEL, and/or, Sandbags; provided that sandbags used as ballast MUST NOT BE DAMAGED in order to avoid sand spillage.

3. **"Considered Inoperative"** The phrase "Considered Inoperative", as used in the Remarks or Exceptions column, means an item must be treated for dispatch, taxi with intent for flight, and flight purposes as though it were inoperative. The item must not be used or operated until the original deferred item is repaired. Additional actions include: documenting the item on the dispatch release (if applicable); placarding; complying with all Remarks or Exceptions, including and (M) and (O) procedures; considering applicable notes; and observing the repair category.

4. "**(CAL)**" Identifies a system, component, or function, listed in the "Item" Column, for which the repair category and/or number required for dispatch is determined by Caribbean Airlines and differs from that required by the FAA MMEL. Caribbean Airlines determined dispatch requirements will never be less restrictive than required by the FAA MMEL.

5. **"Contaminated Runway"** A runway condition where more than 25% of runway surface area (within reported length and width being used) is covered by frost, ice, and any depth of snow, slush, or water.

6. "**Dash (-) symbol"** Indicates a variable number (quantity) of the item may be installed or required for dispatch. Where a **"-"** is shown in the Number Required for Dispatch column, the note in the Remarks column will sometimes qualify the requirement.

7. **"Day of Discovery"** This is the calendar-day an item malfunction was recorded in the aircraft maintenance record/logbook and is excluded from the interval established by the assigned repair category. See definitions for sub-elements of "Repair Category".

8. **"Deactivated" or "Secured"** When the MEL refers to an item as "deactivated" or "secured" or both, the specified item must be put into an acceptable condition for safe flight. An acceptable method of deactivating or securing will be established by the Department Head Quality and Chief Pilot.

9. **"Deleted" or "Moved"** – **"Deleted"** in the Remarks or Exceptions column indicates the item was previously listed but is no longer addressed by the MEL and is now required to be operative if installed in the aircraft. **"Moved"** in the Remarks or Exceptions column indicates the item was moved within the chapter, to a different chapter in the MEL, or another TTCAA-approved document.

10. **"Dispatch"** The point at which an aircraft first moves under its own power for the purpose of commencing a flight.

11. **"Extended Range" (ER)** Refers to extended range operations of a two-engine airplane which has a type design approval for ER operations and complies with current Regulations and Advisory Circulars as amended.

    NOTE: ER Operations are operations beyond CAL 60 minute circles. Caribbean Airlines does not currently conduct ER Operations.

12. **"Excess items"** Those items that have been installed that are redundant to the requirements of the TTCAR No.7.

13. **"Flight Day"** A twenty (24) hour period (from midnight to midnight) either Co-ordinated Universal Time (UTC) or local time at the aircraft's base, during which at least one flight is initiated for the affected aircraft.

14. **"Flight Hour"** The time from the moment an aircraft leaves the surface of the earth until it touches it at the next point of landing.

15. **"Icing Conditions"** An atmospheric environment that may cause ice to form on the aircraft (structural) or in the engine(s) (induction). Icing conditions may be known or forecast, and may be defined in the AFM.

16. **"Inoperative"** Malfunction of an item to the extent that it does not accomplish its intended purpose or is not consistently functioning normally within its approved operating limit(s) or tolerance(s), or both.

17. **"Inoperative Components of an Inoperative System"** Inoperative components of an inoperative system are usually considered components directly associated with and having no other function than to support that system. Warning/caution systems associated with the inoperative system must be operative unless relief is specifically authorized per the MEL.

18. **"Installed"** See definition for "Number Installed" under "System Page/Column Definitions".

19. **"Instrument Flight Rules" (IFR)** As defined in the appropriate TTCARs.

20. **"Interval"** See definition for "Repair Category".

21. **"Is Not Used"** The phrase "Is Not Used" in the Remarks or Exceptions column for an MEL item may specify that another item is not used. In such cases, crewmembers must not activate, actuate, or otherwise use the referenced item under normal operations. If the item not to be used is located elsewhere in the MEL, it is not necessary to accomplish any (M) procedure(s) associated with the referenced item. However, compliance with operational requirements, and an additional placard must be affixed as close as practical to the control or indicator for the item that is not to be used. This informs crewmembers that an item is not to be used under normal operations.

22. **"Item"** An instrument, equipment, system, component, message, or function that is installed on or exhibited by the aircraft.

23. **"(M)"** symbol indicates a requirement for a specific maintenance procedure which must be accomplished prior to operation with the listed item inoperative. Normally these procedures are accomplished by maintenance personnel; however, other personnel may be qualified and authorized to perform certain functions. Procedures requiring specialized knowledge or skills, or requiring the use of tools or test equipment should be accomplished by maintenance personnel. The satisfactory accomplishment of all maintenance procedures, regardless of who performs them is the responsibility of Caribbean Airlines. Appropriate procedures are contained in the Procedures Section of this MEL.

24. **"N/A"** Not Applicable in the remarks column indicates that a MEL item has been previously assessed as "Not Applicable" to either Caribbean Airlines aircraft configuration or operation.

25. **"Night"** The time between the end of evening civil twilight and the beginning of morning civil twilight, as published in the Air Almanac, converted to local time.

26. **"Nonessential Equipment and Furnishings" (NEF)** NEFs are those items installed on the aircraft as part of the original type certification (TC), supplemental type certificate (STC), engineering order, or other form of alteration that have no effect on the safe operation of flight and would not be required by the applicable certification rules or operational rules. They are those items that, if damaged, inoperative, or missing, have no effect on the aircraft's ability to be operated safely under all operational conditions. These nonessential items may be installed in areas including, but not limited to:
    - Cargo Areas
    - Crew Rest Areas
    - Flight Deck Area
    - Galley Area
    - Lavatories
    - Passenger Compartment
    - Service Areas

27. **"NOTES"** NOTES provide additional information for crewmember or maintenance consideration. Notes are used to identify applicable material, which is intended to assist with compliance, but do not relieve Caribbean Airlines of the responsibility for compliance with all applicable requirements. Notes are not a part of the provisos.

28. **"(O)"** symbol indicates a requirement for a specific operations procedure which must be accomplished in planning for and/or operating with the listed item inoperative. Normally these procedures are accomplished by flight crew; however, other personnel may be qualified and authorized to perform certain functions. The satisfactory accomplishment of all procedures, regardless of who performs them, is the responsibility of Caribbean Airlines. Appropriate procedures are contained in the Procedures Section of this MEL.

29. **"Operative"** An operative item will accomplish its intended purpose and is consistently functioning normally within its design operating limit(s) and tolerance(s). When an MEL item specifies an item must be operative, it is not required to verify the item's operational status. It should be considered operative unless reported or known to be malfunctioning. See definition for Verified Operative.

30. **"Passenger Convenience Items"** Those items related to passenger convenience, comfort or entertainment such as, but not limited to, galley equipment, movie equipment, ash trays, stereo equipment, overhead reading lamps, etc.

31. **"Performance-Based Navigation" (PBN)** See definition of PBN in the expanded section below.

32. **"Placarding"** Each inoperative item must be placarded to inform and remind the crewmembers and maintenance personnel of the equipment condition.

    NOTE: To the extent practical, placards should be located adjacent to the control or indicator for the item affected; however, unless otherwise specified, placard wording and location will be determined by the Department Head Quality in concurrence with the Chief Pilot.

33. **"Procedure"** See definitions for "(M)" and "(O)".

34. **"Repair Category"** Repairs of inoperative systems or components, deferred in accordance with the MEL must be accomplished at or prior to the repair times established by the following letter designators:

    Category A Items in this category shall be repaired within the time interval specified in the remarks column.

    Category B Items in this category shall be repaired within three (3) consecutive calendar days (72 hours), excluding the day the malfunction was recorded in the aircraft maintenance record/logbook.

    For example, if it were recorded at 10 a.m. on January 26th, the three day interval would begin at midnight the 26th and end at midnight the 29th.

    Category C Items in this category shall be repaired within ten (10) consecutive calendar days (240 hours), excluding the day the malfunction was recorded in the aircraft maintenance record/logbook.

    For example, if it were recorded at 10 a.m. on January 26th, the 10 day interval would begin at midnight the 26th and end at midnight February 5th.

    Category D Items in this category shall be repaired within one hundred and twenty (120) consecutive calendar days (2880 hours) but may only be deferred for a maximum of thirty (30) days. The item may then be deferred for a further thirty (30) days up to a maximum of one hundred and twenty (120) days.

35. **"Required"** See definition for "Number Required' under "System Page/Column Definitions".

36. **"Required by TTCAR"** When the MEL contains statements such as "As Required by TTCAR," "Not required by TTCAR," or "Any in excess of those required by TTCAR," the listed item is subject to certain requirements expressed in TTCAR operating rules. The number of items required by applicable TTCAR operating rules must be operative. When the listed item is not required by TTCAR, it may be inoperative for the time specified by repair category.

37. **"System"** The group of directly related components which together performs a specific function. For example: "RPM indication system" would include the RPM indicator, tachometer generator, circuit breaker and associated circuitry. Inoperative items which are components of a system which is inoperative are usually considered components directly associated with and having no other function than to support that system. (Warning/Caution systems associated with the inoperative system must be operative unless relief is specifically authorized per the MEL).

38. **"System Page/Column Definitions"** The MMEL system page is divided into columns that include sequence number, item, repair category, number installed, number required for dispatch, and remarks or exceptions, as well as provision for a vertical change bar.

    A. **Item Number** - This column lists the unique identification for each MEL item. System numbers are based on the Air Transportation Association (ATA) Specification Number 100 and items are numbered sequentially.

    B. **Item** - See definition for "Item".

    C. **Repair Category** - See definition for "Repair Category".

    D. **Number Installed** - This column indicates the number (quantity) of items normally installed in the aircraft. This number represents the aircraft configuration considered in developing this MEL. Should the number be a variable or impractical to exactly determine (e.g., optional equipment, fleet configuration differences, cockpit lighting items, cabin lighting items, cargo restraint components, Flight Data Recorder (FDR) recording parameters), a number is not required and the dash "-" symbol is used instead. Number installed on particular aircraft are incorporated where applicable.

    E. **Number Required for Dispatch** – This column indicates the minimum number (quantity) of items required for operation provided the conditions specified in the Remarks or Exceptions are met.

    F. **Remarks or Exceptions** - This column may be blank, or it may include a statement permitting operation with a specific number of items inoperative. The statement may include a proviso for such operation and appropriate notes.

    G. **Provisos** - A proviso is used to stipulate conditions or limitations that must be complied with for operation with the listed item inoperative.

    H. **"NOTE"** - See definition for "NOTE".

39. **"Takeoff"** The act of beginning a flight in which an aircraft is accelerated from a state of rest to that of flight. For the purposes of MEL relief, this translates to the point at which the pilot physically begins to apply power to initiate the takeoff from the runway or takeoff surface.

40. **"Verified Operative"** When an MEL item specifies an item must be verified operative or checked operative, it is required to check and confirm the item is operative at the interval(s) specified for that MEL item (e.g., verified operative prior to each flight). If no interval is specified, verification is required only at the time of deferral of the original item.

41. "**Vertical Bar" (change bar)** in the margin indicates a change, addition or deletion in the adjacent text for the current revision of that page only. The change bar is dropped at the next revision of that page.

42. **"Visible Moisture"** means an atmospheric environment containing water in any form that can be seen in natural or artificial light, for example, clouds, fog, rain, sleet, hail, or snow.

43. **"Visual Flight Rules" (VFR)** As defined in the appropriate TTCARs. If the Remarks or Exceptions state flight must be completed in VFR, the pilot is precluded from filing an Instrument Flight Rules (IFR) flight plan.

44. **"Visual Meteorological Conditions" (VMC)** The atmospheric environment is such that would allow a flight to proceed under the visual flight rules applicable to the flight. This does not preclude operating under Instrument Meteorological Conditions (IMC).

PERFORMANCE-BASED NAVIGATION (PBN)

AREA NAVIGATION

A method of navigation which permits aircraft operation on any desired flight path within the coverage of ground or space-based navigation aids or within the limits of the capability of self-contained aids, or a combination of these.

NOTE: AREA NAVIGATION INCLUDES PERFORMANCE-BASED NAVIGATION AS WELL AS CLASS II NAVIGATION.

OPERATIONS THAT DO NOT MEET THE DEFINITION OF PERFORMANCE-BASED NAVIGATION.

PERFORMANCE-BASED NAVIGATION

Area navigation based on performance requirements for aircraft operating along an ATS route, on an instrument approach procedure or in a designated airspace.

NAVIGATION SPECIFICATION

A set of aircraft and aircrew requirements needed to support Performance-Based Navigation operations within a defined airspace.

There are two kinds of navigation specification:

RNAV SPECIFICATION

A navigation specification based on area navigation that does not include the requirement for on-board performance monitoring and alerting, designated by the prefix RNAV, e.g. RNAV 5, RNAV 1.

RNP SPECIFICATION

A navigation specification based on area navigation that includes the requirement for on-board performance monitoring and alerting, designated by the prefix RNP, e.g., RNP 4, RNP APCH.

Approach Navigation Specifications cover all segments of the instrument approach. RNP specifications are designated using RNP as a prefix and an abbreviated textual suffix, e.g., RNP APCH or RNP AR APCH. There are no RNAV approach specifications.

Caribbean Airlines is approved to conduct RNAV 1, RNAV 2, RNAV 5, RNAV 10 (RNP 10), and RNP APCH.

In this MEL, PBN Operations means all of the above RNAV Specifications and RNP APCH.

CLASS I NAVIGATION

Class I navigation is any en-route flight operation or portion of an operation that is conducted entirely within the designated Operational Service Volumes (or ICAO equivalents) of ICAO standard airway navigation facilities (VOR, VOR/DME, NDB). Class I navigation also includes en-route flight operations over routes designated with an "MEA GAP" (or ICAO equivalent). En-route flight operations conducted within these areas are defined as "Class I Navigation" operations irrespective of the navigation means used. Class I Navigation includes operations within these areas using pilotage or any other means of navigation which does not rely on the use of the VOR, VOR/DME or NDB.

MEA GAP

MEA is established with a gap in navigation signal coverage.

CLASS II NAVIGATION

Class II Navigation is any en-route flight operation which is not defined as Class I Navigation. Class II Navigation is any en-route flight operation or portion of an en-route operation (irrespective of the means of navigation) which takes place outside (beyond) the designated Operational Service Volume (or ICAO equivalents) of ICAO standard airway navigation facilities (VOR, VOR/DME, NDB). However, Class II navigation does not include en-route flight operations over routes designated with an "MEA GAP" (or ICAO equivalent).

The following is applicable for authorised certificate holders operating under Trinidad & Tobago Civil Aviation Regulations (TTCARs): The TTCARs require that all equipment installed on an aircraft in compliance with Airworthiness Standards and other Operating Rules must be operative. However, the rules also permit the publication of a Minimum Equipment List (MEL) where compliance with certain equipment requirements is not necessary in the interests of safety under all operating conditions. Experience has shown that with the various levels of redundancy designed into aircraft, operation of every system or installed component may not be necessary when the remaining operative equipment can provide an acceptable level of safety.

A Master Minimum Equipment List (MMEL) is developed by the FAA, with participation by the aviation industry, to improve aircraft utilization and thereby provide more convenient and economic air transportation for the public. The FAA approved MMEL includes those items of equipment related to airworthiness and operating regulations and other items of equipment which the Administrator finds may be inoperative and yet maintain an acceptable level of safety by appropriate conditions and limitations; it does not contain obviously required items such as wings, flaps, and rudders. The MMEL is the basis for development of individual operator MELs which take into consideration the operator's particular aircraft equipment configuration and operational conditions. Operator MELs, for administrative control, may include items not contained in the MMEL; however, relief for administrative control items must be approved by the Administrator. An operator's MEL may differ in format from the MMEL, but cannot be less restrictive than the MMEL. The individual operator's MEL, when approved and authorized, permits operation of the aircraft with inoperative equipment.

Equipment not required by the operation being conducted and equipment in excess of TTCARs requirements are included in the MEL with appropriate conditions and limitations. The MEL must not deviate from the Aircraft Flight Manual Limitations, Emergency Procedures or with Airworthiness Directives. It is important to remember that all equipment related to the airworthiness and the operating regulations of the aircraft not listed on the MMEL must be operative.

Suitable conditions and limitations in the form of placards, maintenance procedures, crew operating procedures and other restrictions as necessary are specified in the MEL to ensure that an acceptable level of safety is maintained.

The MEL is intended to permit operation with inoperative items of equipment for a period of time until repairs can be accomplished. It is important that repairs be accomplished at the earliest opportunity. In order to maintain an acceptable level of safety and reliability the MMEL establishes limitations on the duration of and conditions for operation with inoperative equipment. The MEL provides for release of the aircraft for flight with inoperative equipment. When an item of equipment is discovered to be inoperative, it is reported by making an entry in the Aircraft Maintenance Record/Logbook as prescribed by the TTCARs. The item is then either repaired or may be deferred per the MEL or other approved means acceptable to the Administrator prior to further operation. MEL conditions and limitations, do not relieve the operator from determining that the aircraft is in condition for safe operation with items of equipment inoperative.

When these requirements are met, an Airworthiness Release, Aircraft Maintenance Record/Logbook entry, or other approved documentation is issued as prescribed by the TTCARs. Such documentation is required prior to operation with any item of equipment inoperative.

Caribbean Airlines is responsible for exercising the necessary operational control to ensure that an acceptable level of safety is maintained. When operating with multiple inoperative items, the interrelationships between those items and the effect on aircraft operation and crew workload will be considered.

Caribbean Airlines is to establish a controlled and sound repair program including the parts, personnel, facilities, procedures, and schedules to ensure timely repair.

WHEN USING THE MEL, COMPLIANCE WITH THE STATED INTENT OF THE PREAMBLE, DEFINITIONS, AND THE CONDITIONS AND LIMITATIONS SPECIFIED IN THE MEL IS REQUIRED.

Revision No.	Signature	Date
3	Chief Pilot – Captain Gordon Forbes Jr.	2021 Nov. 16
3a	Chief Pilot – Captain Gordon Forbes Jr.	2022 Jan. 21
3b	Chief Pilot – Captain Gordon Forbes Jr.	2022 Feb. 15
3c	Chief Pilot – Captain Gordon Forbes Jr.	2022 Mar. 01
3d	Chief Pilot – Captain Gordon Forbes Jr.	2022 May 04
4	Chief Pilot – Captain Gordon Forbes Jr.	2022 May 10
5a (CAL)	Chief Pilot – Captain Gordon Forbes Jr.	2023 Apr. 03
5b (CAL)	Chief Pilot – Captain Christopher Kelly	2023 Nov. 09
5c (CAL)	Chief Pilot – Captain Christopher Kelly	2024 May 15
6	Chief Pilot – Captain Renè Seow	2025 Feb. 20
6a (CAL)	Chief Pilot – Captain Renè Seow	2025 Jun. 09
6b (CAL)	Chief Pilot -	2026 Jun. 09

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Revision No.	Signature	Date
1	Chief Pilot – Captain Gordon Forbes Jr.	2022 May 04
2	Chief Pilot –	2023 Sep 04

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ITEMS AFFECTING RVSM OPERATIONS

MEL RVSM Restriction	Airframe System
22-01-1	Autopilot Systems
22-11-4	Mode Control Panel Switches
11)	L NAV, HDG HOLD Switches
34-16-1	Altitude Alerting System
34-53-1	ATC Transponders and Automatic Altitude Reporting Systems
34-61-2	Traffic Alert Collision Avoidance System (TCAS II)

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FLY AWAY KIT

For the purpose of this MEL the following are the basic items of a Fly Away kit:

Item	ATA	Description	Qty
1	32	Main Wheel Assembly, Wheel Tools and Axle Protector	1
2	32	Nose Wheel Assembly, Wheel Tools and Axle Protector	1
3	80	Engine Starter and Starter Valve	1
4	80	Packing	1
5	0	Torque Meter	1

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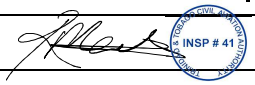
Signature: 	Date: 12 June 2026

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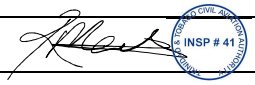
Signature: 	Date: 12 June 2026

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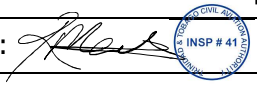
Signature: 	Date: 12 June 2026

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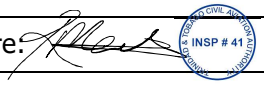
Signature: 	Date: 12 June 2026

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	26-20.5	3	2021 Apr. 30
	26-20.6	3	2021 Apr. 30
	26-22.1	3	2021 Apr. 30
	26-22.2	3	2021 Apr. 30
	26-23.1	3	2021 Apr. 30
	26-23.2	3	2021 Apr. 30
	26-23.3	3	2021 Apr. 30
	26-23.4	3	2021 Apr. 30

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ATA 26 FIRE PROTECTION *(Cont'd)*	26-24.1	3	2021 Apr. 30
	26-24.2	3	2021 Apr. 30
	26-26.1	6	2024 Aug. 20
	26-26.2	6	2024 Aug. 20
ATA 27 FLIGHT CONTROLS	ATA 27-00.1	6	2024 Aug. 20
	ATA 27-00.2	6	2024 Aug. 20
	27-08.1	3	2021 Apr. 30
	27-08.2	3	2021 Apr. 30
	27-21.1	3	2021 Apr. 30
	27-21.2	3	2021 Apr. 30
	27-24.1	3	2021 Apr. 30
	27-24.2	3	2021 Apr. 30
	27-31.1	3	2021 Apr. 30
	27-31.2	3	2021 Apr. 30
	27-31.3	3	2021 Apr. 30
	27-31.4	3	2021 Apr. 30
	27-51.1	3	2021 Apr. 30
	27-51.2	3	2021 Apr. 30
	27-61.1	6	2024 Aug. 20
	27-61.2	6	2024 Aug. 20
	27-61.3	3	2021 Apr. 30
	27-61.4	3	2021 Apr. 30
	27-61.5	3	2021 Apr. 30
	27-61.6	3	2021 Apr. 30
	27-62.1	3	2021 Apr. 30
	27-62.2	5	2022 Jul. 15

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ATA 27 FLIGHT CONTROLS *(Cont'd)*	27-83.1	3	2021 Apr. 30
	27-83.2	3	2021 Apr. 30
	27-88.1	6	2024 Aug. 20
	27-88.2	6	2024 Aug. 20
ATA 28 FUEL	ATA 28-00.1	3	2021 Apr. 30
	ATA 28-00.2	3	2021 Apr. 30
	28-21.1	3	2021 Apr. 30
	28-21.2	3	2021 Apr. 30
	28-21.3	3	2021 Apr. 30
	28-21.4	3	2021 Apr. 30
	28-21.5	3	2021 Apr. 30
	28-21.6	3	2021 Apr. 30
	28-22.1	3	2021 Apr. 30
	28-22.2	3	2021 Apr. 30
	28-22.3	3	2021 Apr. 30
	28-22.4	3	2021 Apr. 30
	28-22.5	3	2021 Apr. 30
	28-22.6	3	2021 Apr. 30
	28-22.7	3	2021 Apr. 30
	28-22.8	3	2021 Apr. 30
	28-22.9	3	2021 Apr. 30
	28-22.10	3	2021 Apr. 30
	28-22.11	3	2021 Apr. 30
	28-22.12	4	2022 May 10
	28-22.13	3	2021 Apr. 30
	28-22.14	3	2021 Apr. 30

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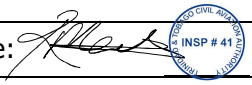
Signature: 	Date: 12 June 2026

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ATA 28 FUEL (*Cont'd)*	28-25.1	3	2021 Apr. 30
	28-25.2	3	2021 Apr. 30
	28-41.1	3	2021 Apr. 30
	28-41.2	3	2021 Apr. 30
	28-41.3	3	2021 Apr. 30
	28-41.4	3	2021 Apr. 30
	28-42.1	3	2021 Apr. 30
	28-42.2	3	2021 Apr. 30
	28-42.3	3	2021 Apr. 30
	28-42.4	5	2022 Jul. 15
	28-43.1	3	2021 Apr. 30
	28-43.2	3	2021 Apr. 30
	28-44.1	3	2021 Apr. 30
	28-44.2	3	2021 Apr. 30
ATA 29 HYDRAULIC POWER	ATA 29-00.1	3	2021 Apr. 30
	ATA 29-00.2	3	2021 Apr. 30
	29-09.1	3	2021 Apr. 30
	29-09.2	3	2021 Apr. 30
	29-18.1	3	2021 Apr. 30
	29-18.2	3	2021 Apr. 30
	29-31.1	3	2021 Apr. 30
	29-31.2	3	2021 Apr. 30
	29-32.1	3	2021 Apr. 30
	29-32.2	3	2021 Apr. 30
	29-33.1	3	2021 Apr. 30
	29-33.2	3	2021 Apr. 30
	29-33.3	3	2021 Apr. 30
	29-33.4	3	2021 Apr. 30

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ATA 29 HYDRAULIC POWER *(Cont'd)*	29-34.1	3	2021 Apr. 30
	29-34.2	3	2021 Apr. 30
	29-34.3	3	2021 Apr. 30
	29-34.4	3	2021 Apr. 30
ATA 30 ICE AND RAIN PROTECTION	ATA 30-00.1	6a (CAL)	2025 Jun. 02
	ATA 30-00.2	3	2021 Apr. 30
	30-11.1	6	2024 Aug. 20
	30-11.2	5	2022 Jul. 15
	30-11.3	5	2022 Jul. 15
	30-11.4	5	2022 Jul. 15
	30-11.5	6	2024 Aug. 20
	30-11.6	5	2022 Jul. 15
	30-11.7	5	2022 Jul. 15
	30-11.8	5	2022 Jul. 15
	30-11.9	5	2022 Jul. 15
	30-11.10	5	2022 Jul. 15
	30-21.1	6	2024 Aug. 20
	30-21.2	6	2024 Aug. 20
	30-21.3	6	2024 Aug. 20
	30-21.4	6	2024 Aug. 20
	30-21.5	6	2024 Aug. 20
	30-21.6	6	2024 Aug. 20
	30-31.1	3	2021 Apr. 30
	30-31.2	3	2021 Apr. 30
	30-31.3	3	2021 Apr. 30
	30-31.4	3	2021 Apr. 30
	30-31.5	3	2021 Apr. 30
	30-31.6	3	2021 Apr. 30
	30-31.7	3	2021 Apr. 30
	30-31.8	3	2021 Apr. 30
	30-31.9	3	2021 Apr. 30
	30-31.10	3	2021 Apr. 30

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ATA 30 ICE AND RAIN PROTECTION *(Cont'd)*	30-41.1	3	2021 Apr. 30
	30-41.2	3	2021 Apr. 30
	30-41.3	3	2021 Apr. 30
	30-41.4	3	2021 Apr. 30
	30-42.1	5	2022 Jul. 15
	30-42.2	3	2021 Apr. 30
	30-42.3	3	2021 Apr. 30
	30-42.4	3	2021 Apr. 30
	30-71.1	3	2021 Apr. 30
	30-71.2	3	2021 Apr. 30
ATA 31 INDICATING/RECORDING SYSTEMS	ATA 31-00.1	6	2024 Aug. 20
	ATA 31-00.2	6	2024 Aug. 20
	31-25.1	3	2021 Apr. 30
	31-25.2	3	2021 Apr. 30
	31-31.1	5c (CAL)	2024 May 06
	31-31.2	5c (CAL)	2024 May 06
	31-31.3	5c (CAL)	2024 May 06
	31-31.4	5c (CAL)	2024 May 06
	31-31.5	5c (CAL)	2024 May 06
	31-31.6	5c (CAL)	2024 May 06
	31-62.1	3	2021 Apr. 30
	31-62.2	3	2021 Apr. 30
	31-62.3	6	2024 Aug. 20
	31-62.4	3	2021 Apr. 30
	31-62.5	6	2024 Aug. 20
	31-62.6	6	2024 Aug. 20
	31-62.7	6	2024 Aug. 20
	31-62.8	6	2024 Aug. 20
	31-62.9	6	2024 Aug. 20
	31-62.10	6	2024 Aug. 20

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ATA 32 LANDING GEAR	ATA 32-00.1	3	2021 Apr. 30
	ATA 32-00.2	3	2021 Apr. 30
	32-30.1	3	2021 Apr. 30
	32-30.2	3	2021 Apr. 30
	32-31.1	6	2024 Aug. 20
	32-31.2	6	2024 Aug. 20
	32-31.3	3	2021 Apr. 30
	32-31.4	3	2021 Apr. 30
	32-31.5	3	2021 Apr. 30
	32-31.6	3	2021 Apr. 30
	32-32.1	3	2021 Apr. 30
	32-32.2	6	2024 Aug. 20
	32-35.1	3	2021 Apr. 30
	32-35.2	3	2021 Apr. 30
	32-41.1	3	2021 Apr. 30
	32-41.2	3	2021 Apr. 30
	32-41.3	6	2024 Aug. 20
	32-41.4	5	2022 Jul. 15
	32-42.1	6	2024 Aug. 20
	32-42.2	3	2021 Apr. 30
	32-42.3	6	2024 Aug. 20
	32-42.4	6	2024 Aug. 20
	32-44.1	6	2024 Aug. 20
	32-44.2	3	2021 Apr. 30
	32-51.1	3	2021 Apr. 30
	32-51.2	3	2021 Apr. 30

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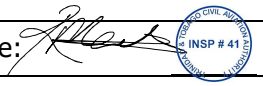
Signature: 	Date: 12 June 2026

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ATA 33 LIGHTS	ATA 33-00.1	6a (CAL)	02 Jun. 25
	ATA 33-00.2	6a (CAL)	02 Jun. 25
	33-11.1	3	2021 Apr. 30
	33-11.2	3	2021 Apr. 30
	33-11.3	3	2021 Apr. 30
	33-11.4	3	2021 Apr. 30
	33-16.1	5	2022 Jul. 15
	33-16.2	3	2021 Apr. 30
	33-21.1	3	2021 Apr. 30
	33-21.2	6	2024 Aug. 20
	33-24.1	3	2021 Apr. 30
	33-24.2	3	2021 Apr. 30
	33-24.3	3	2021 Apr. 30
	33-24.4	3	2021 Apr. 30
	33-31.1	3	2021 Apr. 30
	33-31.2	3	2021 Apr. 30
	33-37.1	3	2021 Apr. 30
	33-37.2	3	2021 Apr. 30
	33-41.1	3	2021 Apr. 30
	33-41.2	3	2021 Apr. 30
	33-42.1	3	2021 Apr. 30
	33-42.2	3	2021 Apr. 30
	33-42.3	4	2022 May 10
	33-42.4	3	2021 Apr. 30
	33-42.5	3	2021 Apr. 30
	33-42.6	3	2021 Apr. 30

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ATA 33 LIGHTS *(Cont'd)*	33-43.1		3	2021 Apr. 30
	33-43.2		3	2021 Apr. 30
	33-43.3		3	2021 Apr. 30
	33-43.4		3	2021 Apr. 30
	33-44.1		3	2021 Apr. 30
	33-44.2		3	2021 Apr. 30
	33-44.3		3	2021 Apr. 30
	33-44.4		3	2021 Apr. 30
	33-45.1		3	2021 Apr. 30
	33-45.2		3	2021 Apr. 30
	33-51.1		3	2021 Apr. 30
	33-51.2		3	2021 Apr. 30
	33-51.3		3	2021 Apr. 30
	33-51.4		6	2024 Aug. 20
	33-51.5		6	2024 Aug. 20
	33-51.6		6	2024 Aug. 20
ATA 34 NAVIGATION	ATA 34-00.1	*	6b (CAL)	2026 Jun. 05
	ATA 34-00.2		6a (CAL)	2025 Jun. 02
	ATA 34-00.3		5a (CAL)	2023 Mar. 23
	ATA 34-00.4		5a (CAL)	2023 Mar. 23
	34-16.1		3	2021 Apr. 30
	34-16.2		3	2021 Apr. 30
	34-23.1		3	2021 Apr. 30
	34-23.2		3	2021 Apr. 30
	34-24.1		3	2021 Apr. 30
	34-24.2		3	2021 Apr. 30
	34-31.1		3	2021 Apr. 30
	34-31.2	*	6b (CAL)	2026 Jun. 05

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ATA 34 NAVIGATION *(Cont'd)*	34-32.1	3	2021 Apr. 30
	34-32.2	3	2021 Apr. 30
	34-33.1	3	2021 Apr. 30
	34-33.2	3	2021 Apr. 30
	34-41.1	3	2021 Apr. 30
	34-41.2	3	2021 Apr. 30
	34-41.3	3	2021 Apr. 30
	34-41.4	3	2021 Apr. 30
	34-41.5	3	2021 Apr. 30
	34-41.6	3	2021 Apr. 30
	34-45.1	3	2021 Apr. 30
	34-45.2	3	2021 Apr. 30
	34-45.3	3	2021 Apr. 30
	34-45.4	3	2021 Apr. 30
	34-49.1	3	2021 Apr. 30
	34-49.2	3	2021 Apr. 30
	34-49.3	3	2021 Apr. 30
	34-49.4	3	2021 Apr. 30
	34-49.5	3	2021 Apr. 30
	34-49.6	3	2021 Apr. 30
	34-49.7	3	2021 Apr. 30
	34-49.8	3	2021 Apr. 30
	34-49.9	6	2024 Aug.20
	34-49.10	6	2024 Aug.20
	34-51.1	3	2021 Apr. 30
	34-51.2	3	2021 Apr. 30

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ATA 34 NAVIGATION *(Cont'd)*	34-53.1	6	2024 Aug. 20
	34-53.2	6	2024 Aug. 20
	34-53.3	6	2024 Aug. 20
	34-53.4	6	2024 Aug. 20
	34-55.1	3	2021 Apr. 30
	34-55.2	3	2021 Apr. 30
	34-57.1	3	2021 Apr. 30
	34-57.2	3	2021 Apr. 30
	34-58.1	3	2021 Apr. 30
	34-58.2	5	2022 Jul. 15
	34-58.3	5	2022 Jul. 15
	34-58.4	5	2022 Jul. 15
	34-61.1	4	2022 May 10
	34-61.2	5	2022 Jul. 15
	34-61.3	3	2021 Apr. 30
	34-61.4	4	2022 May 10
	34-61.5	4	2022 May 10
	34-61.6	3	2021 Apr. 30
	34-61.7	3	2021 Apr. 30
	34-61.8	3	2021 Apr. 30
	34-65.1	3	2021 Apr. 30
	34-65.2	3	2021 Apr. 30
ATA 35 OXYGEN	ATA 35-00.1	6	2024 Aug. 20
	ATA 35-00.2	6	2024 Aug. 20
	35-11.1	3	2021 Apr. 30
	35-11.2	3	2021 Apr. 30

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ATA 35 OXYGEN *(Cont'd)*	35-21.1	3	2021 Apr. 30
	35-21.2	6a (CAL)	2025 Jun. 02
	35-21.3	6	2024 Aug. 20
	35-21.4	6	2024 Aug. 20
	35-21.5	6	2024 Aug. 20
	35-21.6	6	2024 Aug. 20
	35-21.7	6	2024 Aug. 20
	35-21.8	6	2024 Aug. 20
	35-31.1	5	2022 Jul. 15
	35-31.2	5	2022 Jul. 15
ATA 36 PNEUMATIC	ATA 36-00.1	3	2021 Apr. 30
	ATA 36-00.2	3	2021 Apr. 30
	36-11.1	5	2022 Jul. 15
	36-11.2	5	2022 Jul. 15
	36-11.3	3	2021 Apr. 30
	36-11.4	3	2021 Apr. 30
	36-11.5	3	2021 Apr. 30
	36-11.6	3	2021 Apr. 30
	36-11.7	3	2021 Apr. 30
	36-11.8	3	2021 Apr. 30
	36-11.9	5	2022 Jul. 15
	36-11.10	3	2021 Apr. 30
	36-11.11	5	2022 Jul. 15
	36-11.12	6	2024 Aug. 20
	36-11.13	3	2021 Apr. 30
	36-11.14	5	2022 Jul. 15
	36-11.15	5	2022 Jul. 15
	36-11.16	6	2024 Aug. 20
	36-11.17	3	2021 Apr. 30
	36-11.18	3	2021 Apr. 30
	36-11.19	3	2021 Apr. 30
	36-11.20	3	2021 Apr. 30
	36-12.1	6	2024 Aug. 20
	36-12.2	6	2024 Aug. 20

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ATA 36 PNEUMATIC *(Cont'd)*	36-13.1	6	2024 Aug. 20
	36-13.2	6	2024 Aug. 20
	36-13.3	6	2024 Aug. 20
	36-13.4	6	2024 Aug. 20
	36-13.5	6	2024 Aug. 20
	36-13.6	3	2021 Apr. 30
	36-21.1	3	2021 Apr. 30
	36-21.2	5	2022 Jul. 15
ATA 38 WATER/WASTE	ATA 38-00.1	3	2021 Apr. 30
	ATA 38-00.2	3	2021 Apr. 30
	38-10.1	3	2021 Apr. 30
	38-10.2	3	2021 Apr. 30
	38-30.1	3	2021 Apr. 30
	38-30.2	3	2021 Apr. 30
	38-30.3	3	2021 Apr. 30
	38-30.4	3	2021 Apr. 30
ATA 46 INFORMATION SYSTEMS	ATA 46-00.1	5a (CAL)	2023 Mar. 23
	ATA 46-00.2	5a (CAL)	2023 Mar. 23
	46-13.1	6	2024 Aug. 20
	46-13.2	6	2024 Aug. 20
	46-20.1	5	2022 Jul. 15
	46-20.2	5	2022 Jul. 15
	46-20.3	5	2022 Jul. 15
	46-20.4	5	2022 Jul. 15
ATA 47 NITROGEN GENERATION SYSTEM	ATA 47-00.1	3	2021 Apr. 30
	ATA 47-00.2	3	2021 Apr. 30
	47-11.1	3	2021 Apr. 30
	47-11.2	3	2021 Apr. 30

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ATA 49 AIRBORNE AUXILIARY POWER	ATA 49-00.1	3	2021 Apr. 30
	ATA 49-00.2	3	2021 Apr. 30
	49-11.1	6	2024 Aug. 20
	49-11.2	6	2024 Aug. 20
	49-15.1	5	2022 Jul. 15
	49-15.2	5	2022 Jul. 15
	49-15.3	5	2022 Jul. 15
	49-15.4	5	2022 Jul. 15
	49-15.5	5	2022 Jul. 15
	49-15.6	5	2022 Jul. 15
	49-40.1	4	2022 May 10
	49-40.2	3	2021 Apr. 30
	49-40.3	3	2021 Apr. 30
	49-40.4	3	2021 Apr. 30
	49-52.1	6a (CAL)	2025 Jun. 02
	49-52.2	3	2021 Apr. 30
	49-52.3	3	2021 Apr. 30
	49-52.4	3	2021 Apr. 30
	49-70.1	3	2021 Apr. 30
	49-70.2	3	2021 Apr. 30
	49-70.3	3	2021 Apr. 30
	49-70.4	3	2021 Apr. 30
ATA 50 CARGO AND ACCESSORY COMPARTMENT	ATA 50-00.1	3	2021 Apr. 30
	ATA 50-00.2	3	2021 Apr. 30
	50-10.1	6	2024 Aug. 20
	50-10.2	6	2024 Aug. 20
	50-21.1	3	2021 Apr. 30
	50-21.2	3	2021 Apr. 30
	50-21.3	3	2021 Apr. 30
	50-21.4	3	2021 Apr. 30

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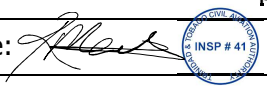
Signature: 	Date: 12 June 2026

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ATA 52 DOORS	ATA 52-00.1	6a (CAL)	2025 Jun. 02
	ATA 52-00.2	6a (CAL)	2025 Jun. 02
	52-11.1	6	2024 Aug. 20
	52-11.2	3	2021 Apr. 30
	52-11.3	6	2024 Aug. 20
	52-11.4	3	2021 Apr. 30
	52-11.5	3	2021 Apr. 30
	52-11.6	3	2021 Apr. 30
	52-22.1	3	2021 Apr. 30
	52-22.2	3	2021 Apr. 30
	52-31.1	3	2021 Apr. 30
	52-31.2	6	2024 Aug. 20
	52-31.3	3	2021 Apr. 30
	52-31.4	3	2021 Apr. 30
	52-31.5	3	2021 Apr. 30
	52-31.6	3	2021 Apr. 30
	52-51.1	6a (CAL)	2025 Jun. 02
	52-51.2	3	2021 Apr. 30
	52-51.3	3	2021 Apr. 30
	52-51.4	3	2021 Apr. 30
	52-51.5	3	2021 Apr. 30
	52-51.6	3	2021 Apr. 30
	52-51.7	3	2021 Apr. 30
	52-51.8	3	2021 Apr. 30
	52-51.9	3	2021 Apr. 30
	52-51.10	3	2021 Apr. 30
	52-71.1	4	2022 May 10
	52-71.2	3	2021 Apr. 30
	52-71.3	3	2021 Apr. 30
	52-71.4	4	2022 May 10
	52-71.5	4	2022 May 10
	52-71.6	6	2024 Aug. 20
	52-71.7	3	2021 Apr. 30
	52-71.8	3	2021 Apr. 30

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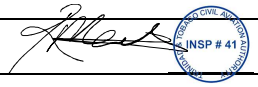
Signature: 	Date: 12 June 2026

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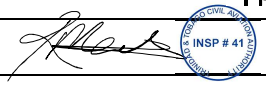
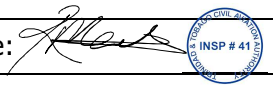
ATA 71 POWER PLANT	ATA 71-00.1	3	2021 Apr. 30
	ATA 71-00.2	3	2021 Apr. 30
	71-11.1	5	2022 Jul. 15
	71-11.2	5	2022 Jul. 15
ATA 73 ENGINE FUEL AND CONTROL	ATA 73-00.1	3	2021 Apr. 30
	ATA 73-00.2	3	2021 Apr. 30
	73-11.1	3	2021 Apr. 30
	73-11.2	3	2021 Apr. 30
	73-21.1	3	2021 Apr. 30
	73-21.2	3	2021 Apr. 30
	73-30.1	3	2021 Apr. 30
	73-30.2	3	2021 Apr. 30
	73-31.1	3	2021 Apr. 30
	73-31.2	3	2021 Apr. 30
	73-34.1	3	2021 Apr. 30
	73-34.2	3	2021 Apr. 30
ATA 74 IGNITION	ATA 74-00.1	3	2021 Apr. 30
	ATA 74-00.2	3	2021 Apr. 30
	74-00.1	5	2022 Jul. 15
	74-00.2	3	2021 Apr. 30
	74-00.3	3	2021 Apr. 30
	74-00.4	3	2021 Apr. 30
ATA 75 BLEED AIR	ATA 75-00.1	3	2021 Apr. 30
	ATA 75-00.2	3	2021 Apr. 30
	75-23.1	3	2021 Apr. 30
	75-23.2	5	2022 Jul. 15
	75-24.1	5	2022 Jul. 15
	75-24.2	5	2022 Jul. 15

Approved by TTCAA	
Signature: 	Date: 12 June 2026

Description	Page	Revision	Date
ATA 77 ENGINE INDICATING	ATA 77-00.1	3	2021 Apr. 30
	ATA 77-00.2	3	2021 Apr. 30
	77-21.1	3	2021 Apr. 30
	77-21.2	3	2021 Apr. 30
	77-31.1	3	2021 Apr. 30
	77-31.2	3	2021 Apr. 30
ATA 78 ENGINE EXHAUST	ATA 78-00.1	3	2021 Apr. 30
	ATA 78-00.2	3	2021 Apr. 30
	78-31.1	6	2024 Aug. 20
	78-31.2	6	2024 Aug. 20
	78-34.1	3	2021 Apr. 30
	78-34.2	6	2024 Aug. 20
	78-36.1	3	2021 Apr. 30
	78-36.2	3	2021 Apr. 30
ATA 79 ENGINE OIL	ATA 79-00.1	3	2021 Apr. 30
	ATA 79-00.2	3	2021 Apr. 30
	79-21.1	3	2021 Apr. 30
	79-21.2	3	2021 Apr. 30
	79-31.1	6	2024 Aug. 20
	79-31.2	6	2024 Aug. 20
	79-35.1	5	2022 Jul. 15
	79-35.2	5	2022 Jul. 15

Approved by TTCAA	
Signature: 	Date: 12 June 2026

Description	Page	Revision	Date
ATA 80 STARTING	ATA 80-00.1	3	2021 Apr. 30
	ATA 80-00.2	3	2021 Apr. 30
	80-11.1	3	2021 Apr. 30
	80-11.2	3	2021 Apr. 30
	80-11.3	6	2024 Aug. 20
	80-11.4	6	2024 Aug. 20
APPENDIX 1 - NONESSENTIAL EQUIPMENT AND FURNISHINGS (NEF) PROGRAM	*Refer to NEF LEP dated*		2023 Nov. 08

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HISTORICAL DATA

The following source publications have been used for the development of the Boeing 737-8 MEL and Procedures.

- FAA Master Minimum Equipment List at Revision No.: 6 dated 2024 July, 22
- Boeing 737 MAX Dispatch Deviations Guide at Revision No.: 6 dated 2024 August, 20

ORGANISATION

This MEL comprises:

- The Caribbean Airlines Boeing 737-8 Minimum Equipment List combined with Procedures.

  The MEL is divided into three sections as follows:

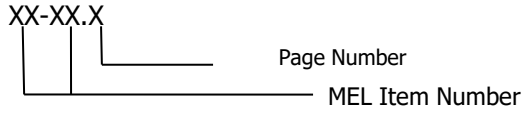
  SECTION 0 - GENERAL

  SECTION 2 - MEL

  SECTION 3 - NEF LISTING

  SECTION 2 - MEL

  This Section contains items from the FAA MMEL and procedures from Boeing Dispatch Deviations Guide. For ease of use, page numbering uses the MEL item numbering as follows:



  For example, MEL item 21-51-01 would be found on 21-51.01.

- Caribbean Airlines Equipment and Furnishings (NEF) Program – Appendix 1.

MEL items have a table of contents for each ATA.

NONESSENTIAL EQUIPMENT AND FURNISHINGS (NEF)

NEF are those items installed on the aircraft as part of the original type certification, supplemental type certificate, or other form of alteration that have no effect on the safe operation of flight and would not be required by the applicable certification rules or operational rules. They are those items that, if damaged, inoperative, or missing, have no effect on the aircraft's ability to be operated safely under all operational conditions. These nonessential items may be installed in areas including, but not limited to:

- Cargo Areas
- Crew Rest Areas
- Flight Deck Area
- Galley Area
- Lavatories
- Passenger Compartment
- Service Areas

ADMINISTRATION

The Chief Pilot is responsible for the administration and authorization of this Manual. This MEL and amendments thereto require TTCAA approval.

This MEL is effective for the following aircraft:

AIRCRAFT NUMBER	REGISTRY NUMBER	SERIAL NUMBER	TABULATION NUMBER
7947	**9Y-CAL**	43385	1B326
8021	**9Y-GUY**	43319	1B327
8052	**9Y-ANT**	62892	1B366
8090	**9Y-BAR**	62893	1B367
8159	**9Y-JAM**	62901	1B368
8167	**9Y-GRN**	62906	1B369
8172	**9Y-SUR**	62908	1B370
8145	**9Y-BAH**	43345	1B328
8238	**9Y-TTO**	63811	1B329
9644	**9Y-CUB**	65286	1B359

REVISIONS

Filing Instructions

Normal Revision

All revisions will include Filing Instructions, Record of Amendment Approval, List of Effective Pages, and Manual Organisation & Amendment.

Consult the List of Effective Pages. The List of Effective Pages determines the correct content of the Manual. On the List of Effective Pages, a New page is identified by the character *'n'*, a Revised page by an asterisk *'*'* and a Deleted page by the character *'d'*. All Amendments shall be numbered in sequence.

Pages may be re-dated without revision bars. This is due to repagination of contents or date change. The List of Effective Pages will reflect the date change, but there will not be a corresponding bar on the revised page.

Temporary Revision

Details of revisions which may be urgently required in the interests of flight safety, or which are supplementary to the MEL, will be promulgated as Temporary Revision printed on yellow paper. They are accompanied by filling instructions, which will be filed at the front of the manual and will be incorporated into the manual at the next revision, which should not exceed a period of twelve (12) months between revisions.

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SYSTEM NO	SYSTEM
21	Air Conditioning
22	Autoflight
23	Communications
24	Electrical Power
25	Equipment/Furnishings
26	Fire Protection
27	Flight Controls
28	Fuel
29	Hydraulic Power
30	Ice and Rain Protection
31	Indicating/Recording Systems
32	Landing Gear
33	Lights
34	Navigation
35	Oxygen
36	Pneumatic
38	Water/Waste
46	Information Systems
47	Nitrogen Generation System
49	Airborne Auxiliary Power
50	Cargo and Accessory Compartments
52	Doors
71	Power Plant
73	Engine Fuel and Control
74	Ignition
75	Bleed Air
77	Engine Indicating
78	Engine Exhaust
79	Engine Oil
80	Starting

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1. General

Distribution of manuals and all approved revisions will be processed by the Document Owner or by the Technical Library in accordance with the detailed distribution list (kept in the Technical Library) and be made available and accessible to the relevant Caribbean Airlines personnel.

(1.1) Access to electronic manuals is provided via the following online portals:

(a) IQM Document Module, where recipients are required to acknowledge receipt.

(b) Office 365 Maintenance & Engineering SharePoint website:
https://caribbbeanairlines.sharepoint.com/sites/MandESharepointPortal

(c) Office 365 Flight Operations SharePoint website:
https://caribbbeanairlines.sharepoint.com/flightops

NOTE: Assigned Numbers are maintained by the Publications Officers in the Technical Library.

This distribution is as follows:

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71	- MIA		X	
72	AIRTECHS (Mr. Carl Cardon) - ANU - SLU - GND - MBJ - SVD - PTP - FDF		X	
73	Reserved			
74	- JFK		X	
75	- YYZ		X	
76	- GEO		X	
77	Reserved			
78	ALPHA-TECH Services - MCO		X	
79	Reserved			

2. Location of Manual and Copy Requirements

The Chief Pilot controls and maintains the B737 Minimum Equipment List (MEL) Doc. No.: BW-B737-MEL-0046 and the Distribution / Access for the B737 Minimum Equipment List. The Technical Library will make available as indicated in paragraph 1, a copy of the B737 Minimum Equipment List (MEL) Doc. No.: BW-B737-MEL-0046, or appropriate portions, to any personnel performing maintenance, flight or ground operations on Caribbean Airlines aircraft. The Chief Pilot identifies the required number of copies to meet distribution requirements.

(1) Each Department and Line Station receiving a manual will be responsible for its security, maintenance and currency. The person revising the manual will follow the instructions included with the revision. The Company's publications platform will be updated as changes are issued.

(2) Request for distribution of the B737 Minimum Equipment List (MEL) Doc. No.: BW-B737 MEL-0046 or additional copies of this manual should be made by written request through the Chief Pilot.

MEL NUMBERING EXAMPLES

The following is an example of the MEL numbering scheme using the MEL item 46-20-01 as a reference.

46-20-01	**Electronic Flight Bag (EFB) Systems**
46-20-01-02	**Class 2 EFB**
46-20-01-02-01	**Data Connectivity**
46-20-01-02-01A	**Alternate Procedures Required**

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Introduction

Status Message Cross Reference List

Maintenance Level Messages

Considerations for Dispatch with Displayed Status Message

Cross Reference List

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Status Message Cross Reference List

A Cross Reference List of status messages and corresponding MEL items is provided in this section.

The Cross Reference List is intended to assist in quickly identifying the proper MEL item to consult for possible dispatch relief. The MEL item references contained in the Cross Reference List are intended to be specific to the system subitem level, although higher level system MEL relief may be utilized where appropriate. A description of the information provided in each column of the Cross Reference List follows.

Message Text — This column alphabetically lists status messages.

Level — Only status level messages are listed.

MEL Item — This column lists the appropriate MEL item(s) to consult for possible dispatch relief with the associated message displayed. Status messages that do not have a specific MEL item to allow dispatch are listed with one of the following cross references:

- N/A — Not Applicable: There is no MEL item listed because the message does not indicate a system failure.
- None — There is no MEL item for the failure condition indicated by this message

Maintenance Level Messages

Maintenance level messages are not included in the Cross Reference List, since they do not affect the airworthiness release of the airplane. Maintenance level messages are provided by the MAX Display System (MDS) and Onboard Maintenance Function (OMF) as follows.

Scheduled Maintenance Task (SMT): These are a special category of messages that are uniquely associated with a Certification Maintenance Requirement (CMR) or a Scheduled Maintenance Task (SMT). They are displayed on a special MFD maintenance page. SMT messages are checked as a result of a specified maintenance task defined by the CMR or maintenance program requirement, which also defines a specific time interval when corrective maintenance action is required. There is no requirement to check SMT messages prior to each flight. SMT messages are not included in the Cross Reference List. SMT repair time limits can be found in AMM 05-00-00/201 and AMM 73-00-00/501.

Onboard Maintenance Function: Onboard Maintenance System processes BITE fault reports into maintenance messages. These are nonessential messages that are displayed on Maintenance Laptops or specific flight deck display formats. Maintenance messages are not used to determine the airworthiness of the airplane. They provide diagnostic information useful in troubleshooting or maintenance planning. There are two general types of OMF maintenance messages: correlated and uncorrelated. Correlated messages are associated with a specific status message or other flight deck effect. Each fault indicated by a status message has at least one correlated maintenance message to assist the maintenance personnel in determining the root cause of the fault.

Uncorrelated messages do not have a corresponding flight deck effect. OMF maintenance messages are not included in the Cross Reference List.

Considerations for Dispatch with Displayed Status Message

Any monitored faults that affect airplane dispatchability will have a flight deck alert or a status level message, with a corresponding correlated maintenance message. System faults displayed at status level should beresolved by MEL compliance or maintenance action prior to engine start. After engine start, flight deck alerts are the primary means of alerting the crew to non-normal conditions or improper configuration. Display of flight deck alerts requires accomplishment by the flight crew of any applicable steps in the appropriate non-normal procedure. Conditions which can be resolved by flight crew action may allow continued normal operation. For system faults, following completion of the non-normal procedure and priorto takeoff, the operator's MEL should be Consulted to determine if relief is available for continued operation.

Cross Reference List

Message	**Level**	**MEL Item**
AFT CARGO LOADER RLY	Status	50-21-01
AIR/GND SYS	Status	None
ALTN NOSE STEERING	Status	None
AOA VANE L	Status	None
AOA VANE R	Status	None
APU DOOR	Status	49-15-01
APU GENERATOR	Status	24-21-01
APU OIL QTY	Status	N/A
AUTOSLAT FAIL	Status	None
AUTOSLAT FAIL SMYD 1	Status	27-83-01
AUTOSLAT FAIL SMYD 2	Status	27-83-01

Message	**Level**	**MEL Item**
BLD CTRL BACKUP PROTECT L	Status	36-11-01-05
BLD CTRL BACKUP PROTECT R	Status	36-11-01-05
BLD CTRL CARD L - ENG DATA	Status	36-11-01-06
BLD CTRL CARD R - ENG DATA	Status	36-11-01-06
BLEED CTRL - ISO VALVE	Status	36-13-02
BLEED CTRL BACKUP L	Status	36-11-01-03
BLEED CTRL BACKUP R	Status	36-11-01-03
BLEED CTRL ICING INHIB L	Status	36-11-01-02
BLEED CTRL ICING INHIB R	Status	36-11-01-02
BLEED CTRL L - ASC HEALTH	Status	36-11-01-04
BLEED CTRL LIGHT OUTPUT L	Status	36-11-05
BLEED CTRL LIGHT OUTPUT R	Status	36-11-05
BLEED CTRL PRESS IND L	Status	36-21-01
BLEED CTRL PRESS IND R	Status	36-21-01
BLEED CTRL PRIMARY L	Status	36-11-01-01

Cross Reference List

Message	**Level**	**MEL Item**
BLEED CTRL PRIMARY R	Status	36-11-01-01
BLEED CTRL R - ASC HEALTH	Status	36-11-01-04
BLEED FAMV L	Status	36-12-01-01
BLEED FAMV R	Status	36-12-01-01
BLEED HPSOV CTRL L	Status	36-11-04-02
BLEED HPSOV CTRL R	Status	36-11-04-02
BLEED HPSOV L	Status	36-11-04-01
BLEED HPSOV R	Status	36-11-04-01
BLEED IASC SOFTWARE CONFIG	Status	None
BLEED PI SENSOR L	Status	36-11-02-02
BLEED PI SENSOR R	Status	36-11-02-02
BLEED PM SENSOR 2 L	Status	36-11-02-01
BLEED PM SENSOR 2 R	Status	36-11-02-01
BLEED PRSOV L	Status	36-11-03
BLEED PRSOV R	Status	36-11-03
BRAKE TEMP SYS	Status	32-46-01

Message	**Level**	**MEL Item**
DISPLAY TOUCH LEFT INBD	Status	31-62-01-01-04
DISPLAY TOUCH RIGHT INBD	Status	31-62-01-01-04
DISPLAY UNIT INTERFACE	Status	31-62-01-01-01 31-62-01-01-02
DISPLAYS CONFIG DATA	Status	None
DISPLAYS SOFTWARE CONFIG	Status	None
DOOR AFT CARGO	Status	52-71-01-02
DOOR AFT ENTRY	Status	52-71-01-01
DOOR AFT SERVICE	Status	52-71-01-01

Cross Reference List

Message	**Level**	**MEL Item**
DOOR AIRSTAIR	Status	52-71-01-03
DOOR EE ACCESS	Status	52-71-01-02
DOOR EMER EXIT PRGM PINS	Status	52-71-02
DOOR EQUIP IND	Status	52-71-01-02
DOOR FWD ACCESS	Status	52-71-01-02
DOOR FWD CARGO	Status	52-71-01-02
DOOR FWD ENTRY	Status	52-71-01-01
DOOR FWD SERVICE	Status	52-71-01-01
DOOR L MID CABIN	Status	52-71-01-05
DOOR L MID CABIN FLTLOCK	Status	52-22-01
DOOR L OVERWING	Status	52-71-01-04
DOOR L OVERWING FLTLOCK	Status	52-22-01
DOOR R MID CABIN	Status	52-71-01-05
DOOR R MID CABIN FLTLOCK	Status	52-22-01
DOOR R OVERWING	Status	52-71-01-04
DOOR R OVERWING FLTLOCK	Status	52-22-01
DPC 1	Status	None
DPC 1 HOT BATTERY	Status	None
DPC 2	Status	None
DPC 2 HOT BATTERY	Status	None
DPC CROSSTALK BUS	Status	None

Message	**Level**	**MEL Item**
ELT SYS	Status	23-24-02
ENG 1 BAI/SBV VALVE	Status	None
ENG 1 CONTROL	Status	None
ENG 1 EEC COOLING SYS	Status	75-23-01
ENG 1 EEC OVERHEAT	Status	None

Cross Reference List

Message	**Level**	**MEL Item**
ENG 1 FAN COMP OVHT	Status	None
ENG 1 FAN COMP OVHT SENSOR	Status	None
ENG 1 FUEL FILTER	Status	None
ENG 1 FUEL FILTER SENSOR	Status	73-34-01-02
ENG 1 FUEL MAIN HX	Status	None
ENG 1 FUEL MAIN HX SENSOR	Status	73-34-01-01
ENG 1 FUEL SPLIT VALVE	Status	None
ENG 1 FUEL STRAINER	Status	None
ENG 1 FUEL STRAINER SENSOR	Status	73-11-01
ENG 1 HPTACC VALVE	Status	75-24-01-01
ENG 1 IGNITER L	Status	74-00-01-01
ENG 1 IGNITER R	Status	74-00-01-02
ENG 1 LPTACC VALVE	Status	None
ENG 1 MTC VALVE	Status	75-23-02
ENG 1 OIL DEBRIS	Status	79-21-01
ENG 1 OIL DEBRIS SENSOR	Status	79-21-01-01
ENG 1 OIL FILTER SENSOR	Status	79-35-01
ENG 1 OIL PRESS SENSOR	Status	None
ENG 1 OIL TEMP SENSOR	Status	None
ENG 1 OVERSPEED PROTECT	Status	None
ENG 1 PSS OVERHEAT	Status	None
ENG 1 REVERSER	Status	78-31-01
ENG 1 TBV	Status	None
ENG 1 VIBRATION MONITOR	Status	77-31-01
ENG 2 BAI/SBV VALVE	Status	None
ENG 2 CONTROL	Status	None
ENG 2 EEC COOLING SYS	Status	75-23-01
ENG 2 EEC OVERHEAT	Status	None

Cross Reference List

Message	Level	MEL Item
ENG 2 FAN COMP OVHT	Status	None
ENG 2 FAN COMP OVHT SENSOR	Status	None
ENG 2 FUEL FILTER	Status	None
ENG 2 FUEL FILTER SENSOR	Status	73-34-01-02
ENG 2 FUEL MAIN HX	Status	None
ENG 2 FUEL MAIN HX SENSOR	Status	73-34-01-01
ENG 2 FUEL SPLIT VALVE	Status	None
ENG 2 FUEL STRAINER	Status	None
ENG 2 FUEL STRAINER SENSOR	Status	73-11-01
ENG 2 HPTACC VALVE	Status	75-24-01-01
ENG 2 IGNITER L	Status	74-00-01-01
ENG 2 IGNITER R	Status	74-00-01-02
ENG 2 LPTACC VALVE	Status	None
ENG 2 MTC VALVE	Status	75-23-02
ENG 2 OIL DEBRIS	Status	79-21-01
ENG 2 OIL DEBRIS SENSOR	Status	79-21-01-01
ENG 2 OIL FILTER SENSOR	Status	79-35-01
ENG 2 OIL PRESS SENSOR	Status	None
ENG 2 OIL TEMP SENSOR	Status	None
ENG 2 OVERSPEED PROTECT	Status	None
ENG 2 PSS OVERHEAT	Status	None
ENG 2 REVERSER	Status	78-31-01
ENG 2 TBV	Status	None
ENG 2 VIBRATION MONITOR	Status	77-31-01

Message	Level	MEL Item
FLAP TE POSN DEGRADED IND	Status	None
FWD CARGO LOADER RLY	Status	50-21-01

Cross Reference List

Message	**Level**	**MEL Item**
GEAR CONTROL	Status	32-31-02B
GEAR CONTROL 1	Status	32-31-02A
GEAR CONTROL 2	Status	32-31-02A
GEAR IND	Status	None
GEAR L SHRINK LINK	Status	32-61-01
GEAR L SHRINK LINK PROX SW	Status	32-61-01
GEAR LEVER	Status	None
GEAR LEVER LATCH	Status	None
GEAR LEVER SW	Status	32-31-03
GEAR R SHRINK LINK	Status	32-61-01
GEAR R SHRINK LINK PROX SW	Status	32-61-01
GEAR WARN ALTIMETER	Status	None
GEAR WARN SYS	Status	None
GEAR WARN THROTTLE SW	Status	None
GEAR XFR VALVE CONTROL	Status	None
GROUND SPOILER CONTROL	Status	None

Message	**Level**	**MEL Item**
MAINTENANCE ENABLE	Status	N/A
MLG L SENSOR	Status	None
MLG R SENSOR	Status	None

Message	**Level**	**MEL Item**
NLG SENSOR	Status	None

Message	**Level**	**MEL Item**
PACK FLOW CTRL SENSOR L	Status	21-51-04
PACK FLOW CTRL SENSOR R	Status	21-51-04

Cross Reference List

Message	**Level**	**MEL Item**
PACK FLOW CTRL VALVE L	Status	21-51-02
PACK FLOW CTRL VALVE R	Status	21-51-02
PACK FLOW PRIMARY L	Status	21-51-02-01
PACK FLOW PRIMARY R	Status	21-51-02-01
PACK IASC COMM L	Status	21-51-08
PACK IASC COMM R	Status	21-51-08
PACK IASC SOFTWARE CONFIG	Status	None
PACK RAM AIR SYSTEM L	Status	21-51-09
PACK RAM AIR SYSTEM R	Status	21-51-09
PACK TEMP CTRL PRIMARY L	Status	21-51-05
PACK TEMP CTRL PRIMARY R	Status	21-51-05
PSEU AIR/GND OVRD ENABLE	Status	N/A
PSEU COMM	Status	None
PSEU CONFIG	Status	None
PSEU FAIL	Status	None
PSEU POWER	Status	None

Message	**Level**	**MEL Item**
REMOTE LIGHT SENSOR	Status	31-62-02

Message	**Level**	**MEL Item**
SPEEDBRAKE LEVER XDCR	Status	None
SPOILER CONTROL CHANNEL	Status	27-61-01-01
SPOILER CONTROL SYSTEM	Status	None
SPOILER CTRL WHEEL XDCR	Status	None
SPOILER DESCENTSPEEDBRAKE	Status	27-61-03

Cross Reference List

Message	**Level**	**MEL Item**
SPOILER DIRECT LIFT CTRL	Status	None
SPOILER LOAD ALLEVIATION	Status	27-61-02
SPOILER PAIR 2/11	Status	27-61-01
SPOILER PAIR 3/10	Status	27-61-01
SPOILER PAIR 4/9	Status	27-61-01
SPOILER PAIR 5/8	Status	None
STALL WARNING SYS	Status	None
STALL WARNING SYS L	Status	None
STALL WARNING SYS R	Status	None

Message	**Level**	**MEL Item**
TAKEOFF CONFIG	Status	None
TAKEOFF WARN FAIL	Status	None
TAT DISAGREE	Status	None
TIRE PRESS SYS	Status	32-49-01
TRIM AIR MOD VLV AFT CAB	Status	21-61-01
TRIM AIR MOD VLV CONT CAB	Status	21-61-01
TRIM AIR MOD VLV FWD CAB	Status	21-61-01

Message	**Level**	**MEL Item**
YAW DAMPER SYS	Status	22-23-01

21-21-01 Ground Conditioned Air Connector Check Valve

21-21-01A *Valve Closed*

21-21-01B *Valve Open*

21-22-01 Flight Deck Foot Heater Systems

21-23-01 Return Air Grille

21-25-01 Recirculation Fans

21-25-01-01 Left Fan

21-25-01-02 Right Fan

21-27-01 Equipment Cooling Supply and Exhaust Fans

21-27-02 Overboard Exhaust Valve

21-27-02A *Valve in SMOKE Position*

21-27-02B *Valve Fully Open*

21-27-03 Equipment Cooling Low Flow Detector Systems

21-27-04 Equipment Cooling Air Filter

21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan

21-31-01 Cabin Pressure Control Systems

21-31-01-01 Automatic Mode

21-31-01-01A *One Inoperative*

21-31-01-01B *Both Inoperative*

21-31-01-02 Manual Mode

21-31-02 Main Outflow Valve

21-31-02-01 Automatic Mode Electronic Actuators/Motors

21-32-01 Positive Pressure Relief Valves

21-32-01A *Valve Closed*

21-32-01B *Valve Open – Unpressurized Flight*

21-33-01 Cabin Rate of Climb Indicator

21-33-01-01 Pressurized Flight

21-33-01-02 Unpressurized Flight

21-33-02 Cabin Differential Pressure Indication

21-33-02-01 Cabin Altitude Indication Operative

21-33-02-02 Unpressurized Flight

21-33-03 Cabin Altitude Indication

21-33-03-01 Cabin Differential Pressure Indication Operative

21-33-03-02 Unpressurized Flight

21-33-04 Cabin Altitude Warning System

21-33-04-01 CABIN ALTITUDE Light

21-33-04-02 High Altitude Warning System – *N/A*

21-33-05 Outflow Valve Position Indicator

21-45-01 Door Area Heater Systems

21-51-01 Air Conditioning Packs

21-51-01-01 Air Conditioning Packs

21-51-01-01A One Pack Inoperative

21-51-01-01B Unpressurized Flight

21-51-02 Pack Flow Control and Shutoff Valves

21-51-02-01 Electronic Flow Control

21-51-02-02 Position Indicator Switch Discrete Signal

21-51-02-02A Failed Open

21-51-02-02B Failed Closed

21-51-03 Fan Bypass Check Valves

21-51-03A Airport OAT Below 80 Deg F (27 Deg C)

21-51-03B Associated Pack Inoperative

21-51-03C One Inoperative, Opposite Pack Operates Normally

21-51-04 Pack Flow Sensors

21-51-05 Pack Temperature Control Valves

21-51-05A Standby Pack Temperature Control Valve Operates Normally

21-51-05B Pack Temperature Control Valve Locked Closed

21-51-05C Associated Pack Not Used

21-51-06 Standby Pack Temperature Control Valves

21-51-07 PACK Light Systems

21-51-08 Pack Control Communication Systems

21-51-09 Pack Ram Air Systems

21-61-01 Pack/Zone Temperature Control Systems

21-61-01A Trim Air Switch Remains OFF

21-61-01B Trim Air Modulating Valve Deactivated CLOSED

21-61-02 ZONE TEMP Lights – *N/A*

21-61-03 Temperature Indications

21-61-03-01 SUPPLY DUCT

21-61-03-02 PASS CABIN

21-61-03-03 PACK

21-61-04 Trim Air Check Valves

21-62-01 Trim Air Pressure Regulating and Shutoff Valves

21-73-01 Ozone Converters – *N/A*

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21-21-01 **Ground Conditioned Air Connector Check Valve**

21-21-01A **Valve Closed**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative closed.

21-21-01 **Ground Conditioned Air Connector Check Valve**

21-21-01B **Valve Open**

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00 01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.

NOTE: Ensure cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments:

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded as such.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log.
4. Traffic, SOC, MCC, Departments to be notified accordingly.

21-22-01 Flight Deck Foot Heater Systems

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided flight deck temperature is acceptable to flight crew.

INTENTIONALLY
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21-23-01 Return Air Grille

Interval	Installed	Required	Procedure
C	89	80	**(M)**

One may be broken or missing provided:

a. Broken or missing grille is located within a designated area station 887-907.
b. Grille is removed and replaced with a blanking plate.

MAINTENANCE (M) PROCEDURES

NOTE: The grille is to be removed (see below if any minor damage, cracks or cosmetic conditions affect the performance of the intended function of the grille.

Remove the affected grille and replace it with a blanking plate (AMM 21-00-00/901).

1. Confirm the inoperative grille is in the designated location, station 887-907.
   NOTE: If two grille assemblies have the same part number, they are exchangeable.
2. Remove the affected grille and replace it with a panel that completely blocks the grille area.

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21-25-01 Recirculation Fans

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

a. OAT remains below 100 degrees F (38 degrees C).

b. Flight is conducted pressurized.

OPERATIONS NOTE

Position associated RECIRC FAN switch(es) to OFF.

21-25-01 Recirculation Fans

21-25-01-01 Left Fan

Interval	Installed	Required	Procedure
C	1	0	

Left fan may be inoperative provided left pack is operating when OAT is above 100 degrees F (38 degrees C).

OPERATIONS NOTE

Set the L RECIRC FAN switch to OFF.

21-25-01 Recirculation Fans

21-25-01-02 Right Fan

DInterval	Installed	Required	Procedure
C	1	0	

Right fan may be inoperative provided:

a. Left pack is operating when OAT is above 100 degrees F (38 degrees C).

b. Flight is conducted pressurized.

OPERATIONS NOTE

Set the R RECIRC FAN switch to OFF.

INTENTIONALLY
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21-27-01 Equipment Cooling Supply and Exhaust Fans

Interval	Installed	Required	Procedure
B	4	3	**(M)**

One fan may be inoperative provided:

a. All remaining fans are verified to operate normally.
b. Both low flow detectors are verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the remaining fans and the two low flow detectors operate normally (AMM 21-00-00/901):

1. Confirm the low-flow detector and the remaining fan operate normally for the system with the inoperative fan:
   A. Set the associated EQUIP COOLING SUPPLY or EXHAUST switch to the operative fan position (NORM or ALTN).
   B. Confirm the associated equipment cooling supply or exhaust OFF light extinguishes.
2. Confirm the fans and the low-flow detector operate normally for the system with two operative fans:
   A. Do these steps when the supply system has both fans operative:
      1) Set the EQUIP COOLING SUPPLY switch to NORM.
      2) Open and collar the P92 panel EQPT CLG SPLY FAN PWR-NORM circuit breaker.
      3) Confirm the equipment cooling supply OFF light illuminates.
      4) Set the EQUIP COOLING SUPPLY switch to ALTN.
      5) Confirm the equipment cooling supply OFF light extinguishes.
      6) Open and collar the P91 panel EQPT CLG SPLY FAN PWR-ALTN circuit breaker.
      7) Confirm the equipment cooling supply OFF light illuminates.
      8) Close the open circuit breakers.
      9) Confirm the equipment cooling supply OFF light extinguishes.
      10) Set the EQUIP COOLING SUPPLY switch to NORM.
   B. Do these steps when the exhaust system has both fans operative:
      1) Set the EQUIP COOLING EXHAUST switch to NORM.
      2) Open and collar the P91 panel EQPT CLG EXH FAN PWR-NORM circuit breaker.
      3) Confirm the equipment cooling exhaust OFF light illuminates.
      4) Set the EQUIP COOLING EXHAUST switch to ALTN.
      5) Confirm the equipment cooling exhaust OFF light extinguishes.
      6) Open and collar the P92 panel EQPT CLG EXH FAN PWR-ALTN circuit breaker.
      7) Confirm the equipment cooling exhaust OFF light illuminates.
      8) Close the open circuit breakers.
      9) Confirm the equipment cooling exhaust OFF light extinguishes.
      10) Set the EQUIP COOLING EXHAUST switch to NORM.

21-27-02 Overboard Exhaust Valve

21-27-02A Valve in SMOKE Position

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations, may be inoperative provided:

a. Overboard exhaust valve is deactivated in the SMOKE position and,

b. Both packs operate normally.

MAINTENANCE (M) PROCEDURES

Verify and deactivate the overboard exhaust valve in the SMOKE position (AMM 21-00-00/901).

1. Put the airplane in the air mode (AMM 32-09-00/201)
2. Set one of the PACK switches to HIGH.
3. Set the R RECIRC FAN switch to OFF.
4. Wait 10 seconds, then confirm the overboard exhaust valve is in the SMOKE position.
5. Open and collar the following P6-4 panel circuit breakers:
   A. A/C OVERBOARD EXHAUST VALVE CONT
   B. A/C OVERBOARD EXH VALVE RECONFIG CONT
6. Return the applicable PACK switch to AUTO.
7. Return the R RECIRC FAN switch to AUTO.
8. Disconnect, cap and stow electrical connector D11880 on the overboard exhaust valve.
9. Close the following P6-4 panel circuit breakers:
   A. A/C OVERBOARD EXHAUST VALVE CONT
   B. A/C OVERBOARD EXH VALVE RECONFIG CONT
10. Confirm the airplane is in ON GROUND mode. If necessary, put the airplane in ON GROUND mode (AMM 32-09-00/201).

21-27-02 Overboard Exhaust Valve
21-27-02B Valve Fully Open

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative in open position provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.

NOTE: Ensure the forward cargo compartment remains empty, or is verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

<u>For Empty Cargo Compartments</u>

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded as such.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1.
4. Traffic, SOC, MCC Departments as applicable to be notified.

21-27-03 Equipment Cooling Low Flow Detector Systems

Interval	Installed	Required	Procedure
B	2	1	**(M) (O)**

May be inoperative provided associated fans (supply or exhaust) are verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the associated fans operates normally (AMM 21-00-00/901).

1. Verify normal operation of the cooling system fans with the inoperative detector.
   A. For the supply system low flow detector inoperative:
      1) Confirm the normal supply fan operates by feeling the selected fan motor for vibration.
      2) Select the alternate supply fan and repeat.
   B. For the exhaust system low flow detector inoperative:
      1) Confirm the normal exhaust fan operates by feeling for air exiting the overboard exhaust vent.
      2) Select the alternate exhaust fan and repeat.
2. For ground crew call horn sounding on the ground with a low flow detector system inoperative (high alarm condition), the associated P18-3 panel EQPT COOLING LOW FLOW DETECT SUPPLY/EXHAUST circuit breaker may be opened and collared.

OPERATIONS (O) PROCEDURES

If both inboard and outboard display units on one side fail, try to restore displays by selecting the other cooling fan in the system with the inoperative detector. The captain's displays are cooled by the supply fans. The first officer's displays are cooled by the exhaust fans.

21-27-04 Equipment Cooling Air Filter

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Equipment cooling system may be operated with filter removed.

MAINTENANCE (M) PROCEDURES

Remove the air filter from the equipment cooling system (AMM 21-00-00/901).

21-27-05 Enhanced E6 Rack Equipment Cooling Exhaust Fan

Interval	Installed	Required	Procedure
D	1	0	**(M)**

May be inoperative provided:

a. Enhanced E6 Rack Equipment Cooling Exhaust Fan is deactivated.
b. All actively cooled and passively cooled equipment located on Enhanced E6 Rack shelves 1 and 3 is deactivated or removed.

MAINTENANCE (M) PROCEDURES

1. Deactivate the Enhanced E6 Rack Equipment Cooling Exhaust Fan (AMM 21-00-00/901) by opening and collaring the following circuit breakers on the P6-5 F/O Electrical System Panel:
   A. AC CB EQUIPMENT COOLING E6 RACK FAN POWER (C4069)
   B. DC CB EQUIPMENT COOLING E6 RACK FAN CONTROL (C4070)
2. Remove all actively cooled and passively cooled equipment located on Enhanced E6 Rack shelves 1 and 3, or deactivate the equipment by opening and collaring the associated circuit breakers on the P6 panel.

21-31-01 Cabin Pressure Control Systems

21-31-01-01 Automatic Mode

21-31-01-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

One automatic mode may be inoperative provided:

a. Manual mode operates normally.
b. Inoperative controller is deactivated.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative cabin pressure controller by opening and collaring the associated P6-4 panel PRESSURIZATION CONTROL AUTO 1 or AUTO 2 circuit breaker (AMM 21-00-00/901).

OPERATIONS (O) PROCEDURES

1. Normal operation of manual mode may be confirmed as follows:
   A. Set the pressurization mode selector to MAN.
   B. Use the outflow valve (toggle) switch to confirm the outflow valve position indicator moves to full open, to full closed and then back to full open.
2. Set the pressurization mode selector to ALTN prior to dispatch.
3. For inflight illumination of the AUTO FAIL light, set the pressurization mode selector to MAN and control cabin pressurization manually.

21-31-01 Cabin Pressure Control Systems

21-31-01-01 Automatic Mode

21-31-01-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Both automatic modes may be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar these P6-4 panel circuit breakers:
   A. PRESSURIZATION CONTROL AUTO 1
   B. PRESSURIZATION CONTROL AUTO 2
   C. A/C OVERBOARD EXH VALVE RECONFIG CONT.

NOTE: Ensure the forward cargo compartment remains empty, or is verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

OPERATIONS (O) PROCEDURES (Cont'd)

For Empty Cargo Compartments

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1.
4. Traffic, SOC, MCC, Departments as applicable to be notified.

21-31-01 Cabin Pressure Control Systems

21-31-01-02 Manual Mode

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is deactivated in full open position or removed.
c. Recirculation fan(s) operates normally.
d. Extended overwater flight is prohibited.
e. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight (AMM 21-00-00/901).

1. Position the outflow valve in the full open position:
   A. Set the parking brake.
   B. Start both engines (AMM 71-00-00/201).
   C. Run the engines for two minutes to allow them to stabilize.
   D. Advance thrust levers as necessary until N1 is greater than 60% and N2 is greater than 89%.
   E. Set the pressurization mode selector to MAN.
   F. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
   G. Open and collar these P6-4 panel circuit breakers:
      1) PRESSURIZATION CONTROL AUTO 1
      2) PRESSURIZATION CONTROL AUTO 2.
   H. Retard thrust levers and shut down both engines.
2. If the outflow valve can not be positioned to the full open position, remove the outflow valve (AMM 21-31-03/401). Cap and stow the electrical connectors.
3. Open and collar these P6-4 panel circuit breakers:
   A. PRESSURIZATION CONTROL MANUAL
   B. PRESSURIZATION CONTROL AUTO 1
   C. PRESSURIZATION CONTROL AUTO 2
   D. PRESSURIZATION CONTROL IND
   E. PRESSURIZATION CONTROL LCD LTG
   F. A/C OVERBOARD EXH VALVE RECONFIG CONT

MAINTENANCE (M) PROCEDURES (Cont'd)

NOTE: Ensure cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.

NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.

2. Extended overwater flight is prohibited. [This is NOT ER Operations. See below]
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1.
4. Traffic, SOC, MCC Departments as applicable to be notified.

"Extended Over Water Operation" means — an operation conducted at a distance of more than 200 nautical miles from land which is suitable for making an emergency landing in a multiengine land plane with the capability of continuing flight with one engine inoperative. TTCAR (No. 7)

21-31-02 Main Outflow Valve

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is deactivated in full open position or removed.
c. Recirculation fan(s) operates normally.
d. Extended overwater flight is prohibited.
e. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar these P6-4 panel circuit breakers:
   A. PRESSURIZATION CONTROL AUTO 1
   B. PRESSURIZATION CONTROL AUTO 2
4. If the outflow valve cannot be positioned to the full open position, remove the outflow valve (AMM 21-31-03/401). Cap and stow the electrical connectors.
5. Open and collar these P6-4 panel circuit breakers:
   A. PRESSURIZATION CONTROL MANUAL
   B. PRESSURIZATION CONTROL IND
   C. PRESSURIZATION CONTROL LCD LTG
   D. A/C OVERBOARD EXH VALVE RECONFIG CONT

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Extended overwater flight is prohibited.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**

OPERATIONS (O) PROCEDURES (Cont'd)

5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1.
4. Traffic, SOC, MCC Departments as applicable to be notified.

"Extended Over Water Operation" means - an operation conducted at a distance of more than 200 nautical miles from land which is suitable for making an emergency landing in a multiengine land plane with the capability of continuing flight with one engine inoperative. TTCAR (No.7)

21-31-02 Main Outflow Valve

21-31-02-01 Automatic Mode Electronic Actuators/Motors

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided manual mode motor operates normally.

21-32-01 **Positive Pressure Relief Valves**

21-32-01A **Valve Closed**

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative closed for pressurized flight.

MAINTENANCE (M) PROCEDURES

Install a blanking plate in place of the inoperative pressure relief valve (AMM 21-00-00/901):

1. Remove the aft bulkhead liner in the aft cargo compartment.
2. Remove the pressure relief valve (AMM 21-32-01/401).
3. Retain the clamp and the gasket.
4. Using the gasket as a template, make a blanking plate from 0.125 - 0.1875 inch (3.17 - 4.76 mm) 2024 T3 aluminum to seal the pedestal. Make sure holes are drilled through the blanking plate to accept locating pins and ambient sense tube.
5. Apply lubricant to both sides of the gasket.
6. Install the gasket, blanking plate, and clamp on the pedestal.
7. Put an airtight cap on the ambient sense tube.
8. Install the aft bulkhead liner in the aft cargo compartment.

21-32-01 Positive Pressure Relief Valves

21-32-01B Valve Open - Unpressurized Flight

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.

b. Outflow valve is positioned to full open position.

c. Recirculation fan(s) operates normally.

d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.

NOTE: Ensure cargo compartments remain empty, or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew

OPERATIONS (O) PROCEDURE S

1. Limit altitude to 10,000 ft. for passenger flights.

   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments:

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log.
4. Traffic, SOC, MCC Departments as applicable to be notified.

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21-33-01 Cabin Rate of Climb Indicator

21-33-01-01 Pressurized Flight

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided both automatic modes of cabin pressure control systems operate normally.

21-33-01 Cabin Rate of Climb Indicator

21-33-01-02 Unpressurized Flight

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.

NOTE: Ensure cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:

A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**

5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments:

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log.
4. Traffic, SOC, MCC Departments as applicable to be notified.

21-33-02 Cabin Differential Pressure Indication

21-33-02-01 Cabin Altitude Indication Operative

Interval	Installed	Required	Procedure
C	1	0	

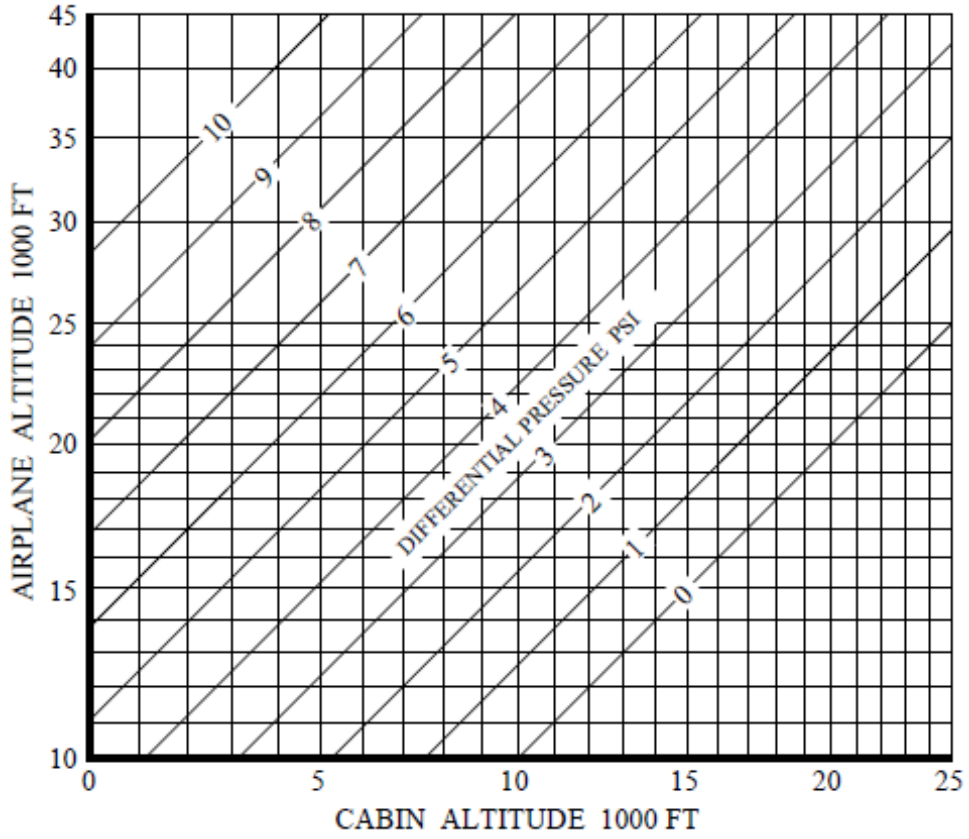
May be inoperative provided:

a. Cabin altitude indication operates normally

b. A chart is provided to crew to convert cabin altitude to differential pressure.

OPERATIONS NOTE

This chart can be used to convert cabin altitude to differential pressure.



21-33-02 Cabin Differential Pressure Indication

21-33-02-02 Unpressurized Flight

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.

NOTE: Ensure cargo compartments remain empty, or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments:

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log.
4. Traffic, SOC, MCC Departments as applicable to be notified.

21-33-03 **Cabin Altitude Indication**

21-33-03-01 **Cabin Differential Pressure Indication Operative**

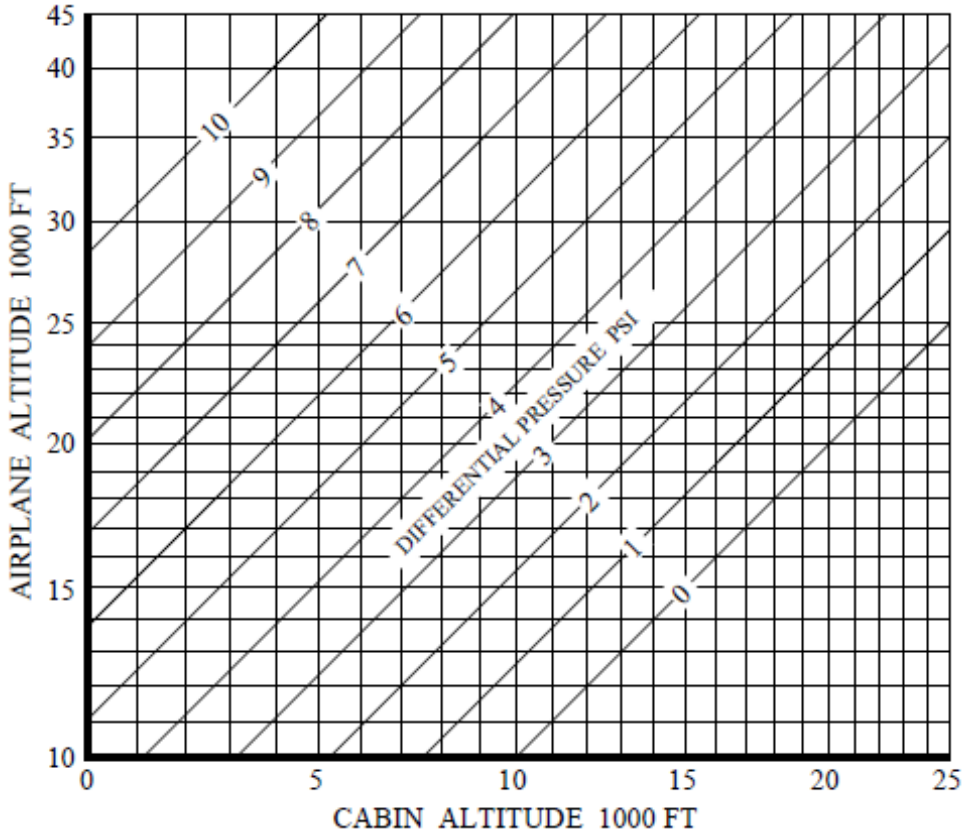
Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Cabin differential pressure indication operates normally.
b. A chart is provided to crew to convert differential pressure to cabin altitude.

OPERATIONS NOTE

This chart can be used to convert differential pressure to cabin altitude:



21-33-03 Cabin Altitude Indication

21-33-03-02 Unpressurized Flight

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.

NOTE: Ensure cargo compartments remain empty, or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments:

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log.
4. Traffic, SOC, MCC Departments as applicable to be notified.

21-33-04 Cabin Altitude Warning System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided flight altitude remains at or below 10,000 ft. MSL.

21-33-04 Cabin Altitude Warning System

21-33-04-01 CABIN ALTITUDE Light

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative provided flight crew performs a briefing on cabin altitude warning indications and procedures before engine start for first flight of day and following any change of either flight crewmember.

OPERATIONS (O) PROCEDURES

Cabin altitude warning indications and procedures must be included in the takeoff briefing conducted by the flight crew before engine start. This briefing should include the following:

Whenever the intermittent warning horn sounds in flight at an airplane flight altitude above 10,000 ft. MSL:

1. Immediately don oxygen masks and set regulators to 100%.
2. Establish crew communications.
3. Do the Cabin Altitude Warning or Rapid Depressurization non-normal checklist.
   Both pilots must verify on the overhead Cabin Altitude panel that the cabin altitude is stabilized below 10,000 ft. before removing oxygen masks.

Whenever the intermittent warning horn sounds on the ground, confirm the airplane is properly configured for takeoff.

21-33-05 Outflow Valve Position Indicator

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided valve is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the outflow valve operates normally (AMM 21-00-00/901):

1. Supply electrical power (AMM 24-22-00).
2. Confirm the following P6-4 panel circuit breakers are closed:
   A. PRESSURIZATION CONTROL LCD LTG
   B. PRESSURIZATION CONTROL AUTO 1
   C. PRESSURIZATION CONTROL AUTO 2
   D. PRESSURIZATION CONTROL MANUAL
   E. PRESSURIZATION CONTROL IND
3. Visually confirm the outflow valve is in the fully open position.
4. Confirm operation of the outflow valve in manual mode:
   A. Set the pressurization mode selector to MAN.
   B. Confirm the MANUAL light illuminates.
   C. Hold the outflow valve switch in the CLOSE position for approximately 30 seconds.
   D. Visually confirm the outflow valve moves to the fully closed position.
   E. Set the pressurization mode selector to AUTO.
   F. Confirm the MANUAL light extinguishes.
   G. Wait approximately 30 seconds.
   H. Visually confirm the outflow moved to the fully open position.

OPERATIONS (O) PROCEDURES

If procedures call for the outflow valve to be positioned in the full open or closed position, hold the outflow valve switch to OPEN or CLOSE for approximately 40 seconds.

INTENTIONALLY
BLANK

21-45-01 Door Area Heater Systems

Interval	Installed	Required	Procedure
D	2	0	**(M)**

May be inoperative deactivated.

MAINTENANCE (M) PROCEDURES

Deactivate inoperative door area heater system (AMM 21-00-00/901).

1. To deactivate the forward entry door and overwing emergency exit hatch area heater systems, open and collar the P91 (power distribution) panel FWD DOOR AREA HEATER circuit breaker.
   NOTE: Opening the circuit breaker deactivates both the forward entry door area heater and all overwing exit hatch heaters.
2. To deactivate the aft entry door area heater system, open and collar the P91 (power distribution) panel AFT DOOR AREA HEATER circuit breaker.

INTENTIONALLY
BLANK

21-51-01 Air Conditioning Packs

21-51-01-01 Air Conditioning Packs

21-51-01-01A One Pack Inoperative

Interval	Installed	Required	Procedure
C	2	1	**(O)**

Except for ER operations, may be inoperative provided flight altitude remains at or below FL 250.

OPERATIONS (O) PROCEDURES

1. Limit altitude to FL 250.
2. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT)**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT)**
3. For galley 4B food cart chiller installed, use only one chiller inflight.
4. Set the associated PACK switch to OFF.
5. Set the ISOLATION VALVE switch to CLOSE after starting engines, or as directed by step 6.
6. If desired, for increased air flow when the flaps are extended (takeoff and landing), use the APU to supply bleed air to the operating pack.
   A. Right pack inoperative – PRIOR to takeoff or landing

      NOTE 1: If anti-ice is required for taxi, perform the steps below just prior to takeoff. If anti-ice is not required for taxi, perform the steps below just after engine start.

      NOTE 2: If additional go-around thrust is desired, perform the steps below.

      1) R PACK switch to OFF.
      2) ISOLATION VALVE switch to CLOSE.
      3) L PACK switch to AUTO.
      4) Engine No. 1 BLEED air switch to OF.
      5) APU BLEED air switch to ON.
      6) Engine No. 2 BLEED air switch to OFF.
      7) Trim air switch to ON.
      8) Wing anti-ice switch to OFF.
         a. The wing anti-ice switch must remain OFF until the bleed air switches are repositioned to ON.

   B. Right pack inoperative - AFTER takeoff or landing

      NOTE: If engine failure occurs after takeoff, do not position engine BLEED air switches ON until reaching 1500 feet or until obstacle clearance has been attained.

      1) Engine No. 2 BLEED air switch to ON.
      2) APU BLEED air switch to OFF.
      3) Engine No. 1 BLEED air switch to ON.
      4) ISOLATION VALVE switch to CLOSE.

OPERATIONS (O) PROCEDURES 21-51-01-01A (Cont'd)

C. Left pack inoperative – PRIOR to takeoff or landing

NOTE 1: If anti-ice is required for taxi, perform the steps below just prior to takeoff. If anti-ice is not required for taxi, perform the steps below just after engine start.

NOTE 2: If additional go-around thrust is desired, perform the steps below.

1) R PACK switch to AUTO.
2) ISOLATION VALVE switch to OPEN.
3) L PACK switch to OFF.
4) Engine No. 1 BLEED air switch to OFF.
5) APU BLEED air switch to ON.
6) Engine No. 2 BLEED air switch to OFF.
7) Trim air switch to ON.
8) Wing anti-ice switch to OFF.
   a. The wing anti-ice switch must remain OFF until the bleed air switches are repositioned to ON.

D. Left pack inoperative – AFTER takeoff or landing

NOTE: If engine failure occurs after takeoff, do not position engine BLEED air switches ON until reaching 1500 feet or until obstacle clearance height has been attained.

1) Engine No. 2 BLEED air switch to ON.

   NOTE: The AIR COND and DUAL BLEED lights may illuminate when the engine No. 2 BLEED air switch is turned ON. If illuminated, they will extinguish after the APU BLEED air switch is turned OFF in the next step.

2) APU BLEED air switch to OFF.
3) Engine No. 1 BLEED air switch to ON.
4) ISOLATION VALVE switch to CLOSE.

21-51-01 Air Conditioning Packs

21-51-01-01 Air Conditioning Packs

21-51-01-01B Unpressurized Flight

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Except for ER operations, both may be inoperative provided:

a. Flight is conducted in an unpressurized configuration.
b. Outflow valve is positioned to full open position.
c. Recirculation fan(s) operates normally.
d. Ensure aft lower cargo compartment remains empty or is verified to contain only empty cargo handling equipment, ballast /(ballast may be loaded in ULDs), and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE (M) PROCEDURES

Configure the airplane for unpressurized flight with the manual mode operational (AMM 21-00-00/901).

1. Manually position the outflow valve in the full open position.
   A. Set the pressurization mode selector to MAN.
   B. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
2. Deactivate (locked closed) the flow control and shutoff valve for the associated pack:
   A. Gain access to the valve in the air conditioning bay.
   B. For the left flow control and shutoff valve, disconnect, cap and stow the D15830 electrical connector.
   C. For the right flow control and shutoff valve, disconnect, cap and stow the D15834 electrical connector.
   D. Confirm the valve is in the closed position:
      1) Pull out the manual control shaft.
      2) Push in the manual control shaft.
3. Open and collar the P6-4 panel A/C OVERBOARD EXH VALVE RECONFIG CONT circuit breaker.
4. Set one of the PACK switches to HIGH and placard.
5. Ensure aft cargo compartment remains empty or is verified to contain only non-combustible and/or non-flammable empty cargo handling equipment, ballast, and/or fly-away kits.
6. Secure cargo compartment after inspection by Flight Crew.
7. For empty aft cargo compartment or when aft cargo compartment is deemed to be unusable the following procedures shall apply:
   A. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
   B. If Cargo compartment is subsequently opened for any reason, repeat step A.
   C. Appropriate entry to be made in the Technical Log Part 1.
   D. Traffic, SOC, MCC Departments as applicable to be notified.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
4. Set the ISOLATION VALVE switch to CLOSE after starting engines.

OPERATIONS (O) PROCEDURES 21-51-01-01B (Cont'd)

5. To improve passenger cabin and crew compartment temperatures:
   A. If possible, pre-condition the cabin to a low temperature prior to dispatch. A low initial cabin temperature will result in a lower peak temperature.
   B. Maintain the highest allowable flight altitude.
   C. Minimize cabin heat by pulling the shades down and turning off unnecessary lights.
   D. Set the RECIRC FAN switch(es) to AUTO.
   E. Use the following table to estimate stabilized temperatures at the end of 30 minutes of flight.

      NOTE 1: It is recommended to limit number of passengers to 45.

      NOTE 2: Prolonged operation at TAT below 10 degrees C / 50 degrees F can cause freezing of unheated water systems located in the aft lower lobe.

      NOTE 3: Operation at TAT above 32 degrees C / 90 degrees F is not recommended due to the resulting high temperatures in the passenger and crew cabins.

TAT Degrees C	No. of Passenger	Passenger Cabin Temperature		Crew Cabin Temperature	
		Degrees F	Degrees C	Degrees F	Degrees C
-1	0	50	10	70	21
-1	45	70	21	75	24
10	0	70	21	90	32
10	45	80	27	95	35
21	0	90	32	110	43
21	45	95	35	110	43
32	0				
32	45				

6. Ensure the aft cargo compartment remains empty or is verified by the Flight Crew to contain only non-combustible and/or non-flammable empty cargo handling equipment, ballast, and/or fly-away kits.

   NOTE 1: Empty cargo compartment or cargo compartment containing only non- combustible and/or non-flammable empty cargo handling equipment, ballast, and/or fly-away kits, must be inspected by flight crew.

   NOTE 2: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1.

21-51-02 Pack Flow Control and Shutoff Valves

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided:

a. Affected valve is deactivated closed.
b. Associated pack is considered inoperative.

NOTE: Also dispatch the airplane using MEL Item 21-51-01.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative flow control and shutoff valve(s) in the closed position (AMM 21-00-00/901).

1. Remove pressure from the pneumatic system.
2. To deactivate the pack flow control and shutoff valve for the left cooling pack, open and collar the P6-4 panel AIR CONDITIONING PACK CONT VALVES L circuit breaker.
3. To deactivate the pack flow control and shutoff valve for the right cooling pack, open and collar the P6-4 panel AIR CONDITIONING PACK CONT VALVES R circuit breaker.
4. Gain access to the inoperative flow control and shutoff valve located in the air conditioning bay.
5. For the left flow control and shutoff valve, disconnect, cap and stow the D15830 electrical connector.
6. For the right flow control and shutoff valve, disconnect, cap and stow the D15834 electrical connector.
7. If the left flow control and shutoff valve limit switch has failed in the OPEN position:
   A. Disconnect the electrical connector D15832.
   B. Insert a jumper cable, STD-15135, between pins 1 and 3.
   C. Confirm there is continuity to airplane ground at pin 3.
   D. Cap and stow connector D15832 with the jumper cable, STD-15135, installed between pins 1 and 3.
8. If the right flow control and shutoff valve limit switch has failed in the OPEN position:
   A. Disconnect the electrical connector D15836.
   B. Insert a jumper cable, STD-15135, between pins 1 and 3.
   C. Confirm there is continuity to airplane ground at pin 3.
   D. Cap and stow connector D15836
9. Push in the manual control shaft. If already in, pull out and push back in to verify that the plunger is in.
10. If the pack flow control and shutoff valve for the left cooling pack was deactivated, close the P6-4 panel AIR CONDITIONING PACK CONT VALVES L circuit breaker.
11. If the pack flow control and shutoff valve for the right cooling pack was deactivated, close the P6-4 panel AIR CONDITIONING PACK CONT VALVES R circuit breaker.
12. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

21-51-02 Pack Flow Control and Shutoff Valves

21-51-02-01 Electronic Flow Control

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Appropriate performance adjustments are applied.

b. Ensure forward and aft lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

NOTE 1: MEL dispatch option 21-51-01-01A may be used as an alternative means of dispatching with one inoperative air conditioning pack that has failed electronic flow control and the other air conditioning pack operating normally with electronic flow control (Reference AMOC 785-21-11462 to AD 2021-16-15).

NOTE 2: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

MAINTENANCE (M) PROCEDURES

1. Maintenance to ensure forward and aft cargo compartments remain empty or are verified with the Load Sheet Agent to contain only non-combustible and/or non-flammable empty cargo handling equipment, ballast, and/or fly-away kits. Appropriate entry to be made in the Log Part 1.

   NOTE: In absence of Maintenance, Flight Crew to check the cargo and verify with the load sheet Agent that the cargo compartment is empty or contains only non-combustible and/or non-flammable empty cargo handling equipment, ballast and/or fly away kits.

2. Secure cargo compartment after inspection by Flight Crew.
3. For empty forward and aft cargo compartments or when cargo compartments are deemed to be unusable the following procedures shall apply:
   A. After inspection by Flight Crew, Engineering to verify cargo compartments are locked out and placarded.
   B. If the Cargo compartment is subsequently opened for any reason, repeat step A.
   C. Appropriate entry to be made in the Technical Log Part 1.
   D. Traffic, SOC, MCC Departments as applicable to be notified.

OPERATIONS (O) PROCEDURES 21-51-02-01 (Cont'd)

NOTE: Failure of the electronic flow control valve torque motor or primary pack flow temperature controller will result in backup pneumatic flow control. Backup pneumatic flow control will result in larger flow transients and airflow noise associated with bleed air pressure changes.

1. Performance limited weights are reduced by the following amount for one or both Electronic Flow Controls inoperative

Takeoff	Enroute Climb	Approach & Landing
1,043 kg	1,270 kg	635 kg

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below - Enroute Diversion Speed Effects.

Diversion Speed	Factor
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

2. Increase flight planning fuel by either incrementing the current fuel flow correction factor by 1.3% or flight planning fuel by 1.4% for each engine with an inoperative engine electronic flow control.
3. Increase engine-out ETOPS critical fuel reserves by an additional 1.1% on top of the flight planning fuel penalty noted above. The all-engine ETOPS critical fuel reserves would be covered by the flight planning fuel penalty noted above.
4. Flight Crew to ensure the forward and aft cargo compartments remain empty and are verified with the Load Sheet Agent/ Maintenance to contain only non-combustible and/or non-flammable empty cargo handling equipment, ballast, and/or fly-away kits.

NOTE 1: In absence of Maintenance, Flight Crew to check the cargo and verify with the load sheet Agent that both cargo compartments are empty or contain only non-combustible and/or non-flammable empty cargo handling equipment, ballast and/or fly away kits.

NOTE 2: Empty cargo compartments or cargo compartments containing only non-combustible and/or non-flammable empty cargo handling equipment, ballast, and/or fly-away kits, must be inspected by Flight Crew.

NOTE 3: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

21-51-02 Pack Flow Control and Shutoff Valves
21-51-02-02 Position Indicator Switch Discrete Signal
21-51-02-02A Failed Open

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative failed open provided both air conditioning packs operate normally.

OPERATIONS NOTE

NOTE: When a position indicator switch discrete signal is failed open, pack ON is signaled to the FMCS and EEC.

1. Normal takeoff, engine BLEED air switches ON:
   A. Normal engine bleed-on takeoff N1 target displayed.
   B. Observe normal engine bleed-on AFM performance limits.
   C. Determine performance based on packs ON.
2. No engine bleed takeoff:
   A. No engine bleed takeoff N1 target displayed.
   B. Observe no engine bleed-on AFM performance limits.
   C. Determine performance based on packs OFF.
3. In-flight:
   A. Normal engine bleed-on N1 target displayed.
   B. Observe normal engine bleed-on AFM performance limits.

21-51-02 Pack Flow Control and Shutoff Valves

21-51-02-02 Position Indicator Switch Discrete Signal

21-51-02-02B Failed Closed

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative failed closed provided:

a. Both air conditioning packs operate normally, and

b. An Unpressurized Takeoff or a No Engine Bleed Takeoff is not conducted.

OPERATIONS (O) PROCEDURES

NOTE: When a position indicator switch discrete signal is failed closed, pack OFF is signaled to the FMCS and EEC.

1. Conducting the FCOM supplementary procedure for an "Unpressurized Takeoff" or a "No Engine Bleed Takeoff" is prohibited.
2. Normal takeoff, engine BLEED air switches ON:
   A. Normal engine bleed-on takeoff N1 target displayed.
   B. Observe normal engine bleed-on AFM performance limits.
   C. Determine performance based on packs ON.
3. In-Flight:
   A. FMC target will represent packs HIGH flow mode for both engines.
   B. Observe A/C HIGH enroute and go-around AFM performance limits.

21-51-03 Fan Bypass Check Valves

21-51-03A Airport OAT Below 80 Deg F (27 Deg C)

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative open or missing provided airport ambient temperature remains below 80 degrees F (27 degrees C).

21-51-03 Fan Bypass Check Valves

21-51-03B Associated Pack Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative open or missing for an associated inoperative pack.

21-51-03 Fan Bypass Check Valves

21-51-03C One Inoperative, Opposite Pack Operates Normally

Interval	Installed	Required	Procedure
D	2	1	

May be inoperative open or missing provided pack associated with remaining fan bypass check valve operates normally.

OPERATIONS NOTE

On ground at high ambient temperature (above 80 Deg F/27 Deg C) and high takeoff thrust setting, using affected AC PACK with fan bypass valve inoperative (Fail Open/Missing) may result in Pack Light ON.

21-51-04 Pack Flow Sensors

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided pack inlet pressure sensor(s) for associated air conditioning pack(s) is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the pack inlet pressure sensor for the associated air conditioning pack(s) operates normally (AMM 21-00-00/901).

1. Confirm that the Scheduled Maintenance Task (SMT) message(s) PACK CONTROL CHANNEL L and/or PACK CONTROL CHANNEL R for the associated air conditioning pack(s) is not annunciated.
2. Disconnect, cap, and stow the electrical connector from the associated Pack Flow Sensor.
3. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

21-51-05 Pack Temperature Control Valves

21-51-05A Standby Pack Temperature Control Valve Operates Normally

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided associated standby pack temperature control valve is verified to operate normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Verify the associated standby pack temperature control valve operates normally.

1. Set the AIR TEMP source selector to SUPPLY DUCT - FWD or SUPPLY DUCT - AFT.
2. Confirm the passenger cabin temperature is stabilized between 70 – 80 degrees F (21 - 27 degrees C).
3. Rotate the temperature selector associated with the selected SUPPLY DUCT zone fully clockwise (W).
4. Observe the air temperature indicator and confirm the temperature increases.
5. Rotate the applicable temperature selector fully counter clockwise (C).
6. Observe the air temperature indicator and confirm the temperature decreases.

21-51-05 Pack Temperature Control Valves

21-51-05B Pack Temperature Control Valve Locked Closed

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Associated pack temperature control valve is locked closed.
b. Associated standby pack temperature control valve is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Lock the affected pack temperature control valve closed (AMM 21-00-00/901).

NOTE: The valve could be stuck in the open position. To help prevent excess hot air from entering into the mix manifold resulting in a hot cabin temperature, the temperature control valve can be manually closed.

1. Gain access to the valve in the associated air conditioning bay.
2. Turn the manual override on the pack temperature control valve until the visual position indicator shows the full closed position.
3. Disconnect, cap and stow the electrical connector from the valve.
4. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Verify the associated standby pack temperature control valve operates normally.

1. Set the AIR TEMP source selector to SUPPLY DUCT - FWD or SUPPLY DUCT - AFT.
2. Confirm the passenger cabin temperature is stabilized between 70 – 80 degrees F (21 - 27 degrees C).
3. Rotate the temperature selector associated with the selected SUPPLY DUCT zone fully clockwise (W).
4. Observe the air temperature indicator and confirm the temperature increases.
5. Rotate the applicable temperature selector fully counterclockwise (C).
6. Observe the air temperature indicator and confirm the temperature decreases.

21-51-05 Pack Temperature Control Valves

21-51-05C Associated Pack Not Used

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated pack is considered inoperative.

NOTE: 1 The airplane must also be dispatched using MEL Item 21-51-01.

NOTE: 2 If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

21-51-06 Standby Pack Temperature Control Valves

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated pack temperature control valve operates normally.

21-51-07 PACK Light Systems

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated pack is considered inoperative.

NOTE: The airplane must also be dispatched using MEL Item 21-51-01.

21-51-08 Pack Control Communication Systems

Interval	Installed	Required	Procedure
C	2	0	

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

21-51-09 Pack Ram Air Systems

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Associated ram air inlet panels and exhaust louvers are deactivated and secured in flight open position.
b. Operations are not conducted on runways covered with slush or on gravel runways.
c. Associated pack is not operated during takeoff or landing on wet runways or runways with standing water.
d. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the associated ram air inlet panels and exhaust louvers and secure them in the flight open position (AMM 21-00-00/901).

1. Open and collar the associated circuit breakers:
   A. AIR CONDITIONING RAM AIR MOD LEFT (RIGHT)
   B. AIR CONDITIONING MISC LOADS LEFT (RIGHT)
2. Open the associated ram air inlet and exit actuator panels.
3. Deactivate the ram air inlet modulating system.
   A. Disconnect and stow the two electrical connectors.
   B. Remove bolt, washers, and nut that connect the actuator rod end to the bellcrank.
   C. Disconnect the actuator rod end from the bellcrank and reinstall and tighten the bolt, washer, and nut in the bellcrank.
   D. Rotate the bellcrank to align the rigging pin holes on the shaft assembly with the holes on the bearing housing support.
   E. Install a bolt (BACB30UU3P36D or equivalent) with a 0.1875 inch (4.7625 mm) shank diameter and 2.2 inch (55.88 mm) shank length through the rigging pin holes. The shank end must be drilled for a cotter pin.
   F. Install a nut (BACN10JD3 or equivalent) and cotter pin (BACP18BC01D06P or equivalent) with a 0.047 inch (1.193 mm) diameter and a 0.75 inch (19.05 mm) length on the bolt.
   G. Use lockwire to safety the actuator rod end to the attached bolt in the bellcrank.
   H. Close the associated ram air inlet actuator panel.
4. Deactivate the ram air exit modulating system using one (A or B) of the following procedures.
   A. With Actuator Removal
      1) Disconnect and stow the two electrical connectors.
      2) Remove bonding jumper from the structure attached to SRADA.
      3) Remove actuator and mounting hardware. Reinstall mounting hardware without actuator. Retain hardware for remounting the replacement actuator.
      4) Remove bolt (BACB30PW4-9), nut (BACN10YR3CM), and washers (NAS1149E0463P and NAS1149E0316R) from the forward most drive link (213A3338-5/-6).

MAINTENANCE (M) PROCEDURES 21-51-09 (Cont'd)

5) Once free, open the exit louvers until the forward most hole in the link assembly lines up with the hole in the forward most drive link arm.
6) Reinstall the hardware removed in step 4A.4 in the same order it was removed.
7) The louvers should be open approximately 56 degrees relative to the airplane skin.
8) Close the associated ram air exit actuator panel.

B. Without Actuator Removal (using SRADA stowage link, part number C21009)

1) Disconnect and stow the two electrical connectors.
2) Disconnect the actuator rod end from the actuator drive arm.
3) Remove bolt (BACB30PW4-9), nut (BACN10YR3CM), and washers (NAS1149E0463P and NAS1149E0316R) from the forward most drive link (213A3338-5/-6).
4) Once free, open the exit louvers until the forward most hole in the link assembly lines up with the hole in the forward most drive link arm.
5) Reinstall the hardware removed in step 4B.3 in the same order it was removed.
6) The louvers should be open approximately 56 degrees relative to the airplane skin.
7) Connect one end of the SRADA stowage link to the actuator drive arm using the hardware removed in step 4B.2.
8) Connect the other end of the SRADA stowage link to the actuator rod end using bolt (BACB30LM4U12), washer (BACW10BP4ACU) under the head of the bolt, washer (BACW10BP4APU) under the nut, and the nut (BACN10YR4CM).
9) Close the associated ram air exit actuator panel.

5. Close the associated circuit breakers:
   A. AIR CONDITIONING RAM AIR MOD LEFT (RIGHT)
   B. AIR CONDITIONING MISC LOADS LEFT (RIGHT)
6. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. During takeoff and landing when operating on wet runways or on runways with standing water:
   A. Set the associated PACK switch(es) to OFF.
   B. Set the bleed air panel ISOLATION VALVE switch to CLOSE.
2. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V_1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**

OPERATIONS (O) PROCEDURES 21-51-09 (Cont'd)

3. Performance limited weights are reduced by the following amount for each inoperative pack ram air system:

TAKEOFF	ENROUTE CLIMB	APPROACH & LANDING
45 kg	**136 kg**	**No Penalty**

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below - Enroute Diversion Speed Effects.

Diversion Speed	Factor
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

4. Increase flight planning fuel by either incrementing the current airplane drag correction factor by 0.2% or flight planning fuel by 0.2% for each system deactivated and secured in flight open position.

21-61-01 Pack/Zone Temperature Control Systems

21-61-01A Trim Air Switch Remains OFF

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided TRIM AIR switch remains OFF.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS NOTE

Use normal procedures to operate the temperature selectors. If a comfortable flight deck temperature cannot be maintained with the CONT CAB zone temperature selector in the normal range, set the CONT CAB zone temperature selector to OFF.

21-61-01 **Pack/Zone Temperature Control Systems**

21-61-01B **Trim Air Modulating Valve Deactivated CLOSED**

Interval	Installed	Required	Procedure
C	3	0	**(M)**

May be inoperative provided associated trim air modulating valve is deactivated CLOSED.

MAINTENANCE (M) PROCEDURES

Deactivate the associated trim air modulating valve in the closed position (AMM 21-00-00/901):

1. Set the associated zone temperature selector to OFF.
2. Gain access to the associated trim air modulating valve.
3. Turn the manual override until the visual position indicator shows the full closed position.
4. Disconnect, cap and stow the electrical connector from the valve.
5. Set the associated zone temperature selector to the normal temperature range.
6. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS NOTE

Use normal procedures to operate the temperature selectors. If a comfortable flight deck temperature cannot be maintained with the CONT CAB zone temperature selector in the normal range, set the CONT CAB zone temperature selector to OFF.

21-61-02 **ZONE TEMP Lights**

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided associated SUPPLY DUCT temperature indication operates normally.

21-61-03 Temperature Indications

21-61-03-01 SUPPLY DUCT

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided associated ZONE TEMP light operates normally.

21-61-03 Temperature Indications

21-61-03-02 PASS CABIN

Interval	Installed	Required	Procedure
C	2	0	

21-61-03 Temperature Indications

21-61-03-03 PACK

Interval	Installed	Required	Procedure
C	2	0	

21-61-04 Trim Air Check Valves

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided associated valve is secured closed.

See also OPERATIONS NOTE on next page.

MAINTENANCE (M) PROCEDURES

Secure the inoperative valve closed (AMM 21-00-00/901).

1. Do one of the following:
   A. Obtain Blank-Off Plate, Trim Air Check Valve (part number A21015-1).
   B. Fabricate a blanking plate with the following specifications:
      1) Materials:
         a. Titanium 6AL-4V per MIL-T-9046 Type 3 Comp C
         b. Inconel 718 per AMS 5596
         c. 301 CRES per MIL-S-5059 1/2, 3/4, or Fullhard
      2) Dimensions:
         a. Diameter - 3.10 (+0.10 or -0.00) inches
         b. Thickness - 0.040 inch
2. Supply electrical power to the airplane (AMM 24-22-00/201).
3. Remove the pressure from the pneumatic system (AMM 36-05-00/201).
4. Set these switches on the P5-10 air conditioning panel to OFF and attach DO-NOT-OPERATE tags:
   A. BLEED 1
   B. BLEED 2
   C. BLEED APU.
5. Gain access to the inoperative trim air check valve.
6. Modify the Ducting:
   A. Loosen the clamp on the forward side of the inoperative trim air check valve.
   B. Move the clamp forward onto the duct.
   C. Install the blanking plate between the duct assembly and the inoperative trim air check valve.
      NOTE: Install on the downstream side of the inoperative trim air check valve.
   D. Move the clamp backward onto the check valve.
   E. Tighten the nut on the clamp to 55 pound-inches (6.2 newton-meters).
      NOTE: The clamp must be evenly located on the valve and duct flanges.

MAINTENANCE (M) PROCEDURES 21-61-04 (Cont'd)

7. Perform a Leakage Test:
   A. Remove the DO-NOT-OPERATE tags from the BLEED 1, BLEED 2 and BLEED APU switches. Set the switches in their usual positions.
   B. Supply pneumatic power (AMM 36-05-00/201).
   C. Set the associated PACK switch to AUTO.
   D. Set the TRIM AIR switch to OFF.
   E. Check valve flange for leakage.
      1) Diffused leakage is acceptable.
      2) Jet blast leakage must be repaired by joint or clamp realignment.

OPERATIONS NOTE

A blanking plate installed on one of the two trim air check valves does not affect normal pack operation.

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21-62-01 Trim Air Pressure Regulating and Shutoff Valve

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative deactivated closed.

MAINTENANCE (M) PROCEDURES

Deactivate the trim air pressure regulating and shutoff valve closed (AMM 21-00-00/901).

1. Set the TRIM AIR switch to OFF.
2. Open and collar the P6-4 panel AIR CONDITIONING TRIM AIR PRESS circuit breaker.
3. Gain access to the trim air pressure regulating and shutoff valve.
4. Disconnect, cap and stow the electrical connector from the valve.
5. Turn the manual override hex on the shutoff valve until the visual position indicator shows the full closed position.

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22-10-01 Autopilot Systems

22-11-03 Mode Control Panel Windows

22-11-03-01 IAS /MACH

22-11-03-02 HEADING

22-11-03-03 VERT SPEED

22-11-03-04 ALTITUDE

22-11-03-05 COURSE

22-11-03-06 Window Lighting

22-11-04 Mode Control Panel Selectors

22-11-04-01 Vertical Speed Thumbwheel (DN, UP)

22-11-04-02 Bank Angle Selector (10, 15, 20, 25, 30)

22-11-05 Mode Control Panel Switches

22-11-05-01 A/P ENGAGE Control Wheel Steering (CWS) Switches (A or B)

22-11-05-02 A/P ENGAGE Command (CMD) Switches (A or B)

22-11-05-03 Autothrottle (A/T) ARM Switch

22-11-05-04 SPEED Switch

22-11-05-05 Flight Director (F/D) Switches

22-11-05-06 IAS/MACH Change Over (C/O) Switch

22-11-05-07 Approach (APP) Switch

22-11-05-08 N1, LNAV, VNAV, LVL CHG, V/S, HDG SEL, ALT HLD, and VOR/LOC Switches

22-11-05-09 SPD INTV and ALT INTV Switches

22-11-05-10 Autopilot DISENGAGE Bar

22-11-06 Mode Control Panel Switch Lights

22-11-06-01 A/P ENGAGE Control Wheel Steering (CWS) Switch Lights

22-11-06-02 A/P ENGAGE Command (CMD) Switch Lights

22-11-06-03 A/T ARM Switch Light

22-11-07 Automatic Landing System (Autoland)

22-11-08 Autoflight Status Annunciator

22-11-08-01 Autopilot (A/P) Disengage Light

22-11-08-02 Autothrottle (A/T) Disengage Light

22-11-08-02A *One Inoperative*

22-11-08-02B *Both Inoperative*

22-11-09 Takeoff/Go-Around (TO/GA) Switches

22-11-09A *One Inoperative*

22-11-09B *Both Inoperative*

22-11-10 Control Wheel Autopilot Disengage Switches

22-11-11 Altitude Alerting System

22-11-11-01 Aural Alert

22-11-11-02 Visual Alert

22-11-12 Flight Director Systems

22-21-01 Mach Trim Systems

22-21-01A *One Inoperative*

22-21-01B *Both Inoperative*

22-23-01 Yaw Damper Function

22-23-01-01 YAW DAMPER Light

22-31-01 Autothrottle System

22-31-02 Thrust Mode Display

22-10-01 Autopilot Systems

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided approach minimums do not require its use.

NOTE: PBN Operations (See Definitions) are available with One Autopilot serviceable.

NOTE: TWO (2) REQUIRED FOR CAT II APPROACH. (CAL)

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22-11-03 Mode Control Panel Windows

22-11-03-01 IAS/MACH

Interval	Installed	Required	Procedure
C	1	0	

22-11-03 Mode Control Panel Windows

22-11-03-02 HEADING

Interval	Installed	Required	Procedure
C	1	0	

22-11-03 Mode Control Panel Windows

22-11-03-03 VERT SPEED

Interval	Installed	Required	Procedure
C	1	0	

22-11-03 Mode Control Panel Windows

22-11-03-04 ALTITUDE

Interval	Installed	Required	Procedure
C	1	0	

22-11-03 Mode Control Panel Windows

22-11-03-05 COURSE

Interval	Installed	Required	Procedure
C	2	0	

22-11-03 Mode Control Panel Windows

22-11-03-06 Window Lighting

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

a. Selected airspeed indications operate normally.
b. Selected heading indications operate normally.
c. Selected vertical speed indications operate normally.
d. Selected altitude indications operate normally.
e. Selected course indications operate normally.

22-11-04 **Mode Control Panel Selectors**

22-11-04-01 **Vertical Speed Thumbwheel (DN, UP)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided procedures do not require its use.

22-11-04 **Mode Control Panel Selectors**

22-11-04-02 **Bank Angle Selector (10, 15, 20, 25, 30)**

Interval	Installed	Required	Procedure
C	1	0	

22-11-05 **Mode Control Panel Switches**

22-11-05-01 **A/P ENGAGE Control Wheel Steering (CWS) Switches (A or B)**

Interval	Installed	Required	Procedure
C	2	0	

22-11-05 **Mode Control Panel Switches**

22-11-05-02 **A/P ENGAGE Command (CMD) Switches (A or B)**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided approach minimums do not require its use.

Dispatch under MEL Item 22-10-01 is also required.

22-11-05 **Mode Control Panel Switches**

22-11-05-03 **Autothrottle (A/T) ARM Switch**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require autothrottle use.

22-11-05 Mode Control Panel Switches
22-11-05-04 SPEED Switch

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require autothrottle use.

22-11-05 Mode Control Panel Switches
22-11-05-05 Flight Director (F/D) Switches

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided approach minimums do not require flight director use.

NOTE: 1 F/D is required for PBN Operations (See Definitions).

NOTE: TWO (2) F/D REQUIRED FOR CAT II APPROACH. (CAL)

Dispatch under MEL 22-11-12 Flight Director Systems is also required.

22-11-05 Mode Control Panel Switches
22-11-05-06 IAS/MACH Change Over (C/O) Switch

Interval	Installed	Required	Procedure
C	1	0	

22-11-05 Mode Control Panel Switches

22-11-05-07 Approach (APP) Switch

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require autopilot or flight director use.

NOTE: REQUIRED FOR CAT II APPROACH. (CAL)

NOTE: Required For GLS, ILS and IAN Approaches.

NOTE: See Operations Manual Part A – General, Section 8, Operating Procedures, Approach Minimums.

Wherever Possible, Fly RNP APCH [See Definitions].

Alternatively, if available, fly LOC, VOR/DME, VOR, or NDB Approaches.

22-11-05 Mode Control Panel Switches

22-11-05-08 N1, LNAV, VNAV, LVL CHG, V/S, HDG SEL, ALT HLD, and VOR/LOC Switches

Interval	Installed	Required	Procedure
C	8	0	

May be inoperative provided enroute operations do not require their use.

NOTE: LNAV and HDG SEL are required for PBN Operations. (See Definition)

22-11-05 Mode Control Panel Switches

22-11-05-09 SPD INTV and ALT INTV Switches

Interval	Installed	Required	Procedure
C	2	0	

22-11-05 Mode Control Panel Switches

22-11-05-10 Autopilot DISENGAGE Bar

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided control wheel autopilot disconnect switches operate normally.

22-11-06 **Mode Control Panel Switch Lights**

22-11-06-01 **A/P ENGAGE Control Wheel Steering (CWS) Switch Lights**

Interval	Installed	Required	Procedure
C	2	0	

22-11-06 **Mode Control Panel Switch Lights**

22-11-06-02 **A/P ENGAGE Command (CMD) Switch Lights**

Interval	Installed	Required	Procedure
C	**2**	**1**	

May be inoperative provided approach minimums do not require use of autopilot.

NOTE: Two (2) Required for CAT II Approach. (CAL)

22-11-06 **Mode Control Panel Switch Lights**

22-11-06-03 **A/T ARM Switch Light**

Interval	Installed	Required	Procedure
C	1	0	

22-11-07 Automatic Landing System (Autoland)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require its use.

22-11-08 Autoflight Status Annunciator
22-11-08-01 Autopilot (A/P) Disengage Light

Interval	Installed	Required	Procedure
C	2	1	

22-11-08 Autoflight Status Annunciator
22-11-08-02 Autothrottle (A/T) Disengage Light
22-11-08-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided approach minimums do not require use of autothrottle.

22-11-08 Autoflight Status Annunciator
22-11-08-02 Autothrottle (A/T) Disengage Light
22-11-08-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided autothrottle is not used.

NOTE: Caribbean Airlines must also comply with MEL item 22-31-01 operational requirements.

22-11-09 Takeoff/Go-Around (TO/GA) Switches

22-11-09A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided approach minimums do not require its use.

NOTE: Two (2) Required for CAT II Approach. (CAL)

22-11-09 Takeoff/Go-Around (TO/GA) Switches

22-11-09B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

a. Operate both thrust levers manually for takeoff and go-around.

b. Do not use Autopilot and Flight Director below Minimum Descent Altitude (MDA) or 500 ft., whichever is higher.

NOTE: Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.

NOTE: Two (2) Required for CAT II Approach. (CAL)

22-11-10 Control Wheel Autopilot Disengage Switches

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided:

a. Mode control panel autopilot DISENGAGE bar operates normally.
b. Autopilot is not used below 1,500 ft. AGL.
c. Approach minimums do not require use of autopilot.

NOTE: Two (2) Required for CAT II Approach. (CAL)

22-11-11 Altitude Alerting System

Interval	Installed	Required	Procedure
A	1	0	**(O)**

May be inoperative provided:

a. Autopilot with altitude hold and altitude capture operates normally.
b. Enroute operations (i.e., RVSM) do not require its use.
c. Airplane does not depart from a designated airport (as listed in the operator's Mel) where repair or replacement can be made. **(See Definition)**
d. Repairs are made within 3 flight days.

OPERATIONS (O) PROCEDURES

NOTE: An operative altitude alerting system is required for RVSM operations.

Use of the following procedures to increase flight crew awareness of altitude is required:

1. Determine that at least one autopilot is operating normally and that altitude hold and altitude capture are operating normally.
2. Use the Autopilot and ALT HOLD for all operations for which it is appropriate.
3. Cross-check Captain's and First Officer's altitude displays upon reaching and departing assigned altitude.
4. Periodically cross-check altitude indications when maintaining an assigned altitude.
5. Pilot Monitoring must call out approaching and departing assigned altitudes.
6. Flight Crew must be aware that the usual alerts for altitude deviations will not occur.

22-11-11 Altitude Alerting System

22-11-11-01 Aural Alert

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Visual alert operates normally.
b. Autopilot with altitude hold and altitude capture operates normally.

22-11-11 Altitude Alerting System

22-11-11-02 Visual Alert

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Aural alert operates normally.
b. Autopilot with altitude hold and altitude capture operates normally.

22-11-12 Flight Director Systems

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided approach minimums do not require its use.
PBN Operations [See Definitions] Not Allowed.
CAT II Approaches NOT Allowed.
See Operation Manual Part A - General, Section 8 – Operating Procedures, Approach Minimums.
Autopilot must be engaged at the Minimum A/P engagement Altitude.

22-11-12A Flight Director Systems (CAL)

Interval	Installed	Required	Procedure
C	2	1	

PBN Operations [See Definitions] are Allowed.
CAT II Approaches Not Allowed.
Autopilot must be engaged at the Minimum A/P engagement Altitude.
For night flight, and/or field conditions less than 3sm visibility or ceiling less than 1,000 ft, serviceable F/D shall be on captain's side.

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22-21-01 Mach Trim Systems

22-21-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided:

a. Associated Mach Trim system is deactivated.
b. Remaining Mach Trim system is verified to operate normally.

NOTE: The airplane must also be dispatched using MEL item 22-11-07.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative Mach Trim system and verify the remaining Mach Trim system is operating normally (AMM 22-00-00/901).

1. Open and collar the associated MACH TRIM AC and MACH TRIM DC circuit breakers on the P18-1 panel (channel A) or P6-2 panel (channel B), as appropriate.
2. Perform the DFCS BITE library test No. 33, Mach Trim, for the remaining system (channel A or channel B) and verify that the test passes.

OPERATIONS NOTE

Deactivation of one mach trim system channel will result in the illumination of the MASTER CAUTION, FLT CONT annunciation, and MACH TRIM FAIL light when performing a Master Caution recall.

22-21-01 Mach Trim Systems

22-21-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Mach trim actuator is verified to be in null/uncommanded elevator position.
b. Appropriate performance adjustments are applied.

NOTE: The airplane must also be dispatched using MEL item 22-11-07.

MAINTENANCE (M) PROCEDURES

1. Verify that the elevator trailing edge is in the correct position (AMM 22-00-00/901).

NOTE 1: A mach trim actuator and mach trim system are required to position the elevator trailing edge if it is out of alignment. An actuator or accessory box that does not operate must be replaced.

NOTE 2: Dispatch with both mach trim systems inoperative is not allowed if the elevator trailing edge is not aligned with the index mark on the fuselage.

A. Supply electrical power
B. Supply hydraulic power.
C. Position the flaps to UP.
D. Put the stabilizer to 4 units of trim.
E. Move the control column to put the elevators in the detent position.
F. Visually check that the elevator trailing edge is aligned within 0.06 inches (1.5 mm) of the index mark on the fuselage.
G. Open and collar the MACH TRIM AC and MACH TRIM DC circuit breakers on the P18-1 panel (channel A) and on the P6-2 panel (channel B).

OPERATIONS (O) PROCEDURES

Do not exceed 280 KIAS/ .82M.

NOTE: Deactivation of both mach trim system channels will result in the illumination of the MASTER CAUTION, FLT CONT annunciation, and MACH TRIM FAIL light on the overhead Flight Controls panel.

22-23-01 Yaw Damper Function

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided YAW DAMPER switch remains OFF.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Set YAW DAMPER switch OFF.
2. Avoid areas of predicted moderate or severe turbulence.
3. Reduce airspeed and/or descend to a lower altitude if turbulence is encountered and passenger comfort is affected.
4. Do not exceed flaps 30.

22-23-01 Yaw Damper Function

22-23-01-01 YAW DAMPER Light

Interval	Installed	Required	Procedure
C	1	0	

22-31-01 Autothrottle System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require its use.

22-31-02 Thrust Mode Display

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided thrust mode limits are observed.

23-11-01 High Frequency (HF) Communication System

23-11-01A *Not Required - N/A*

23-11-01B *Operations Requiring Two LRCS - N/A*

23-11-01-01 HF Data Link

23-11-01-01A *Alternate Procedures Required - N/A*

23-11-01-01B *Procedures Do Not Require Use*

23-12-01 Very High Frequency (VHF) Communication System

23-12-01-01 VHF Data Link

23-12-01-01A *Alternate Procedures Used*

23-12-01-01B *Procedures Do Not Require Use - N/A*

23-12-02 Radio Tuning Panel

23-12-02-01 Switch Lights, Key Lights

23-15-01 Satellite Communication System (SATCOM) *- N/A*

23-24-01 Fixed Emergency Locator Transmitter (ELT)

23-24-01A *Inoperative* ***(CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)***

23-24-01B *Missing (****CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)***

23-24-01C *N/A*

23-24-01D *N/A*

23-24-01-01 Remote ELT Switch (***CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)***

23-24-01-02 ELT Indicator Light (***CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)***

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)

23-24-02-01A *Inoperative* ***(CUB ONLY)***

23-24-02-01B *Missing* ***(CUB ONLY)***

23-24-02-01C *N/A*

23-24-02-01D *N/A*

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)

23-24-02-01A *N/A*

23-24-02-01B *N/A*

23-24-02-01C *N/A*

23-24-02-01D *N/A*

23-24-02-01-01 Remote ELT Switch ***(CUB ONLY)***

23-24-02-01-02 ELT Indicator Switch ***(CUB ONLY)***

23-24-02-02 ADT Trigger Unit (ATU)

23-24-02-02A *Inoperative* ***(CUB ONLY)***

23-24-02-02B *Missing* ***(CUB ONLY)***

23-24-02-02C *N/A*

23-24-02-02D *N/A*

23-25-01 Low Frequency Underwater Locating Device

23-25-01A *N/A*

2*3-25-01B* *Installed on* ***ANT, BAR, JAM, GRN, SUR, CUB***

23-27-01 Aircraft Communications Addressing and Reporting System (ACARS)

23-27-01A *Alternate Procedures Required*

23-27-01B *N/A*

23-27-01-01 FMC Interface Function

23-27-01-01A *Alternate Procedures Required*

23-27-01-01B *N/A*

23-27-01-02 ACARS Printer

23-28-01 Selective Call System (SELCAL)

23-28-01A *Alternate Procedures Required*

23-28-01B *N/A*

23-28-01-01 Channels (Including Lights)

23-28-01-01A *Alternate Procedures Required*

23-28-01-01B *N/A*

23-31-01 Passenger Address (PA) System

23-31-01A *Flight Attendant Alerting System Operates Normally*

23-31-01B *Not Required - N/A*

23-31-01-01 Lavatory Speakers

23-31-01-02 Cabin Speakers

23-31-02 Prerecorded Passenger Announcement System

23-31-02A *Alternate Procedures Required*

23-31-02B *Procedures Do Not Require Use - N/A*

23-41-01 Service Interphone System

23-41-01-01 Flight Deck to Ground/Ground to Flight Deck Function (Includes Interphone Jack Located on External Power Panel)

23-41-01-01A *Flight Interphone Jack Operates Normally*

23-41-01-01B *Flight Interphone Jack Inoperative*

23-42-01 Service (Cabin) Interphone System

23-42-01-01 Flight Deck to Cabin, Cabin to Flight Deck Functions

23-42-01-01A *50% Operate Normally*

23-42-01-01B *Inoperative*

23-42-01-02 Cabin to Cabin Functions

23-42-01-02A *Two Attendant Stations - N/A*

23-42-01-02B *Three Attendant Stations*

23-42-02 Interphone Handset System

23-42-02-01 Flight Deck - *N/A*

23-42-02-02 Cabin

23-42-03 Call System (Audio/Visual Alerting)

23-42-03-01 Flight Deck Call Visual Alerting System

23-42-03-02 Flight Deck Call Audio Alerting System

23-42-03-03 Flight Attendant Visual Alerting System

23-42-03-04 Flight Attendant Audio Alerting System

23-42-04 Attendant Control Panels

23-51-01 Flight Deck Speakers

23-51-02 Audio Control Panels (ACPs)

23-51-02-01 First Officer

23-51-02-02 Observer

23-51-02-03 Switch Lights

23-51-03 Flight Deck Headsets/Headphones

23-51-03-01 Headset Boom Microphones

23-51-03-01A Hand Microphone Operates Normally

23-51-03-01B Not Required - N/A

23-51-03-02 Headset Earphones/Headphones

23-51-03-03 Active Noise Canceling/Reduction Function - *N/A*

23-51-04 Push-To-Talk (PTT) Switches

23-51-04-01 Control Wheel PTT Switches

23-51-04-02 Audio Control Panel (ACP) PTT Switches

23-51-04-03 Glareshield PTT Switches

23-51-04-03A Required By Procedures - N/A

23-51-04-03B Not Required By Procedures

23-51-05 Flight Deck Hand Microphones

23-51-05A Associated Boom Microphone Operates Normally

23-51-05B Procedures Do Not Require Its Use - N/A

23-51-06 Flight Interphone System

23-51-06-01 Flight Deck to Ground/Ground to Flight Deck Function (Includes Interphone Jacks Located on External Power Panel and/or in Nose Wheel Well Interphone Box)

23-51-06-01A Service Interphone Operates Normally

23-51-06-01B Service Interphone Inoperative

23-51-07 Passenger Cabin Medical Communications System - *N/A*

23-71-01 Cockpit Voice Recorder System (CVR)

23-71-01-01 Recorder Independent Power Supply (RIPS)

23-75-01 Flight Deck Door Visual Surveillance System

23-75-01A Flight Deck Door Viewing Port Inoperative

23-75-01B Flight Deck Door Viewing Port Operates Normally

23-75-01C Procedures Do Not Require Use - N/A

23-11-01 High Frequency (HF) Communication System

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative for operations which do not require use of HF Communications.

OPERATIONS NOTE

HF communications radios are not required when conducting operations with VHF Communications available. Two HF communications radios are required for RNAV 10 (RNP 10) Oceanic and Remote Areas Operations. (WATRS Oceanic)

23-11-01 High Frequency (HF) Communication System

23-11-01-01 HF Data Link

23-11-01-01B Procedures Do No Require Use

Interval	Installed	Required	Procedure
D	1	0	

NOTE: Procedures do not require the use of HF Data Link.

OPERATIONS NOTE

HF communications radios are not required when conducting operations with VHF communications available. Two HF communications radios (Voice) are required for RNAV 10 (RNP 10) Oceanic and Remote Areas Operations (WATRS Oceanic). HF Data Link is not required.

23-12-01 Very High Frequency (VHF) Communication System

Interval	Installed	Required	Procedure
A	3	2	

Two transmitters and two independent communications receivers required for all operations.

a. One operative transmitter/receiver system must be powered by standby bus system (should there be a loss of electrical generator system).
b. At least one (1) HF Transmitter / receiver is operational, and
c. Repairs shall be made within twenty-five (25) flight hours.

MAINTENANCE NOTE

Airplanes with ACARS and without Voice Mode Protection (VMP) or ACARS Over AVLC (Aviation VHF Link Control)

1. VHF-3 cannot be used for ATC voice communication when ACARS is operational.
2. If VHF-3 can be tuned with ACARS deactivated, it can be used for dispatch credit.
3. If VHF-3 cannot be tuned with ACARS deactivated, VHF-3 must be switched with an inoperative VHF-1 or VHF-2 transceiver to be used for dispatch credit.

   NOTE: ACARS may be deactivated by opening and collaring the ACARS AC and DC circuit breakers.

   NOTE: See MEL item 23-27 for dispatch with ACARS inoperative.

OPERATION NOTE

VHF radios are interchangeable.

NOTE: The #1 VHF is powered by the standby system.

See MEL item 23-27 for dispatch with ACARS inoperative.

23-12-01 **Very High Frequency (VHF) Communication System**

23-12-01-01 **VHF Data Link**

23-12-01-01A **Alternate Procedures Used**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

OPERATIONS (O) PROCEDURES

1. Use VHF & HF "Voice" to obtain ATIS, weather reports, ATC clearances and to communicate with Company. Communicate with Company.
2. Manually enter the route waypoint winds into the FMC.

23-12-02 Radio Tuning Panel

Interval	Installed	Required	Procedure
C	3	2	

May be inoperative provided:

a. Left radio tuning panel operates normally.

b. Inoperative radio tuning panel remains OFF.

NOTE: Radio tuning panels are interchangeable.

23-12-02 Radio Tuning Panel

23-12-02-01 Switch Lights, Key Lights

Interval	Installed	Required	Procedure
C	45	0	

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23-24-01 Fixed Emergency Locator Transmitter (ELT)

23-24-01A Inoperative (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be inoperative provided:

a. System is deactivated,
b. Repairs are made within 90 consecutive calendar-days, and
c. Dispatch is prohibited on routes over water more than 200 nm from land.

MAINTENANCE (M) PROCEDURES

Deactivate the ELT (AMM 23-00-00/901).

1. Set the ELT (LRU) front panel switch to OFF.

23-24-01 **Fixed Emergency Locator Transmitter (ELT)**

23-24-01B **Missing (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)**

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be missing provided:

a. Placard stating "ELT not installed" is placed in view of the pilot.
b. Repairs are made within 90 consecutive calendar-days.
c. Dispatch is prohibited on routes over water more than 200 nm from land.

MAINTENANCE (M) PROCEDURES

Install a placard stating "ELT not installed" in view of the pilot

23-24-01 Fixed Emergency Locator Transmitter (ELT)

23-24-01-01 Remote ELT Switch (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)

Interval	Installed	Required	Procedure
D	1	0	**(M)**

May be inoperative provided:

a. Remote ELT Switch is deactivated.
b. ELT Switch is placed in the ARMED mode.

MAINTENANCE (M) PROCEDURES

NOTE: If remote ELT switch is failed so that the ELT system is activated, deactivate the ELT system and Dispatch per MEL 23-24-01A or 23-24-01C.

If ELT does not activate with the remote ELT switch set to ON, deactivate the remote ELT switch (AMM 23-00-00/901).

1. Placard the inoperative remote ELT switch.
2. Set the remote ELT switch to the armed position.
   A. Set the remote ELT switch to ARM.
3. Set the ELT (LRU) front panel switch to the armed position.
   A. Set the ELT (LRU) front panel switch to ARM.

23-24-01 Fixed Emergency Locator Transmitter (ELT)

23-24-01-02 ELT Indicator Light (CAL, GUY, ANT, BAR, JAM, GRN, SUR, BAH, TTO)

Interval	Installed	Required	Procedure
D	1	0	

23-24-02 **Aircraft Autonomous Distress Tracking (ADT) System**

23-24-02-01 **Emergency Locator Transmitter - Distress Tracking (ELT-DT)**

23-24-02-01A **Inoperative (CUB ONLY)**

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be inoperative provided:

a. System is deactivated.
b. Repairs are made within 90 consecutive calendar-days.
c. Dispatch is prohibited on routes over water more than 200 nm from land.

MAINTENANCE (M) PROCEDURES

Deactivate the ELT-DT (AMM 23-00-00/901).

1. Gain access to the ELT-DT in the passenger compartment ceiling.
2. For ACR model ELT-DT, set the ELT-DT (LRU) front panel switch to OFF.

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)

23-24-02-01B Missing (CUB ONLY)

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be missing provided:

a. Placard stating "ELT not installed" is placed in view of the pilot.
b. Repairs are made within 90 consecutive calendar-days.
c. Dispatch is prohibited on routes over water more than 200 nm from land.

MAINTENANCE (M) PROCEDURES

Install a placard stating "ELT not installed" in view of the pilot (AMM 23-00-00/901).

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)

23-24-02-01-01 Remote ELT Switch (CUB ONLY)

Interval	Installed	Required	Procedure
D	1	0	**(M)**

May be inoperative provided:

a. Remote ELT Switch is deactivated.

b. ELT-DT Switch is placed in the ARMED mode.

NOTE: The remote ELT switch is meant to refer to the panel in the flight deck and is not to be confused with the remote function panel (RFP) which is a maintenance device provided by the ELT-DT supplier.

MAINTENANCE (M) PROCEDURES

NOTE: If remote ELT switch is failed so that the ELT-DT system is activated, deactivate the ELT-DT system and dispatch per MEL 23-24-02-01A.

If ELT-DT does not activate with the remote ELT switch set to ON, deactivate the remote ELT switch (AMM 23-00-00/901).

1. Placard the inoperative remote ELT switch.
2. Set the remote ELT switch to ARM.
3. Set the ELT-DT (LRU) front panel switch to ARM.

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)

23-24-02-01-02 ELT Indicator Light (CUB ONLY)

Interval	Installed	Required	Procedure
D	1	0	

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-02 ADT Trigger Unit (ATU)

23-24-02-02A Inoperative (CUB ONLY)

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be inoperative provided:

a. System is deactivated.
b. ELT-DT Switch is set to fixed ELT mode.
c. Repairs are made within 90 consecutive calendar-days.

MAINTENANCE (M) PROCEDURES

Deactivate the ATU and set the ELT-DT to fixed ELT mode (AMM 23-00-00/901).

1. Open and collar the P6-3 panel GADSS ATU circuit breaker (C4117, location A1).
2. Gain access to the ATU Lockout toggle switch (S1265) in the aft cargo compartment (J11 junction box, E6 rack).
3. Position the ATU Lockout toggle switch to ATU LOCKOUT to set the ELT-DT to fixed ELT mode.

23-24-02 Aircraft Autonomous Distress Tracking (ADT) System

23-24-02-02 ADT Trigger Unit (ATU)

23-24-02-02B Missing (CUB ONLY)

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be missing provided:

a. ELT-DT is set to fixed ELT mode.

b. Repairs are made within 90 consecutive calendar-days.

MAINTENANCE (M) PROCEDURES

Set the ELT-DT to fixed ELT mode (AMM 23-00-00/901).

1. Open and collar the P6-3 panel GADSS ATU circuit breaker (C4117, location A1).
2. Gain access to the ATU Lockout toggle switch (S1265) in the aft cargo compartment (J11 junction box, E6 rack).
3. Position the ATU Lockout toggle switch to ATU LOCKOUT to set the ELT-DT to fixed ELT mode.

23-25-01 **Low Frequency Underwater Locating Device (ULD)**

23-25-01B **Installed on ANT, BAR, JAM, GRN, SUR, CUB**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative or missing.

NOTE: Aircraft cannot be assigned flight plans that require the aircraft to be more than 400 nm from land suitable for emergency landing. (Long-Range Over-Water Flights as defined in TTCAR No.7).

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23-27-01 Aircraft Communications Addressing and Reporting System (ACARS)

23-27-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	**(O)**

NOTE: Any portion of system that operates normally may be used.

MAINTENANCE NOTE

1. **An inoperative ACARS could cause a CVR fault.**
2. **If datalink is inoperative, refer to MEL item 23-11-01-01, 23-12-01-01, and/or 23-15-01-01. If CVR and/or RIPS is inoperative, refer to MEL item 23-71-01.**

OPERATIONS (O) PROCEDURES

1. Use VHF & HF "Voice" to obtain ATIS, weather reports, ATC clearances and to communicate with Company.
2. Manually enter the route waypoint winds into the FMC

23-27-01 **Aircraft Communications Addressing and Reporting System (ACARS)**

23-27-01-01 **FMC Interface Function**

23-27-01-01A **Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	2	0	**(O)**

NOTE: Any portion of system that operates normally may be used.

OPERATIONS (O) PROCEDURES

1. Use VHF & HF "Voice" to obtain ATIS, weather reports, ATC clearances and to communicate with Company.
2. Manually enter the route waypoint winds into the FMC.

23-27-01 Aircraft Communications Addressing and Reporting System (ACARS)

23-27-01-02 ACARS Printer

Interval	Installed	Required	Procedure
D	1	0	

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23-28-01 **Selective Call System (SELCAL)**

23-28-01A **Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

OPERATIONS (O) PROCEDURES

1. Flight Crew continuously monitors appropriate HF/VHF frequency.

23-28-01 **Selective Call System (SELCAL)**

23-28-01-01 **Channels (Including Lights)**

23-28-01-01A **Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	5	0	**(O)**

OPERATIONS (O) PROCEDURES

1. Flight Crew continuously monitors appropriate HF/VHF frequency.

23-31-01 **Passenger Address (PA) System**

23-31-01A **Flight Attendant Alerting System Operates Normally**

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided:

a. Flight attendant alerting system (audio or visual) operates normally.

NOTE: Any station function(s) that operates normally may be used.

OPERATIONS (O) PROCEDURES

1. Before takeoff the cabin crew shall make a direct voice communication with all passengers in each cabin. The powered megaphone may be used for making these announcements. The flight crew shall be notified when preflight announcements have been completed.
2. In an emergency the flight crew shall notify the cabin crew who shall establish direct voice contact with each passenger. The powered megaphone may be used for making announcements.
3. Refer to Safety and Emergency Procedures Manual (BW-OM-0008) Chapter 4 – Safety Procedures, Chapter 8 – Emergencies for alternate normal and emergency procedures and/or operating restrictions.

23-31-01 Passenger Address (PA) System

23-31-01-01 Lavatory Speakers

Interval	Installed	Required	Procedure
C	3	0	**(O)**

OPERATIONS (O) PROCEDURES

1. Before taxi cabin crew shall check each lavatory to ensure that it is empty and lock the door until airborne and Seat Belt sign is switched off.
2. Whenever Seat Belt sign is switched on in flight cabin crew shall check the lavatory and request passengers return to their seats.
3. Before landing cabin crew again ensure that lavatories are empty and lock doors until after arrival at the gate.

23-31-01 Passenger Address (PA) System

23-31-01-02 Cabin Speakers

Interval	Installed	Required	Procedure
C	94	47	

May be inoperative provided no adjacent cabin speaker pairs (forward to aft) are inoperative.

23-31-02 **Prerecorded Passenger Announcement System**

23-31-02A **Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

OPERATIONS (O) PROCEDURES

1. Before takeoff the cabin attendant shall make a direct voice communication with all passengers in each cabin.
   The PA system or powered megaphone may be used for making these announcements. The flight crew shall be notified when pre-flight announcements have been completed.

23-41-01 **Service Interphone System**

23-41-01-01 **Flight Deck to Ground/Ground to Flight Deck Function (Includes Interphone Jack Located on External Power Panel)**

23-41-01-01A **Flight Interphone Jack Operates Normally**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Service interphone flight deck to ground/ground to flight deck function may be inoperative provided:

a. Flight interphone jack located on external power panel and/or in nose wheel well interphone box operates normally.

OPERATIONS (O) PROCEDURES

1. Use flight interphone for communications with ground crew during ground handling operations (jack is available at the external power receptacle).
2. If communications still cannot be established, use ICAO Hand Signals.
3. At least one Ground Crewman/woman shall maintain eye contact with Captain during all pushing or towing operations.
4. Captain will brief First Officer and Ground Crew on signals to be used. These will include, but are not limited to, hand signals for "chocks out", "chocks in", "brakes off", "brakes on", "cleared to push-back/tow", "stop push-back/tow", "ok to start engine #", "starting engine #", "engine start failed", "return to gate", "engine start complete", "gear pins out", "remove all ground equipment", "ok to taxi", "shut down engines".

23-41-01 **Service Interphone System**

23-41-01-01 **Flight Deck to Ground/Ground to Flight Deck Function (Includes Interphone Jack Located on External Power Panel)**

23-41-01-01B **Flight Interphone Jack Inoperative**

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided ICAO Hand Signals will be used.

OPERATIONS (O) PROCEDURES

1. At least one Ground Crewman/woman shall maintain eye contact with Captain during all pushing or towing operations.
2. Captain will brief First Officer and Ground Crew on signals to be used. These will include, but are not limited to, hand signals for "chocks out", "chocks in", "brakes off", "brakes on", "cleared to push-back/tow", "stop push-back/tow", "ok to start engine #", "starting engine #", "engine start failed", "return to gate", "engine start complete", "gear pins out", "remove all ground equipment", "ok to taxi", "shut down engines".

23-42-01 Service (Cabin) Interphone System

23-42-01-01 Flight Deck to Cabin, Cabin to Flight Deck Functions

23-42-01-01A 50% Operate Normally

Interval	Installed	Required	Procedure
B	3	2	**(O)**

May be inoperative provided:

a. Flight deck to cabin and cabin to flight deck interphone functions operate normally on at least 50% of cabin handsets.
b. Flight deck to cabin and cabin to flight deck interphone functions operate normally at the forward cabin attendant station.

NOTE: Any station function(s) that operates normally may be used.

OPERATIONS (O) PROCEDURES

1. The PA system may be used to communicate between affected stations.
2. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures contained in the Safety and Emergency Procedures Manual (BW-OM-0008) Chapter 8 – Emergencies.

OPERATIONS NOTES

Security Procedures for admittance to/from the flight deck require that the flight deck to cabin and cabin to flight deck interphone functions operate normally at the forward cabin attendant station (BW-OM-0008) Chapter 3 – Security.

23-42-01 **Service (Cabin) Interphone System**

23-42-01-01 **Flight Deck to Cabin, Cabin to Flight Deck Functions**

23-42-01-01B **Inoperative**

Interval	Installed	Required	Procedure
C	3	0	**(O)**

May be inoperative for ferry to repair station provided the flight deck door remains unlocked for the duration of the flight.

NOTE: Any station function(s) that operates normally may be used.

OPERATIONS (O) PROCEDURES

1. The PA system may be used to communicate between affected stations.
2. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures contained in the Safety and Emergency Procedures Manual (BW-OM-0008) Chapter 8 – Emergencies.
3. The flight deck door must remain unlocked for the duration of the flight to allow normal and emergency access to/from the flight deck.

23-42-01 **Service (Cabin) Interphone System**

23-42-01-02 **Cabin to Cabin Functions**

23-42-01-02B **Three Attendant Stations**

Interval	Installed	Required	Procedure
B	3	2	**(O)**

May be inoperative provided cabin to cabin interphone functions operate normally on at least 50% of cabin handsets.

NOTE: Any station function(s) that operates normally may be used.

OPERATIONS (O) PROCEDURES

1. The PA system may be used to communicate between affected stations.
2. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures contained in the Safety and Emergency Procedures Manual (BW-OM-0008) Chapter 8 – Emergencies.

23-42-02 Interphone Handset System

23-42-02-02 Cabin

Interval	Installed	Required	Procedure
B	3	2	**(O)**

May be inoperative provided:

a. 50% of cabin handsets operate normally.
b. The handset at the forward cabin attendant station operates normally.

NOTE 1: An operative handset at an inoperative flight attendant seat shall not be counted to satisfy 50% requirement.

NOTE 2: Any handset functions that operate normally may be used.

OPERATIONS (O) PROCEDURES

1. The PA system may be used to communicate between affected stations.
2. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures contained in the Safety and Emergency Procedures Manual (BW-OM-0008) Chapter 8 – Emergencies.

OPERATIONS NOTE

Security Procedures for admittance to/from the flight deck require that the handset at the forward cabin attendant station operates normally (BW-OM-0008) Chapter 3 – Security.

23-42-03 Call System (Audio/Visual Alerting)

23-42-03-01 Flight Deck Call Visual Alerting System

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

a. Flight deck audio alerting system operates normally.

b. Flight deck audio alerting system differentiates between normal and emergency calls.

23-42-03 Call System (Audio/Visual Alerting)

23-42-03-02 Flight Deck Call Audio Alerting System

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

a. Flight deck visual alerting system operates normally
b. Flight deck visual alerting system differentiates between normal and emergency calls.

OPERATIONS NOTE

To assist the flight deck crew in the recognition of a cabin to flight deck interphone call, the cabin crew may also knock on the flight deck door two times for normal communication or three times for emergency communication. This should be included in the pre-flight briefing between the pilot in command and the cabin crew members.

23-42-03 Call System (Audio/Visual Alerting)

23-42-03-03 Flight Attendant Visual Alerting System

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided:

a. PA system operates normally.
b. If affected visual alerting system is used for lavatory smoke detector alerting, an alternate lavatory smoke detector alert (visual or audio) is installed and operates normally.

NOTE 1: Passenger to Attendant Call System is considered Nonessential Equipment and Furnishing (NEF) item.

NOTE 2: Any visual alerting system function(s) that operates normally may be used.

OPERATIONS (O) PROCEDURES

Alternate procedures will be based on the remaining operative functions of the alerting system or on the use of the PA System. Prior to flight, the pilot in command should brief cabin crew members on the alternate procedures to be used. Discreet and unambiguous phraseology should be used by the flight deck crew via the PA system to alert the cabin crew of the need to communicate via the Interphone System, such as "Interphone Call". This phraseology avoids any conflict with other calls such as "Purser to Cockpit" and "Crew to Stations".

Refer to Safety and Emergency Procedures Manual Doc. No.: BW-OM-0008 Chapter 3 – Security, Chapter 4 – Safety Procedures, Chapter 8 – Emergencies.

23-42-03 Call System (Audio/Visual Alerting)

23-42-03-04 Flight Attendant Audio Alerting System

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided:

a. PA system operates normally.
b. If affected audio alerting system is used for lavatory smoke detector alerting, an alternate lavatory smoke detector alert (visual or audio) is installed and operates normally.

NOTE 1: Passenger to Attendant Call System is considered Nonessential Equipment and Furnishing (NEF) item.

NOTE 2: Any audio alerting system function(s) that operates normally may be used.

OPERATIONS (O) PROCEDURES

Alternate procedures will be based on the remaining operative functions of the alerting system or on the use of the PA System. Prior to flight, the pilot in command should brief cabin crew members on the alternate procedures to be used. Discreet and unambiguous phraseology should be used by the flight deck crew via the PA system to alert the cabin crew of the need to communicate via the Interphone System, such as "Interphone Call". This phraseology avoids any conflict with other calls such as "Purser to Cockpit" and "Crew to Stations".

Refer to Safety and Emergency Procedures Manual Doc. No.: BW-OM-0008 Chapter 3 – Security, Chapter 4 – Safety Procedures, Chapter 8 – Emergencies.

23-42-04 Attendant Control Panels

Interval	Installed	Required	Procedure
C	2	1	**(O)**

NOTE: Any portion of the system that operates normally may be used.

INTENTIONALLY
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23-51-01 Flight Deck Speakers

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided:

a. Procedures do not require their use.

b. Headset or headphones associated with inoperative speaker operate normally and are used.

23-51-02 Audio Control Panels (ACPs)

23-51-02-01 First Officer

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided observer audio control panel operates normally.

23-51-02 Audio Control Panels (ACPs)

23-51-02-02 Observer

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

a. First officer audio control panel operates normally.

b. Repairs are made within 2 flight days.

23-51-02 **Audio Control Panels (ACPs)**

23-51-02-03 **Switch Lights**

Interval	Installed	Required	Procedure
C	26	0	

23-51-03 Flight Deck Headsets/Headphones

Interval	Installed	Required	Procedure
C	3	1	

Either Captain or Fist Officer's headset may be inoperative provided associated flight deck speaker operates normally.

23-51-03 Flight Deck Headsets/Headphones

23-51-03-01 Headset Boom Microphones

23-51-03-01A Hand Microphone Operates Normally

Interval	Installed	Required	Procedure
A	3	2	

May be inoperative provided:

a. Associated hand microphone is installed and operates normally.
b. Repairs are made within 3 flight days.
c. Both the Captain's and First Officer's headset boom microphones are operational.
d. The Observer's seat is NOT occupied by a REQUIRED crew member.

OPERATIONS NOTE

The Captain and First Officer are required to wear operational headset boom microphones at all times during flights below 15,000 ft/FL150. Any required crew member occupying the observer's jumpseat, such as a Check Airman conducting training, must also wear an operational headset boom microphone at all times during flights below 15,000 ft/FL150.

23-51-03 Flight Deck Headsets/Headphones

23-51-03-02 Headset Earphones/Headphones

Interval	Installed	Required	Procedure
C	3	1	

Either Captain or Fist Officer's headset may be inoperative provided associated flight deck speaker operates normally.

23-51-04 Push-To-Talk (PTT) Switches

23-51-04-01 Control Wheel PTT Switches

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided:

a. Inoperative switch is either verified failed open or is deactivated.
b. Associated audio control panel PTT switch operates normally.

MAINTENANCE (M) PROCEDURES

Verify the inoperative control wheel PTT switch is failed open or deactivate the PTT switch (AMM 23-00-00/901).

1. Control wheel PTT switch is electrically failed open if there is no sidetone during a trial transmission not using the associated PTT switch.
2. Use the audio control panel PTT switch to verify R/T and/or I/C communications.
3. For PTT switch deactivation, disconnect, cap and stow the ground wire of the control wheel PTT switch.

23-51-04 Push-To-Talk (PTT) Switches

23-51-04-02 Audio Control Panel (ACP) PTT Switches

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided:

a. Inoperative switch is verified failed open.
b. Associated control wheel PTT switch operates normally.

MAINTENANCE (M) PROCEDURES

Verify the inoperative flight crew audio control panel PTT switch is failed open (AMM 23-00-00/901).

1. Audio control panel PTT switch is electrically failed open if there is no sidetone during a trial transmission not using the associated PTT switch.
2 Use the control wheel PTT switch to verify communications can be established.

23-51-04 Push-To-Talk (PTT) Switches

23-51-04-03 Glareshield PTT Switches

23-51-04-03B Not Required By Procedures

Interval	Installed	Required	Procedure
D	2	0	**(M)**

May be inoperative provided inoperative switch is verified failed open or is deactivated.

MAINTENANCE (M) PROCEDURES

Verify the inoperative glareshield panel PTT switch is failed open (AMM 23-00-00/901).

1. Glareshield panel PTT switch is electrically failed open if there is no sidetone during a trial transmission not using the associated PTT switch.
2. Use the control wheel PTT switch to verify communications can be established.
3. For PTT switch deactivation, disconnect, cap and stow the ground wire of the glareshield PTT switch.

23-51-05 **Flight Deck Hand Microphones**

23-51-05A **Associated Boom Microphone Operates Normally**

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative or missing provided associated boom microphone operates normally.

23-51-06 **Flight Interphone System**

23-51-06-01 **Flight Deck to Ground/Ground to Flight Deck Function (Includes Interphone Jacks Located on External Power Panel and/or in Nose Wheel Well Interphone Box)**

23-51-06-01A **Service Interphone Operates Normally**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Flight interphone flight deck to ground/ground to flight deck function may be inoperative provided external power service interphone jack operates normally.

a. Service interphone jack located on external power panel operates normally.

OPERATIONS (O) PROCEDUREs

1. Use service interphone for communications with ground crew during ground handling operations (jack is available at the external power receptacle).
2. If communications still cannot be established, use ICAO Hand Signals.
3. At least one Ground Crewman/woman shall maintain eye contact with Captain during all pushing or towing operations.
4. Captain will brief First Officer and Ground Crew on signals to be used. These will include, but are not limited to, hand signals for "chocks out", "chocks in", "brakes off", "brakes on", "cleared to push-back/tow", "stop push-back/tow", "ok to start engine #", "starting engine #", "engine start failed", "return to gate", "engine start complete", "gear pins out", "remove all ground equipment", "ok to taxi", "shut down engines"

23-51-06 **Flight Interphone System**

23-51-06-01 **Flight Deck to Ground/Ground to Flight Deck Function (Includes Interphone Jacks Located on External Power Panel and/or in Nose Wheel Well Interphone Box)**

23-51-06-01B **Service Interphone Inoperative**

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided ICAO Hand Signals will be used.

OPERATIONS (O) PROCEDURES

1. At least one Ground Crewman/woman shall maintain eye contact with Captain during all pushing or towing operations.
2. Captain will brief First Officer and Ground Crew on signals to be used. These will include, but are not limited to, hand signals for "chocks out", "chocks in", "brakes off", "brakes on", "cleared to push-back/tow", "stop push-back/tow", "ok to start engine #", "starting engine #", "engine start failed", "return to gate", "engine start complete", "gear pins out", "remove all ground equipment", "ok to taxi", "shut down engines".

INTENTIONALLY
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23-71-01 Cockpit Voice Recorder System (CVR)

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be inoperative provided:

a. Flight Data Recorder (FDR) operates normally.
b. Recorder Independent Power Supply (RIPS) is deactivated.
c. Repairs are made within 3 flight days.

MAINTENANCE (M) PROCEDURES

Deactivate RIPS (AMM 23-00-00/901):

1. At least 15 minutes before departure, open and collar the P18-2 VOICE RCDR/RIPS circuit breaker.

23-71-01 Cockpit Voice Recorder System (CVR)

23-71-01-01 Recorder Independent Power Supply (RIPS)

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided RIPS battery is removed.

MAINTENANCE (M) PROCEDURES

Remove the RIPS battery (AMM 23-00-00/901).

1. Open and collar the P18-2 VOICE RCDR/RIPS circuit breaker.
2. Disconnect the J1 connector from the RIPS.
3. Remove battery from RIPS.
4. Cover battery pack connector on the RIPS charger assembly with an electrical connector cap.
5. Secure electrical connector cap to charger assembly with tape.
6. Connect the J1 connector to the RIPS.
7. Close the P18-2 VOICE RCDR/RIPS circuit breaker.

23-75-01 Flight Deck Door Visual Surveillance System

23-75-01A Flight Deck Door Viewing Port Inoperative

Interval	Installed	Required	Procedure
A	1	0	**(O)**

May be inoperative and components may be missing provided:

a. The pilot in command is satisfied that sufficient portions of the system remain operative for the flight deck crew to view the external area around the flight deck door.
b. Repairs are made within 3 flight days.

NOTE: Any portion of the system which operates normally may be used.

OPERATIONS (O) PROCEDURES

Where a person wishes to exit the flight crew compartment, such person, prior to unlocking the door shall:

1. Verify with the cabin crew that the adjacent lavatory is not occupied by a passenger.
2. Visually determine the exact circumstances existing outside the door, particularly on high risk flights or where unknown passengers are within easy access of the flight crew compartment door.
3. Confirm that cabin crew is positioned to block passenger access to the door area when it is unlocked.

23-75-01 Flight Deck Door Visual Surveillance System

23-75-01B Flight Deck Door Viewing Port Operates Normally

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative and components may be missing provided the flight deck door viewing port is installed and operates normally.

NOTE: Any portion of the system which operates normally may be used.

OPERATIONS (O) PROCEDURES

Where a person wishes to exit the flight crew compartment, such person, prior to unlocking the door shall:

1. Verify with the cabin crew that the adjacent lavatory is not occupied by a passenger.
2. Visually determine the exact circumstances existing outside the door, particularly on high risk flights or where unknown passengers are within easy access of the flight crew compartment door.
3. Confirm that cabin crew is positioned to block passenger access to the door area when it is unlocked.

24-00-01 Frequency Indications (CPS FREQ)

24-00-02 AC Voltage Indications (AC VOLTS)

24-00-03 AC Ammeter (AC AMPS)

24-00-04 DC Ammeter Indications (DC AMPS)

24-00-04-01 Without AUX BAT – *N/A*

24-00-04-02 With AUX BAT

24-00-05 DC Voltmeter Indications (DC Volts)

24-00-05-01 Without AUX BAT - *N/A*

24-00-05-02 With AUX BAT

24-11-01 Engine Driven Generator Systems

24-11-02 Engine Driven Generator DRIVE Lights

24-11-03 Engine Driven Generator Low Oil Pressure Switch

24-21-01 APU Generator System

24-28-01 GEN OFF BUS Lights

24-28-02 APU GEN OFF BUS Light

24-30-01 ELEC Light

24-31-01 BAT DISCHARGE Light

24-31-02 TR UNIT Light

24-41-01 External Power System

INTENTIONALLY
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24-00-01 Frequency Indications (CPS FREQ)

Interval	Installed	Required	Procedure
C	7	1	**(O)**

May be inoperative provided indication is verified to operate normally for STBY PWR position.

OPERATIONS (O) PROCEDURES

Verify normal operation of the frequency indication with the associated AC selector in the STBY PWR position using the FCOM Supplementary Procedures - Electrical - Standby Power Test.

24-00-02 AC Voltage Indications (AC VOLTS)

Interval	Installed	Required	Procedure
C	7	1	**(O)**

May be inoperative provided indication is verified to operate normally for STBY PWR position.

OPERATIONS (O) PROCEDURES

Verify normal operation of the AC VOLTS indication with the associated AC selector in the STBY PWR position using the FCOM Supplementary Procedures - Electrical - Standby Power Test.

24-00-03 AC Ammeter (AC AMPS)

Interval	Installed	Required	Procedure
C	7	0	

May be inoperative provided associated GEN OFF BUS lights operate normally.

24-00-04 DC Ammeter Indications (DC AMPS)

24-00-04-02 With AUX BAT

Interval	Installed	Required	Procedure
C	8	2	**(O)**

May be inoperative provided:

a. Battery positions operate normally.
b. Procedures do not require its use.

OPERATIONS (O) PROCEDURES

Verify normal operation of the DC AMPS indication with the associated DC selector in the BAT and AUX BAT positions, using the FCOM Supplementary Procedures - Electrical - Standby Power Test.

24-00-05 **DC Voltmeter Indications (DC Volts)**

24-00-05-02 **With AUX BAT**

Interval	Installed	Required	Procedure
C	8	3	**(O)**

May be inoperative provided:

a. Battery positions operate normally.
b. STBY PWR position operates normally.

OPERATIONS (O) PROCEDURES

Verify normal operation of the DC VOLTS indication with the associated DC selector in the BAT, AUX BAT, and STBY PWR positions, using the FCOM Supplementary Procedures - Electrical - Standby Power Test.

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24-11-01 Engine Driven Generator Systems

Interval	Installed	Required	Procedure
B	2	1	**(M) (O)**

Except for ER operations, may be inoperative provided:

a. Associated engine driven generator is disconnected.
b. APU generator operates normally and is used to supply associated buses throughout flight.

MAINTENANCE (M) PROCEDURES

NOTE: During the authorized repair interval, if the associated engine is operated for more than 50 hours with the engine driven generator disconnected, then the engine driven generator should be removed for shop inspection prior to returning the airplane to service.

Disconnect the engine driven generator and remove the lamp from appropriate lights (AMM 24-00-00/901).

1. Disconnect the engine driven generator:
   A. To reduce potential damage to the associated engine driven generator, it is recommended the generator's oil level be confirmed normal.
   B. Supply external power or APU generator power to the airplane (AMM 24-22-00/201).
   C. Start the associated engine.
   D. Set the AC meters selector on the AC and DC Metering Panel to the associated GEN position.
   E. Lift the DISCONNECT switch guard.
   F. Push the DISCONNECT switch to the DISCONNECT position when the engine is at or above idle speed.
   G. Close the DISCONNECT switch guard.
   H. Confirm the DRIVE light illuminates.
   I. Confirm these AC meter indications on the P-5 panel:
      1) AC VOLTS indication shows 0 volts.
      2) CPS FREQ indication shows 0.
   J. Stop the engine.
2. Remove the lamp from these associated lights:
   A. Generator DRIVE light on the Generator Drive and Standby Power Panel.
   B. GEN OFF BUS light on Ground Power Panel and Bus Switching Panel.
3. Service APU oil level to full. APU oil quantity is serviced prior to deferral and ONCE EACH FLIGHT DAY (logbook entry required).

OPERATIONS (O) PROCEDURE

NOTE: With the generator disconnected, the ELEC light will illuminate during Master Caution recall and will extinguish when reset.

1. Use the APU generator to power the bus associated with the inoperative engine driven generator system.
2. Increase flight planning fuel by either incrementing the current airplane drag correction factor by 0.9% or flight planning fuel by 0.9% plus 45 kg per hour of APU usage in flight and 109 kg per hour of APU usage on the ground.

24-11-02 Engine Driven Generator DRIVE Lights

Interval	Installed	Required	Procedure
C	2	0	

24-11-03 Engine Driven Generator Low Oil Pressure Switch

Interval	Installed	Required	Procedure
C	2	0	

INTENTIONALLY

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24-21-01 APU Generator Systems

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided both engine driven generator systems operate normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

INTENTIONALLY
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24-28-01 GEN OFF BUS Lights

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided associated generator AC AMPS indication operates normally.

OPERATIONS NOTE

If desired, the associated AC AMPS indication can be used to determine when the generator does not supply power to the generator bus.

24-28-02 APU GEN OFF BUS Light

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. APU generator CPS FREQ indication operates normally.
b. APU generator AC AMPS indication operates normally.

OPERATIONS NOTE

1. With the AC meters selector in APU GEN, the CPS FREQ indication can be used to confirm proper APU generator rotational speed.
2. With APU running and either APU GEN switch selected ON, and the AC meters selector in APU GEN, the AC AMPS indication can be used to confirm the APU generator is supplying a generator bus.

24-30-01 ELEC Light

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative off provided:

a. Standby Power Test is accomplished once each flight day.
b. Battery charger is verified to operate normally.

OPERATIONS (O) PROCEDURES

Verify the battery charger operates normally, and once each flight day, confirm the standby power system operates normally.

1. Once each flight day, do the Flight Crew Operations Manual (FCOM) Supplementary Procedures - Electrical - Standby Power Test.
2. For dual batteries installed, verify the battery changers operates normally.
   A. Supply AC electrical power.
   B. Set the DC meters selector to AUX) BAT.
   C. Set the STANDBY POWER switch to BAT.
   D. Wait 5 seconds.
   E. Set the STANDBY POWER switch to AUTO.
   F. Confirm the DC AMPS indication is greater than or equal to 10 amps.
   G. Set the DC meters selector to BAT.
   H. Confirm the DC AMPS indication is greater than or equal to 10 amps.
   I. If the DC AMPS indication is less than 10 Amps, set the STANDBY POWER switch to BAT.
      1) Wait 5 seconds.
      2) Set the STANDBY POWER switch to AUTO.
      3) Confirm the DC AMPS indication is greater than or equal to 10 amps.
   J. Set the DC meters selector to the normal position.

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24-31-01 BAT DISCHARGE Light

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

If desired, a discharging battery can be confirmed by the following:

1. Set the DC indication selector to BAT.
2. Confirm the DC AMPS indication is negative.
3. Set the DC indication selector to AUX BAT.
4. Confirm the DC AMPS indication is negative.

24-31-02 TR UNIT Light

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided the transformer rectifiers are verified to operate normally.

OPERATIONS (O) PROCEDURES

Verify normal operation of the TRs.

1. Set the DC meters selector to TR1.
2. Confirm the DC AMPS indication is greater than 5 amps.
3. Set the DC meters selector to TR2.
4. Confirm the DC AMPS indication is greater than 5 amps.
5. Set the DC meters selector to TR3.
6. Confirm the DC VOLTS indication is 28 ± 6 volts.

24-41-01 External Power System

Interval	Installed	Required	Procedure
C	1	0	

NOTE: Any portion of system that operates normally may be used.

INTENTIONALLY

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25-00-01 Nonessential Equipment & Furnishings (NEF)

25-11-01 Flight Crew Seats

25-11-01-01 Recline Mechanism

25-11-01-02 Vertical Adjustment

25-11-01-03 Armrests

25-11-01-04 Lumbar/Thigh Supports

25-11-01-05 Headrests

25-11-02 Observer Seats

25-11-02-01 First Observer Seat (Including Associated Equipment)

25-11-02-01A Passenger Seat Available

25-11-02-01B Second Observer Seat Available - N/A

25-11-02-01C Associated Safety Equipment Acceptable

25-11-02-02 Second Observer Seat (Including Associated Equipment) - *N/A*

25-20-01 FASTEN SEAT BELT WHILE SEATED Signs or Placards

25-22-01 Passenger Seats (Includes all Configurations and Locations)

25-22-01-01 Positioning Controls for Taxi, Takeoff, and Landing (TTL) (Mechanical and/or Electrical)

25-22-01-01A Seat Secured in the TTL Position

25-22-01-01B Seat Immovable in the TTL Position

25-22-01-02 Underseat Baggage Restraining System

25-22-01-03 Armrests

25-22-01-03-01 Armrests with Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls

25-22-01-03-02 Armrests without Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls

25-22-01-04 Seat Belt Air Bag Restraint Systems - *N/A*

25-22-01-05 Delethalization Pads - *N/A*

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)

25-25-01-01 Required Flight Attendant Seats

25-25-01-01A Passengers Carried

25-25-01-01B No Passengers Carried

25-25-01-02 Excess Flight Attendant Seats

25-28-01 Storage Bins/Cabin, Galley, and Lavatory Storage Compartments/Closets

25-28-01A Secured Closed

25-28-01B Door Removed or Retractable Door Secured Open

25-28-01C Failed Closed

25-28-01-01 Multi Latch/Quarter-Turn Lug Installations

25-28-01-02 Storage Compartment Key Locks - *N/A*

25-31-01 Galley/Cabin Waste Receptacle Access Doors/Covers

25-40-01 Exterior Lavatory Door Ashtrays

25-40-01-01 Airplanes With Multiple Exterior Lavatory Door Ashtrays Installed

25-40-01-01A 50% Missing or Inoperative

25-40-01-01B More than 50% Missing or Inoperative

25-40-01-02 Airplanes With Only One Exterior Lavatory Door Ashtray Installed - *N/A*

25-40-02 Lavatory Waste Container Flapper/Access Doors

25-62-01 Flotation Equipment (Crew and Passengers)

25-63-01 Emergency Evacuation Signal System - *N/A*

25-63-02 Emergency Locator Transmitter (ELT) (Survival)

25-64-01 Megaphones

25-64-01A Passengers Carried - N/A

25-64-01B No Passengers Carried

25-40-01-02 Tamper Seals or Tags - *N/A*

25-64-02 Portable Flashlights / Flashlight Holders

25-64-02A Required by TTCAR

25-64-02B Not Required by TTCAR N/A

25-64-02-01 Tamper Seals or Tags - *N/A*

25-64-03 Emergency Medical Equipment

25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment

25-64-03-01A *Required BY TTCAR*

25-64-03-01B *1 in Excess of TTCAR Requirement*

25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment - *N/A*

25-64-03-03 Automatic External Defibrillators (AED) and/or Associated Equipment - *N/A*

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25-00-01 Nonessential Equipment & Furnishings (NEF)

Interval	Installed	Required	Procedure
D	37	0	

May be inoperative, damaged, or missing provided that item(s) is deferred in accordance with operator's NEF deferral program. NEF program, procedures, and processes must be outlined in operator's appropriate document. (M) and (O) procedures, if required, must be available to flight crew and included in operator's appropriate document.

NOTE: Exterior lavatory door ash trays are not considered NEF items.

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25-11-01 Flight Crew Seats

25-11-01-01 Recline Mechanism

Interval	Installed	Required	Procedure
A	2	0	**(M)**

May be inoperative provided:

a. Seat is secured in a position acceptable to affected crewmember.
b. Repairs are made within two flight days.

MAINTENANCE (M) PROCEDURES

Secure seat in an acceptable position for the affected crewmember (AMM 25-00-00/901).

1. Perform a functional test to verify the seat back cannot be moved.
2. Verify that full travel of the seat on its track is not impaired.

25-11-01 Flight Crew Seats

25-11-01-02 Vertical Adjustment

Interval	Installed	Required	Procedure
A	2	0	**(M)**

May be inoperative provided:

a) Seat is secured in a position acceptable to affected crewmember.
b) Repairs are made within two (2) flight days.

MAINTENANCE (M) PROCEDURES

Secure seat in an acceptable position for the affected crewmember (AMM 25-00-00/901).

1 . Perform a functional test to verify the seat cannot be moved in a vertical direction.
2. If necessary, secure the seat in a fixed vertical position by disconnecting the vertical adjustment operating cable or removing the vertical adjustment pushrod mechanism, as appropriate.

25-11-01 **Flight Crew Seats**

25-11-01-03 **Armrests**

Interval	Installed	Required	Procedure
B	4	0	**(M)**

May be inoperative in up position or removed provided seat is acceptable to affected crewmember.

a) Affected armrest is stowed in retracted position or removed.
b) Seat is acceptable to affected crewmember.

MAINTENANCE (M) PROCEDURES

Secure an inoperative armrest in the up position or remove from seat (AMM 25-00-00/901).

25-11-01 **Flight Crew Seats**

25-11-01-04 **Lumbar/Thigh Supports**

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided seat is acceptable to affected crewmember.

25-11-01 Flight Crew Seats

25-11-01-05 Headrests

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative or missing provided seat is acceptable to affected crewmember.

25-11-02 **Observer Seats**

25-11-02-01 **First Observer Seat (Including Associated Equipment)**

25-11-02-01A **Passenger Seat Available**

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

a. A passenger seat in passenger cabin is made available to a TTCAA inspector for performance of official duties.

b. Repairs are made within two (2) flight days.

NOTE: For inoperative observer audio control panel (ACP), reference MEL item 23-51-02.

25-11-02 Observer Seats

25-11-02-01 First Observer Seat (Including Associated Equipment)

25-11-02-01C Associated Safety Equipment Acceptable

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

a. Required minimum safety equipment (safety belt and oxygen) is available.
b. Seat is acceptable to a TTCAA inspector for performance of official duties.
c. Repairs are made within two (2) flight days.

NOTE 1: These provisos are intended to provide for occupancy of above seats by a TTCAA inspector when minimum safety equipment (oxygen and safety belt) is functional and inspector determines conditions to be acceptable.

NOTE 2: Pilot-in-Command will determine if minimum safety equipment is functional for other persons authorized to occupy any observer seat(s).

NOTE: For inoperative observer audio control panel (ACP), reference MEL item 23-51-02.

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25-20-01 FASTEN SEAT BELT WHILE SEATED Signs or Placards

Interval	Installed	Required	Procedure
C	56	28	

One or more signs or placards may be illegible or missing provided a legible sign or placard is visible from each occupied passenger seat.

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25-22-01 Passenger Seats (Includes all Configurations and Locations)

Interval	Installed	Required	Procedure
D	160	0	

May be inoperative provided:

a. Seat does not restrict access to any emergency exit, egress route, or main aisle.

b. Affected seat(s) is blocked and placarded "DO NOT OCCUPY".

NOTE 1: A seat with an inoperative or missing seat belt or shoulder harness is considered inoperative.

NOTE 2: Affected seat(s) may include seat(s) behind and/or adjacent outboard seats.

NOTE 3: Inoperative seats do not affect required number of flight attendants.

25-22-01 **Passenger Seats (Includes all Configurations and Locations)**

25-22-01-01 **Positioning Controls for Taxi, Takeoff, and Landing (TTL) (Mechanical and/or Electrical)**

25-22-01-01A **Seat Secured in the TTL Position**

Interval	Installed	Required	Procedure
D	160	0	**(M)**

May be inoperative and seat occupied provided seat is secured in the taxi, takeoff, and landing (TTL) position.

NOTE: Any position controls that operate normally may be used.

MAINTENANCE (M) PROCEDURES

Secure the associated seat back in the full upright position (AMM 25-00-00/901).

RECLINE MECHANISM INOPERATIVE, SEAT IS SECURED IN UPRIGHT POSITION:

Refer to the seat manufacturer's information for securing seat in upright position.

SEAT INOPERATIVE OTHER THAN UPRIGHT POSITION:

1. Secure seat in breakover position with a strap or rope such that seat cannot be moved during flight.
   OR
2. If seat is stuck in the recline position:
   A. Verify that emergency exit or access to the aisle is not blocked.
   B. Block off row of seats aft of inoperative seat and placard the entire row DO NOT OCCUPY.

25-22-01 **Passenger Seats (Includes all Configurations and Locations)**

25-22-01-01 **Positioning Controls for Taxi, Takeoff, and Landing (TTL) (Mechanical and/or Electrical)**

25-22-01-01B **Seat Immovable in the TTL Position**

Interval	Installed	Required	Procedure
D	160	0	

May be inoperative and seat occupied provided seat is immovable in the taxi, takeoff, and landing (TTL) position.

NOTE: Any position controls that operate normally may be used.

25-22-01 Passenger Seats (Includes all Configurations and Locations)

25-22-01-02 Under Seat Baggage Restraining System

Interval	Installed	Required	Procedure
C	56	0	**(O)**

May be inoperative provided:

a. Baggage is not stowed under seat with inoperative restraining system.
b. Associated seat is placarded "DO NOT STOW BAGGAGE UNDER THIS SEAT".

OPERATIONS (O) PROCEDURES

Ensure cabin crew is notified of the inoperative restraining system and that baggage is not to be placed under the associated seat.

25-22-01 **Passenger Seats (Includes all Configurations and Locations)**

25-22-01-03 **Armrests**

25-22-01-03-01 **Armrests with Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls**

Interval	Installed	Required	Procedure
D	148	0	**(M)**

May be inoperative or missing and seat occupied provided:

a. Armrest does not restrict access to any emergency exit, egress route, or main aisle.
b. If Armrest with seat control is missing or removed, seat is secured in taxi, takeoff, and landing (TTL) position.

MAINTENANCE (M) PROCEDURES

If required, secure the associated seat in the full upright position (AMM 25-00-00/901).

25-22-01 **Passenger Seats (Includes all Configurations and Locations)**

25-22-01-03 **Armrests**

25-22-01-03-02 **Armrests without Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls**

Interval	Installed	Required	Procedure
D	12	0	

May be inoperative or missing and seat occupied provided it does not restrict access to any emergency exit, egress route, or main aisle.

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)

25-25-01-01 Required Flight Attendant Seats

25-25-01-01A Passengers Carried

Interval	Installed	Required	Procedure
B	3	2	**(M) (O)**

One seat position or assembly (dual position) may be inoperative provided:

a. Forward jump set is considered inoperative once inboard seat is defective in any way or as determined by b, c, d, and e.
b. Affected seat or seat assembly is not occupied.
c. Flight attendant(s) displaced by inoperative seat(s) occupies either an adjacent flight attendant seat or passenger seat which is most accessible to inoperative seat(s), so as to most effectively perform assign duties.
d. Alternate procedures are established and used as published in crewmembers manuals.
e. Folding type seat stows automatically or is secured in retracted position.
f. Passenger seat assigned to flight attendant is placarded "FOR FLIGHT ATTENDANT USE ONLY".

NOTE 1: An automatic folding seat that will not stow automatically is considered inoperative.

NOTE 2: A seat position with an inoperative or missing restraint system is considered inoperative.

NOTE 3: Individual operators, when operating with inoperative seats, will consider locations and combinations of seats to ensure that proximity to exits and distribution requirements are met.

NOTE 4: If one side of a dual seat assembly is inoperative and a flight attendant is displaced to adjacent seat, adjacent seat must operate normally.

MAINTENANCE (M) PROCEDURES

For automatic stowing inoperative, remove or secure the inoperative seat in the stowed position (AMM 25-00-00/901).

OPERATIONS (O) PROCEDURES

The flight attendant assigned to the affected seat must occupy a passenger seat as close to, or closer than, the nearest seated passenger to the associated exit.

The flight attendant must be able to reach the assigned emergency exit in essentially the same time as from the normally assigned seat, i.e., a 2 or 3 second time difference is considered "essentially the same time".

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)

25-25-01-01 Required Flight Attendant Seats

25-25-01-01B No Passengers Carried

Interval	Installed	Required	Procedure
C	3	0	**(M) (O)**

May be inoperative provided:

a. No passengers are carried.
b. Folding type seat stows automatically or is secured in the retracted position.
c. Affected seat position or seat assembly is not occupied.

NOTE 1: An automatic folding seat that will not stow automatically is considered inoperative.

NOTE 2: A seat position with an inoperative or missing restraint system is considered inoperative.

MAINTENANCE (M) PROCEDURES

For automatic stowing inoperative, secure seat in the stowed position (AMM 25-00-00/901).

OPERATIONS (O) PROCEDURES

1. No passengers may be carried.
2. The flight attendant assigned to the affected seat must occupy a passenger seat as close to the associated exit. The flight attendant must be able to reach the assigned emergency exit in essentially the same time as from the normally assigned seat, i.e., a 2 or 3 second time difference is considered "essentially the same time".

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)

25-25-01-02 Excess Flight Attendant Seats

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided:

a. Affected seat position or seat assembly is not occupied.
b. Folding type seat stows automatically or is secured in the retracted position.

NOTE 1: An automatic folding seat that will not stow automatically is considered inoperative.

NOTE 2: A seat position with an inoperative or missing restraint system is considered inoperative.

MAINTENANCE (M) PROCEDURES

For automatic stowing function inoperative, remove or secure the inoperative seat in the stowed position (AMM 25-00-00/901).

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25-28-01 Storage Bins/Cabin, Galley, and Lavatory Storage Compartments/Closets

25-28-01A Secured Closed

Interval	Installed	Required	Procedure
C	40	0	**(M)**

May be inoperative provided:

a. Procedures are established to secure affected bin, compartment, or closet in closed position.
b. Affected bin, compartment, or closet is prominently placarded "DO NOT USE".
c. Any emergency equipment located in affected bin, compartment, or closet is considered inoperative.
d. Affected bin, compartment, or closet is not used for storage of any items except for those permanently affixed.

NOTE 1: For overhead bins, if no partitions are installed, entire overhead bin is considered inoperative.

NOTE 2: Proviso is not intended to preclude crewmember inspections.

NOTE: Also dispatch the airplane using the appropriate MEL item for any associated emergency equipment.

MAINTENANCE (M) PROCEDURES

Secure the bin, compartment or closet closed using appropriate procedures (AMM 25-00-00/901).

25-28-01 Storage Bins/Cabin, Galley, and Lavatory Storage Compartments/Closets

25-28-01B Door Removed or Retractable Door Secured Open

Interval	Installed	Required	Procedure
C	40	0	**(M) (O)**

May be inoperative provided:

a. For non-retractable doors, affected door is removed.
b. For retractable doors, affected door is removed or secured in retracted (fully open) position.
c. Affected bin, compartment, or closet is prominently placarded "DO NOT USE".
d. Affected bin, compartment, or closet is not used for storage of any items except for those prominently affixed.
e. Passengers are briefed that affected bin, compartment, or closet is not used.

NOTE 1: For overhead bins, if no partitions are installed, entire overhead bin is considered inoperative.

NOTE 2: Any emergency equipment located in affected bin, compartment, or closet (permanently affixed) is available for use.

MAINTENANCE (M) PROCEDURES

Remove the associated door or secure it in the retracted position and placard the bin, compartment or closet (AMM 25-00-00/901).

OPERATIONS (O) PROCEDURES

1. Ensure cabin crew members are advised of inoperative bins, compartment or closet.
2. Instruct cabin crew members to brief passengers that the associated bin, compartment or closet is not to be used.

25-28-01 Storage Bins/Cabin, Galley, and Lavatory Storage Compartments/Closets

25-28-01C Failed Closed

Interval	Installed	Required	Procedure
C	40	0	

May be inoperative in closed position provided:

a. Affected bin, compartment, or closet is prominently placarded "DO NOT USE".
b. Any emergency equipment located in affected bin, compartment, or closet is considered inoperative.
c. Location placarding for any emergency equipment stored in affected bin, compartment, or closet is removed or obscured.
d. Security personnel must be able to verify the contents of any affected bin, compartment, or closet.

25-28-01 Storage Bins/Cabin, Galley, and Lavatory Storage Compartments/Closets

25-28-01-01 Multi Latch/Quarter-Turn Lug Installations

Interval	Installed	Required	Procedure
C	63	0	

One latch/lug per compartment may be inoperative provided:

a. Remaining latch(es)/lug(s) on affected compartment operates normally.
b. If affected compartment is used for a galley cart, cart remains empty.

25-31-01 Galley/Cabin Waste Receptacle Access Doors/Covers

Interval	Installed	Required	Procedure
C	3	0	**(M) (O)**

May be inoperative provided the container is empty and the access is secured to prevent waste introduction into the compartment.

MAINTENANCE (M) PROCEDURES

Confirm the associated waste container is empty and secure the inoperative access door/cover in a manner that will prevent waste from being deposited in the compartment (AMM 25-00-00/901).

OPERATIONS (O) PROCEDURES

Ensure that sufficient galley/cabin waste receptacles are available to accommodate all waste that may be generated on the flight.

1. If required, use empty garbage bags for any extra waste disposal.
2. Tie off garbage bags and secure in EMPTY meal cart for landing.

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25-40-01 **Exterior Lavatory Door Ashtrays**

25-40-01-01 **Airplanes With Multiple Exterior Lavatory Door Ashtrays Installed**

25-40-01-01A **50% Missing or Inoperative**

Interval	Installed	Required	Procedure
A	3	2	

Up to and including 50% may be missing or inoperative for 10 days.

NOTE: Crew lavatories are included in total aircraft exterior lavatory door ashtray count.

25-40-01 **Exterior Lavatory Door Ashtrays**

25-40-01-01 **Airplanes With Multiple Exterior Lavatory Door Ashtrays Installed**

25-40-01-01B **More Than 50% Missing or Inoperative**

Interval	Installed	Required	Procedure
A	3	0	

More than 50% may be missing or inoperative for 3 days.

NOTE: Crew lavatories are included in total aircraft exterior lavatory door ashtray count.

25-40-02 Lavatory Waste Container Flapper/Access Doors

Interval	Installed	Required	Procedure
C	3	0	**(M)**

May be inoperative provided:

a. Associated waste container is empty and access is secured to prevent waste introduction into the waste container.
b. Associated lavatory is used only by crewmembers.
c. Associated lavatory is entrance door is locked closed and placarded "INOPERATIVE – DO NOT ENTER".

NOTE: These provisions are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M) PROCEDURES

Secure the associated waste container to prevent waste from being introduced (AMM 25-00-00/901).

1. Empty the associated lavatory waste receptable.
2. Secure the flapper/access door in a manner that will prevent waste from being deposited into the receptacle.
3. Close and lock the associated lavatory door.

OPERATIONS NOTE

Lavatory door should remain locked closed, except for use or inspection by crewmembers.

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25-62-01 Flotation Equipment (Crew and Passengers)

Interval	Installed	Required	Procedure
D	169	2	

May be inoperative or missing provided the following distribution is maintained:

a. Flight Deck - 1 each for the Captain, First Officer and any occupant of the Observer's Seat.
b. Cabin Attendant Seats - 1 for each occupant of a Cabin Attendant Seat.
c. Passenger Seats - 1 for each occupant of a Passenger Seat.

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25-63-02 Emergency Locator Transmitter (ELT) (Survival)

Interval	Installed	Required	Procedure
D	2	1	**(M)**

One (1) Portable ELT may be inoperative provided the Fixed (Automatic) ELT is operational.

MAINTENANCE (M) PROCEDURES

Deactivate the Portable ELT (Ref. AMM 25-64-00).

1. Place the switch on the Portable ELT to the 'OFF' position.
2. Install an 'INOPERATIVE´ / 'INOP' Decal adjacent to the affected ELT.

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25-64-01 **Megaphones**

25-64-01B **No Passengers Carried**

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative or missing provided:

a. No passengers are carried.

OPERATIONS (O) PROCEDURES

1. No passengers may be carried.
2. Crew members may communicate via the interphone system, PA system or using shouted commands.

25-64-02 **Portable Flashlights / Flashlight Holders**

25-64-02A **Required by TTCAR**

Interval	Installed	Required	Procedure
C	8	0	

May be inoperative or removed provided:

a. Crewmember assigned to the affected position has an equivalent operative flashlight readily available.
b. Inoperative flashlight remains in a certified location or is removed from the aircraft.
c. Location placarding is removed or obscured.

25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01A Required by TTCAR

Interval	Installed	Required	Procedure
A	3	1	**(O)**

If more than one is required by TTCAR, only one of the required FAKs may be incomplete or removed provided:

a. The FAK is labeled or placarded in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
b. Location placarding is removed or obscured.
c. Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an Emergency Medical Service (EMS) operation is not considered part of the normal complement of equipment. No MEL relief applies to that equipment and TTCAR maintenance and inspection requirements do not apply.

OPERATIONS (O) PROCEDURES

Each inoperative or incomplete FAK must be sealed and clearly labeled to identify the missing or inoperative components, as appropriate.

NOTE: An opened FAK can be used on the return journey to base. However, at the end of the flight:

1. **Cabin crew must ensure that the FAK is resealed with another red seal and secured with the combination lock.**
2. **Report on the FAK usage and lock installation must be entered in the Technical Log Part 2 of 2.**

TTCAR Regulation No. 7, Regulation 45 (2) First Aid and Emergency Medical Kit

The number of first aid kits required on an aircraft under sub-regulation (1), shall be in proportion to the number of passenger seats on an aircraft as outlined in Part B of Schedule 3, refer to table below.

PART B

Number of passenger seats installed	Number of first-aid kits required
1 - 100	1
101 - 200	2
201 - 300	3

25-64-03 **Emergency Medical Equipment**

25-64-03-01 **First Aid Kit (FAK) and/or Associated Equipment**

25-64-03-01B **1 in Excess of TTCAR Requirement**

Interval	Installed	Required	Procedure
D	3	2	**(O)**

One first aid kit may be incomplete, missing, or inoperative provided:

a. FAK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
b. Repairs or replacements are made within 120 days.

OPERATIONS (O) PROCEDURES

Each inoperative or incomplete FAK must be sealed and clearly labelled to identify the missing or inoperative components, as appropriate.

NOTE: An opened FAK can be used on the return journey to base. However, at the end of the flight:

1. **Cabin crew must ensure that the FAK is resealed with another red seal and secured with the combination lock.**
2. **Report on the FAK usage and lock installation must be entered in the Technical Log Part 2 of 2.**

TTCAR Regulation No. 7, Regulation 45 (2) First Aid and Emergency Medical Kit

The number of first aid kits required on an aircraft under sub-regulation (1), shall be in proportion to the number of passenger seats on an aircraft as outlined in Part B of Schedule 3, refer to table below.

PART B

Number of passenger seats installed	Number of first-aid kits required
1 - 100	1
101 - 200	2
201 - 300	3

26-10-01 Fire Warning System

26-10-01-01 BELL CUTOUT Switch

26-10-01-02 Bell Cutout Function of Master Fire Warning Light

26-11-01 Engine Overheat and Fire Detection Systems

26-11-02 Engine Start Lever Fire Indication Lights

26-14-01 Lavatory Smoke Detection Systems

26-14-01-01 Lavatory Smoke Detector SELF TEST Switch

26-14-01-02 Flight Deck LAVATORY SMOKE Light

26-14-01-02A Alternate Procedures Required

26-14-01-02B Procedures Do Not Require Its Use

26-15-01 APU Fire Detection System

26-15-01-01 APU DET INOP Light

26-15-01-02 External Warning Horn/Warning Light

26-16-01 Lower Cargo Compartment Fire Detection Systems

26-16-01-01 Fwd/Aft Detection Loops

26-16-01-02 DETECTOR FAULT Light

26-18-01 Wheel Well Fire Detection System

26-18-01A Brake Temperature Monitoring System Operates Normally

26-18-01B Brake Temperature Monitoring System Inoperative

26-18-01-01 Detection Loops

26-18-02 Wing-Body Overheat Detection System

26-18-02-01 Left Loop

26-18-02-02 Right Loop

26-18-02-03 Wing-Body Overheat Test System

26-19-01 Equipment Cooling Smoke Detection System

26-19-01-01 Smoke Detectors

26-19-01-02 EQUIP SMOKE Light

26-20-01 Engine and APU Fire Extinguisher Discharge Lights

26-20-01-01 Engine Fire Extinguisher Discharge Lights

26-20-01-02 APU Fire Extinguisher Discharge Light

26-20-02 Engine/APU Fire Extinguisher Test System

26-20-02-01 APU Fire Extinguisher Squib Test Circuits

26-20-02-02 APU Fire Extinguisher Test Light (Squib Light)

26-22-01 APU Fire Extinguisher System

26-22-02 APU Fire Shutoff System

26-23-01 Lower Cargo Compartment Fire Suppression System

26-23-01-01 DISCH Light

26-23-01-02 Extinguisher Bottle Pressure Switches

26-23-01-03 EXT Light (FWD and AFT) Systems

26-24-01 Lavatory Fire Extinguisher Systems

26-24-01A Lavatory Smoke Detection System Operates Normally

26-24-01B Lavatory Smoke Detection System Inoperative

26-26-01 Portable Fire Extinguishers

26-10-01 Fire Warning System

26-10-01-01 BELL CUTOUT Switch

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided the Bell Cutout Function of both Master Fire Warning Lights operate normally.

26-10-01 **Fire Warning System**

26-10-01-02 **Bell Cutout Function of Master Fire Warning Light**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided the BELL CUTOUT Switch operates normally.

26-11-01 Engine Overheat and Fire Detection Systems

Interval	Installed	Required	Procedure
C	4	2	**(O)**

Except for ER operations beyond 120 minutes, one loop (A or B) per engine may be inoperative provided remaining operative loops are verified to operate normally once each flight day.

OPERATIONS (O) PROCEDURES

Verify normal operation of the remaining operative loops once each flight day using the FCOM Supplementary Procedure - Fire and Overheat System Test with an Inoperative Loop.

26-11-02 Engine Start Lever Fire Indication Lights

Interval	Installed	Required	Procedure
A	2	0	**(O)**

May be inoperative provided:

a. Engine No.1 and Engine No. 2 fire handle switch lights function normally prior to engine start for each flight.
b. Repairs are made within 3 flight days.

OPERATIONS (O) PROCEDURES

In addition to the Fire and Overheat System Test which is required to be performed every flight day, perform the following prior to engine start for each flight:

Fault/Inoperative (FAULT/INOP) and Overheat/Fire (OVHT/FIRE) Test Switch - Hold to OVHT/FIRE, verify that the Engine No. 1 and Engine No. 2 fire switches stay illuminated.

26-14-01 Lavatory Smoke Detection Systems

Interval	Installed	Required	Procedure
C	3	0	**(M) (O)**

For each lavatory, lavatory smoke detection system may be inoperative provided:

a. Lavatory waste receptacle is empty.
b. Associated lavatory door is locked closed and placarded "INOPERATIVE - DO NOT ENTER".
c. Lavatory is used only by crewmembers.

NOTE: These provisos are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M) PROCEDURES

Empty the associated lavatory waste container and close, lock and placard the lavatory door (AMM 26-00-00/901).

OPERATIONS (O) PROCEDURES

Ensure the associated lavatory door remains locked except for use or inspection by crew members.

26-14-01 Lavatory Smoke Detection Systems

26-14-01-01 Lavatory Smoke Detector SELF TEST Switch

Interval	Installed	Required	Procedure
C	3	0	**(M)**

May be inoperative provided associated lavatory smoke detector is verified to operate normally.

MAINTENANCE (M) PROCEDURES

For the associated smoke detector, do the Lavatory Smoke Detection – Smoke Test (AMM 26-14-00/501).

26-14-01 Lavatory Smoke Detection Systems

26-14-01-02 Flight Deck LAVATORY SMOKE Light

26-14-01-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. Lavatory smoke detection systems operate normally.
b. Alternate procedures are established and used.

OPERATIONS (O) PROCEDURES

Alternate procedures must be established and used.

Cockpit Crew will be informed of lavatory status after each Cabin Crew 30 Minute Lavatory Check.

26-14-01 Lavatory Smoke Detection Systems

26-14-01-02 Flight Deck LAVATORY SMOKE Light

26-14-01-02B Procedures Do Not Require Its Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

26-15-01 APU Fire Detection System

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided:

a. APU is not used.
b. Both engine driven generator systems operate normally.
c. At least one engine bleed air shutoff valve (PRSOV) operates normally.

NOTE: Dispatch is not allowed if the APU is required by other procedures.

MAINTENANCE NOTE

To prevent nuisance/false warnings, the P6 panel FIRE PROTECTION DETECTION APU circuit breaker can be opened and collared. With this circuit breaker open, the MASTER CAUTION lights and the OVHT/DET system annunciator light do not illuminate during the Fault/Inop test. The APU DET INOP light will remain illuminated and the MASTER CAUTION lights and the OVHT/DET system annunciator light will illuminate during a system annunciator panel recall.

OPERATIONS NOTE

1. If the P6 panel FIRE PROTECTION DETECTION APU circuit breaker has been opened and collared to prevent nuisance/false fire warnings:
   A. The APU DET INOP light will remain illuminated.
   B. The MASTER CAUTION and OVHT/DET system annunciator lights will not illuminate during the Fault/Inop test part of the overheat and fire protection panel check.
   C. The MASTER CAUTION lights and the OVHT/DET system annunciator light will illuminate during a system annunciator panel recall.

26-15-01 APU Fire Detection System

26-15-01-01 APU DET INOP Light

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. Light is extinguished.

b. APU Fire Detection system is verified to operate normally before each APU start.

OPERATIONS (O) PROCEDURES

NOTE: During the daily FCOM check of the overheat and fire protection panel, if the APU DET INOP light does not illuminate (when the TEST switch is held to FAULT/INOP), but all other indications are normal when the TEST switch is held to FAULT/INOP and OVHT/FIRE, the APU fire detection system is operating normally.

Before each APU start do the FCOM check of the overheat and fire protection panel and confirm all indications are normal except that the APU DET INOP light does not illuminate (when the TEST switch is held to FAULT/INOP).

26-15-01 APU Fire Detection System

26-15-01-02 External Warning Horn/Warning Light

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided flight deck APU overheat/fire protection panel is continuously monitored during ground APU operation.

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26-16-01 Lower Cargo Compartment Fire Detection Systems

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided procedures are established and used to ensure associated compartment or zone remains empty or is verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE NOTE

Ensure applicable cargo compartment(s) remain empty or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments

When any cargo compartment is deemed to be unusable the following procedures shall apply:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1 of 2.
4. Traffic, SOC, MCC Departments as applicable to be notified.

26-16-01 Lower Cargo Compartment Fire Detection Systems

26-16-01-01 Fwd/Aft Detection Loops

Interval	Installed	Required	Procedure
C	4	2	**(O)**

One loop (A or B) in each compartment may be inoperative provided remaining loop is verified to operate normally.

OPERATIONS (O) PROCEDURES

Verify the remaining loop operates normally.

1. Set the cargo fire panel FWD and AFT DET SELECT switches to A.
2. Press and hold the cargo fire panel TEST switch.
3. Note whether the following indications occur within 4 seconds.
   A. Fire bell sounds.
   B. Both master FIRE WARN lights illuminate.
   C. Cargo fire panel FWD and AFT warning lights illuminate.
   D. Cargo fire panel DISCH light illuminates.
   E. Cargo fire panel EXT FWD and EXT AFT lights illuminate.

   NOTE: If one or more smoke detectors in the selected loop is inoperative, the cargo fire panel DETECTOR FAULT light will illuminate and the associated FWD or AFT warning light will remain extinguished for the cargo compartment with the inoperative detector.
4. Release the cargo fire panel TEST switch. The DETECTOR FAULT light remains illuminated if a loop failure is detected in either the forward or aft cargo compartment. Otherwise all indications will clear within half a second.
5. Set the cargo fire panel FWD and AFT DET SELECT switches to B and repeat Steps 2 through 4.
6. Confirm all of the indications in Step 3 occurred for at least one of the A and B loops for each cargo compartment.
7. Set the cargo fire panel DET SELECT switches as required so that only operative loops are selected and at least one loop is selected for each compartment.

26-16-01 Lower Cargo Compartment Fire Detection Systems

26-16-01-02 DETECTOR FAULT Light

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided cargo fire panel TEST switch is used to check for cargo fire detection system faults before each flight.

OPERATIONS (O) PROCEDURES

Before each flight, use the cargo fire panel TEST switch to determine the status of the lower cargo compartment fire detection systems. If faults are found, advise maintenance personnel.

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26-18-01 Wheel Well Fire Detection System

26-18-01A Brake Temperature Monitoring System Operates Normally

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Wheel well fire detection system is deactivated.
b. Brake temperature monitoring system (BTMS) operates normally.

MAINTENANCE (M) PROCEDURES

Deactivate the wheel well fire detection system using one of the following procedures (AMM 26-00-00/901).

1. **Option 1:** Deactivate WHEEL WELL light and wheel well fire detection system input to fire warning system.
   A. Remove the two light bulbs from the WHEEL WELL light on the overheat and fire protection panel (P8-1) and placard the light.
   B. Placard the Overheat/Fire (OVHT/FIRE) TEST switch located on the overheat and fire protection panel (P8-1) to indicate the wheel well fire detection system is inoperative.
   C. Disconnect the fire bell.
      1) Remove Flight Control Computer B to get access to the wiring for the fire bell (AMM TASK 22-11-33-000-801).
      2) Disconnect the wires from D742, pin 16 and pin 25 on the E1-4 shelf.
      3) Cap and stow the wires.
      4) Reinstall Flight Control Computer B (AMM TASK 22-11-33-400-801).
      5) Placard the master FIRE WARN lights on the P7 panel to indicate that the fire warning associated with the wheel well is inoperative.
2. **Option 2:** Deactivate the detection elements of the wheel well fire detection system.
   NOTE: When in dual loop operations, i.e., "normal mode", both MWW LOOP A and MWW LOOP B sensors are required to be in a state other than STANDBY in order annunciate a wheel well fire. If only one sensor is faulted, i.e., MWW LOOP A or MWW LOOP B, it is recommended to enter "SINGLE LOOP CONFIGURATION" (AMM 26-00-00/901, DDG 26-18-01) to maintain wheel well fire detection system coverage rather than disabling the wheel well fire detection system entirely.
   A. Disconnect the terminal lugs A and B on both sides of the M270 and M3068 detection elements.
   B. Perform the OVHT/Fire warning test (AMM 26-18-00). All tests should pass except for the wheel well fire function with the disabled element(s).
   C. Stow the wiring and terminal lugs on either side of M270 and M3068 detection elements.

26-18-01 Wheel Well Fire Detection System

26-18-01B Brake Temperature Monitoring System Inoperative

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Wheel well fire detection system is deactivated.
b. Landing gear remains extended for 10 minutes after takeoff.
c. Appropriate performance adjustments are applied.

NOTE: In case of engine failure after V1, landing gear should be retracted until takeoff obstacles are cleared.

MAINTENANCE (M) PROCEDURES

Deactivate the wheel well fire detection system using one of the following procedures (AMM 26-00-00/901).

1. **Option 1:** Deactivate WHEEL WELL light and wheel well fire detection system input to fire warning system.
   A. Remove the two light bulbs from the WHEEL WELL light on the overheat and fire protection panel (P8-1) and placard the light.
   B. Placard the Overheat/Fire (OVHT/FIRE) TEST switch located on the overheat and fire protection panel (P8-1) to indicate the wheel well fire detection system is inoperative.
   C. Disconnect the fire bell.
      1) Remove Flight Control Computer B to get access to the wiring for the fire bell (AMM TASK 22-11-33-000-801).
      2) Disconnect the wires from D742, pin 16 and pin 25 on the E1-4 shelf.
      3) Cap and stow the wires.
      4) Reinstall Flight Control Computer B (AMM TASK 22-11-33-400-801).
      5) Placard the master FIRE WARN lights on the P7 panel to indicate that the fire warning associated with the wheel well is inoperative.
2. **Option 2:** Deactivate the detection elements of the wheel well fire detection system.
   NOTE: When in dual loop operations, i.e., "normal mode", both MWW LOOP A and MWW LOOP B sensors are required to be in a state other than STANDBY in order annunciate a wheel well fire. If only one sensor is faulted, i.e., MWW LOOP A or MWW LOOP B, it is recommended to enter "SINGLE LOOP CONFIGURATION" (AMM 26-00-00/901, DDG 26-18-01) to maintain wheel well fire detection system coverage rather than disabling the wheel well fire detection system entirely.
   A. Disconnect the terminal lugs A and B on both sides of the M270 and M3068 detection elements.
   B. Perform OVHT/Fire warning test (AMM 26-18-00). All tests should pass except for the wheel well fire function with the disabled element(s).
   C. Stow the wiring and terminal lugs on either side of M270 and M3068 detection elements.

OPERATIONS (O) PROCEDURES

1. Increase flight planning fuel by 726 kg.
2. After takeoff, leave the landing gear extended for 10 minutes.
3. **In case of engine failure after V1, landing gear should be retracted after takeoff until takeoff obstacles are cleared.**
4. It is not necessary to apply landing gear extended takeoff performance adjustments when dispatching under this item.

26-18-01 Wheel Well Fire Detection System

26-18-01-01 Detection Loops

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided wheel well detection system is put in a single loop configuration.

MAINTENANCE (M) PROCEDURES

Put the wheel well fire detection system in a single loop configuration (AMM 26-00-00/901).

1. Gain access to the wheel well fire detection system compartment overheat detection module (CODM).
2. Perform a CODM local test, LOC TEST, and note the indication of a fault for the main wheel well, MWW, sensor loops.
3. Confirm the CODM configured to a single loop configuration:
   A. Verify SINGLE LOOP OPS appears on the front panel display of the CODM.
   B. Open the following P6-2 panel circuit breakers:
      1) FIRE PROTECTION DETECTION ENG 1
      2) FIRE PROTECTION DETECTION ENG 2
      3) FIRE PROTECTION DETECTION APU
   C. On the overheat and fire protection panel, hold the TEST switch in OVHT/FIRE.
   D. Confirm the following indications:
      1) The fire warning bell sounds.
      2) The master FIRE WARN lights illuminate.
      3) The WHEEL WELL light illuminates.
4. Release the TEST switch.
5. Confirm the following:
   A. The fire warning bell cancels.
   B. The master FIRE WARN lights extinguish.
   C. The WHEEL WELL light extinguishes.
6. Close the open circuit breakers.

26-18-02 Wing-Body Overheat Detection System

26-18-02-01 Left Loop

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Except for ER operations, may be inoperative provided:

a. Only right pack and engine bleed is used for pressurization.
b. Left pack is considered inoperative.
c. Use of APU is prohibited except for engine start.
d. ISOLATION VALVE switch remains CLOSED and left engine BLEED air switch remains OFF for all operations except engine start.
e. Airplane is not operated in known or forecast icing conditions.

NOTE: Also dispatch the airplane using MEL Item 21-51-01.

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known icing or forecast icing conditions.
2. Except for engine start, do not use the APU for pneumatic air supply and/or electrical power.
3. After engine start:
   A. Set the ISOLATION VALVE switch to CLOSE.
   B. Set the APU switch to OFF.
   C. Set the Engine No. 1 BLEED air switch to OFF.
   D. Set the L PACK switch to OFF.
   E. Set the R PACK switch as desired.
4. **Don’t do MEL Item 21-51-01 (O) procedure step for increasing air flow when the flaps are extended (takeoff and landing).**

26-18-02 Wing-Body Overheat Detection System

26-18-02-02 Right Loop

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Except for ER operations, may be inoperative provided:

a. Only left pack and left engine or APU bleed air is used for pressurization.
b. Right pack is considered inoperative.
c. ISOLATION VALVE switch remains CLOSED and right engine BLEED air switch remains OFF for all operations except engine start.
d. Airplane is not operated in known or forecast icing conditions.

NOTE: Also dispatch the airplane using MEL Item 21-51-01.

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known icing or forecast icing conditions.
2. After engine start:
   A. Set the ISOLATION VALVE switch to CLOSE.
   B. Set the Engine No. 2 BLEED air switch to OFF.
   C. Set the R PACK switch to OFF.
   D. Set the L PACK switch as desired.
3. Do not set the L PACK to HIGH for takeoff and landing with the engine bleed on.
4. If doing DDG Item 21-51-01 (O) procedure step for increasing air flow when the flaps are extended (takeoff and landing), while doing the Supplementary procedure, do the following:
   A. Keep the R PACK switch OFF.
   B. Keep the ISOLATION VALVE switch in CLOSE.
   C. Keep the Engine No. 2 BLEED air switch OFF.

26-18-02 Wing-Body Overheat Detection System

26-18-02-03 Wing-Body Overheat Test System

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided operative detector loops are verified to operate normally once each flight day.

MAINTENANCE (M) PROCEDURES

Once each flight day, verify each operative detector loop operates normally (AMM 26-00-00/901).

1. For each operative detector loop, apply heat from an appropriate heat source to a detector element in the loop. Limit the heat the test device produces to no greater than 450°F (232°C).
2. For the left detector loop, when a detector element is heated, confirm the following lights illuminate:
   A. Left WING-BODY OVERHEAT light on the bleed air controls and indicators panel (P5)
   B. The MASTER CAUTION lights
   C. The AIR COND light on the glareshield P7
3. For the right detector loop, when a detector element is heated, confirm the following lights illuminate:
   A. Right WING-BODY OVERHEAT light on the bleed air controls and indicators panel (P5)
   B. The MASTER CAUTION lights
   C. The AIR COND light on the glareshield P7

26-19-01 Equipment Cooling Smoke Detection System

26-19-01-01 Smoke Detectors

Interval	Installed	Required	Procedure
C	2	1	**(M)**

One smoke detector may be inoperative provided:

a. Inoperative smoke detector is deactivated.
b. Remaining smoke detector is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify normal operation of the remaining smoke detector (AMM 26-00-00/901).

1. Verify E/E Cooling supply system smoke detector operates normally:
   A. Access the J23 BOX in the E/E Bay.
   B. Push and then release the EQUIP COOL SMK DET TEST button (S1204) on the forward panel of the J23 BOX.
   C. Confirm the green EQUIP COOL SMK DET SPLY light (L1293) illuminates on the forward panel of the J23 BOX.
2. Verify E/E Cooling exhaust system smoke detector operates normally:
   A. Access the J23 BOX in the E/E Bay.
   B. Push and then release the EQUIP COOL SMK DET TEST button (S1204) on the forward panel of the J23 BOX.
   C. Confirm the green EQUIP COOL SMK DET EXH light (L1294) illuminates on the forward panel of the J23 BOX.

Deactivate the inoperative smoke detector (AMM 26-00-00/901).

1. To deactivate the E/E Cooling supply system smoke detector, open and collar the following F/O Electrical System Panel, P6-5:
   A. EQPT COOL SMOKE DET SUPPLY.
2. To deactivate the E/E Cooling exhaust system smoke detector, open and collar the following CAPT Electrical System Panel, P18-3:
   A. EQPT COOL SMOKE DET EXHAUST.

26-19-01 Equipment Cooling Smoke Detection System

26-19-01-02 EQUIP SMOKE Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided E/E Cooling smoke detectors are verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify normal operation of both E/E Cooling smoke detectors (AMM 26-00-00/901).

1. Access the J23 BOX in the E/E Bay.
2. Push and then release the EQUIP COOL SMK DET TEST button (S1204) on the forward panel of the J23 BOX.
3. Confirm both the green EQUIP COOL SMK DET SPLY light (L1293) and EQUIP COOL SMK DET EXH light (L1294) illuminate on the forward panel of the J23 BOX.

26-20-01 Engine and APU Fire Extinguisher Discharge Lights

26-20-01-01 Engine Fire Extinguisher Discharge Lights

Interval	Installed	Required	Procedure
C	2	0	

26-20-01 Engine and APU Fire Extinguisher Discharge Lights

26-20-01-02 APU Fire Extinguisher Discharge Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided APU fire extinguisher bottle is verified to have an adequate charge before each flight.

MAINTENANCE (M) PROCEDURES

Before each flight, verify APU fire extinguisher bottle has an adequate charge (AMM 26-00-00/901).

1. Gain access to the APU fire extinguisher bottle pressure gauge.
2. Verify that the pressure gauge indicates a pressure greater than 400 psig.

26-20-02 Engine/APU Fire Extinguisher Test System

Interval	Installed	Required	Procedure
C	3	0	**(M)**

May be inoperative provided:

a. Failure is verified to be in squib test circuit.
b. For each inoperative squib test circuit, associated squib circuit is verified to operate normally once each flight day.

MAINTENANCE (M) PROCEDURES

For each inoperative squib test circuit, verify the failure is in the squib test circuit by verifying the squib circuit operates normally once each flight day (AMM 26-00-00/901).

NOTE 1: USE THE CORRECT LAMP WHEN MAKING THE SQUIB TESTER IF AN INCORRECT LAMP IS USED, THE RESULTING HIGH ELECTRICAL CURRENT CAN CAUSE THE SQUIB TO FIRE. THIS WILL RELEASE THE EXTINGUISHING AGENT AND CAN CAUSE INJURY TO PERSONNEL.

NOTE 2: THE P6-2 CIRCUIT BREAKER PANEL CONTAINS HIGH VOLTAGE/CURRENTS THAT CAN CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT. TO PREVENT INJURY, RESTRICT PERSONNEL FROM FIRE BOTTLE AREA WHILE PERFORMING TEST.

1. Build a squib test lamp with pigtail leads that end in pin contacts. Use only a 28 volt, 40 milliamp lamp (COM-5826) in a suitable holder with jumper cable (STD-6286 or STD-15135).
2. Confirm the squib circuit operates normally, for each inoperative squib test circuit:
   A. For engine circuits:
      1) Open the following P6-2 circuit breakers:
         a. FIRE PROTECTION EXTINGUISHERS RIGHT
         b. FIRE PROTECTION EXTINGUISHERS LEFT
         c. FIRE PROTECTION EXTINGUISHERS ALTN R
         d. FIRE PROTECTION EXTINGUISHERS ALTN L
      2) Gain access to the P6-2 circuit breaker panel.
      3) Use the following table to determine how to connect the test lamp to test the appropriate circuit.

Bottle	Squib No.	Squib Plug No.	C/B Open and Test Lamp Connections
Left	1	D582, Pin 4	Across C296 Left Ext Bottle
Left	1	D582, Pin 6	Across C1021 Left Ext Bottle
Left	2	D1322, Pin 4	Across C296 Left Ext Bottle
Left	2	D1322, Pin 6	Across C1021 Left Ext Bottle
Right	1	D584, Pin 4	Across C297 Right Ext Bottle
Right	1	D584, Pin 6	Across C1022 Right Ext Bottle
Right	2	D1324, Pin 4	Across C297 Right Ext Bottle
Right	2	D1324, Pin 6	Across C1022 Right Ext Bottle

      4) Do the following procedure for each inoperative squib test circuit:
         a. Connect the test lamp across the terminals of the appropriate circuit breaker for the squib test circuit being tested.
         b. Push the button on the override switch and pull the appropriate Engine 1 or Engine 2 fire switch up.
         c. Rotate the appropriate fire switch left or right for the bottle being tested.
         d. Confirm the test lamp illuminates.

MAINTENANCE (M) PROCEDURE 26-20-02 (Cont'd)

4) Do the following procedure for each inoperative squib test circuit: (Cont'd)
   e. Rotate the appropriate fire switch back to the center position and push it down.
   f. Remove the test lamp from the circuit breaker.
5) Close the open circuit breakers.

B. For the APU circuit:
1) Open the P6-2 panel FIRE PROTECTION EXTINGUISHERS APU circuit breaker.
2) Gain access to the P6-2 circuit breaker panel.
3) Connect the test lamp across the terminals of the circuit breaker.
4) Push the button on the override switch and pull the APU fire switch up.
5) Rotate the APU fire switch, left or right, for the circuit being tested.
6) Confirm the test lamp illuminates.
7) Rotate the APU fire switch back to the center position and push it down.
8) Remove the test lamp from the circuit breaker.
9) Close the open circuit breaker.

26-20-02 Engine/APU Fire Extinguisher Test System

26-20-02-01 APU Fire Extinguisher Squib Test Circuits

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative provided remaining APU squib test circuit is verified to operate normally once each flight day.

OPERATIONS (O) PROCEDURES

Once each flight day, verify remaining APU squib test circuit operates normally.

1. Substitute Step 2 in the following Flight Crew Operations Manual Procedures, in the EXTINGUISHER TEST switch check:
   A. Normal Procedures - Amplified Procedures - Preflight Procedure - First Officer.
   B. Supplementary Procedures - Electrical - Electrical Power Up.
2. For the APU extinguisher test light, confirm the light illuminates when the EXT TEST switch is held in either Position 1 or Position 2.

NOTE: The light will not illuminate for one of the test positions.

26-20-02 Engine/APU Fire Extinguisher Test System

26-20-02-02 APU Fire Extinguisher Test Light (Squib Light)

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided:

a. APU is not used.
b. Both engine driven generator systems operate normally.
c. At least one of bleed system PRSOVs operates normally.

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26-22-01 APU Fire Extinguisher System

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided:

a. APU is not used.
b. Both engine driven generator systems operate normally.
c. At least one of bleed system PRSOVs operates normally.

26-22-02 APU Fire Shutoff System

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided:

a. APU is not used.
b. Both engine driven generator systems operate normally.
c. At least one of bleed system PRSOVs operates normally.

26-23-01 Lower Cargo Compartment Fire Suppression System

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided procedures are established and used to ensure lower cargo compartments or zones remain empty or are verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away Kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

MAINTENANCE NOTE

Ensure cargo compartments remain empty, or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Lower Cargo Compartments:

Inspect all lower cargo compartments to verify they remain empty or contain only empty cargo handling equipment, ballast, and/or Fly Away kits:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1of 2.
4. Traffic, SOC, MCC Departments as applicable to be notified.

26-23-01 Lower Cargo Compartment Fire Suppression System

26-23-01-01 DISCH Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided operative extinguisher bottles are verified to have an adequate charge once each flight day.

MAINTENANCE (M) PROCEDURES

Once each flight day, verify each operative cargo fire extinguisher bottle has an adequate charge (AMM 26-00-00/901).

1. Build a test lamp (28 volts DC/40 milliamps) with pigtail leads terminating in pin contacts in a suitable holder or obtain a digital multimeter.
2. Remove the cargo fire panel.
3. Disconnect electrical connector D12760 from the panel.
4. For Bottle No. 1:
   A. Connect the test lamp or digital multimeter between the airplane side of connector D12760, Pin 48 (+28 volt source for Bottle No. 1) and Pin 28 (pressure switch ground).
   B. Confirm the test lamp remains extinguished or the digital multimeter indicates an open circuit.
5. For Bottle No. 2:
   A. Connect the test lamp or digital multimeter between the airplane side of connector D12760, Pin 45 (+28 volt source for Bottle No. 2) and Pin 28 (pressure switch ground).
   B. Confirm the test lamp remains extinguished or the digital multimeter indicates an open circuit.
6. Reinstall the cargo fire panel.

26-23-01 Lower Cargo Compartment Fire Suppression System

26-23-01-02 Extinguisher Bottle Pressure Switches

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided associated extinguisher bottle is verified to have an adequate charge once each flight day.

MAINTENANCE (M) PROCEDURES

Deactivate the pressure switch and, once each flight day, verify the associated fire extinguisher bottle has an adequate charge (AMM 26-00-00/901).

1. Gain access to the associated cargo fire extinguisher bottle.
2. Cap and stow the associated bottle's pressure switch electrical connector.
3. Verify the bottle has an adequate charge:
   A. Remove the extinguisher bottle.
   B. Weigh the extinguisher bottle on a scale that is accurate to within +/- 0.1 pound.
   C. Confirm the extinguisher bottle weight is within tolerance.
      1) For P/N 473957-4 and 475976-1 extinguisher bottles, weight must be within 0.25 lb (0.11 kg) of the service weight stamped on the bottle's identification plate.
      2) For P/N 475975-1 extinguisher bottle, weight must be within 0.13 lb (0.06 kg) of the service weight stamped on the bottle's identification plate.
   D. Reinstall the extinguisher bottle.

26-23-01 **Lower Cargo Compartment Fire Suppression System**

26-23-01-03 **EXT Light (FWD and AFT) Systems**

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided:

a. Failure is verified to be in squib light circuit.
b. Both associated squib circuits are verified to operate normally once each flight day.

MAINTENANCE (M) PROCEDURES

Verify the failure is in the squib light circuit and that both associated squib circuits operate normally (AMM 26-00-00/901).

1. Build a test lamp (28 volts DC/40 milliamps) with pigtail leads terminating in pin contacts in a suitable holder.
   NOTE: Any suitable lamp can be used.
2. Remove the cargo fire panel.
3. Disconnect electrical connector D12760 from the panel.
   NOTE: TO PREVENT INJURY, RESTRICT PERSONNEL FROM EXTINGUISHER BOTTLE AREA DURING SQUIB CIRCUIT TEST. LIMIT CURRENT TO 40 MILLIAMPS MAXIMUM DURING TEST. USE TEST LAMP (28 VOLT, 40 MILLIAMPS ONLY). DO NOT USE A MULTIMETER.
4. Connect the test lamp between the appropriate pins, on the airplane side of connector D12760, as determined from the table below.

CARGO FIRE BOTTLE SQUIB TEST TABLE

Bottle	Squib	Squib Plug No.	TEST LAMP CONNECTION	
			Pin No. (+28 Vdc Source)	Pin No. (Ground Path)
1	FWD	D16578	D12760, Pin 48	D12760, Pin 26
1	AFT	D16580	D12760, Pin 48	D12760, Pin 24
2	FWD	D16584	D12760, Pin 45	D12760, Pin 12
2	AFT	D16586	D12760, Pin 45	D12760, Pin 23

5. For both of the associated squib circuits (for an inoperative EXT light system), confirm the test lamp illuminates when the appropriate pins are connected.
6. Reinstall the cargo fire panel.

26-24-01 **Lavatory Fire Extinguisher Systems**

26-24-01A **Lavatory Smoke Detection System Operates Normally**

Interval	Installed	Required	Procedure
C	3	0	

For each lavatory, lavatory fire extinguisher system may be inoperative provided associated lavatory smoke detection system operates normally.

26-24-01 Lavatory Fire Extinguisher Systems

26-24-01B Lavatory Smoke Detection System Inoperative

Interval	Installed	Required	Procedure
C	3	0	**(M) (O)**

For each lavatory, lavatory fire extinguisher system may be inoperative provided:

a. Lavatory waste receptacle is empty.
b. Associated lavatory door is locked closed and placarded "INOPERATIVE - DO NOT ENTER".
c. Lavatory is used only by crewmembers.

NOTE: These provisos are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M) PROCEDURES

Empty the associated lavatory waste container and close, lock and placard the lavatory door (AMM 25-00-00/901).

OPERATIONS (O) PROCEDURES

Ensure the associated lavatory door remains locked except for use or inspection by crew members.

26-26-01 Portable Fire Extinguishers

Interval	Installed	Required	Procedure
D	5	4	**(M)**

May be inoperative or removed provided:

a. Inoperative fire extinguisher remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
b. Location placarding is removed or obscured.
c. Required distribution is maintained (1 in cockpit, 3 in cabin).

NOTE: Inoperative fire extinguishers, removed from a certified location or removed from the aircraft, are subject to dangerous goods regulations.

MAINTENANCE (M) PROCEDURES

Configure the airplane for operation with an inoperative portable fire extinguisher (AMM 26-00-00/901).

1. Stow the out-of-service extinguisher in a place (possibly in a bag) where it will not be used by mistake, until it can be removed from the airplane.
2. Ensure required distribution of serviceable extinguishers is maintained throughout the airplane.

NOTE: The required distribution as per TTCAR [(No. 7) Instruments and Equipment] is:

a) One in the Flight Deck, and

b) Three in the Passenger Compartment of an aircraft having a seating capacity of 61 through 200, conveniently located and uniformly distributed throughout the compartment.

INTENTIONALLY

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27-08-01 Control Surface Position Indication System

27-21-01 Rudder Trim Indication

27-21-02 STBY RUD ON Light

27-24-01 Wheel to Rudder Interconnect System (WTRIS)

27-31-01 FEEL DIFF PRESS Light System

27-51-01 Flap Load Relief Function

27-61-01 Flight Spoiler Systems

27-61-01-01 Spoiler Control Channels

27-61-02 Maneuver Load Alleviation (MLA) Function

27-61-03 Emergency Descent Spoilers (EDS) Function

27-62-01 Auto Speedbrake System

27-62-02 SPEEDBRAKES EXTENDED Light

27-83-01 Autoslat Systems

27-83-02 AUTO SLAT FAIL Light System

27-88-01 Leading Edge Flap/Slat Position Light Systems

27-88-01-02 Leading Edge Indications for Slats 1, 2, 7, and 8

INTENTIONALLY

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27-08-01 Control Surface Position Indication System

Interval	Installed	Required	Procedure
C	1	0	

INTENTIONALLY

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27-21-01 Rudder Trim Indicator

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Rudder trim actuator is verified to operate normally.
b. Rudder trim is verified to be centered before each departure.

MAINTENANCE (M) PROCEDURES

Verify the rudder trim actuator operates normally (AMM 27-00-00/901).

1. Pressurize rudder hydraulic systems A and B.
2. Verify normal operation of the rudder trim actuator:
   A. Turn the rudder trim knob to the NOSE LEFT position.
   B. Confirm the captain's left rudder pedal moves forward and right rudder pedal moves aft.
   C. Turn the rudder trim knob to the NOSE RIGHT position.
   D. Confirm the captain's left rudder pedal moves aft and right rudder pedal moves forward.

OPERATIONS (O) PROCEDURES

Verify rudder trim is centered before each departure.

1. Turn the rudder trim knob in the appropriate direction until the captain's rudder pedals are aligned with each other.
2. Confirm the rudder is centered (in the neutral position). If not, use rudder trim until rudder is centered.

27-21-02 STBY RUD ON Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Rudder is verified to operate normally on hydraulic system A and B independently.
b. Standby hydraulic pump is verified to operate normally.
c. Rudder force fight monitor is deactivated.

MAINTENANCE (M) PROCEDURES

Verify the rudder operates normally independently on hydraulic system A and B, verify the standby hydraulic pump operates normally and deactivate the rudder force fight monitor (AMM 27-00-00/901).

1. Verify the rudder operates normally independently on hydraulic system A and B.
   A. Pressurize hydraulic systems A and B.
   B. Set the FLT CONTROL A switch to ON and the FLT CONTROL B switch to OFF.
   C. While holding the tiller to prevent tire scrub, confirm normal rudder pedal forces by pushing the captain's left rudder pedal forward until it touches the stop and hold.
   D. Have a ground observer confirm the rudder travels left.
   E. While continuing to hold the tiller, release the left rudder pedal and confirm normal rudder pedal forces by pushing the captain's right rudder pedal forward until it touches the stop and hold.
   F. Have a ground observer confirm the rudder travels right.
   G. While continuing to hold the tiller, release the captain's right rudder pedal.
   H. Have a ground observer confirm the rudder returns to neutral.
   I. Set the FLT CONTROL B switch to ON and the FLT CONTROL A switch to OFF.
   J. Repeat steps C through H.
2. Verify standby hydraulic pump is operating normally by doing one of the following:
   A. First option:
      1) Set either the FLT CONTROL A or B switch to STBY RUD.
      2) Confirm the standby hydraulic pump can be heard turning on.
   B. Second option:
      1) Set the FLT CONTROL A switch to STBY RUD and the FLT CONTROL B switch to OFF.
      2) Repeat steps 1C through 1H.
3. Deactivate the rudder force fight monitor.
   A. Open and collar P6-2 panel FORCE FIGHT MON circuit breaker.
   B. Confirm STBY RUD ON light is extinguished after circuit breaker is opened.

27-24-01 Wheel to Rudder Interconnect System (WTRIS)

Interval	Installed	Required	Procedure
C	1	0	

INTENTIONALLY

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27-31-01 FEEL DIFF PRESS Light System

Interval	Installed	Required	Procedure
B	1	0	**(M)**

May be inoperative provided elevator feel system is verified to operate normally once each flight day.

MAINTENANCE (M) PROCEDURES

Verify the elevator feel system operates normally once each flight day using one of the following procedures (AMM 27-00-00/901).

1. Option 1 (recommended method): Verify the elevator feel system operates normally using the MFD FLIGHT CONTROL display to measure elevator position.
   A. Pressurize the elevator hydraulic Systems A and B.
   B. Open P6-2 panel ELEV FEEL LIMIT circuit breaker.
   C. Use the stabilizer trim switches to set the stabilizer to 4 units of trim.
   D. Set the FLT CONTROL A and B switches to OFF.
   E. Confirm the horizontal stabilizer "B" dimension is 39.17 ± 0.01 inches.
   F. Set the Mach trim actuator in null position.
   G. Set the PSEU to ground mode.
   H. Do the Pitot System Leak Test.
   I. Set the FLT CONTROL A and B switches to ON.
   J. Pressurize the pitot system to 2.47 ± 0.05 psi (315 ± 1 knots).
      NOTE: Do not exceed a test pressure of 5.0 psi (437 knots) during test.
   K Select the FLIGHT CONTROL DISPLAY on an MFD.
   L. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.
      NOTE: Ensure the control columns are not disturbed from this position during the next step.
   M. Verify the right elevator position is at the neutral position of 4.0 ± 0.3 degrees. Record the right elevator position value shown on the MFD.
   N. Set the FLT CONTROL B switch to OFF and leave the FLT CONTROL A switch in ON.
   O. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly aft until the MFD displays a right elevator position of 2.0 degrees less than the neutral position determined in Step M. Verify the force required is within the range indicated in the table below.
      NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.
   P. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.
   Q. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly forward until the MFD displays a right elevator position of 2.0 degrees greater than the neutral position determined in
      Step M. Verify the force required is within the range indicated in the table below.

MAINTENANCE (M) PROCEDURES 27-31-01 (Cont'd)

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

Model	Elevator Position	Column Force
-8	2.0 degrees less than MFD neutral value (column aft, trailing edge up)	37.5 ± 10.0 lbs
	2.0 degrees more than MFD neutral value (column forward, trailing edge down)	40.5 ± 11.5 lbs

R. Set the FLT CONTROL A switch to OFF and the FLT CONTROL B switch to ON.

S. Repeat steps O-Q.

T. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

U. Verify the pitot system is pressurized to 2.47 ± 0.05 psi (315 ± 1 knots).

V. Close P6-2 panel ELEV FEEL LIMIT circuit breaker and wait 15 seconds.

W. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly aft until the MFD displays a right elevator position of 2.0 degrees less than the neutral position determined in Step M. Note the force required.

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

X. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

Y. Open P6-2 panel ELEV FEEL LIMIT circuit breaker and wait 20 seconds.

Z. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly aft until the MFD displays a right elevator position of 2.0 degrees less than the neutral position determined in Step M. Note the force required.

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

AA. Verify that the difference in the forces noted in steps W and Z is at least 20.0 lbs.

AB. Return the captain's control column to the neutral position.

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MAINTENANCE (M) PROCEDURES 27-31-01 (Cont'd)

AC. Restore the airplane to its usual condition:

1) Make sure that all tools used during testing are removed.
2) Remove seals from the drain holes in each vertical fin pitot probe.
3) Remove the pressure gage lines from systems A and B drain lines.
4) Install the drain plugs to the systems A and B drain lines.
5) Remove regulated air pressure lines from both vertical fin pitot probes.
6) Close P6-2 panel ELEV FEEL LIMIT circuit breaker.
7) Restore airplane to normal hydraulic configuration.
8) Reset the PSEU to normal mode.

2. Option 2: Verify the elevator feel system operates normally using masking tape on the tail cone to measure elevator position.

A. Pressurize the elevator hydraulic Systems A and B.
B. Open P6-2 panel ELEV FEEL LIMIT circuit breaker.
C. Use the stabilizer trim switches to set the stabilizer to 4 units of trim.
D. Set the FLT CONTROL A and B switches to OFF.
E. Confirm the horizontal stabilizer "B" dimension is 39.17 ± 0.01 inches.
F. Set the Mach trim actuator in null position.
G. Set the PSEU to ground mode.
H. Do the Pitot System Leak Test.
I. Set the FLT CONTROL A and B switches to ON.
J. Pressurize the pitot system to 2.47 ± 0.05 psi (315 ± 1 knots).

NOTE: Do not exceed a test pressure of 5.0 psi (437 knots) during test.

K. Apply masking tape on the tail cone along the direction of the right elevator trailing edge travel.

NOTE: Elevator travel measuring tool may be used in lieu of tape method.

L. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

NOTE: Ensure the control columns are not disturbed from this position during the next step.

M. Mark the actual elevator trailing edge neutral position on the masking tape.
N. Mark 1.20 inches above and below the neutral position mark on the masking tape.
O. Set the FLT CONTROL B switch to OFF and leave the FLT CONTROL A switch in ON.
P. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly aft until the elevator reaches the 1.20 inch mark above the neutral position. Verify the force required is within the range indicated in the table below.

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

Q. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.
R. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly forward until the elevator reaches the 1.20 inch mark below the neutral position. Verify the force required is within the range indicated in the table below.

MAINTENANCE (M) PROCEDURES 27-31-01 (Cont'd)

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

Model	Elevator Position	Column Force
-8	1.20 ± 0.05 inches up	37.5 ± 10.0 lbs
	1.20 ± 0.05 inches down	40.5 ± 11.5 lbs

S. Set the FLT CONTROL A switch to OFF and the FLT CONTROL B switch to ON.

T. Repeat steps P-R.

U. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

V. Verify the pitot system is pressurized to 2.47 ± 0.05 psi (315 ± 1 knots).

W. Close P6-2 panel ELEV FEEL LIMIT circuit breaker and wait 15 seconds.

X. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly aft until the elevator reaches the 1.20 inch mark above the neutral position. Note the force required.

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

Y. Move the captain's control column lightly forward and aft to put the elevator system in the neutral position.

Z. Open P6-2 panel ELEV FEEL LIMIT circuit breaker and wait 20 seconds.

AA. With a push/pull type spring scale applied to the center of the control wheel, move the captain's control column slowly aft until the elevator reaches the 1.20 inch mark above the neutral position. Note the force required.

NOTE: Ensure the column is not moved beyond the target elevator positions and then back in the direction of neutral prior to taking force readings. If a column overshoot occurs, return the column to neutral and repeat the step. Additionally, ensure the column is not held in place while reading column force; simultaneously read the column force at the moment when the column reaches the correct position.

AB. Verify that the difference in the forces noted in steps X and AA is at least 20.0 lbs.

AC. Return the captain's control column to the neutral position.

AD. Restore the airplane to its usual condition:

1) Make sure that all tools used during testing are removed.
2) Remove all masking tape.
3) Remove seals from the drain holes in each vertical fin pitot probe.
4) Remove the pressure gage lines from systems A and B drain lines.
5) Install the drain plugs to the systems A and B drain lines.
6) Remove regulated air pressure lines from both vertical fin pitot probes.
7) Close P6-2 panel ELEV FEEL LIMIT circuit breaker.
8) Restore airplane to normal hydraulic configuration.
9) Reset the PSEU to normal mode.

27-51-01 Flap Load Relief Function

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided appropriate performance adjustments are applied.

OPERATIONS (O) PROCEDURES

1. For takeoff:
   A. When using Flaps 25, gross weight may not exceed 80,422 kg.
   B. When using Flaps 15, gross weight may not exceed 79,968 kg.
   C. When using Flaps 10, gross weight may not exceed 78,879 kg.
2. For landing:
   A. When using Flaps 40, gross weight may not exceed 45,359 kg.
   B. When using Flaps 30, gross weight may not exceed 47,174 kg.
   C. When using Flaps 15, gross weight may not exceed 62,006 k

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27-61-01 Flight Spoiler Systems

Interval	Installed	Required	Procedure
C	8	6	**(M) (O)**

Except for spoiler pair 5/8, one symmetrical pair may be inoperative provided:

a. Inoperative spoiler pair is deactivated in the retracted position.
b. All other spoilers are verified to operate normally.
c. Antiskid system operates normally.
d. Use of flaps 1 for takeoff is prohibited.
e. Flight altitude remains at or below FL 375.
f. Crosswinds for takeoff and landing do not exceed 25 knots.
g. Crosswinds for a zero crab landing technique do not exceed 13 knots.
h. Appropriate performance adjustments are applied.

NOTE: The flight spoilers include spoiler pairs 2/11, 3/10, 4/9 and 5/8. Flight spoiler pair 5/8 must remain operative.

MAINTENANCE (M) PROCEDURES

Deactivate the affected spoiler pair in the retracted position and verify all remaining spoilers operate normally (AMM 27-00-00/901).

1. Deactivate the affected spoiler pair and secure it in the retracted position:
   A. Extend TE flaps to the Flaps 40 position to allow access to spoiler actuators (PCUs) (AMM 27-51-00/201).
   B. Remove power from hydraulic systems A and B (AMM 29-11-00/201).
   C. Open the following P6-2 panel circuit breakers:
      1) FLIGHT CONTROL SHUTOFF VALVES SPOILER
      2) GROUND SPOILER CONTROL
   D. Open the following P6-5 panel circuit breakers:
      1) SPOILER CONTROL ELEC PWR A
      2) SPOILER CONTROL ELEC PWR B
   E. Operate the manual release cam on the bottom of the PCU and manually move the spoiler panel to the up position.
   F. Disconnect, cap and stow the electrical connector from the PCU.
   G. Manually move the spoiler panel to the stowed position.
   H. Close the following P6-5 panel circuit breakers:
      1) SPOILER CONTROL ELEC PWR A
      2) SPOILER CONTROL ELEC PWR B

MAINTENANCE (M) PROCEDURES 27-61-01 (Cont'd)

I. Close the following P6-2 panel circuit breakers:
   1) FLIGHT CONTROL SHUTOFF VALVES SPOILER
   2) GROUND SPOILER CONTROL

J. Pressurize hydraulic systems A and B (AMM 29-11-00/201).
K. Retract TE flaps (AMM 27-51-00/201).
L. Confirm the deactivated spoiler panels remain in the stowed position.

2. Once each flight day, verify all other spoilers operate normally.
   A. Use the Onboard Maintenance Function (OMF) to perform the ground test for the flight spoilers (AMM 27-61-00/201).
   NOTE: This will extinguish the SPOILERS light on the P5-3 panel.
3. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).
   NOTE: Only one spoiler pair (2/11, 3/10, or 4/9) may be deactivated.

OPERATIONS (O) PROCEDURES

1. Use of flaps 1 for takeoff is prohibited.
2. Limit altitude to FL 375.
3. Do not exceed 25 knot crosswinds for takeoff and landing, and do not exceed 13 knot crosswinds for a zero crab landing technique.
4. Apply the following adjustments:
   A. For dry runway takeoff, reduce the performance limited takeoff weight by 3,357 kg. The minimum takeoff field length required is 3,700 ft.
   B. For wet runway takeoff, reduce the performance limited takeoff weight by 3,357 kg. The minimum takeoff field length required is 4,900 ft.
   C. For landing, base landing performance on manual speedbrakes and reduce the performance limited landing weight by 6,940 kg.

   NOTE: The flight spoilers include spoiler pairs 2/11, 3/10, 4/9 and 5/8. Flight spoiler pair 5/8 must remain operative. Spoiler pair 2/11 or 3/10 inoperative has a takeoff weight limitation of 71,214 kg.

27-61-01 Flight Spoiler Systems

27-61-01-01 Spoiler Control Channels

Interval	Installed	Required	Procedure
C	3	2	

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

27-61-02 Maneuver Load Alleviation (MLA) Function

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided appropriate performance adjustments are applied.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Apply maximum takeoff weight limitation of 47,627 kg.

27-61-03 Emergency Descent Spoilers (EDS) Function

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Flight spoiler systems operate normally.

b. Flight altitude remains at or below FL 300.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

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27-62-01 Auto Speedbrake System

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Auto speedbrake system is deactivated.
b. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the auto speedbrake system (AMM 27-00-00/901).

1. Open and collar the P6-2 panel AUTO SPEED BRAKE circuit breaker.

OPERATIONS (O) PROCEDURES

1. Base landing performance on manual speedbrakes.
2. Extend speedbrakes manually for rejected takeoff or landing.
   A. For rejected takeoff:
      1) Simultaneously close the thrust levers, disengage the autothrottles and apply maximum manual wheel brakes or verify operation of RTO autobrakes.
      2) Manually raise the SPEED BRAKE lever.
      3) Apply the maximum amount of reverse thrust consistent with conditions.
   B. For landing:
      1) Simultaneously close the thrust levers and apply manual wheel brakes or verify operation of autobrakes.
      2) Manually raise the SPEED BRAKE lever.
      3) Apply reverse thrust consistent with conditions.
3. The autobrakes may not disarm when the speedbrake lever is stowed. Autobrakes can be disarmed by positioning the AUTO BRAKE select switch to OFF, applying brakes manually or advancing the thrust

27-62-02 SPEEDBRAKES EXTENDED Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided speedbrakes are verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the speedbrakes operate normally (AMM 27-00-00/901).

1. Confirm wheel chocks are in place.
2. Set the speedbrake lever in the DOWN detent.
3. Visually confirm flight and ground spoilers are all down.
4. Set the speedbrake lever in the FLIGHT DETENT.
5. Visually confirm ground spoilers and flight spoilers deploy.
6. Set the speedbrake lever in the DOWN detent.
7. Visually confirm flight and ground spoilers are all down.
8. Confirm that stabilizer trim is in the green band.
9. Set the flap lever to a takeoff position.
10. Release the parking brake.

For Single Weather Radar

11. Open P6-1 panel WEATHER RADAR RT circuit breaker.

For Dual Weather Radar

12. Open the following circuit breakers:
    A. P6-1 panel WEATHER RADAR RT
    B. P18-3 panel WEATHER RADAR RT RIGHT

    NOTE: Make sure that the circuit breaker(s) for the Weather Radar System are opened. The forward movement of the thrust lever can cause the automatic operation of the system. The operation of this system can cause serious injury to persons and damage to equipment in the area of nose radome.

13. With engines not running, advance both thrust levers beyond 60 degrees.
14. Confirm the takeoff configuration warning horn does not sound.
15. Retard the thrust levers to idle.
16. Return the airplane to its normal configuration, i.e., set parking brake and retract flaps.
17. If the SPEEDBRAKES EXTENDED light remains illuminated while on the ground, extinguish it by removing the light bulb.

For Single Weather Radar

18. Close P6-1 panel WEATHER RADAR RT circuit breaker.

For Dual Weather Radar

19. Close the following circuit breakers:
    A. P6-1 panel WEATHER RADAR RT
    B. P18-3 panel WEATHER RADAR RT RIGHT

27-83-01 Autoslat Systems

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative provided:

a. Remaining autoslat system is verified to operate normally.

b. AUTO SLAT FAIL light system operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

At the time of deferral, verify the remaining autoslat system operates normally.

1. Press and release a system annunciator panel on the glareshield and confirm these lights illuminate:
   A. FLT CONT
   B. AUTO SLAT FAIL
   C. MASTER CAUTION
2. Press and release a MASTER CAUTION light and confirm these lights extinguish:
   A. FLT CONT
   B. AUTO SLAT FAIL
   C. MASTER CAUTION

27-83-02 AUTO SLAT FAIL Light System

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided autoslat systems are verified to operate normally once every 2 flight days.

MAINTENANCE (M) PROCEDURES

Verify autoslat systems operate normally once every two flight days (AMM 27-00-00/901).

1. Do the EXISTING FAULTS test on the SMYD BITE and confirm the display shows NO FAULTS after pushing the YES switch.

27-88-01 Leading Edge Flap/Slat Position Light Systems

27-88-01-02 Leading Edge Indications for Slats 1, 2, 7, and 8

Interval	Installed	Required	Procedure
C	12	11	**(M) (O)**

One leading edge slat indication light on overhead annunciator panel may be inoperative provided:

a. All remaining indications on leading edge devices annunciator panel are verified to operate normally.
b. Stall warning operation of both systems is verified to operate normally.
c. Associated leading edge slat position is visually verified by flight crew before each takeoff and landing.
d. Maximum speed is limited to 300 KIAS at/below FL 200 or 0.65 Mach above FL 200.
e. Maximum altitude is limited to FL300.

MAINTENANCE (M) PROCEDURES

Verify remaining indications on the leading-edge devices annunciator panel operate normally and verify both stall warning systems operate normally (AMM 27-00-00/901).

1. Cycle the flaps.
2. Confirm the remaining lights on the P5 leading edge devices annunciator panel operate normally during flap extension and retraction.
3. Determine which TE flap position(s) result in inoperative LE indication.
4. Set the flap lever to the position that produces the faulty indication.
5. Confirm the leading edge and trailing edge devices are in the correct position.
6. Verify normal operation of both stall warning systems:

   NOTE: If the inoperative slat indication occurs at more than one flap position, repeat the procedure to verify normal operation of both stall warning systems at Flaps 0, Flaps 1 and Flaps 10.

   A. Confirm hydraulic system B pressure is between 2900 and 3100 psi.
   B. Push and hold the P5 stall warning test panel STALL WARNING TEST NO. 1 and NO. 2 switches.
   C. Confirm the control column stick shakers operate normally.
   D. If both stick shakers fail to operate normally, open and collar the P18-2 panel STALL WARN ASYM MODE circuit breaker and then repeat steps B and C.

OPERATIONS (O) PROCEDURES

1. Visually verify the leading-edge slat position for the inoperative indication before each takeoff and landing.
2. Limit airspeed as follows:
   A. At or below FL 200, limit airspeed to 300 KIAS.
   B. Above FL 200, limit airspeed to 0.65 Mach.
   C. Limit altitude to FL300 to ensure a 0.3g margin to initial buffet.

   NOTE 1: Minimum maneuvering speeds will be a maintained at 0.65 Mach up to FL300 at 83,000 kg and at the most limiting CG of 5% MAC (Normal Cruise Altitude Limits Due to Buffet in the Performance Chapter/Section of the Airplane Flight Manual).

   NOTE 2: For failure that indicates slat is not retracted, VNAV will limit the speed target to 230 KIAS or the minimum flaps up maneuver speed, whichever is higher, unless speed intervention is used.

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28-21-01 Pressure Fueling System

28-21-01-01 Refuel Panel Fuel Quantity Indications

28-21-01-02 Fueling Manifold Check Valves

28-21-01-03 Fueling Shutoff Valves

28-21-01-04 Refuel Panel Fueling Power Control Switch

28-22-01 Main Tank Boost Pumps

28-22-02 Center Tank Boost Pumps

28-22-02A One Pump Inoperative

28-22-02B Both Pumps Inoperative

28-22-03 Center Tank Fuel Boost Pump Automatic Shutoff System

28-22-03A Center Tank Boost Pump Considered Inoperative

28-22-03B Center Tank Remains Empty

28-22-03C Center Tank Fuel Used

28-22-04 Fuel Scavenge System

28-22-04A Valve Closed

28-22-04B Valve Open With Center Tank Fuel

28-22-04C Valve Open With Center Tank Empty

28-22-05 Crossfeed VALVE OPEN Light

28-22-06 SPAR VALVE CLOSED Lights

28-22-07 Fuel Shutoff Valve Battery and Charger

28-25-01 APU DC Fuel Boost Pump

28-25-02 APU Fuel Shutoff Valve

28-41-01 Fuel Quantity Indication Systems

28-41-01-01 Main Tanks

28-41-01-02 Center Tank

28-41-01-02A Center Tank Remains Empty

28-41-01-02B Both Center Tank Fuel Pumps Operate Normally

28-41-02 Total Fuel Quantity Indication

28-42-01 Main Tank LOW PRESSURE Light Systems

28-42-01-01 LOW PRESSURE Lights

28-42-02 Center Tank LOW PRESSURE Light Systems

28-42-02-01 LOW PRESSURE Lights

28-43-01 Fuel Temperature Indicating System

28-44-01 Fuel Measuring Sticks

28-21-01 Pressure Fueling System

Interval	Installed	Required	Procedure
C	1	0	**(M)**

NOTE: MEL item 28-21-01 relief may not be applied to defer maintenance of an inoperative system or indication for which an MEL 28-21-01 subitem exists.

MAINTENANCE (M) PROCEDURES

Individual components of the system which operate normally can be used. Use alternate refuelling procedures as per (AMM 12-11-00/301) and (BW-AFM-0001).

28-21-01 **Pressure Fueling System**

28-21-01-01 **Refuel Panel Fuel Quantity Indications**

Interval	Installed	Required	Procedure
C	3	0	**(M)**

MAINTENANCE (M) PROCEDURES

Use AMM 28-00-00/901 as the alternate procedures for refueling the associated fuel tank.

28-21-01 Pressure Fueling System

28-21-01-02 Fueling Manifold Check Valves

Interval	Installed	Required	Procedure
C	3	0	**(M)**

May be inoperative provided associated fueling shutoff valve is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the associated fueling shutoff valve operates normally (AMM 28-00-00/901).

1. Push the test button and confirm refuel lights illuminate and extinguish.

28-21-01 **Pressure Fueling System**

28-21-01-03 **Fueling Shutoff Valves**

Interval	Installed	Required	Procedure
C	3	0	**(M)**

May be inoperative provided:

a. Inoperative fueling shutoff valve(s) is verified to be sealed closed.
b. Fueling receptacle does not leak.

MAINTENANCE (M) PROCEDURES

Refueling the associated fuel tank manually opening and closing the inoperative fueling shutoff valve (AMM 28-00-00/901, Task 28-00-00-040-814) and BW-AFM-0001.

1. Confirm the planned fuel load for the associated tank does not exceed the tank capacity.
2. Manually open the inoperative fuel shutoff valve.
   NOTE: Overfill protection is inoperative. The pressure fueling system will not stop the refuel operation at a set fuel quantity.
3. Manually close the inoperative fuel shutoff valve at the desired fuel quantity.

28-21-01 Pressure Fueling System

28-21-01-04 Refuel Panel Fueling Power Control Switch

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative off provided refuel panel indicator test switch operates normally in FUEL DOOR SWITCH BYPASS position.

INTENTIONALLY
BLANK

28-22-01 Main Tank Boost Pumps

Interval	Installed	Required	Procedure
C	4	3	**(M) (O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Inoperative pump is deactivated.
b. Appropriate minimum fuel quantities are maintained in associated tank.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative fuel pump (AMM 28-00-00/901).

1. For left main tank aft pump inoperative, open and collar P92 panel, FUEL BOOST PUMP TANK 1 AFT.
2. For left main tank forward pump inoperative, open and collar P91 panel, FUEL BOOST PUMP TANK 1 FWD.
3. For right main tank aft pump inoperative, open and collar P91 panel, FUEL BOOST PUMP TANK 2 AFT.
4. For right main tank forward pump inoperative, open and collar P92 panel, FUEL BOOST PUMP TANK 2 FWD.

OPERATIONS (O) PROCEDURES

1. For main tank aft pump inoperative, the minimum fuel load in each main tank for takeoff is 3402 kg and for landing is 1134 kg.
2. For main tank forward pump inoperative, the minimum fuel load in each main tank for takeoff is 2177 kg and for landing is 817 kg.

28-22-02 Center Tank Boost Pumps

28-22-02A One Pump Inoperative

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative with center tank fueled provided:

a. Inoperative pump is deactivated.
b. Center tank quantity indication operates normally.
c. LOW PRESSURE light for remaining center fuel tank pump operates normally.
d. With center tank fueled, fuel quantity remaining in main tanks is adequate to reach a suitable airport if remaining center pump fails at any time.
e. Center tank fuel is accounted for in the airplane weight and balance in the event center tank fuel cannot be used.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative fuel pump (AMM 28-00-00/901).

1. For left center tank pump inoperative, open and collar P91 panel FUEL BOOST PUMP CTR TANK LEFT.
2. For right center tank pump inoperative, open and collar P92 panel FUEL BOOST PUMP CTR TANK RIGHT.

OPERATIONS (O) PROCEDURES

1. Ensure fuel quantity in main tanks is adequate to reach a suitable airport if remaining center pump fails at any time.
2. Adjust the zero fuel weight of the airplane by including the weight of center tank fuel, or adjust the maximum zero fuel weight by subtracting the weight of center tank fuel.
3. Account for the effect on airplane balance (compare to limitations) for the event that the center tank fuel cannot be used.
4. Use the following fuel configuration:
   A. For takeoff:
      1) Set the CROSSFEED selector to the closed position.
      2) Set the AFT and FWD main tank FUEL PUMPS switches to ON for all operating main tank pumps.
      3) Set the operating center tank FUEL PUMPS switch to OFF.

      NOTE: Fuel CONFIG alert may be displayed with fuel in the center tank.
   B. After takeoff:
      1) Set the operating center tank FUEL PUMPS switch to ON if the center tank contains usable fuel.
      2) Set the CROSSFEED selector to the open position.
   C. When the operating center tank pump LOW PRESSURE light illuminates:
      1) Set the operating center tank FUEL PUMPS switch to OFF.
      2) Set the CROSSFEED selector to the closed position.

      NOTE: Prior to the center tank LOW PRESSURE light illuminating, a fuel imbalance between the main tanks may be indicated as a result of differences in fuel pump output pressures when operating with the crossfeed valve open.

OPERATIONS (O) PROCEDURES 28-22-02A (Cont'd)

4. Use the following fuel configuration:
   D. For landing:
      1) Set the operating center tank FUEL PUMPS switch to OFF whether or not the center tank contains usable fuel.
      2) Set the CROSSFEED selector to the closed position whether or not the center tank contains usable fuel.

28-22-02 Center Tank Boost Pumps

28-22-02B Both Pumps Inoperative

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Inoperative pumps are deactivated.
b. Center tank quantity indication operates normally.
c. Center tank fuel is considered unusable and is accounted for in the airplane weight and balance.

NOTE: AFM fuel loading and usage limitations are for usable fuel.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative fuel pumps (AMM 28-00-00/901).

1. For left center tank pump inoperative, open and collar P91 panel FUEL BOOST PUMP CTR TANK LEFT.
2. For right center tank pump inoperative, open and collar P92 panel FUEL BOOST PUMP CTR TANK RIGHT.

OPERATIONS (O) PROCEDURES

NOTE 1: Center tank fuel is considered unusable. For fuel in center tank, FMC fuel prediction for INSUFFICIENT FUEL alert on CDU will not be accurate.

NOTE 2: For fuel in center tank, the fuel CONFIG alert may be displayed.

1. For fuel in center tank:
   A. Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
   B. Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

28-22-03 **Center Tank Fuel Boost Pump Automatic Shutoff System**

28-22-03A **Center Tank Boost Pump Considered Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated center tank fuel boost pump is considered inoperative.

NOTE: The airplane must also be dispatched using MEL Item 28-22-02.

28-22-03 Center Tank Fuel Boost Pump Automatic Shutoff System

28-22-03B Center Tank Remains Empty

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided center tank remains empty.

MAINTENANCE NOTE

Set the associated CTR FUEL PUMP switch to OFF.

28-22-03 **Center Tank Fuel Boost Pump Automatic Shutoff System**

28-22-03C **Center Tank Fuel Used**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative with center tank fueled provided:

a. Both center tank pump LOW PRESSURE light systems operate normally.
b. Center tank fuel quantity indication system operates normally.
c. Center tank boost pumps are not ON unless personnel are available in flight deck to monitor LOW PRESSURE lights.
d. For ground operations, center tank boost pump switches are not ON unless center tank fuel quantity exceeds 453 kg, except when defueling or transferring fuel.
e. Center tank boost pumps are OFF for takeoff if center tank fuel is less than 2,300 kg with airplane readied for initial taxi.
f. Center tank fuel pumps may be repositioned ON when established in cruise flight if more than 453 kg of fuel remain in center tank.
g. Both center tank boost pumps are selected OFF when center tank fuel quantity reaches 453 kg of fuel during climb or cruise.
h. Both center tank boost pumps are selected OFF when center tank fuel quantity reaches 1,400 kg of fuel during descent or landing.
i. Both center tank fuel boost pumps are selected OFF at first indication of fuel pump low pressure.
j. Defueling with passengers on board is prohibited.

28-22-04 Fuel Scavenge System

28-22-04A Valve Closed

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided fuel scavenge shutoff valve is closed.

28-22-04 Fuel Scavenge System

28-22-04B Valve Open With Center Tank Fuel

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Scavenge shutoff valve is open.
b. Left main tank forward boost pump is not used.
c. Left main tank aft boost pump operates normally.
d. Appropriate minimum fuel quantities are maintained in left main tank.

OPERATIONS (O) PROCEDURES

1. Set the left main tank FWD FUEL PUMPS switch OFF.
2. Use MEL Item 28-22-01 (O) procedure.

28-22-04 **Fuel Scavenge System**

28-22-04C **Valve Open With Center Tank Empty**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Scavenge shutoff valve is open.

b. Center tank remains empty.

28-22-05 Crossfeed VALVE OPEN Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations, may be inoperative provided:

a. Crossfeed valve is verified to operate normally.
b. Fuel quantity indications for both main tanks operate normally.

MAINTENANCE (M) PROCEDURES

Verify crossfeed valve operates normally (AMM 28-00-00/901).

1. Set crossfeed valve selector to open position.
2. Confirm crossfeed valve manual override lever moves to open position.
3. Set crossfeed valve selector to closed position.
4. Confirm crossfeed valve manual override lever moves to closed position.

28-22-06 SPAR VALVE CLOSED Lights

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided associated valve is verified to operate normally once each flight day.

MAINTENANCE (M) PROCEDURES

Verify that the associated spar valve operates normally (AMM 28-00-00/901).

1. On the P10 control stand, confirm the associated engine start levers are in CUTOFF position.
2. Confirm that the ENGINE START 1 and 2 switches are in the OFF position.
3. Open and collar the following circuit breakers:
   A. P18-2 panel ENGINE 1 START VALVE
   B. P6-2 panel ENGINE 2 START VALVE
4. Set the applicable engine start lever to the IDLE position.
5. Visually confirm associated spar valve override lever moved to the OPEN position.
6. On the P8-1 Fire Control Panel, push the applicable manual override button
   CAUTION: DO NOT TURN THE FIRE HANDLE. YOU WILL CAUSE THE FIRE BOTTLES TO RELEASE THEIR CONTENTS IF YOU TURN THE HANDLE.
7. On the P8-1 Fire Control Panel, pull the applicable Engine Fire Handle.
8. Visually confirm that the associated spar valve override lever moved to the CLOSED position.
9. Push the Engine Fire Handle to the normal position.
10. Visually confirm that the associated spar valve override lever moved to the OPEN position.
11. Set the applicable engine start lever to the CUTOFF position.
12. Visually confirm that the associated spar valve override lever moved to the CLOSED position.
13. Close the following circuit breakers:
    A. P18-2 panel ENGINE 1 START VALVE
    B. P6-2 panel ENGINE 2 START VALVE

28-22-07 Fuel Shutoff Valve Battery and Charger

Interval	Installed	Required	Procedure
D	1	0	**(M)**

May be inoperative deactivated.

MAINTENANCE (M) PROCEDURES

Deactivate the fuel shutoff valve battery and charger (AMM 28-00-00/901).

1. Open and collar the P6-3 panel FUEL SHUTOFF VALVES PWR PACK circuit breaker.

INTENTIONALLY

BLANK

28-25-01 APU DC Fuel Boost Pump

Interval	Installed	Required	Procedure
D	1	0	**(M)**

May be inoperative provided pump is deactivated.

MAINTENANCE (M) PROCEDURES

Deactivate the APU DC fuel boost pump (AMM 28-00-00/901).

1. Open and collar P6 panel AUX POWER UNIT FUEL BOOST PUMP circuit breaker

28-25-02 APU Fuel Shutoff Valve

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations, may be inoperative provided:

a. Valve is deactivated closed.
b. APU is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL Item 49-11-01.

MAINTENANCE (M) PROCEDURES

Deactivate the APU fuel shutoff valve (AMM 28-00-00/901):

1. Open and collar the following P6 panel circuit breakers:
   A. AUX POWER UNIT CONT
   B. APU FIRE SW POWER.
2. Gain access to APU shutoff valve.
3. Disconnect, cap and stow the electrical connector.
4. Rotate the manual override handle to the closed position.
5. Lockwire the manual override handle in the closed position.

28-41-01 Fuel Quantity Indication Systems

28-41-01-01 Main Tanks

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

Except for ER operations, may be inoperative provided:

a. Fuel quantity in associated tank is verified by an acceptable procedure (refer to **Maintenance (M) Procedures**).
b. Both fuel pumps in associated tank operate normally.
c. Fuel flow indicators operate normally.
d. Center tank fuel quantity indication system operates normally.
e. FMC FUEL is initialized with known total fuel quantity.

NOTE: Inoperative flight deck fuel quantity indications can affect refuel panel fuel quantity indications. See MEL item 28-21-01-01.

MAINTENANCE (M) PROCEDURES

Verify the fuel quantity in the associated main tank (AMM 28-00-00/901).

OPERATIONS (O) PROCEDURES

NOTE 1: Total fuel quantity indication may be blank or show incorrect values.

NOTE 2: The FUEL DISAGREE and fuel IMBAL alerts may be inoperative. The fuel LOW alert for the associated tank may be inoperative.

1. Select FMC CALCULATED fuel quantity on CDU FUEL PROGRESS page 5/5.
2. Operate the flight with fuel crossfeed valve(s) closed unless directed otherwise.

28-41-01 **Fuel Quantity Indication Systems**

28-41-01-02 **Center Tank**

28-41-01-02A **Center Tank Remains Empty**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. Center tank remains empty.

b. Both main tank fuel quantity indication systems operate normally.

NOTE: Inoperative flight deck fuel quantity indications can affect refuel panel fuel quantity indications. See MEL item 28-21-01-01.

OPERATIONS (O) PROCEDURES

NOTE 1: Total fuel quantity indication may be blank or show incorrect values.

NOTE 2: The FUEL DISAGREE alert may be inoperative.

NOTE 3: The fuel CONFIG alert may not be displayed when appropriate.

1. Select FMC CALCULATED fuel quantity on CDU FUEL PROGRESS page 5/5.

28-41-01 **Fuel Quantity Indication Systems**

28-41-01-02 **Center Tank**

28-41-01-02B **Both Center Tank Fuel Pumps Operate Normally**

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

Except for ER operations, may be inoperative provided:

a. Fuel quantity in center tank is verified by an acceptable procedure (refer to **Maintenance (M) Procedures**).
b. Both center tank fuel pumps operate normally.
c. Fuel flow indicators operate normally.
d. Both main tank fuel quantity indication systems operate normally.
e. FMC FUEL is initialized with known total fuel quantity.

NOTE: Inoperative flight deck fuel quantity indications can affect refuel panel fuel quantity indications. See MEL item 28-21-01-01.

MAINTENANCE (M) PROCEDURES

Verify the fuel quantity in the center tank (AMM 28-00-00/901).

OPERATIONS (O) PROCEDURES

NOTE 1: Total fuel quantity indication may be blank or show incorrect values.

NOTE 2: The FUEL DISAGREE alert may be inoperative.

NOTE 3: The fuel CONFIG alert may not be displayed when appropriate.

1. Select FMC CALCULATED fuel quantity on CDU FUEL PROGRESS page 5/5.

28-41-02 Total Fuel Quantity Indication

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Except for ER operations, may be inoperative provided:

a. Fuel flow indicators operate normally.

b. FMC FUEL is initialized with known total fuel quantity.

OPERATIONS (O) PROCEDURES

NOTE: FUEL DISAGREE alert will be inoperative.

1. Select FMC CALCULATED fuel quantity on CDU FUEL PROGRESS page 5/5.

28-42-01 Main Tank LOW PRESSURE Light Systems

Interval	Installed	Required	Procedure
C	4	3	

May be inoperative provided associated pump is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 28-22-01.

28-42-01 Main Tank LOW PRESSURE Light Systems

28-42-01-01 LOW PRESSURE Lights

Interval	Installed	Required	Procedure
C	4	3	**(M)**

May be inoperative provided:

a. MASTER CAUTION lights and FUEL system annunciator light are verified to operate normally.
b. Both pumps in associated tank operate normally.

MAINTENANCE (M) PROCEDURES

Verify the MASTER CAUTION lights and FUEL system annunciator light operate normally for a pump low pressure condition in the associated main tank (AMM 28-00-00/901).

1. Reset the MASTER CAUTION lights by pressing either Master Caution light module.
2. Set both fuel boost pump switches for the associated tank to OFF.
3. Confirm the MASTER CAUTION lights and FUEL system annunciator panel light illuminate.
4. If the MASTER CAUTION lights and FUEL system annunciator panel light illuminate, no further maintenance steps are required.

28-42-02 Center Tank LOW PRESSURE Light Systems

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated pump is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 28-22-02.

28-42-02 Center Tank LOW PRESSURE Light Systems

28-42-02-01 LOW PRESSURE Lights

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. MASTER CAUTION lights and FUEL system annunciator light are verified to operate normally.
b. Both center tank pumps operate normally.

MAINTENANCE (M) PROCEDURES

Verify the MASTER CAUTION lights and FUEL system annunciator light operate normally for a pump low pressure condition in the center tank (AMM 28-00-00/901).

NOTE: Perform the following procedure for the center tank fuel boost pump(s) associated with the inoperative LOW PRESSURE light(s).

1. On the P5-2 fuel system module, set the FUEL PUMPS CTR L and R switches to the OFF position.
2. On the P5 overhead panel, confirm that the ENGINE START 1 and 2 switches are in the OFF position.
3. On the P10 control stand, confirm that the Engine Start Levers are in the CUTOFF position.
4. For an inoperative left center tank fuel boost pump LOW PRESSURE light:
   A. Open and collar the P91 panel FUEL BOOST PUMP CTR TANK LEFT circuit breaker.
   B. Reset the MASTER CAUTION lights by pressing either MASTER CAUTION light.
   C. Verify that the Captain's and First Officer's MASTER CAUTION lights go off.
   D. Verify that the FUEL annunciator light goes off on the Captain's system annunciator light panel.
   E. Set the FUEL PUMPS CTR L switch to the ON position.
      NOTE: There will be a delay of approximately 10 seconds before the lights come on.
   F. Verify the MASTER CAUTION lights and FUEL system annunciator panel light illuminate.
   G. Set the FUEL PUMPS CTR L switch to the OFF position.
   H. Close P91 panel FUEL BOOST PUMP CTR TANK LEFT circuit breaker.
5. For an inoperative right center tank fuel boost pump LOW PRESSURE light:
   A. Open and collar the P92 panel FUEL BOOST PUMP CTR TANK RIGHT circuit breaker.
   B. Reset the MASTER CAUTION lights by pressing either MASTER CAUTION light.
   C. Verify that the Captain's and First Officer's MASTER CAUTION lights go off.
   D. Verify that the FUEL annunciator light goes off on the Captain's system annunciator light panel.
   E. Set the FUEL PUMPS CTR R switch to the ON position.
      NOTE: There will be a delay of approximately 10 seconds before the lights come on.
   F. Verify the MASTER CAUTION lights and FUEL system annunciator panel light illuminate.
   G. Set the FUEL PUMPS CTR R switch to the OFF position.
   H. Close the P92 panel FUEL BOOST PUMP CTR TANK RIGHT circuit breaker.

OPERATIONS (O) PROCEDURES

When dispatching with one or both center tank fuel pump LOW PRESSURE lights inoperative and center tank fuel is to be used, position the two CTR FUEL PUMPS switches to OFF when both the MASTER CAUTION lights and the FUEL system annunciator panel light illuminate as a result of center tank fuel depletion.

28-43-01 Fuel Temperature Indicating System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided total air temperature (TAT) is used as an indication of fuel temperature.

INTENTIONALLY

BLANK

28-44-01 Fuel Measuring Sticks

Interval	Installed	Required	Procedure
C	16	0	

May be inoperative or missing provided fuel quantity is determined by the fuel quantity indication systems.

INTENTIONALLY
BLANK

29-09-01 Hydraulic Reservoir Pressurization System Sources

29-09-02 Hydraulic Reservoir Pressurization System Wheel Well Air Pressure Gauge

29-18-01 Hydraulic Reservoir Fill System

29-31-01 Hydraulic Systems A and B Pressure Indications

29-32-01 Electric Hydraulic Pump Overheat Indication Systems

29-33-01 Hydraulic Systems A and B Flight Deck Quantity Indications

29-33-02 Standby System Low Quantity Indication System

29-33-03 Hydraulic Reservoir Wheel Well Quantity Indicators

29-34-01 Engine Driven Hydraulic Pump Low Pressure Indication Systems

29-34-02 Engine Hydraulic Pump Low Pressure Indication Systems

29-34-03 Standby System Pump Low Pressure Indication

INTENTIONALLY

BLANK

29-09-01 Hydraulic Reservoir Pressurization System Sources

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided it is verified hydraulic reservoirs can be pressurized.

MAINTENANCE (M) PROCEDURES

Verify the hydraulic reservoirs can be pressurized (AMM 29-00-00/901).

1. Disconnect the inoperative pressurization supply line at the reservoir restrictor cross fitting located in the wheel well.
2. Install a cap on the reservoir restrictor cross fitting.
3. Install a cap on the disconnected pressurization supply line.
4. Pressurize the pneumatic manifold.
5. Verify the hydraulic reservoir air pressure gauge located on the forward wheel well bulkhead indicates the same pressure as the pneumatic duct pressure indicator on the flight deck forward overhead panel.

29-09-02 Hydraulic Reservoir Pressurization System Wheel Well Air Pressure Gauge

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

If desired, remove indicator and install a suitable plug/cap at the tube end of the air pressure gauge port to prevent air leakage.

29-18-01 Hydraulic Reservoir Fill System

Interval	Installed	Required	Procedure
C	1	0	

INTENTIONALLY

BLANK

29-31-01 Hydraulic Systems A and B Pressure Indications

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative provided:

a. Associated system pressure is verified to be normal before each departure.
b. Associated system pump LOW PRESSURE indications operate normally.
c. Associated flight deck quantity indication operates normally.

OPERATIONS (O) PROCEDURES

Verify the associated system pressure is normal before each departure.

1. For System A, confirm the Flight Control Panel LOW PRESSURE light for System A extinguishes when the hydraulic system is pressurized.
2. For System B:
   A. With Systems A and B depressurized, confirm the brake pressure indicator shows less than 2800 psi. If not, apply and release brakes until pressure drops below 2800 psi.
   B. Set the Flight Control Panel FLT CONTROL B and SPOILER B switches to OFF.
   C. Pressurize System B hydraulics with electric hydraulic pump switch ELEC 1.
   D. Confirm pressure on the brake pressure indicator increases to between 2800 psi and 3200 psi.
   E. Set the Flight Control Panel SPOILER B switch to ON.
   F. Confirm the Flight Control Panel LOW PRESSURE light for System B is illuminated.
   G. Set the Flight Control Panel FLT CONTROL B switch to ON.
   H. Confirm the Flight Control Panel LOW PRESSURE light for System B extinguishes.

INTENTIONALLY
BLANK

29-32-01 Electric Hydraulic Pump Overheat Indication Systems

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated pump low pressure indication operates normally.

INTENTIONALLY

BLANK

29-33-01 Hydraulic Systems A and B Flight Deck Quantity Indications

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided:

a. Associated system quantity is verified to be normal before each departure.
b. Associated system pressure indication operates normally.
c. Associated system pump LOW PRESSURE indications operate normally.

MAINTENANCE (M) PROCEDURES

Verify the associated system hydraulic reservoir quantity is normal before each departure (AMM 29-00-00/901).

1. Configure the airplane with:
   A. Flaps/leading edge devices retracted.
   B. Spoilers down.
   C. All flight controls are in neutral position.
   D. Thrust reversers stowed.
2. Confirm fluid quantity is normal at the hydraulic quantity indicator mounted on the associated reservoir.

OPERATIONS NOTE

For System B flight deck quantity indication inoperative, nose wheel steering may not be available if hydraulic System A power is lost.

29-33-02 Standby System Low Quantity Indication System

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Standby system quantity is verified to be normal before each departure.
b. Standby system pump low pressure indication operates normally.

MAINTENANCE (M) PROCEDURES

Verify System B hydraulic reservoir quantity is normal before each departure (AMM 29-00-00/901).

1. Configure the airplane with:
   A. Flaps/leading edge devices retracted.
   B. Spoilers down.
   C. All flight controls are in neutral position.
   D. Thrust reversers stowed.
2. Confirm the gauge mounted on the System B reservoir indicates more than refill or confirm the System B flight deck quantity indication indicates more than refill.
3. Disconnect, cap and stow the low level quantity switch electrical connector.

29-33-03 Hydraulic Reservoir Wheel Well Quantity Indicators

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated flight deck quantity indicator operates normally.

INTENTIONALLY

BLANK

29-34-01 Engine Driven Hydraulic Pump Low Pressure Indication Systems

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided:

a. Associated pump pressure is verified to be normal before each departure.
b. Associated system flight deck quantity indication operates normally.
c. Associated system electric hydraulic pump low pressure indication operates normally.

MAINTENANCE NOTE

For associated pressure switch inoperative, disconnect, cap, and stow the electrical connector (AMM 29-34-11/401).

OPERATIONS (O) PROCEDURES

Verify associated engine driven hydraulic pump pressure is normal before each departure.

1. With both system pumps depressurized, verify the associated Flight Control LOW PRESSURE light is illuminated.
2. Dry motor or start the associated engine.
3. Set the associated engine hydraulic pump switch (ENG 1 or ENG 2) to ON.
4. Set the associated system electric hydraulic pump switch (ELEC 1 or ELEC 2) to OFF.
5. Verify the associated Flight Control LOW PRESSURE light extinguishes.
6. Confirm the associated system pressure is between 2800 psi and 3200 psi.

29-34-02 Electric Hydraulic Pump Low Pressure Indication Systems

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided:

a. Associated pump pressure is verified to be normal before each departure.
b. Associated system flight deck quantity indication operates normally.
c. Associated system engine driven hydraulic pump low pressure indication operates normally.

MAINTENANCE NOTE

For associated pressure switch inoperative, disconnect, cap, and stow the electrical connector (AMM 29-34-21/401).

OPERATIONS (O) PROCEDURES

Verify associated electric hydraulic pump pressure is normal before each departure.

1. With both system pumps depressurized, verify the associated Flight Control LOW PRESSURE light is illuminated.
2. Power the associated generator bus (Generator Bus No. 1 or No. 2).
3. Set the associated electric hydraulic pump switch (ELEC 1 or ELEC 2) to ON.
4. Set the associated system engine hydraulic pump switch (ENG 1 or ENG 2) to OFF.
5. Verify the associated Flight Control LOW PRESSURE light extinguishes.
6. Confirm the associated system pressure is between 2800 psi and 3200 psi.

29-34-03 Standby System Pump Low Pressure Indication

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Standby pump pressure is verified to be normal before each departure.
b. Standby system low quantity indication operates normally.
c. System B hydraulic pumps operate normally.

MAINTENANCE (M) PROCEDURES

Verify the standby system pump pressure is normal before each departure (AMM 29-00-00/901).

1. Get ground clearance to extend the leading edge flaps.
2. Set the flight control panel ALTERNATE FLAPS master switch to ARM.
3. Momentarily set the flight control panel ALTERNATE FLAPS position switch to DOWN.
4. On the LE DEVICES annunciator panel, confirm the lights indicate all leading edge devices are fully extended within approximately one minute.
5. Set both the flight control panel ALTERNATE FLAPS master and position switches to OFF.
6. Set the hydraulic panel HYD PUMPS B ELEC 1 switch to ON.
7. Confirm the HYD PUMPS B ELEC 1 LOW PRESSURE light extinguishes.
8. Leave the HYD PUMPS B ELEC 1 switch on until the flaps are up.

INTENTIONALLY

BLANK

30-11-01 Wing Anti-Ice Valves

30-11-01A Valve Closed

30-11-01B Valve Open

30-11-02 Wing VALVE Lights

30-11-03 Control Stand Wing Anti-Ice Switches

30-11-03A Inoperative Closed

30-11-03B Inoperative Open

30-21-01 Engine (Cowl) Anti-Ice Valves

30-21-01A Valve Closed

30-21-02 Engine Anti-Ice COWL VALVE Position Lights and TAI Indications

30-21-02A Associated Valve Operates Normally

30-21-02B One Per Engine Inoperative

30-21-02C Associated Valve Considered Inoperative

30-21-03 COWL ANTI-ICE Lights

30-21-03A Not Operated In Icing Conditions

30-31-01 Pitot Probe Heaters

30-31-01-01 Captain and First Officer Pitot Heaters

30-31-01-02 Auxiliary Pitot Heater (Right Lower Probe)

30-31-02 Total Air Temperature Probe Heater

30-31-03 Angle of Attack Sensor Heaters

30-31-04 Elevator Pitot Probe Heaters

30-31-05 Pitot and Temperature Probe Heater Lights

30-31-05-01 Pitot Probe Heater Lights

30-31-05-02 Total Air Temperature (TAT) Probe Heater Light

30-31-05-02A Associated Heater Verified To Operate Normally

30-31-05-02B Associated Heater Inoperative

30-31-06 Alpha Vane Heater Light Systems

30-31-06A *Associated Heater Verified To Operate Normally*

30-31-06B *Associated Heater Inoperative*

30-41-01 Electrically Heated Windshields

30-41-01-01 No. 1 Windows

30-41-01-02 No. 2 Windows

30-41-01-03 No. 3 Window Heat Systems

30-41-02 Windshield Defog System

30-42-01 Windshield Wiper Systems

30-42-01-01 Park Function

30-42-01-02 Intermittent Speed Function

30-42-01-03 Low Speed Function

30-42-01-04 High Speed Function

30-42-01-04A *One Inoperative*

30-42-01-04B *Both Inoperative*

30-71-01 Drain Mast Heaters

30-81-01 Ice Detection System - *N/A*

30-11-01 **Wing Anti-Ice Valves**

30-11-01A **Valve Closed**

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Affected valve is secured closed.
b. Airplane is not operated in known or forecast icing conditions.

MAINTENANCE (M) PROCEDURES

Secure the affected valve closed (AMM 30-00-00/901).

1. Deactivate the leading edge slats (AMM 27-81-00/201).
2. Open the following P18-3 panel circuit breakers:
   A. ANTI-ICE & RAIN WING ANTI-ICE VALVE
   B. ANTI-ICE & RAIN WING ANTI-ICE VALVE-L
   C. ANTI-ICE & RAIN WING ANTI-ICE VALVE-R
3. Gain access to the inoperative wing anti-ice valve.
4. Disconnect, cap and stow the electrical connector from the affected wing anti-ice valve.
5. Ensure the arrow on the visual position indicator points to the CLOSED position.
6. Install the lock pin in the lock hole on the applicable wing anti-ice valve in the CLOSED position.
   NOTE: To avoid damage, tighten lock pin by hand prior to applying torque with a wrench.
7. Torque the manual lock bolt to 19-21 in-lb (2.1-2.4 N-m).
8. Close the open circuit breakers.
9. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known or forecast icing conditions.
2. The associated L/R VALVE light will be illuminated.

30-11-01 Wing Anti-Ice Valves

30-11-01B Valve Open

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative secured open provided:

a. Except for engine start, associated manifold is depressurized when outside air temperature is above 50 degrees F, (10 degrees C).
b. Associated engine bleed thrust limits are followed, WING ANTI-ICE switch is set to ON position when use of the wing anti-ice system is required, and the associated manifold is pressurized.
c. Air conditioning and pressurization requirements are followed when one manifold is depressurized.
d. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Prepare the airplane for flight with wing anti-ice valve secured open (AMM 30-00-00/901).

1. For dispatch with left valve open:
   A. To open the left valve when APU bleed air will be used for engine start, do these steps:
      1) Deactivate the leading edge slats (AMM 27-81-00/201).
      2) Open the following P18-3 panel circuit breakers:
         a. ANTI-ICE & RAIN WING ANTI-ICE VALVE
         b. ANTI-ICE & RAIN WING ANTI-ICE VALVE-L
      3) Gain access to the left wing anti-ice valve.
      4) Disconnect, cap and stow the electrical connector from the left wing anti-ice valve.
         NOTE: Disconnecting the left wing anti-ice valve's electrical connector will cause the associated wing anti-ice L VALVE light to illuminate.
      5) Move the visual position indicator on the left wing anti-ice valve to the CLOSED position.
      6) Ensure all personnel are cleared from the right engine danger area.
      7) Start the right engine (AMM 71-00-00/201).
      8) Set the ISOLATION VALVE switch to CLOSE and the APU BLEED air switch to OFF to depressurize the left pneumatic duct.
      9) Loosen the manual lock bolt from the stowed position on the valve.
      10) After the pneumatic duct is depressurized, move the visual position indicator on the left wing anti-ice valve to the OPEN position.
      11) Install the manual lock bolt to the lock position on the valve.
      12) Tighten the manual lock bolt to 19-21 in-lb (2.1-2.4 N-m).
      13) Activate the leading edge slats (AMM 27-81-00/201).
      14) Close the open circuit breakers.
      15) When personnel have been cleared from the engine danger area, inform the flight deck the left engine may be started (using engine crossbleed).

MAINTENANCE (M) PROCEDURES 30-11-01B (Cont'd)

NOTE: For subsequent dispatches after initial MMEL dispatch option 30-11-01B deferral with the left wing anti-ice valve secured open and using APU bleed air for engine start, repeat steps 1.A.1 through 1.A.15.

B. To open the left valve when ground service air will be used for engine start, do these steps:

1) Deactivate the leading edge slats (AMM 27-81-00/201).
2) Open the following P18-3 panel circuit breakers:
   a. ANTI-ICE & RAIN WING ANTI-ICE VALVE
   b. ANTI-ICE & RAIN WING ANTI-ICE VALVE-L
3) Gain access to the left wing anti-ice valve.
4) Disconnect, cap and stow the electrical connector from the left wing anti-ice valve.

NOTE: Disconnecting the left wing anti-ice valve's electrical connector will cause the associated wing anti-ice L VALVE light to illuminate.

5) Loosen the manual lock bolt from the stowed position on the valve.
6) Move the visual position indicator on the left wing anti-ice valve to the OPEN position.
7) Install the manual lock bolt to the lock position on the valve.
8) Tighten the manual lock bolt to 19-21 in-lb (2.1-2.4 N-m).
9) Close the open circuit breakers.
10) Activate the leading edge slats (AMM 27-81-00/201).
11) When personnel have been cleared from the engine danger area, inform the flight deck the right engine may be started (isolation valve closed) and then the left engine may be started (using crossbleed start).

NOTE: Personnel should take special care to remain well clear of the right engine inlet hazard area when disconnecting the start cart after right engine start

NOTE: For subsequent dispatches after initial MEL dispatch option 30-11-01B deferral with the left wing anti-ice valve secured open and using ground service air for engine start, refer to the (O) procedure for guidance.

2. For dispatch with right valve open:

A. To open the right valve when APU bleed air will be used for engine start, do these steps:

1) Deactivate the leading edge slats (AMM 27-81-00/201).
2) Open the following P18-3 panel circuit breakers:
   a. ANTI-ICE & RAIN WING ANTI-ICE VALVE
   b. ANTI-ICE & RAIN WING ANTI-ICE VALVE-R
3) Gain access to the right wing anti-ice valve.
4) Disconnect, cap and stow the electrical connector from the right wing anti-ice valve.

NOTE: Disconnecting the right wing anti-ice valve's electrical connector will cause the associated wing anti-ice R VALVE light to illuminate.

5) Loosen the manual lock bolt from the stowed position on the valve.
6) Move the visual position indicator on the right wing anti-ice valve to the OPEN position.
7) Install the manual lock bolt to the lock position on the valve.
8) Tighten the manual lock bolt to 19-21 in-lb (2.1-2.4 N-m).

MAINTENANCE (M) PROCEDURE S 30-11-01B (Cont'd)

9) Close the open circuit breakers.
10) Activate the leading edge slats (AMM 27-81-00/201).
11) When personnel have been cleared from the engine danger area, inform the flight deck the left engine may be started (isolation valve closed) and then the right engine may be started (using crossbleed start).

NOTE: For subsequent dispatches after initial MEL dispatch option 30-11-01B deferral with the right wing anti-ice valve secured open and using APU bleed air for engine start, refer to the (O) procedure for guidance.

B. To open the right valve when ground service air will be used for engine start, do these steps:
1) Deactivate the leading edge slats (AMM 27-81-00/201).
2) Open the following P18-3 panel circuit breakers:
   a. ANTI-ICE & RAIN WING ANTI-ICE VALVE
   b. ANTI-ICE & RAIN WING ANTI-ICE VALVE-R
3) Gain access to the inoperative wing anti-ice valve.
4) Disconnect, cap and stow the electrical connector from the rightwing anti-ice valve.

NOTE: Disconnecting the right wing anti-ice valve's electrical connector will cause the associated wing anti-ice R VALVE light to illuminate.

5) Move the visual position indicator on the right wing anti-ice valve to the CLOSED position.
6) Ensure all personnel are cleared from the left engine danger area.
7) Start the left engine (AMM 71-00-00/201).
8) Remove the ground air source from the airplane.
9) Set the ISOLATION VALVE switch to CLOSE to depressurize the right pneumatic duct.
10) Loosen the manual lock bolt from the stowed position on the valve.
11) After the pneumatic duct is depressurized, move the visual position indicator on the right wing anti-ice valve to the OPEN position.
12) Install the manual lock bolt to the lock position on the valve.
13) Tighten the manual lock bolt to 19-21 in-lb (2.1-2.4 N-m).
14) Close the open circuit breakers.
15) Activate the leading edge slats (AMM 27-81-00/201).
16) When personnel have been cleared from the engine danger area, inform the flight deck the right engine may be started (using engine crossbleed).

NOTE: For subsequent dispatches after initial MEL dispatch option 30-11-01B deferral with the right wing anti-ice valve secured open and using ground service air for engine start, repeat steps 2.B.1 through 2.B.16.

OPERATIONS (O) PROCEDURES 30-11-01B (Cont'd)

1. The associated L/R VALVE light may be illuminated.
2. Bleed trip off and possible loss of cabin pressure above approximately FL350 may occur.
3. During ground operations where use of the wing anti-ice system is required, the engine bleed manifold associated with the inoperative wing anti-ice valve must be pressurized and the WING ANTI-ICE switch must be set to the ON position.
4. When the left wing anti-ice valve is failed open, do not use APU bleed air for air conditioning purposes on the ground. The APU may only be used for engine start.
5. When the right wing anti-ice valve is failed open, APU bleed air may be used for air conditioning purposes on the ground using only the left AC pack and with the isolation valve closed.
6. With the right wing anti-ice valve secured open and APU used for bleed air, the isolation valve must be closed and the left engine must first be started using the APU. The right engine can then be started using crossbleed.
7. With the left wing anti-ice valve secured open and using ground service air for engine start, the isolation valve must be closed and the right engine must first be started using the start cart. The start cart is then disconnected and the left engine can be started using crossbleed.

   NOTE: Personnel should take special care to remain well clear of the right engine inlet hazard area when disconnecting the start cart after right engine start.
8. The engine BLEED switch for the associated manifold must be set to the OFF position for takeoff and landing, and the ISOLATION VALVE switch must remain in the CLOSE position to maintain isolated bleed sources. If icing conditions are encountered in flight where use of the wing anti-ice system is required, the WING ANTI-ICE switch must be set to the ON position and the associated manifold must be pressurized using either of the following airplane configurations:
   A. Associated engine BLEED switch is set to ON position and ISOLATION VALVE switch is set to CLOSE position.
   B. Associated engine BLEED switch is set to OFF position and ISOLATION VALVE switch is set to OPEN position.
9. When dispatching with a single engine bleed on for takeoff due to one wing anti-ice valve failed open (airplane pressurized):
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
10. Performance limited weights are reduced by the following amount for an inoperative wing anti-ice valve in the open position.

    NOTE: There is no penalty associated with takeoff field length, climb, and obstacle clearance limit weights.

OPERATIONS (O) PROCEDURES 30-11-01B (Cont'd)

Takeoff Fuel-Jettison	**Enroute Climb**	**Approach & Landing**
1,724kg	4,763kg	1,724kg

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum ift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds. Multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below – Enroute Diversion Speed Effects.

Diversion Speed	**Factor**
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

11. Increase flight planning fuel by either incrementing the current fuel flow correction factor by 1.6% or flight planning fuel by 1.8% for one inoperative valve in the open position.
12. Increase engine-out ETOPS critical fuel reserves by an additional 0.4% on top of the flight planning fuel penalty noted above for one inoperative valve in the open position. The all-engine ETOPS critical fuel reserves would be covered by the flight planning fuel penalty noted above.

30-11-02 Wing VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided valve is verified to operate normally before operating in known or forecast icing conditions.

MAINTENANCE (M) PROCEDURES

Verify the associated wing anti-ice valve operates normally (AMM 30-00-00/901):

1. Deactivate the leading edge slats (AMM 27-81-00/201).
2. Supply pressure to the pneumatic system (AMM 36-00-00/201).
3. Gain access to the associated wing anti-ice valve.
4. Set the WING ANTI-ICE switch to the ON position.
5. Confirm the position indicator on the valve indicates OPEN.
6. Set the WING ANTI-ICE switch to the OFF position.
7. Confirm the position indicator on the valve indicates CLOSED.
8. Restore airplane to desired configuration and activate the leading edge slats (AMM 27-81-00/201).

30-11-03 Control Stand Wing Anti-Ice Switches

30-11-03A Inoperative Closed

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative closed.

OPERATIONS (O) PROCEDURES

Set the WING ANTI-ICE switch to OFF on the ground and for takeoff.

NOTE: Wing anti-ice will not be available on the ground. Wing anti-ice will operate normally in-flight.

30-11-03 Control Stand Wing Anti-Ice Switches

30-11-03B Inoperative Open

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative open.

OPERATIONS (O) PROCEDURES

Set the WING ANTI-ICE switch to OFF for takeoff.

NOTE: Wing anti-ice is available on the ground. Wing anti-ice will operate normally in-flight.

INTENTIONALLY

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30-21-01 Engine (Cowl) Anti-Ice Valves

30-21-01A Valve Closed

Interval	Installed	Required	Procedure
C	2	1	**(M)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Affected valve is locked closed.
b. Airplane is not operated in known or forecast icing conditions.

MAINTENANCE (M) PROCEDURES

Lock the affected valve closed (AMM 30-00-00/901).

1. Without automatic ignition:
   Verify the associated ENGINE START switch is in the OFF position.
2. With automatic ignition:
   Verify the associated ENGINE START switch is in the AUTO position.
3. Verify the associated engine start lever is in the CUTOFF position.
4. Open circuit breakers for the associated inoperative engine anti-ice valve:
   A. For left engine anti-ice valve inoperative, open the following circuit breaker on the P18-3 panel:
      1) ANTI-ICE & RAIN ENG 1 & WING CONT
      2) ANTI-ICE & RAIN ENG 1 COWL AI VALVE
   B. For right engine anti-ice valve inoperative, open the following circuit breakers:
      1) ANTI-ICE & RAIN ENG 2 COWL AI VALVE (P18-3 panel)
      2) ANTI-ICE & RAIN ENGINE 2 CONTROL (P6-5 panel)
5. Gain access to the inoperative engine anti-ice valve.
6. Loosen the retaining screw on the manual locking assembly.
7. Move the hex shaft to the CLOSED position.
8. Slide the locking assembly up the shaft into the recess in the cover plate.
9. Tighten the retaining screw.
10. Close the open circuit breakers.

30-21-02 Engine Anti-Ice COWL VALVE Position Lights and TAI Indications

30-21-02A Associated Valve Operates Normally

Interval	Installed	Required	Procedure
C	4	0	**(O)**

May be inoperative provided associated valve is verified to operate normally before each departure.

OPERATIONS (O) PROCEDURES

Before each departure, verify the associated engine anti-ice valve operates normally.

1. Start the associated engine.
2. Wait for the engine to stabilize at idle.
3. Set the engine and APU bleed air switches to OFF.
4. Confirm the associated ENG ANTI-ICE switch is set to OFF.
5. Monitor the associated engine EGT:
   A. Wait for the EGT to stabilize and record the temperature.
   B. Set the associated ENG ANTI-ICE switch to ON.
      NOTE: Cowl TAI operation should be limited to 30 seconds, maximum.
   C. Wait for the EGT to stabilize and record the temperature.
   D. Set the associated ENG ANTI-ICE switch to OFF.
   E. Wait for the EGT to stabilize and record the temperature.
6. Confirm the following changes occurred:
   A. The EGT increased by a minimum of 15 degrees C when the ENG ANTI-ICE switch was set to ON.
   B. The EGT returned to the original idle temperature when the ENG ANTI-ICE switch was set to OFF.
7. Configure the airplane as needed for flight or shut down the engine if required.

30-21-02 **Engine Anti-Ice COWL VALVE Position Lights and TAI Indications**

30-21-02B **One Per Engine Inoperative**

Interval	Installed	Required	Procedure
C	4	2	

One valve position indication (either COWL VALVE light or TAI indication) for each engine may be inoperative.

30-21-02 Engine Anti-Ice COWL VALVE Position Lights and TAI Indications

30-21-02C Associated Valve Considered Inoperative

Interval	Installed	Required	Procedure
C	4	2	

May be inoperative provided associated valve is considered inoperative.

NOTE: Also dispatch the airplane using MEL Item 30-21-01.

30-21-03 COWL ANTI-ICE Lights

30-21-03A Not Operated In Icing Conditions

Interval	Installed	Required	Procedure
C	2	1	

Except for ER operations beyond 120 minutes, may be inoperative provided airplane is not operated in known or forecast icing conditions.

INTENTIONALLY
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30-31-01 Pitot Probe Heaters

30-31-01-01 Captain and First Officer Pitot Heaters

Interval	Installed	Required	Procedure
B	2	1	

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Operations are conducted in Day VMC only.
b. Aux pitot heater operates normally.
c. Airplane is not operated in visible moisture.
d. Airplane is not operated in known or forecast icing conditions.

30-31-01 **Pitot Probe Heaters**

30-31-01-02 **Auxiliary Pitot Heater (Right Lower Probe)**

Interval	Installed	Required	Procedure
B	1	0	

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Captain and First Officer pitot heaters operate normally.

b. Airplane is not operated in known or forecast icing conditions.

30-31-02 Total Air Temperature Probe Heater

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations beyond 120 minutes, may be inoperative provided airplane is not operated in known or forecast icing conditions.

30-31-03 Angle of Attack Sensor Heaters

Interval	Installed	Required	Procedure
C	2	0	

Except for ER operations beyond 120 minutes, may be inoperative provided airplane is not operated in known or forecast icing conditions.

30-31-04 Elevator Pitot Probe Heaters

Interval	Installed	Required	Procedure
B	2	1	

Except for ER operations beyond 120 minutes, may be inoperative provided airplane is not operated in known or forecast icing conditions.

30-31-05 Pitot and Temperature Probe Heater Lights

30-31-05-01 Pitot Probe Heater Lights

Interval	Installed	Required	Procedure
B	5	0	**(M)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Associated heater is verified to operate normally.
b. Airplane is not operated in known or forecast icing conditions.

MAINTENANCE (M) PROCEDURES

Verify the associated pitot probe heater operates normally (AMM 30-00-00/901).

1. Ensure covers are removed from all air data probes and vanes.
2. Station a ground observer at the associated probe heater and establish communications.
3. Supply electrical power to the airplane (AMM 24-22-00/201).
4. Set the associated PROBE HEAT switch to ON.
5. Confirm that heat radiation can be detected from the associated probe. Do not touch the probe.
   NOTE: TEMPERATURES ARE HIGH ENOUGH TO CAUSE BURNS. DO NOT LEAVE HEATER ON LONGER THAN REQUIRED. HEATER LIFE MAY BE UNNECESSARILY SHORTENED.
6. Set the associated PROBE HEAT switch to AUTO.

30-31-05 **Pitot and Temperature Probe Heater Lights**

30-31-05-02 **Total Air Temperature (TAT) Probe Heater Light**

30-31-05-02A **Associated Heater Verified to Operate Normally**

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided associated heater is verified to operate normally before each departure.

MAINTENANCE (M) PROCEDURES

Verify the TAT probe heater operates normally prior to each departure (AMM 30-00-00/901).

1. Ensure covers are removed from all air data probes and vanes.
2. Station a ground observer at the associated probe heater and establish communications.
3. Supply electrical power to the airplane (AMM 24-22-00/201).
4. Set the associated PROBE HEAT switch to ON.
5. For the aspirated TAT probe, momentarily press/hold the TAT TEST switch.
6. Confirm that heat radiation can be detected from the associated probe. Do not touch the probe.

   NOTE: TEMPERATURES ARE HIGH ENOUGH TO CAUSE BURNS. DO NOT LEAVE HEATER ON LONGER THAN REQUIRED. HEATER LIFE MAY BE UNNECESSARILY SHORTENED.
7. Set the associated PROBE HEAT switch to AUTO.

30-31-05 **Pitot and Temperature Probe Heater Lights**
30-31-05-02 **Total Air Temperature (TAT) Probe Heater Light**
30-31-05-02B **Associated Heater Inoperative**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided associated heater is inoperative.

NOTE: Airplane must also be dispatched using MEL Item 30-31-02.

30-31-06 Alpha Vane Heater Light Systems

30-31-06A Associated Heater Verified To Operate Normally

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided associated heater function is verified to operate normally before each departure.

MAINTENANCE (M) PROCEDURES

Verify the associated alpha vane (AOA sensor) heater operates normally prior to each departure (AMM 30-00-00/901).

1. Ensure covers are removed from all air data probes and vanes.
2. Station a ground observer at the associated vane heater and establish communications.
3. Supply electrical power to the airplane (AMM 24-22-00/201).
4. Set the associated PROBE HEAT switch to ON.
5. Confirm that heat radiation can be detected from the associated vane. Do not touch the vane.

NOTE: TEMPERATURES ARE HIGH ENOUGH TO CAUSE BURNS. DO NOT LEAVE HEATER ON LONGER THAN REQUIRED. HEATER LIFE MAY BE UNNECESSARILY SHORTENED.

6. Set the associated PROBE HEAT switch to AUTO.

30-31-06 Alpha Vane Heater Light Systems

30-31-06B Associated Heater Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated heater is considered inoperative.

NOTE: Also dispatch the airplane using MEL Item 30-31-03.

30-41-01 Electrically Heated Windshields

30-41-01-01 No. 1 Windows

Interval	Installed	Required	Procedure
C	2	1	

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Associated FWD WINDOW HEAT switch remains OFF.
b. Airplane is not operated in known or forecast icing conditions.
c. Both No. 2 Window heaters operate normally.
d. Windshield defog system operates normally.
e. Airspeed is limited to 250 KIAS below 10,000 feet MSL.

30-41-01 Electrically Heated Windshields

30-41-01-02 No. 2 Windows

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided:

a. Associated SIDE WINDOW HEAT switch remains OFF.
b. Both No. 1 window heaters operate normally.
c. Windshield defog system operates normally.
d. Airspeed is limited to 250 KIAS below 10,000 ft. MSL.

NOTE: Associated No. 3 window heat system will be inoperative when SIDE WINDOW HEAT switch is OFF. Also dispatch the airplane using MEL Item 30-41-01-03 if applicable.

30-41-01 **Electrically Heated Windshields**

30-41-01-03 **No. 3 Window Heat Systems**

Interval	Installed	Required	Procedure
D	2	0	

30-41-02 Windshield Defog System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided electrically heated windshields for No. 1 and No. 2 windows operate normally.

30-42-01 Windshield Wiper Systems

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

a. Airplane is not operated in precipitation within 5 statute miles of airport of takeoff or intended landing.

b. Approach minimums do not require its use.

MAINTENANCE NOTE

If wiper blade obstructs forward vision, do these AMM tasks:

1. Windshield Wiper Arm Removal (AMM 30-42-31/401).
2. Windshield Wiper Arm Installation (AMM 30-42-31/401).

NOTE: It is only necessary to put the wiper arm in the parked position and to apply the appropriate downward force on the wiper blade. The steps related to the sweep of the wiper arm are not necessary.

30-42-01 Windshield Wiper Systems

30-42-01-01 Park Function

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative for all flight conditions provided blade(s) can be positioned in a location that will not obstruct forward vision.

30-42-01 Windshield Wiper Systems

30-42-01-02 Intermittent Speed Function

Interval	Installed	Required	Procedure
D	2	0	

30-42-01 Windshield Wiper Systems

30-42-01-03 Low Speed Function

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided both high speed functions operate normally.

30-42-01 **Windshield Wiper Systems**
30-42-01-04 **High Speed Function**
30-42-01-04A **One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided both low speed functions operate normally.

30-42-01 **Windshield Wiper Systems**
30-42-01-04 **High Speed Function**
30-42-01-04B **Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided both low speed functions operate normally and rain intensity is less than moderate.

30-71-01 Drain Mast Heaters

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided water supply to associated components is secured off.

MAINTENANCE (M) PROCEDURES

Turn off the water supply to the associated components and deactivate the inoperative drain mast heaters (AMM 30-00-00/901).

1. Close the associated water supply shutoff valves (lavatory, galley, or fountain).
2. Deactivate the inoperative drain mast heater(s):
   A. For an inoperative forward drain mast heater, disconnect, cap and stow wire 0243-20 from terminal block TB2201.
   B. For an inoperative aft drain mast heater, disconnect, cap and stow wire 0244-20 from terminal block TB2201.

   NOTE: Alternatively, both drain mast heaters may be deactivated by opening and collaring the following P-18 panel circuit breakers:

   A. HEATERS DRAIN MAST GND
   B. HEATERS DRAIN MAST AIR

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31-25-01 Clock Switches

31-31-01 Flight Data Recorder System (FDR)

31-31-01A Not Required by TTCAR - N/A

31-31-01B Required by TTCAR

31-31-01-01 FDR Recording Parameters Required by TTCAR

31-31-01-02 FDR Recording Parameters not Required by TTCAR

31-31-02 Aircraft Condition Monitoring System (ACMS) Function

31-62-01 MAX Display System (MDS)

31-62-01-01 Display Units (DU)

31-62-01-01-01 Outboard DU

31-62-01-01-02 Inboard DU

31-62-01-01-03 Manual BRT/CONTRAST Controls

31-62-02 Remote Light Sensor (RLS) System

31-62-03 Engine Display Control SELECTORS

31-62-04 Electronic Flight Instrument System (EFIS) Controls Panels

31-62-04-01 Minimums (MINS) Reference Selectors

31-62-04-01A Both Inoperative

31-62-04-01B One Inoperative

31-62-04-02 MAP Switches

31-62-04-02-01 STA (Switches)

31-62-04-02-02 WPT (Waypoint)

31-62-04-02-03 ARPT (Airport)

31-62-04-02-04 DATA

31-62-04-02-05 POS (Position)

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31-25-01 Clock Switches

Interval	Installed	Required	Procedure
C	2	1	

INTENTIONALLY

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31-31-01 Flight Data Recorder System (FDR)

31-31-01B Required by TTCAR

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

a. Cockpit Voice Recorder (CVR) operates normally.

b. Airplane is not dispatched from a designated airport as listed in the NOTE below unless:

   1) FDR failure occurs after pushback but prior to takeoff, or

   2) FDR repair was attempted but was not successful.

c. In those cases where repair is attempted but not successful, aircraft may be dispatched on a flight or series of flights until next designated airport where repair must be accomplished prior to dispatch.

d. Repairs are made within 3 flight days.

NOTE: THE DESIGNATED AIRPORT FOR THIS MEL ITEM IS TTPP (POS).

31-31-01 Flight Data Recorder System (FDR)

31-31-01-01 FDR Recording Parameters Required by TTCAR

Interval	Installed	Required	Procedure
A	58	55	

Up to three recording parameters may be inoperative provided (Non-Mandatory):

a. Cockpit Voice Recorder (CVR) operates normally.
b. Repairs are made within 20 calendar days.

NOTE: FLIGHT RECORDER SYSTEMS SHALL BE CONSIDERED UNSERVICEABLE IF THERE IS A SIGNIFICANT PERIOD OF POOR QUALITY DATA, UNINTELLIGIBLE SIGNALS, OR IF ONE OR MORE OF THE MANDATORY PARAMETERS ARE NOT RECORDED CORRECTLY.

REFER TO TABLE 1 FOR TTCAA MANDATORY& NON-MANDATORY PARAMETERS.

TABLE 1 - TTCAA MANDATORY & NON-MANDATORY PARAMETERS

Mandatory	Non-Mandatory		
Time (UTC when available, otherwise relative time count or GPS time sync)	Trailing edge flap or/and cockpit control selection	Selected speed (all pilot selectable modes of operation)	Operational stall protection,stick shaker and pusher activation
Pressure-altitude	Leading edge flap or/and cockpit control selection	Selected Mach (all pilot selectable modes of operation)	Primary navigation system reference (GNSS, INS, VOR/DME, MLS, Loran C, localiser glideslope)
Indicated airspeed or calibrated airspeed	Thrust reverser position	Selected vertical speed (all pilot selectable modes of operation)	Ice detection
Heading (primary flight crew reference)	Ground spoiler/speed brake selection (Selection and position)	Selected heading (all pilot selectable modes of operation)	Engine warning each engine vibration
Normal acceleration	Autopilot/auto throttle/AFCS mode and engagement status	Selected flight path (all pilot selectable modes of operation) [course/DSTRK, path angle, final approach path (IRNAV/IAN)]	Engine warning each engine vibration engine over temperature
Pitch attitude	Radio altitude	Selected Decision Height	Engine warning each engine oil pressure low
Roll attitude	Vertical beam deviation (ILS/GPS/GLS glide path, MLS elevation, IRNAV/IAN vertical deviation)	EFIS display format (pilot, co-pilot)	Engine warning each engine oil pressure low engine over speed
Radio transmission Keying	Horizontal beam deviation (ILS/GPS/GLS localiser, MLS azimuth,IRNAV/IAN lateral deviation)	Multi-function/engine/ alerts display format	Yaw Trim Surface Position
Power on each engine	GPWS (ground proximity warning system) /TAWS/GCAS status (selection of terrain display mode including pop-up display status) and (terrain alerts, both cautions and warnings, and advisories)	AC electrical bus status	Roll Trim Surface Position
Outside air temperature	Angle of attack	DC electrical bus status	Yaw or sideslip angle
Longitudinal acceleration	Hydraulics, each system (low pressure)	Engine bleed valve position	De-icing and/or anti-icing systems selection
Lateral acceleration	Navigation data (latitude/longitude, ground speed and drift angle)	APU bleed valve position	Hydraulic pressure (each system)
Pilot input and/or control surface position-primary controls (pitch, roll, yaw)	Landing gear and gear selector position	Computer failure	Loss of cabin pressure
Pitch trim position	Groundspeed	Engine thrust command	Cockpit trim control input Position Pitch
Marker beacon passage	Additional engine parameters (EPR, N1, indicated vibration level, N2, EGT, fuel flow, fuel cut-off lever position, N3)	Engine thrust target	Cockpit trim control input Position Roll
Master warning	TCAS/ACAS (traffic alert and collision avoidance system)	Computed centre of gravity	Cockpit trim control input Position Yaw
Each NAV frequency selection	Windshear warning	Fuel quantity in CG trim tank	All cockpit flight control input forces (control wheel, control column, rudder pedal)
Air/ground status	Selected barometric setting (pilot, co-pilot)	Head up display in use	Event marker
Brakes (left and right brake pressure, left and right brake pedal position)	Selected altitude (all pilot selectable modes of	Para visual display on/off	Date
			ANP or EPE or EPU

31-31-01 **Flight Data Recorder System (FDR)**

31-31-01-02 **FDR Recording Parameters Not Required by TTCAR**

Interval	Installed	Required	Procedure
A	34	0	

May be inoperative provided repairs are made prior to completion of next heavy maintenance visit.

31-31-02 Aircraft Condition Monitoring System (ACMS) Function

Interval	Installed	Required	Procedure
D	1	0	

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31-62-01 MAX Display System (MDS)

31-62-01-01 Display Units (DU)

31-62-01-01-01 Outboard DU

Interval	Installed	Required	Procedure
A	2	1	**(M) (O)**

May be inoperative provided:

a. Associated PFD/MFD display selector switch remains in the INBD position.
b. Opposite PFD/MFD display selector switch is verified to operate normally.
c. Captain's and first officer's engine display control selectors operate Normally.
d. Engine display is available and selected on opposite inboard display.
e. ISFD operates normally.
f. EFIS control panels operate normally.
g. Inboard display units operate normally.
h. Repairs are made within 2 flight days.

MAINTENANCE (M) PROCEDURES

NOTE: For an inoperative outboard DU, the associated side's Remote Light Sensor (RLS) will also be inoperative.

1. Verify the opposite PFD/MFD display selector switch operates normally (AMM 31-00-00/901).
2. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Verify that engine indications can be manually selected to either Captain's or First Officer's inboard DU.
2. Ensure the associated PFD/MFD display selector switch remains in the INBD position.

31-62-01 **MAX Display System (MDS)**

31-62-01-01 **Display Units (DU)**

31-62-01-01-02 **Inboard DU**

Interval	Installed	Required	Procedure
A	2	1	**(M) (O)**

May be inoperative provided:

a. Associated PFD/MFD display selector switch remains in the OUTBD position.
b. Opposite PFD/MFD display selector switch is verified to operate normally.
c. Captain's and First Officer's engine display control selectors operate normally.
d. Engine display is available and selected on opposite inboard display.
e. ISFD operates normally.
f. EFIS Control Panels operate normally.
g. Outboard display units operate normally.
h. Repairs are made within 2 flight days.

MAINTENANCE (M) PROCEDURES

1. Verify the opposite PFD/MFD display selector switch operates normally (AMM 31-00-00/901).
2. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Verify that engine indications can be manually selected to either Captain's or First Officer's inboard DU.
2. Ensure the associated PFD/MFD display selector switch remains in the OUTBD position.

31-62-01 **MAX Display System (MDS)**

31-62-01-01 **Display Units (DU)**

31-62-01-01-03 **Manual BRT/CONTRAST Controls**

Interval	Installed	Required	Procedure
C	4	0	

Brightness function (outer control) may be inoperative for each display unit provided remote light sensor (RLS) system operates normally and display brightness is acceptable to flight crew.

NOTE: Automatic adjustment of the display brightness for the DUs is provided by the RLS system.

31-62-02 Remote Light Sensor (RLS) System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided manual display brightness/contrast controls operate normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

31-62-03 Engine Display Control SELECTORS

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided both FMCs and both CDU/MCDUs operate normally.

31-62-04 Electronic Flight Instrument System (EFIS) Control Panels

31-62-04-01A Minimums (MINS) Reference Selectors

Interval	Installed	Required	Procedure
C	2	0	

Both may be inoperative provided:

a. CAT II/III approaches are NOT allowed, and

b. Dispatch weather minimums at the destination and alternate(s) are no less than a ceiling of 1500' and a visibility of 3 sm /5 km from 1 hour before to 1 hour after the estimated time of arrival.

NOTE: MEL 31-62-04-01 is applicable to the complete minimums (MINS) control selector knob assembly which includes the outer rotary two-position switch, the spring-loaded middle rotary slew selector, and inner radio minimums reset (RST) switch.

OPERATIONS NOTES

1. GPWS height callouts for "APPROACHING MINIMUMS" at DA/MDA plus 80 feet and "MINIMUMS" at DA/MDA will **NOT** be available.
2. A portion of an Instrument Approach Procedure may be used to establish the aircraft in a position from which a Visual Approach and Landing can be made.

31-62-04 Electronic Flight Instrument System (EFIS) Control Panels

31-62-04-01B Minimums (MINS) Reference Selectors

Interval	Installed	Required	Procedure
C	2	1	**(O)**

One may be inoperative provided:

a. CAT II/III approaches are NOT allowed, and

b. The DISPLAYS CONTROL PANEL SWITCH on the forward overhead panel is set to the operative side in order to set the DA/MDA prior to commencing the approach.

NOTE: MEL 31-62-04-01 is applicable to the complete minimums (MINS) control selector knob assembly which includes the outer rotary two-position switch, the spring-loaded middle rotary slew selector, and inner radio minimums reset (RST) switch.

OPERATIONS (O) PROCEDURES

1. For an **INOPERATIVE First Officer's Minimums Selector** set the DISPLAYS CONTROL PANEL SWITCH on the forward overhead panel to **BOTH ON 1** and set the DA/MDA using the Captain's Minimums selector prior to commencing the approach.
2. For an **INOPERATIVE Captain's Minimums Selector** set the DISPLAYS CONTROL PANEL SWITCH on the forward overhead panel to **BOTH ON 2** and set the DA/MDA using the First Officer's Minimums selector prior to commencing the approach.

NOTE 1: After setting the DA/MDA, the DISPLAYS CONTROLS PANEL SWITCH may be returned to the NORMAL position to restore all remaining EFIS Control functions to the affected side.

NOTE 2: GPWS height callouts for "APPROACHING MINIMUMS" at DA/MDA plus 80 feet and "MINIMUMS" at DA/MDA should be available.

31-62-04 **Electronic Flight Instrument System (EFIS) Control Panels**
31-62-04-02 **MAP Switches**
31-62-04-02-01 **STA (Station)**

Interval	Installed	Required	Procedure
C	2	1	

31-62-04 **Electronic Flight Instrument System (EFIS) Control Panels**
31-62-04-02 **MAP Switches**
31-62-04-02-02 **WPT (Waypoint)**

Interval	Installed	Required	Procedure
C	2	1	

31-62-04 **Electronic Flight Instrument System (EFIS) Control Panels**
31-62-04-02 **MAP Switches**
31-62-04-02-03 **ARPT (Airport)**

Interval	Installed	Required	Procedure
C	2	1	

31-62-04 **Electronic Flight Instrument System (EFIS) Control Panels**
31-62-04-02 **MAP Switches**
31-62-04-02-04 **DATA**

Interval	Installed	Required	Procedure
C	2	1	

31-62-04 **Electronic Flight Instrument System (EFIS) Control Panels**
31-62-04-02 **MAP Switches**
31-62-04-02-05 **POS (Position)**

Interval	Installed	Required	Procedure
C	2	1	

INTENTIONALLY

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32-30-01 Landing Gear Actuation System

32-31-01 Landing Gear Selector Bypass Valve

32-31-02 Landing Gear Control Systems

32-31-02A One Inoperative

32-31-02B Both Inoperative

32-31-03 Landing Gear Lever Input Systems

32-32-01 Main Landing Gear Uplock Springs

32-32-02 Main Landing Gear Frangible Fitting

32-35-01 Landing Gear Manual Extension System

32-41-01 Hydraulic Brake Pressure Indication System

32-41-01-01 Wheel Well Brake Accumulator Gauge

32-41-01-02 Flight Deck Brake Pressure Indicating System

32-41-02 Gear Retraction Braking System

32-42-01 Antiskid System

32-42-02 Alternate Antiskid Valves

32-42-03 Automatic Brake System

32-44-01 Parking Brake Valve

32-44-02 Parking Brake Light

32-44-02-01 Flight Deck Parking Brake Light

32-44-02-02 External Parking Brake Light – *N/A*

32-46-01 Brake Temperature Monitoring System – *N/A*

32-49-01 Tire Pressure Indication System – *N/A*

32-51-01 Rudder Pedal Nose Wheel Steering System

32-51-02 Nose Wheel Steering Switch

32-71-01 Two-Position Tail Skid - *N/A*

INTENTIONALLY

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32-30-01 Landing Gear Actuation System

Interval	Installed	Required	Procedure
B	1	0	**(M) (O)**

May be inoperative provided:

a. Landing gear is secured in down position.
b. Airplane is dispatched in accordance with AFM Landing Gear Extended appendix.

MAINTENANCE (M) PROCEDURES

Secure the landing gear in the down position and set VMO/MMO to gear down speed limits (AMM 32-00-00/901).

1. Install landing gear downlock pins in nose and main landing gear.
2. Set alternate VMO/MMO select switch in main equipment center to ALTN position.

OPERATIONS (O) PROCEDURES

1. The airplane must be dispatched using the AFM Landing Gear Extended appendix.
2. Determine flight planning fuel adjustments using performance data generated with the Gear Down Dispatch option in the Boeing performance software. (Dispatchers must select the LGE option in the NavBlue electronic flight planning software).

0PERATIONS NOTES

1. Extended over-water operation is not authorized for airplanes operating under this appendix.
2. VMO = 270 KIAS from sea level up to FL320, above FL320 MMO = M0.73; The airspeed indicator and Mach-airspeed warning switch must be reprogrammed for VMO and MMO.
3. RVSM operations prohibited – Flights to be planned and flown below FL290.
4. In the event of Cabin Pressurization - Target speed - 270 KIAS
5. The conditions for severe turbulent air penetration are unchanged except that the recommended airspeed for flight in severe turbulence is VREF40 + 70 KIAS.
6. Flight Management Computer System (FMCS)
   A. The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn and thrust-limited maximum altitude, and compute overly shallow descent path.
   B. To eliminate erroneous displays the flight crew should enter only gross weight data on the PERF INIT page of the Control Display Unit (CDU). Omission of the cost index and cruise altitude entries on the PERF INIT page will render the VNAV function unavailable during flight. As a result, the following information will not be provided: VNAV guidance and speed schedules, trip fuel and ETA predictions, optimum and maximum altitude data, step climb and top of descent predictions, and the VNAV descent guidance path.
   C. The gross weight entry allows the FMCS takeoff and approach speed schedules to be generated. In addition, the flap maneuver speed and VREF speed bugs will be available for display on the primary flight display speed tape. Except for VNAV, normal autopilot and autothrottle modes will remain available for use during the flight, as will the LNAV mode.
7. The flap retraction speed schedule is unchanged except as follows: Final Segment Climb Speed: VREF40 + 55 KIAS.

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32-31-01 Landing Gear Selector Bypass Valve

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided deactivated in normal position.

a. Landing gear selector bypass valve is deactivated in the normal position.
b. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the landing gear selector valve bypass valve in the normal position. (AMM 32-00-00/901).

1. Install landing gear downlock pins in nose and main landing gear.
2. Open and collar P6-3 panel LANDING GEAR ALTN EXTEND SOL circuit breaker.
3. Confirm the bypass valve is in the normal position:
   A. Push and hold the LOCK OVRD button.
   B. Momentarily set the landing gear lever in the UP position.
   C. Confirm the main landing gear downlock actuator tries to extend.
   D. Return the landing gear lever to the DN position.
   E. Release the LOCK OVRD button.
4. Remove landing gear downlock pins.

OPERATIONS (O) PROCEDURES

1. Maximum performance limited landing weight reduced by 18,506 kg.
2. For landing use either:
   A. Flaps 15 and VREF 15 + 10
   B. Flaps 30 and VREF 30 + 10
3. If all landing gear fail to extend when a manual gear extension is attempted per the Non-Normal Checklist Manual Gear Extension, use the following procedure to extend the landing gear.

   NOTE: Manual gear extension using this procedure may require up to 10 minutes to accomplish. Do not accomplish this procedure unless there is sufficient fuel to extend the gear and then complete an approach, with appropriate reserves. If sufficient fuel is not available, accomplish the Non-Normal Checklist PARTIAL OR ALL GEAR UP LANDING.

   A. Verify NOSE WHEEL STEERING switch is NORM.

      NOTE Do not use alternate nose wheel steering.

   B. Set the system A FLT CONTROL switch to STBY RUD.
   C. Set the system A hydraulic pump switches to OFF.
   D. Cycle the speed brake lever until system A hydraulic pressure is below 500 psi.

      NOTE 1: **It may take a number of cycles to reduce the system A pressure to below 500 psi.**

      NOTE 2: Do not pressurize hydraulic system A inflight or on the ground and do not use alternate nose wheel steering.

      NOTE 3: **The engine No. 1 thrust reverser will deploy and retract at a slower rate and some thrust asymmetry can be anticipated during thrust reverser deployment.**

OPERATIONS (O) PROCEDURES 32-31-01 (Cont'd)

NOTE 4: The following items will be inoperative:

1) Ground spoilers.
2) Flight spoilers 2, 4, 9, 11.
3) Autopilot A.
4) Normal nose wheel steering.
5) Alternate brakes.
6) Two-position tail skid (if installed).

E. Pull the manual gear extension handles.

NOTE 1: 270K/0.82M maximum speed.

NOTE 2: Do not wait for an indication that a landing gear is down and locked before pulling the next handle.

F. Continue to cycle the speed brake lever to relieve hydraulic pressure as the gear extends.

NOTE 1: It may take a number of cycles and up to 4 minutes until all three landing gear indicator lights are illuminated.

NOTE 2: With the LANDING GEAR lever in the UP position, the red landing gear indicator lights will stay illuminated.

G. If all landing gear are not down and locked, accomplish the Non-Normal Checklist PARTIAL OR ALL GEAR UP LANDING.

H. If all landing gear are down and locked, plan for a flaps 15 landing.

1) Set VREF 15 + 10 knots.
2) Set the GROUND PROXIMITY FLAP INHIBIT switch to FLAP INHIBIT.
3) Set the GROUND PROXIMITY GEAR INHIBIT switch to GEAR INHIBIT.
4) Set the GROUND PROXIMITY RUNWAY INHIBIT switch to INHIBIT (if the overrun warning system and GROUND PROXIMITY RUNWAY INHIBIT switch are installed).

32-31-02 Landing Gear Control Systems

32-31-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

32-31-02 Landing Gear Control Systems

32-31-02B Both Inoperative

Interval	Installed	Required	Procedure
B	2	0	**(M) (O)**

May be inoperative provided:

a. Landing gear is secured in down position.
b. Airplane is dispatched in accordance with AFM Landing Gear Extended appendix.

MAINTENANCE (M) PROCEDURES

Secure the landing gear in the down position and set VMO/MMO to gear down speed limits (AMM 32-00-00/901).

1. Install landing gear downlock pins in nose and main landing gear.
2. Set alternate VMO/MMO select switch in main equipment center to ALTN position.
3. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. The airplane must be dispatched using the AFM Landing Gear Extended appendix.
2. Determine flight planning fuel adjustments using performance data generated with the Gear Down Dispatch option in the Boeing performance software. (Dispatchers must select the LGE option in the NavBlue electronic flight planning software).

0PERATIONS NOTES

1. Extended over-water operation is not authorized for airplanes operating under this appendix.
2. VMO = 270 KIAS from sea level up to FL320, above FL320 MMO = M0.73; The airspeed indicator and Mach-airspeed warning switch must be reprogrammed for VMO and MMO.
3. RVSM operations prohibited – Flights to be planned and flown below FL290.
4. In the event of Cabin Pressurization - Target speed - 270 KIAS
5. The conditions for severe turbulent air penetration are unchanged except that the recommended airspeed for flight in severe turbulence is VREF40 + 70 KIAS.
6. Flight Management Computer System (FMCS)
   A. The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn and thrust-limited maximum altitude, and compute overly shallow descent path.
   B. To eliminate erroneous displays the flight crew should enter only gross weight data on the PERF INIT page of the Control Display Unit (CDU). Omission of the cost index and cruise altitude entries on the PERF INIT page will render the VNAV function unavailable during flight. As a result, the following information will not be provided: VNAV guidance and speed schedules, trip fuel and ETA predictions, optimum and maximum altitude data, step climb and top of descent predictions, and the VNAV descent guidance path.
   C. The gross weight entry allows the FMCS takeoff and approach speed schedules to be generated. In addition, the flap maneuver speed and VREF speed bugs will be available for display on the primary flight display speed tape. Except for VNAV, normal autopilot and autothrottle modes will remain available for use during the flight, as will the LNAV mode.
7. The flap retraction speed schedule is unchanged except as follows: Final Segment Climb Speed: VREF40 + 55 KIAS.

32-31-03 Landing Gear Lever Input Systems

Interval	Installed	Required	Procedure
C	2	1	

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

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32-32-01 Main Landing Gear Uplock Springs

Interval	Installed	Required	Procedure
B	4	3	**(M)**

One spring on one main gear uplock mechanism may be missing.

MAINTENANCE (M) PROCEDURES

Remove the affected main landing gear uplock spring (AMM 32-00-00/901).

32-32-02 Main Landing Gear Frangible Fitting

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be broken or missing provided:

a. Fitting is replaced with a hydraulic cap assembly.
b. Landing gear remains extended for 2 minutes after takeoff.
c. Takeoff performance is based on landing gear extended.

MAINTENANCE (M) PROCEDURES

Remove the broken fitting and replace with hydraulic cap assembly (AMM 32-00-00/901).

1. Remove the broken frangible fitting and nut.
2. Replace with a standard BACC14AD06JL or equivalent hydraulic cap assembly.

OPERATIONS (O) PROCEDURES

1. Increase flight planning fuel by 91 kg.
2. Base takeoff performance on landing gear extended.
3. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.

32-35-01 Landing Gear Manual Extension System

Interval	Installed	Required	Procedure
B	1	0	**(M) (O)**

May be inoperative provided:

a. Landing gear is secured in down position.
b. Airplane is dispatched in accordance with AFM Landing Gear Extended appendix.

MAINTENANCE (M) PROCEDURES

Secure the landing gear in the down position and set VMO/MMO to gear down speed limits (AMM 32-00-00/901).

1. Install landing gear downlock pins in nose and main landing gear.
2. Set alternate VMO/MMO select switch in main equipment center to ALTN position.

OPERATIONS (O) PROCEDURES

1. The airplane must be dispatched using the AFM Landing Gear Extended appendix.
2. Determine flight planning fuel adjustments using performance data generated with the Gear Down Dispatch option in the Boeing performance software. (Dispatchers must select the LGE option in the NavBlue electronic flight planning software).

0PERATIONS NOTES

1. Extended over-water operation is not authorized for airplanes operating under this appendix.
2. VMO = 270 KIAS from sea level up to FL320, above FL320 MMO = M0.73; The airspeed indicator and Mach-airspeed warning switch must be reprogrammed for VMO and MMO.
3. RVSM operations prohibited – Flights to be planned and flown below FL290.
4. In the event of Cabin Pressurization - Target speed - 270 KIAS
5. The conditions for severe turbulent air penetration are unchanged except that the recommended airspeed for flight in severe turbulence is VREF40 + 70 KIAS.
6. Flight Management Computer System (FMCS)
   A. The Flight Management Computer System (FMCS) does not contain special provisions for operation with landing gear extended. As a result, the FMCS may generate inappropriate enroute speed schedules, display non-conservative predictions of fuel burn and thrust-limited maximum altitude, and compute overly shallow descent path.
   B. To eliminate erroneous displays the flight crew should enter only gross weight data on the PERF INIT page of the Control Display Unit (CDU). Omission of the cost index and cruise altitude entries on the PERF INIT page will render the VNAV function unavailable during flight. As a result, the following information will not be provided: VNAV guidance and speed schedules, trip fuel and ETA predictions, optimum and maximum altitude data, step climb and top of descent predictions, and the VNAV descent guidance path.
   C. The gross weight entry allows the FMCS takeoff and approach speed schedules to be generated. In addition, the flap maneuver speed and VREF speed bugs will be available for display on the primary flight display speed tape. Except for VNAV, normal autopilot and autothrottle modes will remain available for use during the flight, as will the LNAV mode.
7. The flap retraction speed schedule is unchanged except as follows: Final Segment Climb Speed: VREF40 + 55 KIAS.

INTENTIONALLY
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32-41-01 Hydraulic Brake Pressure Indication System

32-41-01-01 Wheel Well Brake Accumulator Gauge

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided flight deck brake pressure indicating system operates normally.

32-41-01 Hydraulic Brake Pressure Indication System

32-41-01-02 Flight Deck Brake Pressure Indicating System

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Wheel well brake accumulator gauge operates normally.
b. Brake accumulator charge is verified normal once each flight day.

MAINTENANCE (M) PROCEDURES

Verify brake accumulator charge once each flight day (AMM 32-00-00/901).

1. Chock wheels and release the parking brake.
2. Depressurize hydraulic systems A and B.
3. Apply and release the brakes (approximately 3 seconds between each application) 10 times.
4. Confirm the wheel well brake accumulator gauge pressure indication is within limits.
5. Pressurize hydraulic systems A and B.
6. Confirm the wheel well brake accumulator gauge pressure indication is approximately 3000 psi.
7. Set the parking brake.

32-41-02 Gear Retraction Braking System

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. Landing gear remains extended for 2 minutes after takeoff.

b. Takeoff performance is based on landing gear extended.

OPERATIONS (O) PROCEDURES

1. Increase flight planning fuel by 91kg.
2. Base takeoff performance on landing gear extended. **(Refer to Onboard Performance Tool (OPT))**
3. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.

INTENTIONALLY
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32-42-01 Antiskid System

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative for Ferry to Repair Station provided:

a. Inoperative antiskid channels are deactivated,
b. Autobrake system is not used,
c. Appropriate performance adjustments for antiskid inoperative are applied,
d. Dispatch is prohibited to/ from contiguous US Airports not included in the published FAA Domestic Notice on 5G C-Band Mitigated Airports,
e. Dispatch to runways less than 7500 ft (2290 m) is not allowed, and
f. CAT II/III approaches are not allowed.

NOTES: 1. Antiskid must be operative for normal operations.
2. Dispatch is allowed to/from the following contiguous US destination and alternate airports currently authorized in CAL's FAA issued Operations Specifications: KMIA, KFLL, KMCO, KJFK, KTPA, KBOS, KPHL. For flights planned to other contiguous US Airports, Flight Dispatch must verify they are listed as FAA 5G C-Band Mitigated Airports.

MAINTENANCE (M) PROCEDURES

NOTE: An operative antiskid channel should not be deactivated. Antiskid channels that are faulted, must be deactivated.

Deactivate the inoperative antiskid system channels (AMM 32-00-00/901).

1. For inboard antiskid channel inoperative, open and collar the P6-3 panel LANDING GEAR ANTISKID INBD circuit breaker.
2. For outboard antiskid channel inoperative, open and collar the P6-3 panel LANDING GEAR ANTISKID OUTBD circuit breaker.

OPERATIONS (O) PROCEDURES

NOTE 1: Reduced thrust takeoff using the assumed temperature method is not permitted.

NOTE 2: Takeoff on wet runways is not permitted.

1. Base takeoff and landing performance on AFM antiskid inoperative. **(Refer to Onboard Performance Tool)**
2. Set autobrake system OFF.
3. Operate speed brakes manually.
4. Use AFM antiskid inoperative braking procedure for rejected takeoffs and landings, including the following guidance.
   A. Use minimum braking consistent with runway length and conditions to reduce the possibility of a tire blowout.
   B. Do NOT apply the brakes until the nose wheel is on the ground and the speedbrakes have been manually deployed.
   C. Brake initially using light steady pedal pressure. Increase pressure as ground speed decreases. Do NOT pump the brakes.
5. Prior to landing, set the GROUND PROXIMITY RUNWAY INHIBIT switch to INHIBIT (if the overrun warning system and GROUND PROXIMITY RUNWAY INHIBIT switch are installed).

OPERATIONS NOTES:

REVIEW BEFORE DEPARTURE:

QRH: Non Normal Checklist – **ANTISKID INOPERATIVE.** (This is a Landing Checklist)

32-42-02 Alternate Antiskid Valves

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided:

a. Alternate brake system is verified to operate normally, and

b. Dispatch is prohibited to/from contiguous US Airports not included in the published FAA Domestic Notice on 5G C-Band Mitigated Airports.

NOTE: Dispatch is allowed to/from the following contiguous US destination and alternate airports currently authorized in CAL's FAA issued Operations Specifications: KMIA, KFLL, KMCO, KJFK, KTPA, KBOS, KPHL. For flights planned to other contiguous US Airports, Flight Dispatch must verify they are listed as FAA 5G C-Band Mitigated Airports.

MAINTENANCE (M) PROCEDURES

Verify alternate brake system operates normally (AMM 32-00-00-901).

1. Ensure wheels are chocked.
2. Release parking brake.
3. Pressurize hydraulic System A
4. Depressurize hydraulic System B.
5. Depress and release the brake pedals several times.
6. Confirm the brake wear indicator pin movement during the brake applications.

32-42-03 Automatic Brake System

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative deactivated provided dispatch is prohibited to/from contiguous US Airports not included in the published FAA Domestic Notice on 5G C-Band Mitigated Airports.

NOTE: Dispatch is allowed to/from the following contiguous US destination and alternate airports currently authorized in CAL's FAA issued Operations Specifications: KMIA, KFLL, KMCO, KJFK, KTPA, KBOS, KPHL. For flights planned to other contiguous US Airports, Flight Dispatch must verify they are listed as FAA 5G C-Band Mitigated Airports.

MAINTENANCE (M) PROCEDURES

Deactivate the autobrake system (AMM 32-00-00/901).

1. Set the AUTO BRAKE select switch to OFF.
2. For AUTO BRAKE DISARM light extinguished, no further deactivation procedures are required.
3. For the AUTO BRAKE DISARM light illuminated:
   A. Depressurize hydraulic system B (AMM 29-11-00).
   B. Open the following P6-3 panel circuit breakers:
      1) LANDING GEAR AUTOBRAKE BITE CONT 1
      2) LANDING GEAR AUTOBRAKE BITE CONT 2
   C. In the main gear wheel well, on the ceiling, locate the autobrake pressure control module (V122) and deactivate the system by one of the following methods:
      1) METHOD 1
         a. Disconnect, cap and stow the electrical connectors from the autobrake pressure control module.
         b. Remove the autobrake pressure control module and cap the supply pressure line.
         c. Connect the brake pressure line and the return line with a 3/8 inch section of tubing or flexible hose.
         NOTE: This prevents hydraulically locking the brake shuttle valves in the brakes off position.
      2) METHOD 2
         a. Disconnect, cap and stow the electrical connectors from the autobrake pressure control module.
         b. Disconnect and plug the supply pressure line and cap the supply pressure port on the module.
   D. Close the open circuit breakers.
   E. Repressurize the hydraulic systems (AMM 29-11-00) and check the vicinity of the autobrake pressure control module for leaks.

OPERATIONS NOTE

Set AUTO BRAKE select switch OFF.

32-44-01 Parking Brake Valve

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative for Ferry provided:

a. Parking brake valve is deactivated in closed position.
b. Antiskid system is considered inoperative.
c. Autobrake system is not used.
d. Appropriate performance adjustments for antiskid inoperative are applied.

NOTE: The airplane must also be dispatched using MEL item 32-42-01.

MAINTENANCE (M) PROCEDURES

Deactivate the parking brake valve in the closed position and deactivate the antiskid system (AMM 32-00-00/901).

1. Deactivate the parking brake valve in the closed position:
   A. Install landing gear downlock pins in nose and main landing gear.
   B. With electrical power on airplane and wheels chocked, release the parking brake.
   C. Open the P6-3 panel LANDING GEAR PARKING BRAKE circuit breaker.
   D. Gain access to the parking brake valve and disconnect, cap and stow the electrical connector at the valve.
   E. Manually position the override lever on the valve to the fully closed (POS 2) position.
   F. Close the P6-3 panel LANDING GEAR PARKING BRAKE circuit breaker.
   G. Remove landing gear downlock pins.
2. Deactivate the antiskid system by opening and collaring P6-3 panel circuit breakers:
   A. LANDING GEAR ANTISKID INBD
   B. LANDING GEAR ANTISKID OUTBD
3. Confirm the ANTISKID INOP light is on.
4. Set AUTO BRAKE select switch OFF.

OPERATIONS (O) PROCEDURES

NOTE 1: The parking brake operates normally with the parking brake valve inoperative closed.

NOTE 2: Reduced thrust takeoff using the assumed temperature method is not permitted.

NOTE 3: Takeoff on wet runways is not permitted.

1. Base takeoff and landing performance on AFM antiskid inoperative. **(Refer to Onboard Performance Tool (OPT))**
2. Set autobrake system OFF.
3. Operate speed brakes manually.
4. Use AFM antiskid inoperative braking procedure.
   A. Use minimum braking consistent with runway length and conditions to reduce the possibility of a tire blowout.
   B. Do NOT apply the brakes until the nose wheel is on the ground and the speedbrakes have been manually deployed.
   C. Brake initially using light steady pedal pressure. Increase pressure as ground speed decreases. Do NOT pump the brakes.

32-44-02 **Parking Brake Light**

32-44-02-01 **Flight Deck Parking Brake Light**

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided parking brake shutoff valve is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify the parking brake shutoff valve operates normally (AMM 32-00-00/901).

1. Ensure wheels are chocked.
2. Release parking brake.
3. Confirm the ANTISKID INOP light is extinguished.
4. Confirm parking brake shutoff valve (wheel well aft bulkhead) moves to the open position (POS 1).
5. Set the parking brake.
6. Confirm parking brake shutoff valve moves to the closed position (POS 2).

32-51-01 Rudder Pedal Nose Wheel Steering System

Interval	Installed	Required	Procedure
A	1	0	**(M) (O)**

May be inoperative provided:

a. Rudder pedal nose wheel steering is deactivated disengaged.
b. Tiller is verified to operate normally.
c. Taxi, takeoffs, and landings are made by pilot with access to an operating tiller.
d. Approach minimums do not require its use.
e. Airplane is not operated on runways contaminated with standing water, slush, or snow.
f. Repairs are made within 3 flight days.

MAINTENANCE (M) PROCEDURES

Deactivate the rudder pedal nose wheel steering system in the disengage position and verify tiller operates normally (AMM 32-00 00/901).

1. Put the airplane in the air mode.
2. Wait minimum of 5 seconds.
3. Open and collar the P6-3 panel LANDING GEAR NOSE GEAR STEER circuit breaker.
4. Return the airplane to the ground mode.
5. Pressurize hydraulic system A.
6. Operate the rudder pedals and confirm the nose gear does not turn.
7. Operate the rudder pedals and confirm the rudder operates normally.
8. Operate the tiller and confirm the nose gear steering operates normally.
9. Depressurize hydraulic system A.

OPERATIONS (O) PROCEDURES

The flight crewmember with access to the tiller will be flying. Use the following procedures.

NOTE: Caution must be used when using the nose wheel steering tiller (wheel) above 20 knots to avoid over-controlling the nose wheels resulting in possible loss of directional control.

Takeoff:

The pilot flying should steer the airplane by manipulating the nose wheel steering tiller until the rudder becomes effective between 40 and 60 kts. The pilot monitoring should assist by keeping the wings level and applying a light forward pressure on the control column until the tiller is no longer required for steering.

Landing:

The pilot flying should use the rudder and nose wheel steering tiller as required during the landing roll. The pilot monitoring should assist by keeping the wings level and applying a light forward pressure on the control column.

32-51-02 Nose Wheel Steering Switch

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Landing gear transfer valve is verified to operate normally.
b. Rudder pedal nose wheel steering operates normally.

MAINTENANCE (M) PROCEDURES

Verify the landing gear transfer valve operates normally (AMM 32-00-00/901):

1. Pressurize Hydraulic System A.
2. Use the Onboard Network System (ONS) to verify landing gear transfer valve operates normally.
3. Depressurize Hydraulic System A.

OPERATIONS NOTE

Nose wheel steering will not be available if Hydraulic System A power is lost.

33-11-01 Flight Compartment and Instrument Light System

33-11-02 System Annunciator Lights (Glareshield Panel)

33-11-02A Light for Operative System

33-11-02B Light for Inoperative System

33-11-03 Main Panel Light Controls

33-16-01 Master Lights Test and Dim System

33-16-01-01 Master and Individual Light's Test Function

33-16-01-02 Dim Function

33-21-01 Cabin Interior Illumination

33-21-01-01 Without Photoluminescent Emergency Escape Path Marking System – *N/A*

33-21-01-02 With Photoluminescent Emergency Escape Path Marking System

33-24-01 Passenger Lighted Information Signs and Notice System

33-24-01-01 Individual Passenger Lighted Information Signs

33-24-01-01A PA System Inoperative

33-24-01-01B PA System Operates Normally

33-24-01-02 Aural Tone System

33-24-01-03 Flight Deck Automatic Function

33-25-01 Sterile Flight Compartment Light System – *N/A*

33-31-01 Wheel Well Lights

33-37-01 Lower Cargo Compartment Light Systems

33-37-01-01 Forward Compartment Light Lens

33-37-01-02 Aft Compartment Light Lens

33-41-01 Wing Illumination Lights

33-42-01 Landing Light LED Array Systems

33-42-01A One Inoperative Per Side

33-42-01B Runway Turnoff Lights Operate Normally

33-42-01C Day Operations

33-42-01-01 Alternate Flash Function

33-42-02 Taxi Light LED Array Systems

33-42-03 Runway Turnoff Light LED Array Systems

33-43-01 Position Light LED Modules

33-43-01A *Day Operations*

33-43-01B *All Operations*

33-43-02 Winglet Aft Marker Lights

33-43-02A *Alternate Procedures Required*

33-43-02B *Procedures Do Not Require Use*

33-44-01 Red ANTI-COLLISION Lights

33-44-01A *All Operations*

33-44-01B *Day Operations*

33-44-02 White Anti-Collision STROBE Lights

33-44-02A *All Operations*

33-44-02B *Day Operations*

33-45-01 Logo Light System

33-51-01 Interior Emergency Lighting System

33-51-01-01 Aisle Lights

33-51-01-02 Flight Deck Exit Light

33-51-02 Exterior Emergency Lighting System

33-51-03 Floor Proximity Emergency Escape Path Marking System

33-51-03-01 Seat-Mounted and Monument-Mounted LED and Incandescent Lighting System

33-51-03-01-01 Floor Aisle Marker Lights – *N/A*

33-51-03-01-02 Exit Indicator Lamps/Bulbs

33-51-03-02 Photoluminescent Lighting System

33-51-03-02-01 Photoluminescent Strips

33-51-03-02-02 Exit Indicator Lamps/Bulbs

33-11-01 Flight Compartment and Instrument Lighting System

Interval	Installed	Required	Procedure
C	22	8	

Individual lights may be inoperative provided:

a. Remaining lighting system lights are sufficient to clearly illuminate all required instruments, controls, and other devices for which they are provided.
b. Remaining lighting system lights are positioned so that direct rays are shielded from flight crewmembers' eyes.
c. Lighting configuration and intensity is acceptable to the flight crew.

NOTE 1: Individual button/switch lights and/or annunciations/indications are excluded from this relief.

33-11-02 System Annunciator Lights (Glareshield Panel)

33-11-02A Light for Operative System

Interval	Installed	Required	Procedure
C	12	11	**(O)**

May be inoperative for an operating system.

OPERATIONS (O) PROCEDURES

1. Confirm the system indicator lights on the associated panel are operating normally.
2. If the MASTER CAUTION light illuminates with none of the annunciator lights illuminated, assume the system with the annunciator light placarded INOP is affected.
3. Just prior to descent, check the associated panel for any light indications.

33-11-02 System Annunciator Lights (Glareshield Panel)

33-11-02B Light for Inoperative System

Interval	Installed	Required	Procedure
C	12	0	

May be inoperative for an associated inoperative system.

33-11-03 Main Panel Light Controls

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

a. Associated switch remains in OFF position.
b. Associated display unit brightness/contrast controls operate normally.
c. Light intensity is acceptable to flight crew.

33-16-01 Master Lights Test and Dim System

33-16-01-01 Master and Individual Light's Test Function

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Master test and individual light's press-to-test functions may be inoperative provided intended function of associated light is verified to operate normally once each flight day.

OPERATIONS (O) PROCEDURES

For each light with an inoperative test function, verify the light system operates normally by one of the following methods:

1 If the master test function fails to illuminate a light, press the affected light and confirm the press-to-test feature causes the light to illuminate.

2. If the individual press-to-test feature fails to illuminate a light, perform the appropriate operational or system test and confirm the affected light illuminates as intended.

33-16-01 System Annunciator Lights (Glareshield Panel)

33-16-01-02 Dim Function

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

a. Master TEST and BRT functions operate normally.
b. Except during light test, switch is set to BRT.
c. Light intensity is acceptable to flight crew.

33-21-01 Cabin Interior Illumination

33-21-01-02 With Photoluminescent Emergency Escape Path Marking System

Interval	Installed	Required	Procedure
C	60	48	

Individual lights may be inoperative provided:

a. Sufficient lighting remains for cabin attendants to perform their duties.
b. Remaining lighting is sufficient to charge photoluminescent emergency escape path marking system.

OPERATIONS NOTE

For the Boeing Sky Interior with the photoluminescent escape path marking system installed, the following minimum interior lighting requirements apply.

1. A maximum of 20% of the ceiling LED lights may be inoperative provided opposite and adjacent lights are operative.
   A. The LED direct lights and cove lights at the forward and aft ends of the passenger cabin are NOT counted as ceiling lights because these lights are not critical for charging the photoluminescent strips.
   B. For counting purposes, ceiling lights, bullnose ceiling lights, wash only ceiling lights, and (if installed) Center Overhead Stowage (COS) lights are counted. Ceiling lights are mounted on top of the stow bins, and COS lights are installed on the optional COS boxes.
      1) Each ceiling light or bullnose ceiling light has two element assemblies, a white 'cross bin' element assembly and a color wash element assembly. The cross-bin element assembly used on each bullnose ceiling lights has a section 16 inches or less that is intentionally left dark, and placarded accordingly.
      2) Each wash-only ceiling light has one wash element assembly.
      3) Each Center Overhead Stowage (COS) light has one element assembly.
   C. Long light assemblies noted in the table below that have only a short section inoperable/dark may be counted as operable, provided the conditions noted are met.

Ceiling Light Assemblies and Minimum Assembly Lengths

Light Assembly	Element Assembly	Minimum Assembly Length
COS Light Assembly	COS Element Assembly	40 Inches
Wash-only Ceiling Light Assembly	Wash Element Assembly	40 Inches
Ceiling Light Assembly	Wash Element Assembly	40 Inches
	Cross-bin Element Assembly	40 Inches
Bullnose Ceiling Light Assembly	Wash Element Assembly	40 Inches
	Cross-bin Element Assembly	55 Inches

      1) The inoperable/dark section on each element is less than 16 inches in length.
      2) A cross-bin element assembly with one intentionally dark section may have one additional inoperative/dark section of less than 16 inches and still be counted as operable.

OPERATIONS NOTE (Cont'd)

2. Sidewall lights may be inoperative using one of the following options.

A. Option 1: A maximum of 40% of the sidewall LED lights may be inoperative provided the following conditions are met.

NOTE 1: Each overwing exit light should be counted as a sidewall LED light, both in determining the percentage of inoperative sidewall LED lights and in counting the number of sidewall LED lights in a string. This note refers to the cabin lighting feature of the overwing exit light, not the emergency down lighting feature.

NOTE 2: A string is a group of lights connected in series and powered by one electrical power drop.

1) One string containing a maximum of 7 adjacent sidewall LED lights may be inoperative.
2) In addition to one inoperative string containing a maximum of 7 adjacent sidewall LED lights, additional sidewall LED lights may be inoperative provided inoperative lights are not adjacent to or directly opposite from other inoperative lights.

B. Option 2: Greater than 40% of sidewall LED lights may be inoperative provided that the following criteria is followed:

1) For photoluminescent escape path marking system charging, set lighting scene to White Bright.
2) Charge photoluminescent escape path marking system in accordance with approved procedure for Crossbin/ COS/Direct W1/W2 High charging tables.

NOTE 1: Refer to FCOM (Chapter 1 Airplane General, Emergency Equipment, Doors, Windows - System Description – Lighting Systems - Emergency Lighting) or other approved source document for charging guidance.

NOTE 2: More than one string of sidewall LED lights, adjacent sidewall LED lights, and/or opposite sidewall LED lights may also be inoperative provided the above criteria for Option 2 is followed.

33-24-01 Passenger Lighted Information Signs and Notice System

33-24-01-01 Individual Passenger Lighted Information Signs

33-24-01-01A PA System Inoperative

Interval	Installed	Required	Procedure
C	59	0	**(M)**

May be inoperative provided:

a. Associated passenger seat or lavatory is not occupied from which a passenger lighted information sign is not readily legible.
b. Associated seat or lavatory is blocked and placarded "DO NOT OCCUPY".

NOTE: These conditions are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M) PROCEDURES

Block and placard each associated seat or lavatory (AMM 33-00-00/901).

NOTE: In tests conducted at Boeing for the FAA, a sign was considered "readily legible" if enough of the sign could be seen to identify it. For example, NO SMOK and FASTEN SEAT B is considered readily legible.

1. If required, install tapes or ropes of conspicuous contrasting colors to block access to unusable seats prior to boarding passengers.
2. Install conspicuous signs or placards in appropriate locations to indicate seats or lavatories that are not to be occupied.
3. For an associated lavatory, close and lock the lavatory door.

33-24-01 **Passenger Lighted Information Signs and Notice System**

33-24-01-01 **Individual Passenger Lighted Information Signs**

33-24-01-01B **PA System Operates Normally**

Interval	Installed	Required	Procedure
C	59	0	**(O)**

May be inoperative and associated passenger seat or lavatory may be occupied provided:

a. PA system operates normally.
b. PA system is used to notify passengers and cabin crew when associated sign(s) is placed on or off.

OPERATIONS (O) PROCEDURES

Use the passenger address (PA) system to notify passengers and cabin crew when associated signs are placed on or off by annunciating "PASSENGER SIGNS ON" or "PASSENGER SIGNS OFF".

33-24-01 Passenger Lighted Information Signs and Notice System

33-24-01-02 Aural Tone System

Interval	Installed	Required	Procedure
C	1	0	

33-24-01 Passenger Lighted Information Signs and Notice System

33-24-01-03 Flight Deck Automatic Function

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. Manual control function operates normally.
b. Alternate procedures are established and used.

OPERATIONS (O) PROCEDURES

Before Flight Captain will brief Cabin Crew; confirm procedures per Caribbean Airlines SOP's.
The pilot in command shall notify Cabin Crew / passengers when seat belts must be fastened, smoking is prohibited and passengers should return to cabin from lavatories.

33-31-01 Wheel Well Lights

Interval	Installed	Required	Procedure
C	5	0	

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33-37-01 Lower Cargo Compartment Light Systems

Interval	Installed	Required	Procedure
C	17	0	

33-37-01 Lower Cargo Compartment Light Systems
33-37-01-01 Forward Compartment Light Lens

Interval	Installed	Required	Procedure
C	7	6	

One may be broken or missing.

33-37-01 Lower Cargo Compartment Light Systems
33-37-01-02 Aft Compartment Light Lens

Interval	Installed	Required	Procedure
C	10	0	

Any number may be broken or missing.

INTENTIONALLY

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33-41-01 Wing Illumination Lights

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided ground deicing procedures do not require their use.

OPERATIONS (O) PROCEDURES

Utilize alternate lighting source for wing inspections as necessary.

When de-icing or anti-icing is required. A post de-icing check of the critical surfaces is always required. Normally, this check will be accomplished by the company personnel who supervised the de-icing, and once accomplished, must be reported to the captain via aircraft interphone.

In airports where deicing/anti-icing will be performed at a central deicing facility, the designated "iceman" on duty or qualified ground personnel at this facility will inform aircraft flight crew that the aircraft is clean.

The only acceptable decision making criterion, for takeoff without a pre-takeoff contamination inspection, is the shorter time within the applicable holdover timetable cell. This means that after the initial inspection as above, to continue taxi for takeoff beyond the shorter time within the applicable holdover timetable cell, a pre-takeoff contamination inspection is required to be performed by cockpit crew.

Consequently, when the wing illumination lights are inoperative, but would be required for a pre-takeoff contamination inspection, hold over times are limited as per paragraph above. For example, where the applicable holdover timetable cell reads 1:30 – 2:45; holdover time is limited to 1:30.

INTENTIONALLY
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33-42-01 Landing Light LED Array Systems

33-42-01A One Inoperative Per Side

Interval	Installed	Required	Procedure
C	4	2	

One LED array may be inoperative on each side.

NOTE: There is an inboard LED array and an outboard LED array inside the strakelet on each wing. These same lights are also used for the taxi lights. Taxi lights may also be affected (See MEL 33-42-02).

33-42-01 Landing Light LED Array Systems

33-42-01B Runway Turnoff Lights Operate Normally

Interval	Installed	Required	Procedure
C	4	2	

Both LED arrays on one side may be inoperative provided the Runway Turnoff Light on the same side operates normally.

NOTE: There is an inboard LED array and an outboard LED array inside the strakelet on each wing. These same lights are also used for the taxi lights. Taxi lights may also be affected (see MEL 33-42-02).

NOTE: A Runway Turnoff Light (RTO) may be considered to operate normally when the RTO assembly contains one inoperative LED.

33-42-01 Landing Light LED Array Systems

33-42-01C Day Operations

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided operations are not conducted at night.

NOTE: There is an inboard LED array and an outboard LED array inside the strakelet on each wing. These same lights are also used for the taxi lights. Taxi lights may also be affected (See MEL 33-42-02).

33-42-01 Landing Light LED Array Systems

33-42-01-01 Alternate Flash Function

Interval	Installed	Required	Procedure
D	1	0	

33-42-02 Taxi Light LED Array Systems

Interval	Installed	Required	Procedure
C	4	0	

NOTE: There is an inboard LED array and an outboard LED array inside the strakelet on each wing. These same lights are also used for the landing lights. Landing lights may also be affected (See MEL item 33-42-01).

33-42-03 Runway Turnoff Light LED Array Systems

Interval	Installed	Required	Procedure
C	2	0	

33-43-01 Position Light LED Modules

33-43-01A Day Operations

Interval	Installed	Required	Procedure
C	6	0	

May be inoperative provided operations are not conducted from sunset to sunrise.

33-43-01 **Position Light LED Modules**

33-43-01B **All Operations**

Interval	Installed	Required	Procedure
C	6	4	

May be inoperative provided:

a. One red wing tip LED module operates normally.
b. One green wing tip LED module operates normally.
c. Both white tail position LED modules operate normally.

33-43-02 Winglet Aft Marker Lights

33-43-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O) PROCEDURES

Wing Walkers required for Taxi/Tow/Push-back To/From Gate Position. Use Extra Caution during Taxi on Ramps and Taxiways.

33-43-02 Winglet Aft Marker Lights

33-43-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided procedures do not require its use.

33-44-01 **Red ANTI-COLLISION Lights**

33-44-01A **All Operations**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided white anti-collision STROBE lights operate normally.

33-44-01 **Red ANTI-COLLISION Lights**

33-44-01B **Day Operations**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

a. At least one white anti-collision STROBE light operates normally.

b. Operations are not conducted at night.

33-44-02 White Anti-Collision STROBE Lights

33-44-02A All Operations

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided red ANTI-COLLISION lights operate normally.

33-44-02 **White Anti-Collision STROBE Lights**

33-44-02B **Day Operations**

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided:

a. At least one red ANTI-COLLISION light operates normally.

b. Operations are not conducted at night.

33-45-01 Logo Light System

Interval	Installed	Required	Procedure
D	1	0	

INTENTIONALLY
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33-51-01 Interior Emergency Lighting System

33-51-01-01 Aisle Lights

Interval	Installed	Required	Procedure
C	17	15	**SEE NOTE**

NOTE: Light assemblies installed above aisle may be inoperative provided no two adjacent (opposite side) light assemblies are inoperative.

33-51-01 Interior Emergency Lighting System

33-51-01-02 Flight Deck Exit Light

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided operations are not conducted at night.

33-51-02 Exterior Emergency Lighting System

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided operations are not conducted at night.

33-51-03 Floor Proximity Emergency Escape Path Marking System

33-51-03-01 Seat-Mounted and Monument-Mounted LED and Incandescent Lighting System

33-51-03-01-02 Exit Indicator Lamps/Bulbs

Interval	Installed	Required	Procedure
C	68	16	

May be inoperative provided at least two lamps/bulbs in each exit indicator operate normally.

33-51-03 Floor Proximity Emergency Escape Path Marking System

33-51-03-02 Photoluminescent Lighting System

33-51-03-02-01 Photoluminescent Strips

Interval	Installed	Required	Procedure
C	41	20	

A maximum of 5 inches of any length of photoluminescent strip may be inoperative.

NOTE 1: For Standard Photoluminescent Strip - There can be no more than 5.0 inches of any length of photoluminescent strip between 15.0 inches and 72.0 inches that will not give out light.

NOTE 2: For Colored, Narrow, and SSUL Narrow Photoluminescent Strip - There can be no more than 5.0 inches of any length of photoluminescent strip between 15.0 inches and 79.0 inches that will not give out light.

33-51-03 Floor Proximity Emergency Escape Path Marking System

33-51-03-02 Photoluminescent Lighting System

33-51-03-02-02 Exit Indicator Lamps/Bulbs

Interval	Installed	Required	Procedure
C	20	12	

May be inoperative provided at least three lamps/bulbs in each exit indicator operate normally.

34-16-01 Mach/Airspeed Warning Systems

34-23-01 Standby Magnetic Compass

34-24-01 Integrated Standby Flight Display (ISFD)

34-24-01-01 Attitude Indication

34-24-01-02 Approach Mode

34-24-01-03 Heading Display

34-24-01-04 Switch Lights

34-24-01-05 Dedicated Battery

34-31-01 Instrument Landing Systems (ILS)

34-31-02 Global Navigation Satellite Landing System (GLS) ***(CUB ONLY)***

34-32-01 Marker Beacon System

34-33-01 Radio Altimeter Systems

34-36-01 Head-Up Display (HUD) System *- N/A*

34-41-01 Weather Radar System

34-41-01A Dual Channel Configuration – One Inoperative

34-41-01B GPWS Reactive Windshear (Mode 7) Operative

34-41-01C GPWS Reactive Windshear (Mode 7) Inoperative

34-41-01-01 Predictive Windshear (PWS) Function

34-41-01-01A GPWS Reactive Windshear (Mode 7) Operative

34-41-01-01B GPWS Reactive Windshear (Mode 7) Inoperative

34-41-01-02 Auto Tilt Function

34-45-01 Traffic Alert and Collision Avoidance System (TCAS)

34-45-01-01 Resolution Advisory (RA) Display Systems

34-45-01-01A Inoperative For Non-Flying Pilot

34-45-01-01B Procedures Do Not Require Use

34-45-01-02 Traffic Alert (TA) Display Systems

34-45-01-03 Audio Function

34-45-01-04 Airspace Selection Function - *N/A*

34-49-01 Ground Proximity Warning System (GPWS)

34-49-01-01 GPWS Function

34-49-01-01-01 Modes 1-4

34-49-01-01-02 Excessive Deviation Below Glideslope (Mode 5)

34-49-01-01-03 Advisory Altitude/Bank Angle Callout (Mode 6)

34-49-01-01-03A Required by TTCAR - N/A

34-49-01-01-03B Not Required by TTCAR - N/A

34-49-01-01-04 Reactive Windshear (RWS) Detection (Mode 7)

34-49-01-01-04A Predictive Windshear Inoperative

34-49-01-01-04B Predictive Windshear Operative

34-49-01-01-05 Test Mode

34-49-01-02 Terrain Awareness Function

34-49-01-03 Terrain Displays

34-49-01-03A One Inoperative

34-49-01-03B Both Inoperative

34-49-01-04 Runway Awareness and Advisory System (RAAS) and / or Overrun Warning (ORW) System

34-51-01 VOR Systems

34-53-01 ATC Transponder/Automatic Altitude Reporting Systems

34-53-01A *One Inoperative*

34-53-01B *Both Inoperative*

34-53-01-01 Elementary and Enhanced Downlink Aircraft Reportable Parameters Not Required by TTCAR – *N/A*

34-53-01-02 ADS-B Out Extended Squitter

34-53-01-02A *ATC Authorization Required*

34-53-01-02B *Required by US Regulations*

34-55-01 Distance Measuring Equipment (DME) Systems

34-57-01 Automatic Direction Finder (ADF) Systems

34-58-01 Global Positioning System (GPS)

34-58-01A *One Inoperative*

34-58-01B *Both Inoperative*

34-61-01 Flight Management Computer System (FMCS)

34-61-01-01 Computer

34-61-01-01A *1 Computer Operates Normally*

34-61-01-01B *All Computers Inoperative*

34-61-01-02 Navigation Databases

34-61-01-03 CDU/MCDU

34-61-01-03A *One Inoperative*

34-61-01-03B *Both Inoperative*

34-61-01-03-01 Switch Lights/Key Lights

34-61-01-04 FMC Alert Lights

34-61-01-04A *One Inoperative*

34-61-01-04B *Both Inoperative*

34-65-01 Vertical Situation Display (VSD) System

INTENTIONALLY

BLANK

34-16-01 Mach/Airspeed Warning Systems

Interval	Installed	Required	Procedure
C	2	1	

INTENTIONALLY

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34-23-01 Standby Magnetic Compass

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

a. Integrated standby flight display (ISFD) heading display operates normally.

b. Airplane is operated with Dual Independent Navigation Capability and under Positive Radar Control by ATC on the en route portion of the flight.

NOTE: RNAV 10 (RNP 10) Oceanic and Remote Areas Operations [WATRS Oceanic], NOT Allowed.

INTENTIONALLY

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34-24-01 Integrated Standby Flight Display (ISFD)

34-24-01-01 Attitude Indication

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

a. Operations are conducted in Day VMC only.
b. Operations are not conducted into known or forecast over-the-top conditions.

34-24-01 Integrated Standby Flight Display (ISFD)

34-24-01-02 Approach Mode

Interval	Installed	Required	Procedure
C	1	0	

34-24-01 Integrated Standby Flight Display (ISFD)

34-24-01-03 Heading Display

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided standby magnetic compass operates normally.

34-24-01 **Integrated Standby Flight Display (ISFD)**

34-24-01-04 **Switch Lights**

Interval	Installed	Required	Procedure
C	5	0	

34-24-01 **Integrated Standby Flight Display (ISFD)**

34-24-01-05 **Dedicated Battery**

Interval	Installed	Required	Procedure
C	1	0	

34-31-01 Instrument Landing Systems (ILS)

Interval	Installed	Required	Procedure
D	2	1	

One may be inoperative provided it is not powered by a Standby Bus.
ILS No 1 Must be Serviceable. See **FCOM Vol. 2, Chapter 6, Electrical, STBY Pwr**.
One ILS receiver is required for dispatch for ILS Category I Landing Minimums.
2 ILS required for CAT II Approaches.

OPERATIONS NOTE

1. **For one ILS inoperative, SINGLE CH will be displayed in the flight mode annunciation autopilot status.**
2. **When dispatched with an inoperative R-ILS, manually tune an ILS frequency on the left Navigation Control Panel (P8) prior to takeoff to prevent display of the GPWS INOP light in flight.**

NOTE: The Digital Flight Control System cannot use VOR/ILS data that is not shown on the displays. Thus, when the Autopilot (A/P) system is engaged the VHF Navigation Control must match the primary Autopilot system that is ENGAGED for proper ILS/VOR operations: i.e., (CMD A uses VHF NAV 1 for control and CMD B uses VHF NAV 2 for control).

34-31-02 Global Navigation Satellite Landing System (GLS) (CUB ONLY)

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided GLS approaches are not conducted.

NOTE: CARIBBEAN AIRLINES FLIGHT CREW ARE NOT AUTHORIZED TO CONDUCT GLS APPROACHES.

OPERATIONS NOTE

NO AUTOLAND will be displayed in the flight mode annunciation autopilot status on fail operational equipped airplanes.

For one GLS inoperative, SINGLE CH will be displayed in the flight mode annunciation autopilot status.

When dispatched with an inoperative R-GLS, manually tune a GLS frequency on the left Navigation Control Panel (P8) prior to takeoff to prevent display of the GPWS INOP light in flight.

34-32-01 Marker Beacon System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require its use.

INTENTIONALLY

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34-33-01 Radio Altimeter Systems

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Inoperative radio altimeter is deactivated.
b. Approach minimums or operating procedures do not require its use.
c. Associated autopilot is not used for approach and landing.
d. Autothrottle is not used for approach and landing.
e. Associated flight director is not used for approach and landing.

NOTE: If arming LNAV on the ground with one radio altimeter inoperative, the flight directors and autopilot should be controlled by the FCC on the same side as the valid radio altimeter (i.e. the first flight director and/or autopilot to be engaged must be receiving valid radio altitude data).

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative radio altimeter (AMM 34-00-00/901).
For radio altimeter inoperative in position 1:

NOTE: Deactivation of a radio altimeter in position 1 will cause the MAINT light to illuminate and the STALL WARNING SYS L and STALL WARNING SYS R status messages to annunciate, which will prevent dispatch. If the radio altimeter transceiver in position 2 is operating normally, it may be possible to clear the associated active status messages and extinguish the MAINT light by swapping the positions of the radio altimeter transceivers such that the inoperative radio altimeter transceiver is moved to position 2. If the issue can be isolated to an inoperative radio altimeter transceiver in position 1, then it may be swapped to position 2 per the following steps.

1. If necessary, erase latched STALL WARNING SYS L and STALL WARNING SYS R status messages by performing Fault Interrogation for a Maintenance Message (AMM 31-65-00/201).
2. Perform the Low Range Radio Altimeter (LRRA) System - Operational Test (AMM 34-33-00/501).
   A. If the Onboard Maintenance Function (OMF) only shows maintenance message 34-33001 as ACTIVE, proceed with swapping the inoperative radio altimeter transceiver from position 1 to position 2.
   B. If the OMF shows INACTIVE for maintenance message 34-33001 or if other radio altimeter maintenance messages are ACTIVE, then additional troubleshooting is required and MEL dispatch is not permitted.
3. Move the inoperative radio altimeter transceiver from position 1 to position 2 and the operational radio altimeter transceiver from position 2 to position 1 (AMM 34-33-21/401).
4. Open and collar associated P6-1 panel circuit breaker for the radio altimeter in position 2.
5. Verify that the MAINT light is extinguished and that the STALL WARNING SYS L and STALL WARNING SYS R status messages are not active.

NOTE: If the MAINT light illuminates and the STALL WARNING SYS L and STALL WARNING SYS R status messages are active after completing the deactivation steps, additional troubleshooting is required and MEL dispatch is not permitted.

For radio altimeter inoperative in position 2:

1. Open and collar associated P6-1 panel Radio Altimeter circuit breaker.

OPERATIONS (O) PROCEDURES 34-33-01 (Cont'd)

1. Confirm that weather minimums or operating procedures are not dependent upon its use.
2. Do not use the associated autopilot, associated flight director, or autothrottle for approach and landing.

   NOTE 1: The FCC engages LNAV when the radio altimeters indicate that the airplane is 50 ft above ground level (AGL). Dispatching with an inoperative radio altimeter system may generate an erroneous radio height indication of 50 ft AGL and may cause the LNAV FMA to flash when armed on the ground.

   NOTE 2: The RA failure flag will annunciate on the associated pilot Primary Flight Display (PFD) while the radio altimeter is deactivated.
3. If remaining radio altimeter fails:
   A. AFDS (both sides) will limit bank angle to 8 degrees in all roll modes.
   B. Use of the Autopilot/Flight Director System (AFDS) is at the discretion of flight crew. AFDS may not:
      1) Command sufficient bank angle to execute proper departure and/or approach maneuvers.
      2) Make enroute course changes within airspace limitations.

34-41-01 **Weather Radar System**

34-41-01A **Dual Channel Configuration - One Operative**

Interval	Installed	Required	Procedure
D	2	1	

34-41-01 **Weather Radar System**

34-41-01B **GPWS Reactive Windshear (Mode 7) Operative**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. Ground proximity warning system (GPWS) reactive windshear alert (mode 7) operates normally.
b. Aircraft is not operated at night or in instrument meteorological conditions in an area where a thunderstorm or other potentially hazardous weather condition, which may be detectable with an airborne weather radar, may be expected to occur along the route.

OPERATIONS (O) PROCEDURES

1. Aircrew must review Windshear Avoidance and Windshear recovery procedures (FCOM and QRH).
2. Review Weather at Departure, Destination and All Alternates for possible Windshear Conditions.
3. Flight crew must brief on use of all airplane systems available to monitor altitude and flight path (Autopilot, Autoland, ILS, FMCS) and flight crew review of Minimum Enroute Altitudes (MEA's), approach altitude call outs, and vertical situation display (VSD).
4. Flight crew must increase awareness of airplane configuration, altitude and flight path.

34-41-01 **Weather Radar System**

34-41-01C **GPWS Reactive Windshear (Mode 7) Inoperative**

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided aircraft is not operated at night or in instrument meteorological conditions in an area where a thunderstorm or other potentially hazardous weather condition, which may be detectable with an airborne weather radar, may be expected to occur along the route.

OPERATIONS (O) PROCEDURES

1. Aircrew must review Windshear Avoidance and Windshear recovery procedures (FCOM and QRH).
2. Review Weather at Departure, Destination and All Alternates for possible Windshear Conditions.
3. Flight crew must brief on use of all airplane systems available to monitor altitude and flight path (Autopilot, Autoland, ILS, FMCS) and flight crew review of Minimum Enroute Altitudes (MEA's), approach altitude call outs, and vertical situation display (VSD).
4. Flight crew must increase awareness of airplane configuration, altitude and flight path.

34-41-01 **Weather Radar System**

34-41-01-01 **Predictive Windshear (PWS) Function**

34-41-01-01A **GPWS Reactive Windshear (Mode 7) Operative**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided ground proximity warning system (GPWS) reactive windshear alert (mode 7) operates normally.

OPERATIONS (O) PROCEDURES

1. Aircrew must review Windshear Avoidance and Windshear recovery procedures (FCOM and QRH).
2. Review Weather at Departure, Destination and All Alternates for possible Windshear Conditions.
3. Flight crew must brief on use of all airplane systems available to monitor altitude and flight path (Autopilot, Autoland, ILS, FMCS) and flight crew review of Minimum Enroute Altitudes (MEA's), approach altitude call outs, and vertical situation display (VSD).
4. Flight crew must increase awareness of airplane configuration, altitude and flight path.

34-41-01 **Weather Radar System**

34-41-01-01 **Predictive Windshear (PWS) Function**

34-41-01-01B **GPWS Reactive Windshear (Mode 7) Inoperative**

Interval	Installed	Required	Procedure
B	1	0	**(O)**

OPERATIONS (O) PROCEDURES

1. Aircrew must review Windshear Avoidance and Windshear recovery procedures (FCOM and QRH).
2. Review Weather at Departure, Destination and All Alternates for possible Windshear Conditions.
3. Flight crew must brief on use of all airplane systems available to monitor altitude and flight path (Autopilot, Autoland, ILS, FMCS) and flight crew review of Minimum Enroute Altitudes (MEA's), approach altitude call outs, and vertical situation display (VSD).
4. Flight crew must increase awareness of airplane configuration, altitude and flight path.

34-41-01 Weather Radar System

34-41-01-02 Auto Tilt Function

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided manual tilt function operates normally.

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34-45-01 Traffic Alert and Collision Avoidance System (TCAS)

Interval	Installed	Required	Procedure
B	1	0	**(M)**

May be inoperative provided:

1. System is deactivated and secured.
2. Flight is not conducted in RVSM Airspace (FL290 – FL410).

MAINTENANCE (M) PROCEDURES

Deactivate the TCAS system (AMM 34-00-00/901).

1. Open and collar TCAS circuit breaker.

34-45-01 **Traffic Alert and Collision Avoidance System (TCAS)**

34-45-01-01 **Resolution Advisory (RA) Display Systems**

34-45-01-01A **Inoperative For Non-Flying Pilot**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative on non-flying pilot side.

34-45-01 **Traffic Alert and Collision Avoidance System (TCAS)**

34-45-01-01 **Resolution Advisory (RA) Display Systems**

34-45-01-01B **Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided:

1. Traffic Alert (TA) visual display and audio functions operate normally.
2. TA only mode is selected by crew.
3. Flight is not conducted in RVSM Airspace (FL290 – FL410).

OPERATIONS (O) PROCEDURES

Observe operative indications and advisories. Respond as appropriate.

34-45-01 Traffic Alert and Collision Avoidance System (TCAS)

34-45-01-02 Traffic Alert (TA) Display Systems

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided:

1. RA visual display and audio functions operate normally.
2. Flight is not conducted in RVSM Airspace (FL290 – FL410).

OPERATIONS (O) PROCEDURES

Observe operative indications and advisories. Respond as appropriate.

34-45-01 Traffic Alert and Collision Avoidance System (TCAS)

34-45-01-03 Audio Function

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided flight is not conducted in RVSM Airspace (FL290 – FL410).

34-49-01 **Ground Proximity Warning System (GPWS)**

34-49-01-01 **GPWS Function**

Interval	Installed	Required	Procedure
A	1	0	**(O)**

May be inoperative provided repairs are made within 2 flight days.

NOTE: GPWS Warning Is Required For Cat II Approach.

OPERATIONS (O) PROCEDURES

Flight Crew must use procedures to increase crew awareness of airplane configuration, altitude and flight path.
They should include flight crew use of all systems available (autopilot, autoland, ILS, and FMC), approach altitude callouts, and vertical situation display (VSD).
Flight Crew must review Minimum Enroute Altitudes (MEA's) and altitude callouts.
Review GPWS and GPWS Maneuver in QRH Non-Normal Maneuvers Section.
Pilot Monitoring should monitor flight path during takeoff, approach, and landing, and alert the pilot flying if any of the following conditions exist.

1. Excessive descent rate.
2. Excessive terrain closure rate.
3. Altitude loss after takeoff or go-around.
4. Unsafe terrain clearance while not in the landing mode.
5. Deviation from glideslope.

34-49-01 Ground Proximity Warning System (GPWS)
34-49-01-01 GPWS Function
34-49-01-01-01 Modes 1-4

Interval	Installed	Required	Procedure
A	4	0	**(O)**

May be inoperative provided repairs are made within 2 flight days.

NOTE: GPWS Modes 1 Thru 4 Are Required For CAT II Approach.

OPERATIONS (O) PROCEDURES

Flight Crew must use procedures to increase crew awareness of airplane configuration, altitude and flight path.
They should include flight crew use of all systems available (Autopilot, Autoland, ILS, and FMC), approach altitude callouts, and vertical situation display (VSD).
Flight Crew must review Minimum Enroute Altitudes (MEA's) and altitude callouts.
Review GPWS and GPWS Maneuver in QRH Non-Normal Maneuvers Section.
Pilot Monitoring should monitor flight path during takeoff, approach, and landing, and alert the pilot flying if any of the following conditions exist.

1. Excessive descent rate.
2. Excessive terrain closure rate.
3. Altitude loss after takeoff or go-around.
4. Unsafe terrain clearance while not in the landing mode.
5. Deviation from glideslope.

34-49-01 Ground Proximity Warning System (GPWS)
34-49-01-01 GPWS Function
34-49-01-01-02 Excessive Deviation Below Glide Slope (Modes 5)

Interval	Installed	Required	Procedure
B	1	0	

34-49-01 Ground Proximity Warning System (GPWS)
34-49-01-01 GPWS Function
34-49-01-01-03 Advisory Altitude/Bank Angle Callout (Mode 6)

Interval	Installed	Required	Procedure
B	1	0	**(O)**

OPERATIONS (O) PROCEDURES

Flight Crew must use procedures to increase flight crew awareness of airplane altitude and flight path.
They should include flight crew use of all systems available (autopilot, autoland, ILS, and FMC), approach altitude callouts, and vertical situation display (VSD).
Pilot Monitoring to exercise extra vigilance of flight path, attitude and altitude. Use Standard Calls to alert Pilot Flying of any parameter approaching normal limits.
If necessary, on Approach and Landing, Pilot Monitoring will call out "FIVE HUNDRED", "ONE HUNDRED" and "FIFTY", "FORTY", "THIRTY", "TWENTY", "TEN" at the appropriate radio altimeter indications.

34-49-01 **Ground Proximity Warning System (GPWS)**
34-49-01-01 **GPWS Function**
34-49-01-01-04 **Reactive Windshear (RWS) Detection (Mode 7)**
34-49-01-01-04A **Predictive Windshear Inoperative**

Interval	Installed	Required	Procedure
B	1	0	**(O)**

OPERATIONS (O) PROCEDURES

Flight Crew must use procedures to increase flight crew awareness of airplane altitude and flight path. They should include flight crew use of all systems available (Autopilot, Autoland, ILS, and FMC), approach altitude callouts, and vertical situation display (VSD).

Flight crew must review Minimum Enroute Altitudes (MEA's) and altitude callouts.

Review Windshear Recognition, Avoidance and Escape Techniques. Refer to FCOM-1, SP.16. Windshear, and QRH Non-Normal Maneuvers – Windshear.

Pilot Monitoring should closely monitor flight path indications and call out "WINDSHEAR" in case of:

1. IAS change in excess of 15 kts
2. V/S change in excess of 500 ft/min
3. Pitch changes in excess of 5°
4. Glideslope displacement of more than one dot.

34-49-01 **Ground Proximity Warning System (GPWS)**
34-49-01-01 **GPWS Function**
34-49-01-01-04 **Reactive Windshear (RWS) Detection (Mode 7)**
34-49-01-01-04B **Predictive Windshear Operative**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided weather radar system predictive windshear alert mode operates normally.

OPERATIONS (O) PROCEDURES

Flight Crew must use procedures to increase flight crew awareness of airplane altitude and flight path. They should include flight crew use of all systems available (autopilot, autoland, ILS, and FMC), approach altitude callouts, and vertical situation display (VSD).

Flight crew must review Minimum Enroute Altitudes (MEA's) and altitude callouts.

Review Windshear Recognition, Avoidance and Escape Techniques. Refer to FCOM-1, SP.16. Windshear, and QRH Non-Normal Maneuvers – Windshear.

Pilot Monitoring should closely monitor flight path indications and call out "WINDSHEAR" in case of:

1. IAS change in excess of 15 kts
2. V/S change in excess of 500 ft/min
3. Pitch changes in excess of 5°
4. Glideslope displacement of more than one dot.

34-49-01 Ground Proximity Warning System (GPWS)

34-49-01-01 GPWS Function

34-49-01-01-05 Test Mode

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

a. GPWS is considered inoperative.

b. Repairs are made within 2 flight days.

NOTE: Airplane must also be dispatched using MMEL item 34-49-01-01.

34-49-01 Ground Proximity Warning System (GPWS)

34-49-01-02 Terrain Awareness Function

Interval	Installed	Required	Procedure
B	1	0	**(O)**

OPERATIONS (O) PROCEDURES

Flight Crew must use procedures to increase crew awareness of airplane configuration, altitude and flight path.

They should include flight crew use of all systems available (autopilot, autoland, ILS, and FMC), approach altitude callouts, and vertical situation display (VSD).

Flight Crew must review Minimum Enroute Altitudes (MEA's) and altitude callouts.

Review GPWS and GPWS Maneuver in QRH Non-Normal Maneuvers Section.

Pilot Monitoring should use extra vigilance in monitoring flight path during takeoff, approach, landing, and flight over terrain and alert the pilot flying if any of the following conditions exist.

1. Excessive descent rate.
2. Excessive terrain closure rate.
3. Altitude loss after takeoff or go-around.
4. Unsafe terrain clearance while not in the landing mode.
5. Deviation from glideslope.

34-49-01 **Ground Proximity Warning System (GPWS)**
34-49-01-03 **Terrain Displays**
34-49-01-03A **One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

34-49-01 **Ground Proximity Warning System (GPWS)**
34-49-01-03 **Terrain Displays**
34-49-01-03B **Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	

34-49-01 Ground Proximity Warning System (GPWS)

34-49-01-04 Runway Awareness and Advisory System (RAAS) and/or Overrun Warning (ORW) System

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided the GROUND PROXIMITY RUNWAY INHIBIT switch is set to INHIBIT.

OPERATIONS (O) PROCEDURES

Set the GROUND PROXIMITY RUNWAY INHIBIT switch to INHIBIT.

INTENTIONALLY
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34-51-01 VOR Systems

Interval	Installed	Required	Procedure
D	2	1	

May be inoperative provided:

1. VHF Nav No 1 is serviceable. (See FCOM2, Chapter 6, Electrical, STBY Pwr.)
2. CAT II Aproaches are not allowed.
3. At least one GPS/GNSS is operative for flights predicated solely on VOR facilities.

OPERATIONS NOTE

The Digital Flight Control System cannot use VOR/ILS data that is not shown on the displays. Thus, when the Autopilot (A/P) system is engaged the VHF Navigation Control must match the primary Autopilot system that ENGAGED for proper ILS/VOR operations; i.e. (CMD A uses VHF NAV 1 for control and CMD B uses VHF NAV 2 for control).

INTENTIONALLY

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34-53-01 ATC Transponder/Automatic Altitude Reporting Systems

34-53-01A One Inoperative

Interval	Installed	Required	Procedure
D	2	1	

OPERATIONS NOTE

1. One is required for Operation in RVSM Airspace.
2. Advise ATC should Remaining Transponder Fail.
3. Ensure that the transponder selector is positioned to the operative side.
4. Whenever available and where practicable, use the autopilot which corresponds with the selected transponder.
5. For ADS-B operations, the left GPS provides data to Transponder 1 containing ADS-B position information and the right GPS provides ADS-B position data to Transponder 2.
6. If the GPS which correlates with the operative Transponder is INOPERATIVE – then ADS-B is Inoperative (See also MEL Item 34-58-01 GPS).
7. ADS-B is required in US Airspace.
8. The flight must be dispatched per MEL Items 34-53-01-02A or 34-53-01-02B.

34-53-01 ATC Transponder/Automatic Altitude Reporting Systems

34-53-01B Both Inoperative

Interval	Installed	Required	Procedure
B	2	0	

May be inoperative provided approval is obtained from ATC facilities having jurisdiction over planned route of flight, prior to dispatch.

MAINTENANCE NOTE

Advise MCC and SOC.

OPERATIONS NOTE

1. SOC must ensure ICAO Flight Plan reflects aircraft status.
2. SOC must obtain Required Permissions/Approvals over Route of Flight – Including Alternate Airports.
3. Flight in RVSM Airspace NOT authorized.
4. If BOTH Transponders are Inoperative, ADS-B is Inoperative.
5. ADS-B is required is US Airspace.
6. Dispatch per MEL Items 34-53-01-02A.

34-53-01 ATC Transponder/Automatic Altitude Reporting Systems

34-53-01-02 ADS-B Out Extended Squitter

34-53-01-02A ATC Authorization Required

Interval	Installed	Required	Procedure
B	2	0	**(O)**

May be inoperative provided prior to flight, authorization is obtained from ATC facilities having jurisdiction over the planned route of flight using an approved authorization process.

NOTE: Any ADS-B Out function that operates normally may be used.

OPERATIONS NOTE

1. ADS-B is required to operate in US Airspace.
2. For one (1) ADS-B inoperative, dispatch using MEL 34-53-01-02B.

OPERATIONS (O) PROCEDURES

If BOTH ADS-B are Inoperative:

1. SOC to evaluate the airspace for the planned operation. ADS-B required for US Operations.
2. If the airspace for the planned operation – including alternates – does not require ADS-B; dispatch is authorized without operative ADS-B.
3. If the airspace for the planned operation requires ADS-B – SOC shall obtain clearance from all ARTCC's required for the operation inclusive of WATRS.
4. Where approvals are obtained, SOC to adjust ICAO Flight Plan Equipment code to reflect Aircraft Status - no operative ADS-B. SOC shall provide operating crew details of all approvals.

34-53-01 **ATC Transponder/Automatic Altitude Reporting Systems**

34-53-01-02 **ADS-B Out Extended Squitter**

34-53-01-02B **Required by US Regulations**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative.

NOTE: Any ADS-B Out function that operates normally may be used.

OPERATIONS NOTE

1. One is required for Operation in RVSM Airspace.
2. Advise ATC should Remaining Transponder Fail.
3. Ensure the transponder selector is positioned to the operative side.
4. Whenever available and where practicable, use the autopilot which corresponds with the selected transponder.
5. For ADS-B operations, the left GPS provides data to Transponder 1 containing ADS-B position information and the right GPS provides ADS-B position data to Transponder 2.
6. If the GPS which correlates with the operative Transponder is INOPERATIVE - then ADS-B is Inoperative (See also MEL Item 34-58-01 GPS).
7. ADS-B is required in US Airspace.

34-55-01 Distance Measuring Equipment (DME) Systems

Interval	Installed	Required	Procedure
D	2	1	

MAINTENANCE NOTE

1. **The inoperative DME may be deactivated by opening and collaring the associated DME circuit breaker on the P6 or P18 panel.**
2. **This is the preferred method for deactivating/inhibiting an inoperative DME.**
3. **The FMC will perform DME/DME position updates using a single operative DME.**

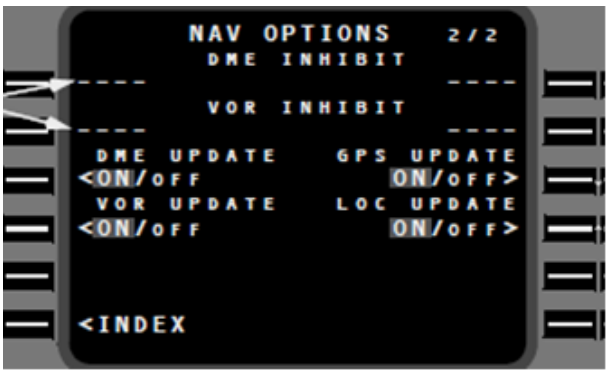
OPERATIONS NOTE

1. **To prevent an inoperative DME from sending invalid position updates to the FMC, DME position updating can be inhibited by setting the DME UPDATE field on the FMC CDU NAV OPTIONS page 2/2 to OFF. LSK3L.**
2. **This is an alternative method to inhibit DME/DME updating.**
3. **This method inhibits all DME/DME updating.**
4. **The method described in the MAINTENANCE NOTE is the preferred method.**
5. **PBN Operations are available. (See Definitions)**

NOTE: WHEN THE DME UPDATE IS "OFF" (LSK3L), VOR-DME AND LOC-DME UPDATES ARE ALSO INHIBITED EVEN IF THE VOR AND LOC UPDATES ARE SELECTED ON. IF THE FMC HASN'T DONE A GPS UPDATE IN THE LAST 5 SECONDS.
THEN THE FMC CAN DO A LOC ONLY UPDATE WITHOUT DME IF THE LOC UPDATE IS SELECTED ON AND THE DME UPDATE IS SELECTED OFF.
BY OPENING AND COLLARING THE CB FOR THE INOPERATIVE DME (DME 1 - PANEL P18-1; DME 2 – PANEL P6-1), AND KEEPING THE DME UPDATE "ON" (LSK3L), VOR-DME AND LOC-DME UPDATING REMAINS AVAILABLE; DME/DME UPDATING IS AVAILABLE FROM THE ONE OPERATIVE DME. (EACH DME RECEIVES AND USES MULTIPLE DME STATIONS).
FOR DME/DME/IRU OPERATIONS, THIS CONFIGURATION WILL ALLOW ENTRY, AND INHIBITING, OF DME STATIONS WHICH ARE UNUSABLE PER NOTAM BY USE OF THE DME INHIBIT FUNCTION. LSK1L AND LSK1R.
THIS IS ESPECIALLY IMPORTANT WHEN GPS IS INOPERATIVE.

Line Select Key Left
LSK1L >>>

LSK3L >>>



Line Select Key Right
<<< LSK1R

INTENTIONALLY

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34-57-01 Automatic Direction Finder (ADF) Systems

Interval	Installed	Required	Procedure
D	2	1	

OPERATIONS NOTE

A flight predicated solely on NDB facilities will require two (2) ADF receivers with automatic bearing presentation in operation when both GPS/GNSS are inoperative.

Dispatch with one serviceable ADF system is allowed provided at least one GPS/GNSS is operative.

INTENTIONALLY
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34-58-01 Global Positioning System (GPS)

34-58-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

NOTE: The associated ADS-B Squitter Transmissions will be inoperative. The airplane must also be dispatched using MEL item 34-53-01-02.

NOTE: One GPS required for flights filed to operate on Q-Routes or Y-Routes.

NOTE: PBN Operations (See Definitions) are allowed.

OPERATIONS NOTE

1. **The associated ADS-B Squitter Transmissions will be inoperative.**
2. **ATC must be advised immediately if remaining GPS fails either while operating on or filed to operate on a Q-Route and/or on a Y-Route. Re-routing and/or Altitude Changes may be necessary.**

   NOTE: The left GPS provides data to Transponder 1 providing ADS-B position information; and the right GPS provides ADS-B position data to Transponder 2. When one GPS is invalid, ensure the Transponder selector is positioned to the side with the valid GPS.

   ADS-B is required to operate in US airspace. Whenever available, use autopilot which corresponds with selected transponder.

34-58-01 Global Positioning System (GPS)

34-58-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Correct UTC time and date are entered manually.

NOTE: The ADS-B Squitter Transmissions will be inoperative. The airplane must also be dispatched using MEL item 34-53-01-02.

MAINTENANCE (M) PROCEDURES

Manually enter the correct UTC time and date from the Airplane Configuration Data (ACD) page (AMM 31-65-00/201).

NOTE: UTC time and date will need to be re-entered each time the MAX Display System is powered down or the Display Processing Computer(s) are swapped out.

OPERATIONS (O) PROCEDURES

1. **The RWY/POS update on the flight management TAKEOFF REF page is active when GPS is inoperative.**
2. **PBN Operations (See Definitions) are allowed, except for flight on Q-Routes or Y-Routes which require one (1) Operative GPS.**
3. **Use Caribbean Airlines Standard Operating Procedures except as noted below.**

SOC must be advised to have the ICAO flight plan reflect No GPS and No ADS-B.

With Both GPS Inoperative, Two IRS are required for RNAV 10 (RNP 10) (See Definitions), Oceanic [WATRS] Operations.

At CDU Preflight Procedure, Aircraft Position Entry on the POS INIT page, Enter the present position on the SET IRS POS line. Use the most accurate Latitude and Longitude. With both GPS Inoperative enter the Gate Position as published on Jeppesen Airport Chart.

NOTE: BOTH PILOTS MUST CONFIRM THE GATE POSITION BEFORE IT IS EXECUTED ON THE FMC. ONE PILOT SHOULD READ THE POSITION AS ENTERED ON THE FMC SCRATCHPAD TO THE PILOT CHECKING THE POSITION AGAINST THE JEPPESEN CHART BEFORE ENTRY INTO THE SET IRS POS BOXES.

In flight, the FMC position is continually updated from the GPS, Navigation Radios, and IRS. Updating priority is based on the availability of valid data from the supporting systems.

FMC position updates from navigation sensor positions are used in the following priority order:

- GPS **(Not available per this MEL Item)**
- Two or more DME stations
- One VOR with a collocated DME
- One localizer and collocated DME
- One localizer

The station identifiers and frequencies of the selected radio navigation aids are displayed on the NAV STATUS page 1/2.

Radio position updating (DME / DME) is very accurate, however, FMC Position will not be updated by Radio Position beyond DME coverage. **Maximum time in RNAV 10 (RNP 10) Airspace without an FMC Radio Position update is 6.2 hours**. Timing starts at the last point at which the FMC position has been Radio updated. Caribbean Airlines RNAV 10 (RNP 10) Operations are all well within this time limit.

OPERATIONS (O) PROCEDURES 34-58-01B (Cont'd)

RNAV SID's, RNAV STAR's, and RNP APCH may be flown when Chart Notes indicate "DME / DME / IRU or GPS required". May also be presented as "RNAV 1 – D/D/I or GNSS required". Pilots must check Chart Notes carefully for "DME / DME RNP-0.30 not authorized", or, "GPS required" Notations which will disqualify the SID, STAR or Approach.

NOTAMS / ATIS must be carefully checked for Navigation Facility outages which will invalidate DME / DME / IRU RNAV operations.

Where a procedure requires a particular VOR / DME / ILS be used, (e.g. For non GNSS equipped aircraft, YSB and YVV DMEs must be operational) these must be manually selected on NAV Status page 1 of 2 in the FMC by tuning the VOR / DME / ILS on VHF NAV Receivers.

The FMC normally rejects the use of navaids that are not suitable for navigation. However, when the aircrew is aware that unreliable navaids exist (by NOTAM, ATIS, ATC, etc.) they should manually exclude these navaids from the FMCs navigation solution. This will prevent the possibility of incorrect position calculations and maximize the FMCs reliability. This is accomplished through the NAV OPTIONS page; DME INHIBIT and VOR INHIBIT fields.

Where "DME / DME / IRU RNAV is not authorized", or "GPS required" notations exist, Standard (Non RNAV) SID's, STAR's and Approaches must be flown.

PM must monitor Raw Data on the ND for all non-precision approaches (VOR, NDB) using Expanded VOR Mode or Center VOR Mode.

Pilots must be particularly aware that Map Shift / Position Shift can occur. The FMC is designed to automatically reject unreliable navaid data during FMC position updating. However, in certain conditions, navaids which are in error may satisfy the reasonableness criteria and provide the FMC with an inaccurate radio position. One of the most vulnerable times is when a radio position update occurs just after takeoff. This is usually manifested in an abrupt heading correction after engaging LNAV. The position shift can be seen on the map which will shift the desired track and runway symbol to a position significantly different from that displayed during ground roll.

NOTE: IF THE FLIGHT CREW OBSERVES EITHER OF THESE INDICATIONS, THE FMC SHOULD BE CAREFULLY MONITORED.

When adequate radio updating is not available, navigation display map mode may display a shift error. This error results in the displayed position of the airplane, route, waypoints, and navigation aids shifted from their actual positions.

An across track, undetected map shift may result in the airplane flying a ground track that is offset from the desired track. An along track, undetected map shift may result in the flight crew initiating altitude changes earlier or later than desired.

In either case, an undetected map shift may compromise terrain or traffic separation.

Map shift errors can be detected by comparing the position of the airplane on the navigation display map mode with data from the ILS, VOR, DME, and ADF systems.

If radio navaids are unavailable, e.g. WATRS Operations.

Monitor FMC position and IRS Positions – Update FMC position if necessary (FCOM-1 – SP.11).

Radio Position updating of the FMC position will resume when in VHF range, approximately 200nm.

Compare FMC position with the IRS position.

OPERATIONS (O) PROCEDURES 34-58-01B (Cont'd)

Use the POS SHIFT page of the FMC CDU. If the two IRS positions are in agreement and the FMC position is significantly different, the FMC position is probably unreliable. The POS SHIFT page may be used to shift FMC position to one of the IRS positions. This is accomplished by line selecting the IRS or radio position and then pressing the EXEC Key. Actual position, can be confirm with ATC radar or visual reference points if available.

Navigate using most accurate information available, continue to monitor FMC position using VOR / ADF raw data displays on non–flying pilot's navigation display. See FCOM-1, SP.11. FMC Navigation Check.

The FMC normally uses DME from two different ground stations to update its position solution. When two DME stations are not available, the FMC reverts to single station radial-DME updating to determine position.

NOTE: One VOR collocated with DME does not provide a particularly accurate position update. When operating into an airport with only one source of DME, e.g., GEO, pilots should consider inhibiting VOR updating. This is achieved on nav options page 2 of 2. Inhibiting VOR updating does not affect DME updating.

Individual stations which must be excluded from a procedure can also be inhibited on this page.

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-01 **Computer**

34-61-01-01A **1 Computer Operates Normally**

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

1. **On airplanes with dual FMC's installed and one FMC inoperative, operating with the FMC Source Select Switch selected to the operational FMC (BOTH ON L or BOTH ON R) is required to accomplish all Flight Management Computing tasks. Primary means RNP RNAV operations up to a 10 minute duration can be accomplished with a single FMC. Long range navigation requires single FMC with dual MCDUs with Alternate Navigation capability.**
2. **RNAV 1, RNAV 2, RNAV 5 (See Definitions) is approved. RNP APCH is approved. Area Navigation (See Definitions) must be conducted in Airspace Navigable by using Conventional Navigation Aids (VOR / DME / ADF / ILS).**
3. **RNAV 10 (RNP 10) is not approved.**
4. **Prior to departure, SOC must ensure ICAO Flight Plan reflects aircraft status; Required Navaids are operational; and any prior permissions have been granted.**

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-01 **Computer**

34-61-01-01B **All Computers Inoperative**

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Except for ER operations, may be inoperative provided:

a. Autothrottle system is deactivated.
b. Enroute operations do not require its use.
c. Appropriate performance adjustments are applied.

NOTE: When dispatching with both FMCS computers inoperative, use of wing anti-ice when OAT is above 37 deg F (3 deg C) may reduce engine life.

MAINTENANCE (M) PROCEDURES

Deactivate the autothrottle system (AMM 34-00-00/901):

1. Reconfigure the M2539 and M2540 Autothrottle System A and B dip switch module positions as follows:
   A. Switch No. 12 - ON
2. Reconfigure the M1992 and M1993 Autothrottle System A and B dip switch module positions as follows:
   A. Switch No. 24 - Change from OFF to ON or from ON to OFF
      NOTE: Switch 24 is an Odd Parity input and changes with each change in switch position.

OPERATIONS (O) PROCEDURES

1. **Determine takeoff performance based on FMC inoperative using the Onboard Performance Tool.**
2. **No PBN Operations. No Area Navigation (See Definitions).**
3. **Fly Conventional SIDs, Airways, STARs and Approaches only. No flight on PBN designated airspace or airways.**
4. **All Navigation must be conducted by use of the MCP using Conventional Navigation Aids (VOR / DME / ADF / ILS).**
5. **Flights over routings predicated on ATC Radar Vectoring Services is Authorized.**
6. **Prior to departure, SOC must ensure ICAO Flight Plan reflects aircraft status; Required Navaids are operational; and any prior permissions have been granted.**
7. **If possible, avoid operations in FAA controlled airspace.**
8. **On airplanes with dual FMC's installed and one FMC inoperative, operating with the FMC Source Select Switch selected to the operational FMC (BOTH ON L or BOTH ON R) is required to accomplish all Flight Management Computing tasks. Primary means RNP RNAV operations up to a 10 minute duration can be accomplished with a single FMC. Long range navigation requires single FMC with dual MCDUs with Alternate Navigation capability. (For Dispatch with 1 FMC operating normally use 34-61-01-01A).**

34-61-01 Flight Management Computer System (FMCS)

34-61-01-02 Navigation Databases

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

a. It is not used in a primary navigation system.
b. The ICAO Flight Plan is updated (as required) to notify ATC of the navigation equipment status of the aircraft.
c. Repairs are made within 10 flight days.

OPERATIONS NOTE

1. **PBN operations are prohibited. All Navigation must be conducted by use of the MCP using Conventional Navigation Aids (ADF/VOR/DME/ILS) and procedures.**
   - **A. Verify status and suitability of navigation facilities used to define route of flight using established procedures.**
   - **B. Verify navigation fixes prior to dispatch with current aeronautical charts.**
   - **C. Manually tune and identify departure/enroute/approach navigation radios.**
2. **Prior to departure, SOC must ensure ICAO Flight Plan reflects aircraft status; Required Navaids are operational; and any prior permissions have been granted.**
3. **Navigation databases must be current for the duration of the flight. If the AIRAC cycle will change during flight, pilots must ensure the accuracy of navigation data, including suitability of navigation facilities used to define the routes and procedures for flight. This can be accomplished by verifying electronic data against textual products. One acceptable means is to compare aeronautical charts (new and old) to verify navigation fixes prior to dispatch. If an amended chart affecting navigation data is published for the procedure, the database must not be used to conduct the procedure.**
4. **An Electronic Flight Bag that is current is an acceptable reference source for navigation database verification.**

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-03 **CDU / MCDU**

34-61-01-03A **One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

1. **RNAV 1, RNAV 2, RNAV 5, is approved.**
2. **RNP APCH (See Definitions) is approved.**
3. **Area Navigation (See Definitions) must be conducted in Airspace Navigable by Conventional Navigation Aids (VOR / DME / ADF / ILS).**
4. **RNAV 10 (RNP 10) is NOT approved.**
5. **Prior to departure, SOC must ensure ICAO Flight Plan reflects aircraft status; Required Navaids are operational; and any prior permissions have been granted.**
6. **On airplanes with dual FMC's installed and one FMC Inoperative, operating with the FMC Source Select Switch selected to the operational FMC (BOTH ON L or BOTH ON R) is required to accomplish all Flight Management Computing tasks. Primary means RNP RNAV operations up to a 10 minute duration can be accomplished with a single FMC. Long range navigation requires single FMC with dual MCDUs with Alternate Navigation capability.**

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-03 **CDU / MCDU**

34-61-01-03B **Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative except for ER operations.

OPERATIONS NOTE

1. **No PBN Operations. (See Definitions)**
2. **No Area Navigation. (See Definitions)**
3. **Fly Conventional SIDs, Airways, STARs, and Approaches only. No flight on PBN designated airspace or airways.**
4. **All Navigation must be conducted by use of the MCP using Conventional Navigation Aids (VOR / DME / ADF / ILS)**
5. **Flights over routings predicated on ATC Radar Vectoring Services is Allowed.**
6. **Prior to departure, SOC must ensure ICAO Flight Plan reflects aircraft status; Required Navaids are operational; and any prior permissions have been granted.**
7. **On airplanes with dual FMC's installed and one FMC inoperative, operating with the FMC Source Select Switch selected to the operational FMC (BOTH ON L or BOTH ON R) is required to accomplish all Flight Management Computing tasks. Primary means RNP RNAV operations up to a 10 minute duration can be accomplished with a single FMC. Long range navigation requires a single FMC with dual MCDUs with Alternate Navigation capability. (For Dispatch with 1 CDU/MCDU inoperative use 34-61-01-03A).**

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-03 **CDU / MCDU**

34-61-01-03-01 **Switch Lights/Key Lights**

Interval	Installed	Required	Procedure
C	72	0	

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-04 **FMC Alert Lights**

34-61-01-04A **One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided FMC is not used for autopilot guidance during approach.

OPERATIONS NOTE

RNP APCH - LNAV / VNAV, RNP APCH – LNAV and Overlay Approaches MUST be MANUALLY Flown.

34-61-01 **Flight Management Computer System (FMCS)**

34-61-01-04 **FMC Alert Lights**

34-61-01-04B **Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided FMC is not used for autopilot guidance.

OPERATIONS NOTE

1. FMC must not be coupled to an Autopilot. With an Autopilot Engaged LNAV and VNAV is prohibited.
2. RNAV 10 (RNP 10) Operations NOT Authorized.

With A/P OFF:

RNAV 1, RNAV 2, RNAV 5 Operations and Area Navigation must be manually flown using F/D only.
With F/D only LNAV and VNAV may be engaged.
RNP APCH - LNAV / VNAV, RNP APCH - LNAV and Overlay Approaches Must be MANUALLY Flown.

With A/P ON:

RNAV 1, RNAV 2, and RNAV 5 Operations and Area Navigation (See Definitions) must be flown using MCP HDG SEL only.
Do Not Engage LNAV and/or VNAV.
Conventional Navigation is available by use of the MCP using VOR/DME/ADF/ILS
Fly Conventional Approaches.
In RVSM Airspace [FL290 and Above] A/P must be ON.
Flights over routings predicated on ATC Radar Vectoring Services is Allowed.
Prior to departure, SOC must ensure ICAO Flight Plan reflects aircraft status; Required Navaids are operational; and any prior permissions have been granted.

34-65-01 Vertical Situation Display (VSD) System

Interval	Installed	Required	Procedure
C	1	0	**(O)**

NOTE: The VSD select button on the EFIS Control Panel is considered part of the VSD System.

OPERATIONS (O) PROCEDURES

Flight crew must brief on use of all airplane systems available to monitor terrain, altitude and flight path (terrain display, autopilot, autoland, ILS, FMCS, Rad Alt.). Flight crew will review Minimum Enroute Altitudes (MEA's) and approach altitude call outs.

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35-11-01 Crew Oxygen System

35-11-01-01 Flight Deck Crew Oxygen Pressure Indicator

35-11-01-02 Overboard Vent and Discharge Indicator Disk

35-11-02 Crew Oxygen External Fill Panel – *N/A*

35-21-01 Passenger Oxygen System

35-21-01A Operation At or Below FL250

35-21-01B Operation At or Below 10,00 Ft. MSL

35-21-02 Oxygen Service Units

35-21-02-01 Passenger Service Units

35-21-02-02 Attendant Service Units

35-21-02-03 Lavatory Service Units

35-21-02-04 Automatic Presentation System

35-21-02-05 Door Latches

35-31-01 Portable Oxygen Bottles or Units (Including Masks and Hoses)

35-31-02 Portable Protective Breathing Equipment (PBE)

INTENTIONALLY
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35-11-01 Crew Oxygen System

35-11-01-01 Flight Deck Crew Oxygen Pressure Indicator

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided crew oxygen supply is verified to be above minimum required before each departure.

MAINTENANCE (M) PROCEDURES

Before each departure verify the oxygen supply is above the minimum required for the flight (AMM 35-00-00/901).

1. Gain access to the crew oxygen cylinder.
2. Confirm the crew oxygen cylinder shutoff valve is open.
3. Confirm the oxygen pressure shown on the gauge of the crew oxygen cylinder is above the minimum required for the flight.

35-11-01 **Crew Oxygen System**

35-11-01-02 **Overboard Vent and Discharge Indicator Disk**

Interval	Installed	Required	Procedure
C	1	0	

May be damaged or missing.

35-21-01 Passenger Oxygen System

35-21-01A Operation At Or Below FL250

Interval	Installed	Required	Procedure
B	1	0	**(O)**

May be inoperative provided:

a. Both air conditioning packs operate normally.
b. Pressurization system operates normally.
c. Flight is not conducted where minimum enroute altitude is above 14,000 ft. MSL.
d. Airplane remains at or below FL 250.
e. Portable oxygen units are provided for 10% of passengers.
f. Passengers are appropriately briefed.

OPERATIONS (O) PROCEDURES

1. Plan flight over route with minimum enroute altitudes (MEA) equal to or lower than 14,000 ft.
2. Maximum flight altitude must not exceed FL 250.
3. Confirm portable oxygen units are available for at least 10% of passengers.
4. Prior to flight, the pilot in command should brief the cabin crew on the following:
   A. In the event of a cabin decompression the flight deck crew will announce "Emergency Descent" via the PA system.
   B. Passenger oxygen masks may not deploy.
   C. If deployed, passenger oxygen masks may not function (Do not use).
   D. Cabin attendants should modify the pre-flight safety demo and brief the passengers accordingly.
   E. In the event of a cabin decompression, cabin crew should make use of the portable oxygen units until advised otherwise by the flight deck crew.
   F. Look for signs of hypoxia amongst the passengers and administer oxygen via the portable oxygen units, as necessary.

OPERATIONS NOTES

The pilot in command and dispatcher should consider the weather conditions and ATC facilities available along the route of flight. If possible, avoid WATRS airspace and maintain a routing with ATC radar services and VHF communications.

35-21-01 **Passenger Oxygen System**

35-21-01B **Operation At Or Below 10,000 Ft. MSL**

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided flight is conducted at or below 10,000 ft. MSL

35-21-02 Oxygen Service Units

35-21-02-01 Passenger Service Units

Interval	Installed	Required	Procedure
B	56	0	**(M)**

May be inoperative provided associated seats are blocked and placarded to prevent occupancy.

MAINTENANCE (M) PROCEDURES

Configure the airplane for dispatch with an inoperative passenger service unit (AMM 35-00-00/901).

1. Close the inoperative oxygen mask door and secure it using adhesive tape.
2. Block the seats with the inoperative units to prevent its use.

35-21-02 Oxygen Service Units

35-21-02-02 Attendant Service Units

Interval	Installed	Required	Procedure
B	3	0	**(M)**

May be inoperative provided associated flight attendant seat is considered inoperative

NOTE: Airplane must also be dispatched using MEL item 25-25-01-01.

MAINTENANCE (M) PROCEDURES

Configure the airplane for dispatch with an inoperative attendant service unit (AMM 35-00-00/901).

1. Close the inoperative oxygen mask door and secure it using adhesive tape.
2. Block the seats with the inoperative units to prevent its use.

35-21-02 Oxygen Service Units

35-21-02-03 Lavatory Service Units

Interval	Installed	Required	Procedure
B	3	0	**(M)**

May be inoperative provided associated lavatory door is locked closed and placarded "INOPERATIVE - DO NOT ENTER".

NOTE: These provisions are not intended to prohibit lavatory inspections by crewmembers.

MAINTENANCE (M) PROCEDURES

Configure the airplane for dispatch with an inoperative lavatory service unit (AMM 35-00-00/901).

1. Close the inoperative oxygen mask door and secure it using adhesive tape.
2. Secure the associated lavatory door closed to prevent passenger use.

35-21-02 Oxygen Service Units

35-21-02-04 Automatic Presentation System

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Manual deployment system is verified to operate normally.
b. Airplane remains at or below FL 300.

MAINTENANCE (M) PROCEDURES

Verify the manual deployment system operates normally by doing the appropriate task in the Passenger Oxygen System Adjustment/Test (AMM 35-00-00/901).

OPERATIONS (O) PROCEDURES

1. Flight altitude is limited to FL 300.
2. Be prepared to deploy passenger masks manually using the PASS OXYGEN Switch.

35-21-02 Oxygen Service Units

35-21-02-05 Door Latches

Interval	Installed	Required	Procedure
B	62	0	**(M) (O)**

Automatic opening feature of door latch(es) may be inoperative unlatched provided:

a. Associated door is taped closed.
b. Associated oxygen service unit operates normally.
c. Flight remains at or below FL 250.
d. Affected passengers occupying seats with an inoperative door latch are briefed on alternate oxygen mask procedure.

MAINTENANCE (M) PROCEDURES

Confirm oxygen is available for the associated masks and tape the door closed (AMM 35-00-00/901).

1. Confirm the associated generator has not been expended. The color band will be black if the generator has been expended and the generator must be replaced.
2. Tape the door closed. Fashion the tape so that a non adhesive "handle", approximately 4 inches long, is left pendant. This "handle" will be used in case of need by the passenger to strip the securing tape down, thus allowing the door to fall open.

OPERATIONS (O) PROCEDURES

Brief affected passengers on the alternate procedure for oxygen mask deployment.

INTENTIONALLY
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35-31-01 Portable Oxygen Bottles or Units (Including Masks and Hoses)

Interval	Installed	Required	Procedure
D	4	3	**(M)**

One may be inoperative or missing provided:

a. A serviceable portable oxygen bottle is maintained in the Forward cabin and one in the Aft cabin.
b. Location placarding is removed or obscured if the unit at a specific storage location is inoperative or missing.
c. Inoperative unit remains in a certified location until replaced, serviced, or removed at the next available maintenance facility.

NOTE: Inoperative portable oxygen bottles or units, removed from a certified location or removed from the aircraft, are subject to dangerous good regulations.

MAINTENANCE (M) PROCEDURES

Configure the airplane for operation with one inoperative portable oxygen dispensing unit (AMM 35-00-00/901).

1. A serviceable portable oxygen bottle is maintained in the Forward cabin and one in the Aft cabin.
2. Location placarding is removed or obscured if the unit at a specific storage location is inoperative or missing.
3. If the unit at a specific storage location is inoperative or missing, remove or obscure the associated location placarding for the affected unit.

OPERATIONS NOTE

1. **During flight when crew portable oxygen is administered to a passenger, use of the gaseous bottle should be discontinued when the gauge pressure reads approximately 1,100 p.s.i. This will ensure that each serviceable bottle has reserve oxygen available in case of a depressurization requirement.**
2. **Smoking is prohibited when oxygen is being administered.**
3. **Relevant speech to be made by Purser.**

35-31-02 Portable Protective Breathing Equipment (PBE)

Interval	Installed	Required	Procedure
D	7	5	

Any in excess of those required by TTCAR may be inoperative or missing provided:

a. Inoperative PBE remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
b. Location placarding is removed or obscured.
c. Required distribution is maintained.

NOTE: Inoperative PBEs, removed from a certified location or removed from the aircraft, are subject to dangerous goods regulations.

OPERATIONS NOTE

One is Required in the Cockpit. (TTCAR 7)
Minimum of Four [4] in the Passenger Cabin. (TTCAR 7)

TTCAR 7 – 47(1)

(b) has sufficient portable protective breathing equipment to protect the eyes, nose and mouth of all required cabin crew members on board the aircraft
Minimum Cabin Crew 160 Pax = 4.

TTCAR 7 – 47

(5) An air operator shall ensure that portable breathing equipment is installed, provided or located at or adjacent to each. required hand fire extinguisher.

36-11-01 Engine Bleed Air Control Systems

36-11-01-01 Primary Control Card

36-11-01-02 Anti-Ice Inhibit Function

36-11-01-03 Backup Control Card

36-11-01-03A ASC Health Monitor Operates Normally

36-11-01-03B PRSOV Is Not Used

36-11-01-04 Air Supply Controller (ASC) Bleed System Health Monitor

36-11-01-04A Both Control Cards Operate Normally

36-11-01-04B PRSOV Is Not Used

36-11-01-05 Bleed Control Backup Protection Functions

36-11-01-05A Primary Control Card Operates Normally

36-11-01-05B Associated PRSOV Is Not Used

36-11-01-06 Bleed Control Engine Data

36-11-02 Engine Bleed Air Control Systems

36-11-02-01 Manifold Pressure (PM2) Sensors

36-11-02-02 Intermediate Pressure (PI) Sensors

36-11-03 Pressure Regulating and Shutoff Valve (PRSOV) Systems

36-11-03-01 Pressure Regulating and Shutoff Valves

36-11-03-01A One Inoperative

36-11-03-01B Both Inoperative

36-11-04 High Pressure Shutoff Valve (HPSOV) Systems

36-11-04-01 High Pressure Shutoff Valves

36-11-04-02 HPSOV Control

36-11-05 Engine BLEED Lights

36-11-05A One Inoperative

36-11-05B Both Inoperative

36-11-06 Duct Vent Valve

36-11-06A *Valve Closed*

36-11-06B *Valve Open*

36-12-01 Fan Air Modulating Valve (FAMV) Systems

36-12-01-01 Fan Air Modulating Valves

36-13-01 Bleed Air Isolation Valve

36-13-02 Bleed Air Isolation Valve Control

36-13-02A *Verification using Visual Position Indicator*

36-13-02B *Verification from Flight Deck*

36-13-03 Ground Pneumatic Valve Control

36-13-03A *Valve Open*

36-13-03B *Valve Closed*

36-21-01 Bleed Air Duct Pressure Indications

36-21-02 Dual Bleed Light System

36-11-01 Engine Bleed Air Control Systems

36-11-01-01 Primary Control Card

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Associated backup card operates normally.
b. Associated high pressure shutoff valve is deactivated closed.
c. A minimum of 60% N1 is maintained on associated engine during flight in icing conditions.
d. Associated air supply controller bleed system health monitor operates normally.
e. Appropriate performance adjustments are applied.

NOTE: If control has not switched to the back-up card, the associated BLEED light will be illuminated and the associated PRSOV will be commanded closed.

MAINTENANCE (M) PROCEDURES

Deactivate the affected High Pressure Shutoff Valve (HPSOV) closed (AMM 36-00-00/901).

1. Set both engine start levers to CUTOFF.
2. Remove pressure from the pneumatic system.
3. Gain access to the associated HPSOV.
4. Manually lock the HPSOV in closed position.
   A. Turn the manual override nut to align the position indicator with the CLOSED position.
   B. Loosen the bolt and push in the lock knob to lock the HPSOV closed.
   C. Tighten the bolt to 19 in-lb (2.1 N-m) – 21 in-lb (2.4 N-m) to hold the lock knob in place.
5. Open and collar the associated engine anti-ice IASC EAI INHIBITS OUT L (or R) circuit breaker on the P6-5 panel.
6. Perform AMM task for Pneumatic Valve Inoperative (AMM 36-00-00/201).
   NOTE: Performing this step will cause the BLEED HPSOV L or BLEED HPSOV R status message to be annunciated.
7. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use the following performance adjustments.

1. Inflight, maintain a minimum of 60% N1 on associated engine during icing conditions.
2. Apply the following additional fuel mileage decrement when operating in known or forecast icing conditions:
   A. 2.0% for trip distances less than or equal to 1000 nautical miles.
   B. 1.0% for trip distances greater than 1000 nautical miles.

36-11-01 Engine Bleed Air Control Systems

36-11-01-02 Anti-Ice Inhibit Function

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Associated high pressure shutoff valve is deactivated closed.
b. A minimum of 60% N1 is maintained on associated engine during flight in icing conditions.
c. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the affected High Pressure Shutoff Valve (HPSOV) closed (AMM 36-00-00/901).

1. Set both engine start levers to CUTOFF.
2. Remove pressure from the pneumatic system.
3. Gain access to the associated HPSOV.
4. Manually lock the HPSOV in closed position.
   A. Turn the manual override nut to alight the position indicator with the CLOSED position.
   B. Loosen the bolt and push in the lock knob to lock the HPSOV closed.
   C. Tighten the bolt to 19 in-lb (2.1 N-m) – 21 in-lb (2.4 N-m) to hold the lock knob in place.
5. Open and collar the associated engine anti-ice IASC EAI INHIBITS OUT L (or R) circuit breaker on the P6-5 panel.
6. Perform AMM task for Pneumatic Valve Inoperative (AMM 36-00-00/201).
   NOTE: Performing this step will cause the BLEED HPSOV L or BLEED HPSOV R status message to be annunciated.
7. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use the following performance adjustments.

1. In flight, maintain a minimum of 60% N1 on associated engine during icing conditions.
2. Apply the following additional fuel mileage decrement when operating in known or forecast icing conditions:
   A. 2.0% for trip distances less than or equal to 1000 nautical miles.
   B. 1.0% for trip distances greater than 1000 nautical miles.

36-11-01 Engine Bleed Air Control Systems

36-11-01-03 Backup Control Card

36-11-01-3A ASC Health Monitor Operates Normally

Interval	Installed	Required	Procedure	
C	2	0		

May be inoperative provided:

a. Associated primary control card operates normally.

b. Associated air supply controller bleed system health monitor operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

36-11-01 Engine Bleed Air Control Systems
36-11-01-03 Backup Control Card
36-11-01-03B PRSOV Is Not Used

Interval	Installed	Required	Procedure
C	2	0	**(O)**

Except for ER operations beyond 120 minutes, may be inoperative provided associated pressure regulating and shutoff valve (PRSOV) is not used.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES
Use MEL Item 36-11-03-01A or 36-11-03-01B (O) Procedures.

36-11-01 Engine Bleed Air Control Systems

36-11-01-04 Air Supply Controller (ASC) Bleed System Health Monitor

36-11-01-04A Both Control Cards Operate Normally

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

a. Associated primary control card operates normally.

b. Associated backup control card operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

36-11-01 **Engine Bleed Air Control Systems**

36-11-01-04 **Air Supply Controller (ASC) Bleed System Health Monitor**

36-11-01-04B **PRSOV Is Not Used**

Interval	Installed	Required	Procedure
C	2	0	**(O)**

Except for ER operations beyond 120 minutes, may be inoperative provided associated pressure regulating and shutoff valve (PRSOV) is not used.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use MEL Item 36-11-03-01A or 36-11-03-01B (O) Procedure.

36-11-01 Engine Bleed Air Control Systems

36-11-01-05 Bleed Control Backup Protection Functions

36-11-01-05A Primary Control Card Operates Normally

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated primary control card operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

36-11-01 Engine Bleed Air Control Systems

36-11-01-05 Bleed Control Backup Protection Functions

36-11-01-05B Associated PRSOV Is Not Used

Interval	Installed	Required	Procedure
C	2	0	**(O)**

Except for ER operations beyond 120 minutes, may be inoperative provided associated pressure regulating and shutoff valve (PRSOV) is not used.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use MEL Item 36-11-03-01A or 36-11-03-01B (O) Procedure.

36-11-01 **Engine Bleed Air Control Systems**

36-11-01-06 **Bleed Control Engine Data**

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Associated high pressure shutoff valve is deactivated closed.
b. A minimum of 60% N1 is maintained on associated engine during flight in icing conditions.
c. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the associated High Pressure Shutoff Valve (HPSOV) closed (AMM 36-00-00/901).

1. Set both engine start levers to CUTOFF.
2. Remove pressure from the pneumatic system.
3. Gain access to the associated HPSOV.
4. Manually lock the HPSOV in the closed position.
   A. Turn the manual override nut to align the position indicator with the CLOSED position.
   B. Loosen the bolt and push in the lock knob to lock the HPSOV closed.
   C. Tighten the bolt to 19 in-lb (2.1 N-m) – 21 in-lb (2.4 N-m) to hold the lock knob in place.
5. Perform AMM task for Pneumatic Valve Inoperative (AMM 36-00-00/201).
   NOTE: Performing this step will cause the BLEED HPSOV L or BLEED HPSOV R status message to be annunciated.
6. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use the following performance adjustments.

1. In flight, maintain a minimum of 60% N1 on associated engine during icing conditions.
2. Apply the following additional fuel mileage decrement when operating in known or forecast icing conditions:
   A. 2.0% for trip distances less than or equal to 1000 nautical miles.
   B. 1.0% for trip distances greater than 1000 nautical miles.

36-11-02 Engine Bleed Air Control Sensors

36-11-02-01 Manifold Pressure (PM2) Sensors

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided:

a. Associated pneumatic pressure indication system is considered inoperative.
b. Associated Manifold Pressure 1 Sensor is verified to operate normally.

NOTE: The airplane must also be dispatched using MEL Item 36-21-01.

MAINTENANCE (M) PROCEDURES

1. Verify associated SMT message BLEED PM SENSOR 1 L/R is not displayed (AMM 36-00-00/901).
2. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

36-11-02 Engine Bleed Air Control Sensors

36-11-02-02 Intermediate Pressure (PI) Sensors

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Associated high pressure shutoff valve is deactivated closed.
b. A minimum of 60% N1 is maintained on associated engine during flight in icing conditions.
c. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the affected high pressure shutoff valve closed (AMM 36-00-00/901).

1. Set both engine start levers to CUTOFF.
2. Remove pressure from the pneumatic system.
3. Gain access to the associated HPSOV.
4. Manually lock the HPSOV in the closed position.
   A. Turn the manual override nut to align the position indicator with the CLOSED position.
   B. Loosen the bolt and push in the lock knob to lock the HPSOV closed.
   C. Tighten the bolt to 19 in-lb (2.1 N-m) - 21 in-lb (2.4 N-m) to hold the lock knob in place.
5. Open and collar the associated engine anti-ice IASC EAI INHIBITS OUT L (or R) circuit breaker on the P6-5 panel.
6. Perform AMM task for Pneumatic Valve Inoperative (AMM 36-00-00/201).
   NOTE: Performing this step will cause the BLEED HPSOV L or BLEED HPSOV R status message to be annunciated.
7. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use the following performance adjustments.

1. In flight, maintain a minimum of 60% N1 on associated engine during icing conditions.
2. Apply the following additional fuel mileage decrement when operating in known or forecast icing conditions:
   A. 2.0% for trip distances less than or equal to 1000 nautical miles.
   B. 1.0% for trip distances greater than 1000 nautical miles.

36-11-03 **Pressure Regulating and Shutoff Valve (PRSOV) Systems**

36-11-03-01 **Pressure Regulating and Shutoff Valves**

36-11-03-01A **One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Valve is secured closed.
b. Airplane is not operated in known or forecast icing conditions.
c. Flight altitude remains at or below FL 250.
d. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Secure the affected pressure regulating and shutoff valve in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position.
2. Deactivate the associated thrust reverser.
3. Gain access to the engine bleed air pressure regulating and shutoff valve.
4. Lock the inoperative valve in the CLOSED position.
   A. Turn the override nut to align the position indicator with the CLOSED position.
   B. Loosen the lock screw/knob bolt and push in the locking knob to lock the valve in the closed position.
   C. Re-tighten the lock screw/knob bolt.
5. Activate the thrust reverser.
6. Activate the leading edge slats.
7. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known or forecast icing conditions.
2. Limit altitude to FL 250.
3. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
4. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Set the associated engine bleed air switch to OFF.
6. For left PRSOV inoperative, operate both packs using the right engine bleed with the isolation valve OPEN.
   NOTE: During ground operation, do not supply both packs using single-engine bleed air due to engine EGT limitation.
7. For right PRSOV inoperative, operate both packs using the left engine bleed with the isolation valve OPEN.
   NOTE: During ground operation, do not supply both packs using single-engine bleed air due to engine EGT limitation.

36-11-03 **Pressure Regulating and Shutoff Valve (PRSOV) Systems**

36-11-03-01 **Pressure Regulating and Shutoff Valves**

36-11-03-01B **Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Valves are secured closed.
b. Airplane is not operated in known or forecast icing conditions.
c. APU bleed air system operates normally.
d. Flight altitude remains at or below 17,000 ft.

MAINTENANCE (M) PROCEDURES

Secure the associated pressure regulating and shutoff valves in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position.
2. Deactivate the associated thrust reverser.
3. Gain access to the engine bleed air pressure regulating and shutoff valve.
4. Lock the inoperative valve in the CLOSED position.
   A. Turn the override nut to align the position indicator with the CLOSED position.
   B. Loosen the lock screw/knob bolt and push in the locking knob to lock the valve in the closed position.
   C. Re-tighten the lock screw/knob bolt.
5. Activate the thrust reverser.
6. Activate the leading edge slats.
7. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known or forecast icing conditions.
2. Limit altitude to 17,000 ft.
3. Set both engine bleed air switches to OFF.
4. Operate the left pack using APU bleed air and the isolation valve switch set to CLOSE**. Refer to Boeing Operations Manual, NO ENGINE BLEED TAKEOFF AND LANDING PROCEDURE.**
5. If the AIR COND system annunciator light and/or both BLEED lights illuminate 45 seconds after the flaps are in the UP position following a takeoff or go around, do not accomplish the BLEED non-normal checklist.

36-11-04 High Pressure Shutoff Valve (HPSOV) Systems

36-11-04-01 High Pressure Shutoff Valves

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Affected valve is deactivated closed.
b. A minimum of 60% N1 is maintained on associated engine during flight in icing conditions.
c. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the affected High Pressure Shutoff Valve (HPSOV) closed (AMM 36-00-00/901).

1. Set both engine start levers to CUTOFF.
2. Remove pressure from the pneumatic system.
3. Gain access to the associated HPSOV.
4. Manually lock the HPSOV in the closed position.
   A. Turn the manual override nut to align the position indicator with the CLOSED position.
   B. Loosen the bolt and push in the lock knob to lock the HPSOV closed.
   C. Tighten the bolt to 19 in-lb (2.1 N-m) – 21 in-lb (2.4 N-m) to hold the lock knob in place.
5. Perform AMM task for Pneumatic Valve Inoperative (AMM 36-00-00/201).
   NOTE: Performing this step may cause the BLEED HPSOV L or BLEED HPSOV R status message to be annunciated.
6. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use the following performance adjustments.

1. In flight, maintain a minimum of 60% N1 on associated engine during icing conditions.
2. Apply the following additional fuel mileage decrement when operating in known or forecast icing conditions:
   A. 2.0% for trip distances less than or equal to 1000 nautical miles.
   B. 1.0% for trip distances greater than 1000 nautical miles.

36-11-04 High Pressure Shutoff Valve (HPSOV) Systems

36-11-04-02 HPSOV Control

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Affected valve is deactivated closed.
b. A minimum of 60% N1 is maintained on associated engine during flight in icing conditions.
c. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate the affected High Pressure Shutoff Valve (HPSOV) closed (AMM 36-00-00/901).

1. Set both engine start levers to CUTOFF.
2. Remove pressure from the pneumatic system.
3. Gain access to the associated HPSOV.
4. Manually lock the HPSOV in the closed position.
   A. Turn the manual override nut to align the position indicator with the CLOSED position.
   B. Loosen the bolt and push in the lock knob to lock the HPSOV closed.
   C. Tighten the bolt to 19 in-lb (2.1 N-m) – 21 in-lb (2.4 N-m) to hold the lock knob in place.
5. Open and collar the associated engine anti-ice circuit breaker IASC EAI INHIBITS OUT L (or R).
5. Open and collar the associated engine anti-ice IASC EAI INHIBITS OUT L (or R) circuit breaker on the P6-5 panel.
6. Perform AMM task for Pneumatic Valve Inoperative (AMM 36-00-00/201).
   NOTE: Performing this step will cause the BLEED HPSOV L or BLEED HPSOV R status message to be annunciated.
7. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Use the following performance adjustments.

1. In flight, maintain a minimum of 60% N1 on associated engine during icing conditions.
2. Apply the following additional fuel mileage decrement when operating in known or forecast icing conditions:
   A. 2.0% for trip distances less than or equal to 1000 nautical miles.
   B. 1.0% for trip distances greater than 1000 nautical miles.

36-11-05 Engine BLEED Lights

36-11-05A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	**(O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Associated engine bleed air switch remains OFF except for engine start.
b. Airplane is not operated in known or forecast icing conditions.
c. Flight altitude remains at or below FL 250.
d. Appropriate performance adjustments are applied.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known or forecast icing conditions.
2. Limit altitude to FL 250.
3. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
4. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. After engine start, set the associated engine bleed air switch to OFF.
6. For left engine bleed light inoperative, operate both packs using the right engine bleed with the isolation valve OPEN.
   NOTE: During ground operation, do not supply both packs using single-engine bleed air due to engine EGT limitation.
7. For right engine bleed light inoperative, operate both packs using the left engine bleed with the isolation valve OPEN.
   NOTE: During ground operation, do not supply both packs using single-engine bleed air due to engine EGT limitation.

36-11-05 Engine BLEED Lights

36-11-05B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	**(O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Engine bleed air switches remain OFF except for engine start.
b. Airplane is not operated in known or forecast icing conditions.
c. APU bleed air system operates normally.
d. Flight altitude remains at or below 17,000 ft.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known or forecast icing conditions.
2. Limit altitude to 17,000 ft.
3. After engine start, set both engine bleed air switches to OFF.
4. Operate the left pack using APU bleed air and the isolation valve switch set to CLOSE. **Refer to Boeing Operations Manual, NO ENGINE BLEED TAKEOFF AND LANDING PROCEDURE.**
5. If the AIR COND system annunciator light and/or both BLEED lights illuminate 45 seconds after the flaps are in the UP position following a takeoff or go around, do not accomplish the BLEED non-normal checklist.

36-11-06 **Duct Vent Valve**

36-11-06A **Valve Closed**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative closed provided that a No Engine Bleed Takeoff is not conducted.

36-11-06 **Duct Vent Valve**

36-11-06B **Valve Open**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative open.

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36-12-01 Fan Air Modulating Valve (FAMV) Systems

36-12-01-01 Fan Air Modulating Valves

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Affected valve is deactivated in full open position.
b. Airplane is not operated in known or forecast icing conditions.
c. Appropriate performance adjustments are applied.

NOTE: Associated thrust reverser system must be considered inoperative if associated thrust reverser inner walls are composed of one Titanium Inner Wall (TIW) half and one Composite Inner Wall (CIW) half. Thrust reversers with Titanium Inner Walls (TIWs) can be identified by part numbers 315A6295-1, -2, -3, and -4, whereas thrust reversers with Composite Inner Walls (CIWs) can be identified by part numbers 315A6295-5 and higher.

MAINTENANCE (M) PROCEDURES

Deactivate the affected Fan Air Modulating Valve (FAMV) in the full open position (AMM 36-00-00/901).

1. For each engine position, review whether thrust reverser inner walls are composed of one Titanium Inner Wall (TIW) half (part numbers 315A6295-1, -2, -3, and -4) and one Composite Inner Wall (CIW) half (part numbers 315A6295-5 and higher).
   A. If either thrust reverser is composed of one TIW half and one CIW half, the associated thrust reverser is must be considered inoperative and the airplane must also be dispatched using MMEL item 78-31-01.
2. Remove pressure from the pneumatic system (AMM 36-00-00/201, Task 36-00-00-860-806).
3. Gain access to the fan air modulating valve.
4. Manually lock the FAMV in the open position.
   A. Remove the locking bolt.
   B. Remove and discard the O-ring.
      NOTE: The O-ring is not required when the pneumatic locking bolt is installed in the locked position.
   C. Install the locking bolt with the end marked "L" visible.
   D. Tighten the locking bolt to 95 in-lb (11 N-m) – 105 in-lb (11.86 N-m).
5. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201, Task 31-65-00-750-801).

OPERATIONS (O) PROCEDURES

1. Performance limited weights are reduced by the following amount for one or both inoperative Fan Air Modulating Valve:

Takeoff	Enroute Climb	Approach & Landing
1,905 kg	3,084 kg	1,179 kg

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below - Enroute Diversion Speed Effects.

Diversion Speed	Factor
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

2. Increase flight planning fuel by either incrementing the current fuel flow correction factor by 0.5% or flight planning fuel by 0.5% for each inoperative valve in the open position.
3. Increase engine-out ETOPS critical fuel reserves by an additional 0.7% on top of the flight planning fuel penalty noted above for one inoperative valve in the open position. The all-engine ETOPS critical fuel reserves would be covered by the flight planning fuel penalty noted above.

36-13-01 Bleed Air Isolation Valve

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations beyond 120 minutes, may be inoperative provided:

a. Valve remains closed except for engine start.
b. Airplane is not operated in known or forecast icing conditions.

MAINTENANCE (M) PROCEDURES

Prepare the airplane for flight with bleed air isolation valve deactivated in closed position (AMM 36-00-00/901).

1. If the position of the bleed air isolation valve can be manually opened and closed, do these steps:
   A. Set both engine BLEED air switches to OFF.
   B. Set the APU BLEED air switch to OFF.
   C. Set both PACK switches to OFF.
   D. Open and collar the P6-4 panel AIR CONDITIONING BLEED AIR VALVE ISOLATION circuit breaker.
   E. Gain access to the bleed air isolation valve.
   F. Remove, cap and stow the connector from the bleed air isolation valve.
      NOTE: It may be necessary to temporarily remove the air conditioning duct segment located in front of keel beam access hole.
   G. Manually move the valve to the closed position using the manual override handle.
   H. Set the BLEED air and PACK switches as required.

   NOTE: The isolation valve must be manually opened for engine start and then manually closed after the engines are started. Personnel should take special care to remain well clear of the engine inlet hazard areas when manually closing the valve after engine start.

2. If the bleed air isolation valve is stuck closed and cannot be opened, do these steps:

leed air isolation valve is stuck closed and cannot be opened, do these steps:

   A. Set both engine BLEED air switches to OFF.
   B. Set the APU BLEED air switch to OFF.
   C. Set both PACK switches to OFF.
   D. Open and collar the P6-4 panel AIR CONDITIONING BLEED AIR VALVE ISOLATION circuit breaker.
   E. Gain access to the bleed air isolation valve.
   F. Remove, cap and stow the connector from the bleed air isolation valve.
      NOTE: It may be necessary to temporarily remove the air conditioning duct segment located in front of keel beam access hole.
   G. When personnel have been cleared from the engine danger area, inform the flight deck that the right engine may be started using a start cart (AMM 71-00-00/201).
      NOTE: Personnel should take special care to remain well clear of the right engine inlet hazard area when disconnecting the start cart after right engine start.
   H. After the start cart is removed and secured out of the right and left engine hazard area, inform the flight deck that the left engine may be started using APU bleed air.

   NOTE: For subsequent dispatches after initial MMEL item 36-13-01 deferral with the bleed air isolation valve stuck closed and using ground service air for engine start, refer to the operations note for guidance.

OPERATIONS NOTE

1. Do not dispatch into known or forecast icing conditions.
2. When the bleed air isolation valve is stuck closed, do these steps:
   A. Do not use APU bleed air for air conditioning purposes on the ground.
   The APU may only be used for engine start.
   B. Start the right engine using a start cart.
   NOTE: Personnel should take special care to remain well clear of the right engine inlet hazard area when disconnecting the start cart after right engine start.
   C. After the start cart has been removed and secured out of the right and left engine hazard area, start the left engine using APU bleed air.

36-13-02 **Bleed Air Isolation Valve Control**

36-13-02A **Verification using Visual Position Indicator**

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided Bleed Air Isolation Valve is verified to operate normally.

MAINTENANCE (M) PROCEDURES

At the time of deferral, verify the Bleed Air Isolation Valve operates normally (AMM 36-00-00/901).

1. Set the ISOLATION VALVE switch to the OPEN position.
2. Confirm the isolation valve visual position indicator moves to the OPEN position.
3. Set the ISOLATION VALVE switch to the CLOSE position.
4. Confirm the isolation valve visual position indicator moves to the CLOSE position.
5. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

36-13-02 Bleed Air Isolation Valve Control

36-13-02B Verification from Flight Deck

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided Bleed Air Isolation Valve is verified to operate normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

At the time of deferral, verify the bleed air isolation valve operates normally by using one of the following procedures.

1. Verification with APU bleed ON prior to engine start:
   A. Set the ISOLATION VALVE switch to the CLOSE position.
   B. Pressurize the left pneumatic manifold.
   C Set the ISOLATION VALVE switch to the OPEN position.
   D. Confirm the right bleed air duct pressure increases to match the left bleed air duct pressure.
   E. Set bleed and air conditioning configuration as required for takeoff.
2. Verification using ground cart prior to engine start:
   A. Set the ISOLATION VALVE switch to the CLOSE position.
   B. Pressurize the right pneumatic manifold.
   C. Set the ISOLATION VALVE switch to the OPEN position.
   D. Confirm the left bleed air duct pressure increases to match the right bleed air duct pressure.
   E. Set bleed and air conditioning configuration as required for takeoff.

36-13-03 Ground Pneumatic Connector Check Valve

36-13-03A Valve Open

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

Except for ER operations beyond 120 minutes, may be inoperative open provided:

a. Right pneumatic manifold remains depressurized after starting right engine.
b. Airplane is not operated in known or forecast icing conditions.
c. Flight altitude remains at or below FL 250.

MAINTENANCE (M) PROCEDURES

Keep the right pneumatic manifold depressurized except for engine start (AMM 36-00-00/901).

1. Use a ground pneumatic source for starting the right engine.
2. Prior to removing the ground pneumatic source, confirm with the flight crew that the right engine bleed air switch is set to OFF and the isolation valve switch is set to CLOSE.
3. Ground pneumatic source will be required at down line stations if right engine is shut down.

OPERATIONS (O) PROCEDURES

1. Do not dispatch into known or forecast icing conditions.
2. Limit altitude to FL 250.
3. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
4. Following engine start, depressurize the right pneumatic duct prior to removing the ground pneumatic source. For flight:
   A. Set the right engine BLEED air switch to OFF.
   B. Set the right PACK switch to OFF.
   C. ISOLATION VALVE switch to CLOSE.
5. If desired, for increased air flow when the flaps are extended (takeoff and landing), use the APU to supply bleed air to the operating pack.
   A. Do the Supplementary Procedure - Air Systems, No Engine Bleed Takeoff and Landing.

      NOTE: Keep the pack switch for the right (inoperative) pack in the OFF position and the ISOLATION VALVE switch in the CLOSE position.

36-13-03 Ground Pneumatic Connector Check Valve
36-13-03B Valve Closed

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative closed.

NOTE: With the ground pneumatic connector check valve inoperative closed, engines cannot be started from a ground source. Use the APU as an air source for starting the engines.

36-21-01 Bleed Air Duct Pressure Indications

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided alternate procedures are established and used.

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Confirmation of minimum recommended pneumatic pressure is not required for engine start.
2. For verification of start valve closure, confirm the associated START VALVE OPEN indication on the display unit has extinguished.

36-21-02 Dual Bleed Light System

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided:

a. APU bleed air is not used in flight.
b. APU bleed air valve is verified closed before each departure.

OPERATIONS (O) PROCEDURES

Prior to takeoff, verify the APU bleed air valve is closed.

1. Set the manifold isolation valve switch to CLOSE.
2. With the APU running, set the APU BLEED air switch to OFF.
3. Set the L PACK switch to OFF.
4. Set the engine No. 1 BLEED air switch to OFF.
5. Confirm the left pneumatic duct pressure decreases to 10 PSI or less.
6. PRIOR to takeoff:
   A. Set the L PACK switch to AUTO.
   B. Set the engine No. 1 BLEED air switch to ON.

38-10-01 Potable Water System

38-10-01A Individual Components Deactivated/Isolated

38-10-01B System Drained and Not Serviced

38-30-01 Lavatory Waste System

38-30-01A Individual Components Deactivated/Isolated

38-30-01B Lavatory Not Used

38-30-01-01 Vacuum Blower

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38-10-01 Potable Water System

38-10-01A Individual Components Deactivated/Isolated

Interval	Installed	Required	Procedure
C	1	1	**(M)**

Individual components may be inoperative provided:

a. Associated components are deactivated or isolated.
b. Associated system components are verified not to have leaks.

NOTE: Any portion of system that operates normally may be used.

MAINTENANCE (M) PROCEDURES

Deactivate or isolate inoperative components and verify associated system components do not have leaks (AMM 38-00-00/901).

38-10-01 **Potable Water System**

38-10-01B **System Drained and Not Serviced**

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. System is drained.
b. Procedures are established to ensure that system is not serviced.

MAINTENANCE (M) PROCEDURES

Drain potable water system and establish procedures to ensure potable water system is not serviced (AMM 38-00-00/901). (Placard the potable water system controls with a **"SEE LOG BOOK sticker"**)

38-30-01 Lavatory Waste System

38-30-01A Individual Components Deactivated/Isolated

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Individual components may be inoperative provided:

a. Associated components are deactivated or isolated.
b. Associated system components are verified not to have leaks.

NOTE: Any portion of system that operates normally may be used.

MAINTENANCE (M) PROCEDURES

Deactivate or isolate inoperative components and verify associated system components do not have leaks (AMM 38-00-00/901).

38-30-01 Lavatory Waste System

38-30-01B Lavatory Not Used

Interval	Installed	Required	Procedure
C	3	0	**(M)**

Associated lavatory system may be inoperative provided:

a. Associated components are deactivated or isolated to prevent leaks.
b. Associated lavatory door is secured closed and placarded "INOPERATIVE - DO NOT ENTER".

NOTE: These provisions are not intended to prohibit inspections by crewmembers.

MAINTENANCE (M) PROCEDURES

Deactivate or isolate inoperative components and lock associated lavatory door (AMM 38-00-00/901).

1. Deactivate or isolate inoperative components.
2. For leaking waste tank, drain the tank. Do not add the precharge chemical to the associated waste tank.
3. Close, lock and placard the associated lavatory door.

38-30-01 **Lavatory Waste System**

38-30-01-01 **Vacuum Blower**

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Vacuum blower is deactivated.
b. Lavatories are not used on the ground or at flight altitudes below 16,000 ft.

MAINTENANCE (M) PROCEDURES

Deactivate the inoperative vacuum blower (AMM 38-00-00/901).

1. Open and collar the P91 panel VACUUM WASTE BLOWER circuit breaker.

OPERATIONS (O) PROCEDURES

1. For inoperative Vacuum Blower:
   A. Lavatory must be locked out on the Ground and below 16,000 ft.
   B. Seat belt sign must remain 'ON' below 16,000 ft.
   C. Lavatory may be used above 16,000 ft.
   D. Flight deck crew will advise the cabin crew when the lavatory may be used.

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46-13-01 Onboard Network System (ONS)

46-*13-01A* *Alternate Procedures Used*

46-13-01B *Procedures Do Not Require Use - (N/A)*

46-20-01 Electronic Flight Bag (EFB) Systems

46-20-01-01 EFB System (Installed EFB System)

46-20-01-01A *Alternate Procedures Required - N/A*

46-20-01-01B *Procedures Do Not Require Use - N/A*

46-20-01-02 Data Connectivity

46-20-01-02A *Alternate Procedures Required*

46-20-01-02B *Procedures Do Not Require Use - N/A*

46-20-01-03 Power Supply / Power Connection

46-20-01-03A *Alternate Procedures Required*

46-20-01-03B *Procedures Do Not Require Use - N/A*

46-20-01-04 Mounting Device

46-20-01-04A *Alternate Procedures Required*

46-20-01-04B *Procedures Do Not Require Use - N/A*

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46-13-01 Onboard Network System (ONS)

46-13-01A Alternate Procedures Used

Interval	Installed	Required	Procedure
C	1	0	**(O)**

NOTE: Any function that operates normally may be used.

NOTE: The Onboard Network System (ONS) includes the Network File Server (NFS), Aircraft Interface Device (AID), Wireless WAN Unit (WWU), and Wireless Access Points (WAPs) components with associated support interfaces.

OPERATIONS (O) PROCEDURES

If the Onboard Network Server (ONS) is inoperable/unserviceable and the Navigation Database (NDB) needs updating, also dispatch using **MEL Item 34-61-01-02 Navigation Databases**.

MAINTENANCE NOTE

If any component/system requires software loaded, then the ONS must be replaced/repaired before dispatched.

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46-20-01 **Electronic Flight Bag (EFB) Systems**

46-20-01-02 **Data Connectivity**

46-20-01-02A **Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	3	0	**(O)**

May be inoperative provided:

1. EFB data is current and will remain current for the duration of each flight sector.
2. Paper charts and documents are be substituted for any EFB that will not remain current for the duration of each flight sector.

NOTE: Paper charts and documents may not be available at stations other than TTPP and MKJP.

OPERATIONS (O) PROCEDURES

Check the EFB database to ensure that data is current and will remain current for the duration of each flight sector or revert to using paper charts and documents.

NOTE: EFB DATA CAN BE MANUALLY UPDATED OR PEDS REPLACED AT TTPP AND MKJP. DISPATCH SHOULD ONLY BE ALLOWED IF RETURN TO ONE OF THESE STATIONS CAN BE REASONABLY ASSURED.

46-20-01 Electronic Flight Bag (EFB) Systems

46-20-01-03 Power Supply / Power Connection

46-20-01-03A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided:

a. Flights beyond 6 hours with no power connection available are not allowed without the use of paper charts and documents.
b. Each PED onboard has a minimum internal battery charge of 80% prior to dispatch.
c. Paper charts and documents may be substituted for each PED with an internal battery charge of less than 80%.

NOTE: Paper charts and documents may not be available at stations other than TTPP and MKJP.

OPERATIONS (O) PROCEDURES

1. If practicable, ensure that all PEDs onboard are fully charged prior to dispatch.
2. Consider switching off affected PEDs when not in use to preserve internal battery charge.
3. Consider dispatching with paper enroute navigation charts onboard to preserve PED battery charge during the cruise portion of the flight.
4. After dispatch, use the spare PED as necessary to substitute any PED with a low battery charge.
5. Use any remaining power connections to charge PEDs on a rotational basis. Use only approved power connections.
6. If possible, ensure a minimum battery charge of 20% for all PEDs in use prior to top of descent.

46-20-01 Electronic Flight Bag (EFB) Systems
46-20-01-04 Mounting Device
46-20-01-04A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

MAINTENANCE (M) PROCEDURES

Secure or remove associated EFB and hardware.

OPERATIONS (O) PROCEDURES – 1 Mounting Device Inoperable

1. When not in use, the affected PED should be stored in a secure location so as not to obstruct any aircraft components such as panels, switches, lights, instruments, displays or flight controls.
2. Ensure that all taxi routes, SIDs, STARs and approach procedures in use are reviewed thoroughly prior to any critical phases of flight.
   NOTE: Critical phases of flight include, but are not limited to, any time the aircraft is moving under its own power below 10,000 ft.
3. Verify that the relevant information pertaining to all SIDs, STARS and IAPs in use are cross-checked with the data displayed on the FMCs and the Navigation Displays.
4. During taxi, the Pilot Monitoring will reference the appropriate taxiway charts and provide detailed guidance to the Pilot Flying. Exercise extreme caution at all times. Should any doubt exist about the location of the aircraft along the taxiway route, STOP IMMEDIATELY, so that the PF may review the appropriate charts.

OPERATIONS (O) PROCEDURES – Both Mounting Devices Inoperable

Revert to using paper charts and documents.

NOTE: PAPER CHARTS AND DOCUMENTS MAY NOT BE AVAILABLE AT STATIONS OTHER THAN TTPP AND MKJP.

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47-11-01 Nitrogen Generation System (NGS)

47-11-01-01 Nitrogen Generation Degraded

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47-11-01 Nitrogen Generation System (NGS)

Interval	Installed	Required	Procedure
A	1	0	**(M)**

May be inoperative provided:

a. NGS shutoff valve is deactivated closed.
b. Repairs are made within 10 flight days.

MAINTENANCE (M) PROCEDURES

Deactivate the NGS shutoff valve in the closed position (AMM 47-00-00/901).

47-11-01 **Nitrogen Generation System (NGS)**

47-11-01-01 **Nitrogen Generation Degraded**

Interval	Installed	Required	Procedure
C	1	0	

49-11-01 Auxiliary Power Unit

49-15-01 APU Air Inlet Door

49-15-01A Door Inoperative in Flight Open Position

49-15-01B Door Inoperative in Closed Position

49-15-01C APU Not Required

49-15-01-01 APU Air Inlet DOOR Light System

49-15-01-01A APU Air Inlet Door Inoperative

49-15-01-01B APU Air Inlet Door Operative

49-40-01 Start Power Unit

49-40-01-01 AC/DC Start Systems

49-40-02 Start Converter Unit

49-40-02-01 Voltage Regulation Function

49-52-01 APU Bleed Air System

49-52-01A Bleed Air Valve Closed

49-52-01B APU Not Used

49-52-02 APU Surge Control System

49-52-02A Surge Control Valve Open

49-52-02B Surge Control Valve Closed

49-70-01 APU FAULT Light

49-70-02 APU LOW OIL PRESSURE Light

49-70-03 APU OVERSPEED Light

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49-11-01 Auxiliary Power Unit

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations, may be inoperative provided:

a. APU switch remains OFF.
b. Both engine driven generator systems operate normally.
c. At least one engine bleed air shutoff valve (PRSOV) operates normally.
d. APU fuel shutoff valve is verified closed.

NOTE: Inflight operations with the APU inlet door not in the closed position and the APU off may reduce APU life (mean time between unscheduled removals).

MAINTENANCE (M) PROCEDURES

Turn off the APU and verify that the APU fuel shutoff valve is closed (AMM 49-00-00/901).

1. Verify APU switch is positioned OFF.
2. Visually inspect the APU air inlet door and verify that it is in a position ranging from closed position to flight open (17 degrees) position.
3. Open and collar the following circuit breakers:
   A. P6-4 panel AUX POWER UNIT CONT
   B. P6-5 panel APU INLET DOOR2.
4. Gain access to the APU fuel shutoff valve through the left main landing gear wheel well.
5. Verify the red manual override lever indicates that the valve is closed.

NOTE: If the APU fuel shutoff valve is not closed, the valve may be inoperative. Refer to MEL item 28-25-02 for instructions on dispatching with an inoperative APU fuel shutoff valve.

NOTE: Aircraft may be operated with the APU removed (AMM 49-11-02, APU SYSTEM - MAINTENANCE PRACTICES - requires APU Deletion Tool Kit C49009-1).

OPERATIONS NOTE

NOTE: Aircraft may be operated with the APU removed in accordance with the AMM provided the resulting changes in weight and balance are accounted for.

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49-15-01 APU Air Inlet Door

49-15-01A Door Inoperative in Flight Open Position

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Door is deactivated in flight open position.
b. Appropriate performance adjustments are applied.

NOTE: Inflight operations with the APU inlet door not in the closed position and the APU off may reduce APU life (mean time between unscheduled removals).

MAINTENANCE (M) PROCEDURES

Deactivate the APU air inlet door in the flight open position (AMM 49-00-00/901).

1. Set the APU switch to OFF.
2. Open the P6-4 panel AUX POWER UNIT CONT circuit breaker and the P6-2 APU FIRE SW POWER circuit breaker.
3. Open and collar the P6-5 panel APU INLET DOOR circuit breaker.
4. Gain access to the APU air inlet door actuator.
5. Deactivate the APU air inlet door in the flight open position.
6. Close the AUX POWER UNIT CONT and APU FIRE SW POWER circuit breakers.
7. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. For APU OFF the performance limited weights are reduced by the following amount:

Takeoff	Enroute Climb	Approach & Landing
136 kg	227 kg	45 kg

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below - Enroute Diversion Speed Effects.

Diversion Speed	Factor
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

OPERATIONS (O) PROCEDURES 49-15-01A (Cont'd)

A. Increase flight planning fuel by either incrementing the current airplane drag correction factor by 1.2% or flight planning fuel by 1.2%.

2. For APU ON there is no penalty on limiting weights:

A. Increase flight planning fuel by either incrementing the current airplane drag correction factor by 0.9% or flight planning fuel by 0.9% plus 45 kg per hour of APU usage in flight and 109 kg per hour of APU usage in flight and 109 kg per hour of APU usage on the ground.

NOTE: The DOOR light will illuminate 165 seconds after the APU Master Switch is set to the ON position. The light may extinguish for 165 seconds if / when the APU shutoff sequence begins, but it will then remain on until power is deliberately removed from the APU ECU.

49-15-01 APU Air Inlet Door

49-15-01B Door Inoperative in Closed Position

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Door is deactivated in closed position.
b. APU is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 49-11-01.

MAINTENANCE (M) PROCEDURES

Deactivate the APU air inlet door in the closed position (AMM 49-00-00/901).

1. Set the APU switch to OFF.
2. Open and collar the P6 panel AUX POWER UNIT CONT, AUX POWER UNIT SCU FAN POWER (prior to incorporation of Boeing Service Bulletin 737-49-1166 or production equivalent), APU FIRE SW POWER and APU INLET DOOR circuit breakers.
3. Open and collar the P91 panel APU START CONV circuit breaker.
4. Gain access to the APU air inlet door actuator.
5. Deactivate the APU air inlet door in the closed position.
6. If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

49-15-01 APU Air Inlet Door

49-15-01C APU Not Required

Interval	Installed	Required	Procedure
C	1	0	**(O)**

Except for ER operations, may be inoperative in any position ranging from closed position to flight open (17 degrees) position provided:

a. APU is considered inoperative.
b. Appropriate performance adjustments are applied.

NOTE 1: Inflight operations with the APU air inlet door not in the closed position and the APU off may reduce APU life (mean time between unscheduled removals).

NOTE 2: Airplane must also be dispatched using MEL item 49-11-01.

NOTE 3: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Performance limited weights are reduced by the following amount:

Takeoff	Enroute Climb	Approach & Landing
136 kg	227 kg	45 kg

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below - Enroute Diversion Speed Effects.

Diversion Speed	Factor
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

2. Increase flight planning fuel by either incrementing the current airplane drag correction factor by 1.2% or flight planning fuel by 1.2%.

49-15-01 **APU Air Inlet Door**

49-15-01-01 **APU Air Inlet DOOR Light System**

49-15-01-01A **APU Air Inlet Door Inoperative**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative for an inoperative APU air inlet door.

49-15-01 APU Air Inlet Door

49-15-01-01 APU Air Inlet DOOR Light System

49-15-01-01B APU Air Inlet Door Operative

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided APU air inlet door is verified to operate normally before each departure.

MAINTENANCE (M) PROCEDURES

Verify the APU air inlet door operates normally before each departure (AMM 49-00-00/901).

1. For APU off:
   A. Set APU to test mode.
   B. On APU maintenance page, confirm the POSITION SW indicates NOT OPEN.
   C. Set Electrical Panel (P5) APU selector ON.
   D. On APU maintenance page, confirm (within 120 seconds) that POSITION SW indicates OPEN.
   E. Set Electrical Panel (P5) APU selector OFF.
   F. On APU maintenance page, confirm (within 120 seconds) that POSITION SW indicates NOT OPEN.
2. For APU on:
   A. Set APU to test mode.
   B. On APU maintenance page, confirm that POSITION SW indicates OPEN.
   C. Set Electrical Panel (P5) APU selector OFF.
   D. On APU maintenance page, confirm (within 120 seconds) that POSITION SW indicates NOT OPEN.
   E. Start the APU.
   F. On APU maintenance page, confirm (within 120 seconds) that POSITION SW indicates OPEN.

49-40-01 Start Power Unit

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations, may be inoperative provided:

a. Start power unit is deactivated.

b. APU is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 49-11-01.

MAINTENANCE (M) PROCEDURES

Deactivate the start power unit (AMM 49-00-00/901).

1. Set the APU switch to OFF.
2. Open and collar the following circuit breakers:
   A. P6-4 panel AUX POWER UNIT CONT
   B. P6-4 panel AUX POWER UNIT SCU FAN POWER (prior to incorporation of Boeing Service Bulletin 737-49-1166 or production equivalent)
   C. P6-2 panel APU FIRE SW POWER
   D. P91 panel APU START CONV
3. Gain access to the start power unit.
4. Disconnect, cap and stow the battery type connector to the start power unit.

49-40-01 Start Power Unit

49-40-01-01 AC/DC Start Systems

Interval	Installed	Required	Procedure
C	2	1	

49-40-02 Start Converter Unit

Interval	Installed	Required	Procedure
C	1	0	**(M)**

Except for ER operations, may be inoperative provided:

a. Start converter unit is deactivated.
b. APU is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 49-11-01.

MAINTENANCE (M) PROCEDURES

Deactivate the start converter unit (AMM 49-00-00/901).

1. Set the APU switch to OFF.
2. Open and collar the following circuit breakers:
   A. P6-4 panel AUX POWER UNIT CONT
   B. P6-4 panel AUX POWER UNIT SCU FAN POWER (prior to incorporation of Boeing Service Bulletin 737-49-1166 or production equivalent)
   C. P6-2 panel APU FIRE SW POWER
   D. P91 panel APU START CONV.

49-40-02 Start Converter Unit

49-40-02-01 Voltage Regulator Function

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided:

a. Both engine driven generator systems operate normally.

b. APU generator is not used for electrical power.

NOTE: APU may be used as a pneumatic source.

49-52-01 APU Bleed Air System

49-52-01A Bleed Air Valve Closed

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. APU bleed air valve is verified to be closed.
b. At least one of bleed system PRSOVs operates normally.

NOTE: APU may be used to provide electrical power.

NOTE: APU will not be able to provide bleed air.

MAINTENANCE (M) PROCEDURES

Verify and deactivate the APU bleed air valve in the closed position (AMM 49-00-00/901).

1. Disconnect, cap, and stow electrical connector P10 on the APU bleed air valve.
2. Set APU BLEED air switch on the P5 overhead panel to OFF.
3. Access the MAX Display System (MDS) APU maintenance page.
4. Set APU ECU Power to TEST.
5. Confirm STATUS CODE: STATUS 2, BIT 2 shows 0.
6. Set APU ECU Power to NORM.

NOTE: Maintenance message 49-52237, "Bleed Air Valve Circuit Shows Open Circuit", will be displayed.

49-52-01 **APU Bleed Air System**

49-52-01B **APU Not Used**

Interval	Installed	Required	Procedure
C	1	0	

Except for ER operations, may be inoperative provided:

a. Both engine driven generator systems operate normally.
b. At least one of bleed system PRSOVs operates normally.
c. APU is not used.

49-52-02 APU Surge Control System

49-52-02A Surge Control Valve Open

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Surge control valve is in open position.
b. At least one of bleed system PRSOVs operates normally.
c. APU bleed air is not used.

NOTE: APU may be used to provide electrical power.

49-52-02 APU Surge Control System

49-52-02B Surge Control Valve Closed

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

a. Surge control valve is in closed position.

b. APU is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 49-11-01.

49-70-01 APU FAULT Light

Interval	Installed	Required	Procedure
C	1	0	

49-70-02 APU LOW OIL PRESSURE Light

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided APU automatic shutdown system is verified to operate normally.

OPERATIONS (O) PROCEDURES

At the time of deferral, verify the APU automatic shutdown system operates normally.

1. Start and shut down the APU normally.

49-70-03 APU OVERSPEED Light

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided APU automatic shutdown system is verified to operate normally.

OPERATIONS (O) PROCEDURES

At the time of deferral, verify the APU automatic shutdown system operates normally.

1. Start and shut down the APU normally.

INTENTIONALLY
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50-10-01 Lower Cargo Compartment Lining Panels and Floor Panels

50-21-01 Automatic Cargo Loading System – *N/A*

50-21-02 Cargo Compartment Restraint Components

50-21-02A Cargo Compartment Used

50-21-02B Cargo Compartment Remains Empty

50-21-02C Cargo Compartment Inoperative

INTENTIONALLY

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50-10-01 Lower Cargo Compartment Lining Panels and Floor Panels

Interval	Installed	Required	Procedure
C	77	0	**(O)**

May be damaged or missing provided procedures are established and used to ensure associated compartment remains empty or is verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away Kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

NOTE: Cargo compartment linings include: sidewall panels; endwall panels; ceiling panels; shrouds; access chute; soft liner; doorway access panels; cargo door panels; blowout panels; floor panels; capstrips; equipment and structure enclosures; fasteners attaching linings that are above the floor level; tape sealing liner edges; and bilge endwall/barrier and sidewall panels.

MAINTENANCE NOTE:

Ensure applicable cargo compartment remains empty, or is verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

For Empty Cargo Compartments

When any cargo compartment is deemed to be unusable the following procedures shall apply:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1.
4. Traffic, SOC, MCC Departments as applicable to be notified.

INTENTIONALLY
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50-21-02 Cargo Compartment Restraint Components

50-21-02A Cargo Compartment Used

Interval	Installed	Required	Procedure
A	50	48	**(M)**

May be inoperative or missing provided:

a. Cargo-loading limits from the OEM Weight and Balance Manual are observed.
b. Repairs are made within 120 consecutive calendar-days.

MAINTENANCE (M) PROCEDURES

Use acceptable cargo loading limits from an approved source (AMM 50-00-00/901).

1. Deactivation procedures for most cargo restraint or cargo guiding equipment are either obvious (e.g. inoperative container stop needs to be released to retracted position or removed to prevent interference with cargo loading) or unnecessary (e.g., a damaged end stop can be left as is). With the exception of the restrictions below, no other procedures are provided.
2. Refer to the Weight and Balance Manual for cargo loading limits when operating with restraint components missing or inoperative. If the number of inoperative or missing restraint components exceeds the allowance in the Weight and Balance Manual, inform Dispatch that the affected cargo compartment cannot be used.

50-21-02 Cargo Compartment Restraint Components

50-21-02B Cargo Compartment Remains Empty

Interval	Installed	Required	Procedure
A	50	48	

May be inoperative or missing provided:

a. Cargo compartment remains empty.

b. Repairs are made within 120 consecutive calendar-days.

50-21-02 Cargo Compartment Restraint Components

50-21-02C Cargo Compartment Inoperative

Interval	Installed	Required	Procedure
A	50	48	

Individual cargo areas may be inoperative provided:

a. Aircraft is operated in accordance with the OEM Weight and Balance Manual.

b. Repairs are made within 120 consecutive calendar-days.

INTENTIONALLY
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52-11-01 Left Main Cabin Door Pressure Stop Fittings

52-11-01-01 Forward Entry Door

52-11-01-02 Aft Door

52-11-02 Entry/Service Door Hold-Open Latch Assemblies

52-11-02-01 Latch Release Lever

52-22-01 Flight Lock System

52-31-01 Lower Cargo Doors Pressure Stop Fittings

52-31-01-01 Pressurized Flight

52-31-01-02 Unpressurized Flight

52-31-02 Lower Cargo Doors Balance Mechanism

52-31-03 Lower Cargo Doors Hold Open Device

52-31-03A *Door Balance Mechanism Operates Normally*

30-31-03B *Cargo Compartment Remains Empty*

52-31-04 Cargo Door Exterior Handle Recess/Hinge Spring Assemblies

52-51-01 Flight Deck Door Automatic Locking System

52-51-01-01 Flight Deck Access Panel System (Keypad, Door Chime)

52-51-01-01-01 LEDs

52-51-01-01-02 Door Bell Mode – *N/A*

52-51-01-01-03 Switch Guard

52-51-01-02 Flight Deck Door LOCK FAIL Light

52-51-01-03 Flight Deck Door AUTO UNLK Light

52-51-01-04 Flight Deck Door Lock Control Selector

52-51-02 Flight Deck Door Dead Bolt

52-51-03 Flight Deck Door Pressure Relief Panels

52-51-04 Flight Deck Door Viewing Port

52-*51-04A* *Alternate Procedures Required*

52-*51-04B* *Electronic Flight Deck Door Visual Surveillance System Operates Normally*

52-61-01 Forward Air Stair - *N/A*

52-71-01 Door Warning Light System

52-71-01-01 Entry and Service Doors

52-71-01-02 Cargo and Access Doors

52-71-01-03 Airstairs Doors – *N/A*

52-71-01-04 Overwing Doors

52-71-01-05 Mid-Cabin Doors (737-8200/-9) – *N/A*

52-71-02 Door Programming Pin Fault

52-11-01 Left Main Cabin Door Pressure Stop Fittings

52-11-01-01 Forward Entry Door

Interval	Installed	Required	Procedure
B	18	17	**(M) (O)**

May be broken or missing provided:

a. There are no visible defects on other fittings for associated door.
b. Pressure differential does not exceed 6.0 psi.
c. Cabin pressure control system AUTO or ALTN control mode operates normally.
d. Alternate procedures are established and used.

NOTE: A door pressure stop consists of two mating stop fittings, one being attached to the door and the other attached to the fuselage. Either or both stop fittings may be broken or missing at a single door pressure stop location.

MAINTENANCE (M) PROCEDURES

Visually inspect remaining fittings on associated door, body frames and sills (if installed) and confirm there are no other defects (AMM 52-00-00/901).

OPERATIONS (O) PROCEDURES

Do the following to ensure a cabin differential pressure of 6.0 psi is not exceeded.

1. Before takeoff, set a landing altitude of 8,000 ft. MSL on the cabin pressure control panel.
   NOTE: Actual landing airport altitude may be selected if flight altitude is limited to 13,000 feet.
2. Monitor cabin differential pressure during climb, as system limits on cabin altitude rate of change may lead to differential pressure exceedance at higher rates of climb.
3. Maintain altitude at or below FL270.
4. Set actual landing airport altitude after initiating descent.
5. If required to hold during descent, it may be necessary to increase the landing altitude in accordance with the differential pressure schedule below.
6. If desired, adjust to a lower cabin altitude by appropriate selection in accordance with the differential pressure schedule below.

OPERATIONS (O) PROCEDURES 52-11-01-01 (Cont'd)

FLIGHT ALTITUDE (FT)	CABIN ALTITUDE (FT) (6.0 PSI PRESSURE DIFFERENTIAL)
14000	100
15000	800
16000	1400
17000	2000
18000	2700
19000	3300
20000	3900
21000	4500
22000	5000
23000	5600
24000	6200
25000	6700
26000	7300
27000	7800
28000	8300

52-11-01 Left Main Cabin Door Pressure Stop Fittings

52-11-01-02 Aft Door

Interval	Installed	Required	Procedure
B	14	13	**(M) (O)**

May be broken or missing provided:

a. There are no visible defects on other fittings for associated door.
b. Pressure differential does not exceed 3.4 psi.
c. Cabin pressure control system AUTO or ALTN control mode operates normally.
d. Alternate procedures are established and used.

NOTE: A door pressure stop consists of two mating stop fittings, one being attached to the door and the other attached to the fuselage. Either or both stop fittings may be broken or missing at a single door pressure stop location.

MAINTENANCE (M) PROCEDURES

Visually inspect remaining fittings on associated door, body frames and sills (if installed) and confirm there are no other defects (AMM 52-00-00/901).

OPERATIONS (O) PROCEDURES

Do the following to ensure a cabin differential pressure of 3.4 psi is not exceeded.

1. Before takeoff, set a landing altitude of 8,000 ft. MSL on the cabin pressure control panel.
   NOTE: Actual landing airport altitude may be selected if flight altitude is limited to 7,000 feet.
2. Monitor cabin differential pressure during climb, as system limits on cabin altitude rate of change may lead to differential pressure exceedance at higher rates of climb.
3. Maintain altitude at or below FL170.
4. Set actual landing airport altitude after initiating descent.
5. If required to hold during descent, it may be necessary to increase the landing altitude in accordance with the differential pressure schedule below.
6. If desired, adjust to a lower cabin altitude by appropriate selection in accordance with the differential pressure schedule below.

OPERATIONS (O) PROCEDURES 52-11-01-02 (Cont'd)

FLIGHT ALTITUDE (FT)	CABIN ALTITUDE (FT) (3.4 PSI PRESSURE DIFFERENTIAL)
7000	0
8000	800
9000	1600
10000	2400
11000	3200
12000	4000
13000	4700
14000	5400
15000	6100
16000	7000
17000	7700
18000	8300

52-11-02 Entry/Service Door Hold-Open Latch Assemblies

52-11-02-01 Latch Release Lever

Interval	Installed	Required	Procedure
C	4	0	

MAINTENANCE NOTE

Operators should select an appropriate procedure/tool to release the hold open latch that meets their operating requirements.

INTENTIONALLY
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52-22-01 Flight Lock System

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. For inoperative flight lock(s), inputs to associated flight deck indications are deactivated.
b. It is verified each associated exit is able to be unlatched and opened before each departure.
c. A person employed by operator is designated to remain seated in passenger seat nearest affected exit when cabin differential pressure is less than 4.0 psi.

MAINTENANCE (M) PROCEDURES

Deactivate affected flight lock input(s) to associated flight deck indications, including the input to the master caution system and, before each departure, verify the associated exit is capable of being unlatched and opened (AMM 52-00-00/901).

1. Verify the associated exit is capable of being unlatched and opened.
2. Open and collar these P6-3 panel circuit breakers.
   A. PSEU PRI
   B. PSEU ALTN
3. Disconnect, cap, and stow the associated PSEU connector wire for the pins associated with the affected exit(s) as indicated in the table below.

AFFECTED OVERWING EXIT	ASSOCIATED CONNECTORS	REMOVE WIRE (CAP & STOW) FROM PIN NUMBER	PIN NUMBERS TO BE JUMPERED
Left Aft	D10986	Pin 20	Pin 40 to Pin 20
Left Forward	D10986	Pin 53	Pin 40 to Pin 53
Left Aft and Left Forward	D10986	Pin 20 and Pin 53	Pin 40 to Pin 20 and Pin 53
Right Aft	D10988	Pin 02	Pin 53 to Pin 02
Right Forward	D10988	Pin 52	Pin 53 to Pin 52
Right Aft and Right Forward	D10988	Pin 02 and Pin 52	Pin 53 to Pin 02 and Pin 52

MAINTENANCE (M) PROCEDURE 52-22-01 (Cont'd)

4. Jumper the applicable pins (indicated in the table above). Using a moisture proof splice, create an extension wire and connect it into the applicable pins.
5. Close these P6-3 panel circuit breakers.
   A. PSEU PRI
   B. PSEU ALTN
6. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Ensure associated door handle is not operated by passengers whenever the cabin differential pressure is less than 4.0 psi.

1. Assign a designated employee, such as a flight attendant, pilot or mechanic, to monitor each emergency exit associated with an inoperative flight lock, when monitoring is required.
2. Have flight crew inform cabin crew and/or designated employee when monitoring is required.

52-31-01 Lower Cargo Doors Pressure Stop Fittings

52-31-01-01 Pressurized Flight

Interval	Installed	Required	Procedure
A	24	22	**(M)**

Anyone may be broken or missing on each door or frame provided:

a. No defects are visible on other fittings for associated door.
b. Cabin pressure controller AUTO mode operates normally.
c. Adjacent stop fittings are inspected within 25 flight sectors.
d. Not more than 50 flights are made before completion of repairs or replacements.

NOTE: A door pressure stop consists of two mating stop fittings, one being attached to the door and the other attached to the fuselage. The number installed shown above is the number of stop locations. For pressurized flight, either or both stop fittings may be broken or missing at a single door pressure stop location.

MAINTENANCE (M) PROCEDURES

Visually inspect remaining fittings for possible defects on affected door, body frames and sills (AMM 52-00-00/901).

NOTE: Repeat the inspection within 25 flight sectors and complete repairs within 50 flight sectors.

52-31-01 Lower Cargo Doors Pressure Stop Fittings

52-31-01-02 Unpressurized Flight

Interval	Installed	Required	Procedure
C	24	20	**(M) (O)**

Two may be broken or missing on each door or frame provided:

a. Flight is conducted in an unpressurized configuration.
b. Recirculation fans operate normally.
c. Procedures are established and used to ensure lower cargo compartments remain empty or are verified to contain only empty cargo handling equipment, ballast (ballast may be loaded in ULDs), and/or Fly Away Kits.

NOTE: See Definition of Ballast in the Introduction. See Fly Away Kit on Page FAK.00-01.1

NOTE: A door pressure stop consists of two mating stop fittings, one being attached to the door and the other attached to the fuselage. The number installed shown above is the number of stop locations. For unpressurized flight, either or both stop fittings may be broken or missing at two door pressure stop locations.

MAINTENANCE (M) PROCEDUREs

Configure the airplane for unpressurized flight with the manual mode operational (AMM 52-00-00/901).

1. Set the pressurization mode selector to MAN.
2. Hold the outflow valve (toggle) switch in the OPEN position until the valve position indicator indicates full open.
3. Open and collar these P6-4 panel circuit breakers:
   A. PRESSURIZATION CONTROL AUTO 1
   B. PRESSURIZATION CONTROL AUTO 2
   C. A/C OVERBOARD EXH VALVE RECONFIG CONT.

MAINTENANCE NOTE

Ensure cargo compartments remain empty, or are verified to contain only empty cargo handling equipment, ballast, and/or Fly Away Kits. Secure cargo compartment after inspection by Flight Crew.

OPERATIONS (O) PROCEDURES

1. Limit altitude to 10,000 ft. for passenger flights.
   NOTE: Non-passenger flights above 10,000 ft. require flight crew to be on oxygen.
2. Except for ditching, keep the outflow valve in the full open position.
3. Use only one pack inflight.
4. When dispatching with one operating pack supplied by engine bleed air on takeoff:
   A. Determine V1(MCG) based on engine bleed for packs OFF. **(Refer to Onboard Performance Tool (OPT))**
   B. Determine takeoff performance based on packs AUTO. **(Refer to Onboard Performance Tool (OPT))**
5. Limit climb and descent rates to 500 FPM to avoid passenger discomfort.
6. EMPTY CARGO COMPARTMENT, OR, CARGO COMPARTENT CONTAINING BALLAST, AND/OR FLY AWAY KITS, MUST BE INSPECTED BY FLIGHT CREW.

OPERATIONS (O) PROCEDURE 52-31-01-02 (Cont'd)

For Empty Cargo Compartments

When any cargo compartment is deemed to be unusable the following procedures shall apply:

1. After inspection by Flight Crew, Engineering to verify cargo compartment is locked out and placarded.
2. If Cargo compartment is subsequently opened for any reason, repeat step 1.
3. Appropriate entry to be made in the Technical Log Part 1 of 2.
4. Traffic, SOC, MCC Departments as applicable to be notified.

52-31-02 Lower Cargo Doors Balance Mechanism

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided a safety hold open device is used when door is in open position.

MAINTENANCE (M) PROCEDURES

While accessing the affected cargo compartment(s), hold the cargo door in the open position either by using the manual strap device located in the affected cargo compartment ceiling or by using another appropriate procedure and/or tool (AMM 52-00-00/901).

52-31-03 Lower Cargo Doors Hold Open Device

52-31-03A Door Balance Mechanism Operates Normally

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated door balance mechanism operates normally.

52-31-03 Lower Cargo Doors Hold Open Device

52-31-03B Cargo Compartment Remains Empty

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided cargo compartment remains empty or items remaining are not accessed.

52-31-04 Cargo Door Exterior Handle Recess/Hinge Spring Assemblies

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative or missing provided the affected cargo door exterior handle is secured in a recessed position flush with the fuselage.

NOTE: This MEL item may be applied to an inoperative exterior handle recess spring/spring assembly and/or to an inoperative flapper door hinge spring assembly.

MAINTENANCE (M) PROCEDURES

Secure the affected cargo door exterior handle in a recessed position flush with the fuselage (AMM 52-00-00/901).

NOTE: Speed Tape may be used to secure affected cargo door exterior handle in a recessed position flush with the fuselage. If speed tape is used, the area where the tape is applied should be cleaned with solvent prior to application, and new speed tape should be applied each time the affected cargo door exterior handle is used to operate the door during the MMEL deferral period.

52-51-01 Flight Deck Door Automatic Locking System

Interval	Installed	Required	Procedure
A	1	0	**(M) (O)**

May be inoperative provided:

a. Automatic locking system is deactivated.
b. Door deadbolt operates normally and is used to lock door.
c. Repairs are made within 2 flight-days.

MAINTENANCE (M) PROCEDURES

Deactivate the automatic locking system (AMM 52-00-00/901).

1. Set the FLIGHT DECK ACCESS SYSTEM switch to OFF (guard extended).
2. Open and collar the P6-3 panel DOOR LOCK circuit breaker.

   NOTE: The LOCK FAIL light will remain extinguished even though the door lock is not operational. The LOCK FAIL light should be placarded appropriately.

OPERATIONS (O) PROCEDURES

1. Use the deadbolt to lock the flight deck door by rotating only the forward deadbolt lever to locked (this allows the key to unlock the door).
2. Entry to the flight deck will be achieved only by interphone request including the password. The cockpit door will not be unlocked unless the password is used.
3. In case of pilot incapacitation, the cabin crew will use the passenger cabin key to unlock the door and gain access to the flight deck.

   NOTE: The passenger cabin key must be kept in a secure location outside the flight deck and must NEVER be used for normal/routine flight deck access.
4. In the event of a cabin disturbance or attempted unauthorized access to the flight deck, rotate both concentric deadbolt levers to the locked (horizontal) position to prevent the passenger cabin key from unlocking the door.

   NOTE: Pilots on the Flight Deck MUST assume that any attempt to enter the flight deck without the use of the correct password is an unauthorized intrusion and so take the necessary action to keep the door locked.

OPERATIONS NOTE

When flight deck door is closed with the flight deck not occupied, make sure the Flight Deck Access System switch is in the OFF position (guard extended).

52-51-01 Flight Deck Door Automatic Locking System

52-51-01-01 Flight Deck Access Panel System (Keypad, Door Chime)

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Keypad is deactivated.
b. The FLIGHT DECK ACCESS SYSTEM switch is placed in the OFF position (guard extended).
c. Door dead bolt operates normally and is used to lock door.

MAINTENANCE (M) PROCEDURES

Deactivate the keypad (AMM 52-00-00/901):

1. Open the P6-3 panel DOOR LOCK circuit breaker.
2. Disconnect the keypad electrical connector at the chime module side and cap and stow it.
3. Close the open circuit breaker.

   NOTE: When the flight deck door is closed with the flight deck not occupied, make sure the FLIGHT DECK ACCESS SYSTEM switch is in the OFF position (guard extended).

OPERATIONS (O) PROCEDURES

1. Place the FLIGHT DECK ACCESS SYSTEM switch in the OFF position (guard extended).
2. Use the deadbolt to lock the flight deck door by rotating only the forward deadbolt lever to locked (this allows the key to unlock the door).
3. Entry to the flight deck will be achieved only by interphone request including the password. The cockpit door will not be unlocked unless the password is used.
4. In case of pilot incapacitation, the cabin crew will use the passenger cabin key to unlock the door and gain access to the flight deck.

   NOTE: The passenger cabin key must be kept in a secure location outside the flight deck and must NEVER be used for normal/routine flight deck access.
5. In the event of a cabin disturbance or attempted unauthorized access to the flight deck, rotate both concentric deadbolt levers to the locked (horizontal) position to prevent the passenger cabin key from unlocking the door.

   NOTE: Pilots on the Flight Deck MUST assume that any attempt to enter the flight deck without the use of the correct password is an unauthorized intrusion and so take the necessary action to keep the door locked.

OPERATIONS NOTE

When the flight deck door is closed with the flight deck not occupied, make sure the FLIGHT DECK ACCESS SYSTEM switch is in the OFF position (guard extended).

52-51-01 **Flight Deck Door Automatic Locking System**
52-51-01-01 **Flight Deck Access Panel System (Keypad, Door Chime)**
52-51-01-01-01 **LEDs**

Interval	Installed	Required	Procedure
C	3	0	**(O)**

May be inoperative provided sufficient functions remain to allow emergency access to the flight deck.

OPERATIONS (O) PROCEDURES

1. Prior to dispatch perform the Flight Deck Door Access System Test in the Supplementary Procedures SP1.1 of the FCOM 1 to determine that sufficient functions remain to allow emergency access to the flight deck using the keypad.
   NOTE: If insufficient functions remain then dispatch using MEL 52-51-01-01.
2. For emergency access to the flight deck, cabin crew will enter the access code into the keypad, and wait approximately 30 seconds for the door to unlock automatically before attempting to open the door.
   NOTE: The emergency access code must NEVER be used for normal/routine flight deck access.

52-51-01 **Flight Deck Door Automatic Locking System**
52-51-01-01 **Flight Deck Access Panel System (Keypad, Door Chime)**
52-51-01-01-03 **Switch Guard**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative or missing provided flight deck door LOCK FAIL light operates normally.

52-51-01 Flight Deck Door Automatic Locking System

52-51-01-02 Flight Deck Door LOCK FAIL Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided automatic lock controls are verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify automatic lock controls are operating normally (AMM 52-00-00/901):

1. With the flight deck door open, supply electrical power on the airplane (AMM 24-22-00/201).
2. Set the FLIGHT DECK ACCESS SYSTEM switch to NORM (guard closed).
3. Set the FLT DK DOOR lock selector to AUTO.
4. Confirm the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you cannot rotate the strike).
5. Enter keypad access code and confirm the door chime sounds.
6. Set the FLT DK DOOR lock selector to DENY.
7. Before the DENY time delay has expired, enter the keypad access code and confirm the door chime does not sound.
8. Set and hold the FLT DK DOOR lock selector in UNLKD.
9. Confirm the electric strike is in the unlocked position (solenoid pin in the electric strike will retract down such that you can rotate the strike).
10. Set the FLT DK DOOR lock selector to AUTO.
11. Confirm the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you cannot rotate the strike).

52-51-01 Flight Deck Door Automatic Locking System

52-51-01-03 Flight Deck Door AUTO UNLK Light

Interval	Installed	Required	Procedure
C	1	0	**(M)**

May be inoperative provided:

a. Automatic lock controls are verified to operate normally.
b. Door chime operates normally.

MAINTENANCE (M) PROCEDURES

Verify automatic lock controls are operating normally (AMM 52-00-00/901):

1. With the flight deck door open, supply electrical power on the airplane (AMM 24-22-00/201).
2. Set the FLIGHT DECK ACCESS SYSTEM switch to NORM (guard closed).
3. Set the FLT DK DOOR lock selector to AUTO.
4. Confirm the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you cannot rotate the strike).
5. Enter keypad access code and confirm the door chime sounds.
6. Set the FLT DK DOOR lock selector to DENY.
7. Before the DENY time delay has expired, enter the keypad access code and confirm the door chime does not sound.
8. Set and hold the FLT DK DOOR lock selector in UNLKD.
9. Confirm the electric strike is in the unlocked position (solenoid pin in the electric strike will retract down such that you can rotate the strike).
10. Set the FLT DK DOOR lock selector to AUTO.
11. Confirm the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you cannot rotate the strike).

52-51-01 Flight Deck Door Automatic Locking System

52-51-01-04 Flight Deck Door Lock Control Selector

Interval	Installed	Required	Procedure
C	1	0	**(M) (O)**

May be inoperative provided:

a. Keypad is deactivated.
b. Automatic lock is verified to operate normally.
c. The FLIGHT DECK ACCESS SYSTEM switch is placed in the OFF position (guard extended).
d. Door dead bolt operates normally and is used to lock door.

MAINTENANCE (M) PROCEDURES

Deactivate the keypad and verify the automatic lock is operating normally (AMM 52-00-00/901):

1. Deactivate the keypad:
   A. Open the P6-3 panel DOOR LOCK circuit breaker.
   B. Disconnect the keypad electrical connector at the chime module side and cap and stow it.
   C. Close the open circuit breaker.

   NOTE: When flight deck door is closed with the flight deck not occupied, make sure the FLIGHT DECK ACCESS SYSTEM switch is in the OFF position (guard extended).
2. Verify the automatic lock is operating normally:
   A. With the flight deck door open, supply electrical power on the airplane (AMM 24-22-00/201).
   B. Set the FLIGHT DECK ACCESS SYSTEM switch to NORM (guard closed).
   C. Confirm the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you cannot rotate the strike).
   D. Set the FLIGHT DECK ACCESS SYSTEM switch to OFF (guard extended).
   E. Confirm the electric strike is in the unlocked position (solenoid pin in the electric strike will retract down such that you can rotate the strike).
   F. Set the FLIGHT DECK ACCESS SYSTEM switch to NORM (guard closed).
   G. Confirm the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you cannot rotate the strike).

OPERATIONS (O) PROCEDURES

1. Place the FLIGHT DECK ACCESS SYSTEM switch in the OFF position (guard extended).
2. Use the deadbolt to lock the flight deck door by rotating only the forward deadbolt lever to locked (this allows the key to unlock the door).
3. Entry to the flight deck will be achieved only by interphone request including the password. The cockpit door will not be unlocked unless the password is used.
4. In case of pilot incapacitation, the cabin crew will use the passenger cabin key to unlock the door and gain access to the flight deck.

   NOTE: The passenger cabin key must be kept in a secure location outside the flight deck and must NEVER be used for normal/routine flight deck access.

OPERATIONS (O) PROCEDURES (Cont'd)

5. In the event of a cabin disturbance or attempted unauthorized access to the flight deck, rotate both concentric deadbolt levers to the locked (horizontal) position to prevent the passenger cabin key from unlocking the door.

   NOTE: Pilots on the Flight Deck MUST assume that any attempt to enter the flight deck without the use of the correct password is an unauthorized intrusion and so take the necessary action to keep the door locked.

OPERATIONS NOTE

When flight deck door is closed with the flight deck not occupied, make sure the FLIGHT DECK ACCESS SYSTEM switch is in the OFF position (guard extended).

52-51-02 Flight Deck Door Dead Bolt

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided flight deck door automatic locking system operates normally.

52-51-03 Flight Deck Door Pressure Relief Panels

Interval	Installed	Required	Procedure
A	2	0	

May be inoperative provided:

a. Panels are in latched position.
b. Repairs are made within 2 flight days.

52-51-04 **Flight Deck Door Viewing Port**

52-51-04A **Electronic Flight Deck Door Visual Surveillance System Inoperative**

Interval	Installed	Required	Procedure
	1	0	

NOTE: Refer to MEL Item 23-75-01A for dispatch relief.

52-51-04 **Flight Deck Door Viewing Port**

52-51-04B **Electronic Flight Deck Door Visual Surveillance System Operates Normally**

Interval	Installed	Required	Procedure
C	1	0	**(O)**

May be inoperative provided an electronic flight deck door visual surveillance system is installed and operates normally.

OPERATIONS (O) PROCEDURES

Where a person wishes to exit the flight crew compartment, such person, prior to unlocking the door shall:

1. Verify with the cabin crew that the adjacent lavatory is not occupied by a passenger.
2. Visually determine the exact circumstances existing outside the door, particularly on high risk flights or where unknown passengers are within easy access of the flight crew compartment door.
3. Confirm that cabin crew is positioned to block passenger access to the door area when it is unlocked.

INTENTIONALLY
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52-71-01 Door Warning Light System

52-71-01-01 Entry and Service Doors

Interval	Installed	Required	Procedure
C	4	0	**(M) (O)**

May be inoperative provided associated door is verified closed and latched before each departure.

NOTE: If two or more entry/service door warning lights are inoperative due to failed door sensors, all flight lock systems are considered inoperative.

MAINTENANCE (M) PROCEDURES

If desired, deactivate the faulty door indication(s).

1. To deactivate a faulty forward entry door indication, do these steps (AMM 52-00-00/901):
   A. Remove wire W2216-0012-24 at the connector D1406 pin 8 on P5-20. Cap and stow the disconnected wire.
   B. Confirm that the FWD ENTRY door light remains off while the door is open.
   C. Run PSEU self-test.
      NOTE: With the FWD ENTRY door light disconnected, PSEU self-test will fail and report the DOOR FWD ENTRY status message.
   D. Confirm that the forward entry door is closed and latched.
2. To deactivate a faulty aft entry door indication, do these steps (AMM 52-00-00/901):
   A. Remove wire W2216-0012-24 at the connector D1406 pin 5 on P5-20. Cap and stow the disconnected wire.
   B. Confirm that the AFT ENTRY door light remains off while the door is open.
   C. Run PSEU self-test.
      NOTE: With the AFT ENTRY door light disconnected, PSEU self-test will fail and report the DOOR AFT ENTRY status message.
   D. Confirm that the aft entry door is closed and latched.
3. To deactivate a faulty forward service entry door indication, do these steps (AMM 52-00-00/901):
   A. Remove wire W2216-0003-24 at the connector D1406 pin 2 on P5-20. Cap and stow the disconnected wire.
   B. Confirm that the FWD SERVICE door light remains off while the door is open.
   C. Run PSEU self-test.
      NOTE: With the FWD SERVICE door light disconnected, PSEU self-test will fail and report the DOOR FWD SERVICE status message.
   D. Confirm that the forward service door is closed and latched.
4. To deactivate a faulty aft service door indication, do these steps (AMM 52-00-00/901):
   A. Remove wire W2216-0004-24 at the connector D1406 pin 3 on P5-20. Cap and stow the disconnected wire.
   B. Confirm that the AFT SERVICE door light remains off while the door is open.
   C. Run PSEU self-test.
      NOTE: With the AFT SERVICE door light disconnected, PSEU self-test will fail and report the DOOR AFT SERVICE status message.
   D. Confirm that the aft service door is closed and latched.

MAINTENANCE (M) PROCEDURE52-71-01-01 (Cont'd)

5. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Verify the associated door is closed and latched before each departure.

1. Confirm the following where applicable:
   A. Exit exterior is flush with the fuselage skin.
   B. The exterior handle is flush with the door skin.
   C. The upper and lower gates are closed and faired with the fuselage skin.
   D. The interior door liner is faired with the cabin wall.
   E. The inside handle is rotated to the latched position.

52-71-01 Door Warning Light System

52-71-01-02 Cargo and Access Doors

Interval	Installed	Required	Procedure
C	3	0	**(M) (O)**

May be inoperative provided associated door is verified closed and latched before each departure, including pushing inward against the door.

MAINTENANCE (M) PROCEDURES

If desired, deactivate the faulty door indication(s).

1. To deactivate a faulty forward or aft cargo door indication switch, do these steps (AMM 52-00-00/901):
   A. Disconnect ground wire from the forward or aft cargo door indication switch (AMM 52-71-00/201). Secure disconnected wire.
   B. Visually inspect the affected door(s) to ensure that door is closed and latched.
   C. Confirm that the affected (FWD or AFT) CARGO door warning and MASTER CAUTION lights are on after the affected door is closed and latched.
   D. Press and release MASTER CAUTION light to reset.
2. To deactivate a faulty forward access or electronic equipment access door indication switch, do these steps (AMM 52-00-00/901):
   A. Disconnect ground wire from the forward access or electronic equipment access door indication switch (AMM 52-71-00/201). Secure disconnected wire.
   B. Visually inspect the affected door(s) to ensure that door is closed and latched.
   C. Confirm that the EQUIP door warning and MASTER CAUTION lights are on after the affected door is closed and latched.
   D. Press and release MASTER CAUTION light to reset.
3. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Verify the associated door is closed and latched before each departure.

1. Confirm the following where applicable:
   A. Exit exterior is flush with the fuselage skin.
   B. The exterior handle is rotated to the latched position and the handle is flush with the door skin.
2. Push inward against the associated door to verify that it is closed and latched.

52-71-01 Door Warning Light System

52-71-01-04 Overwing Doors

Interval	Installed	Required	Procedure
C	4	0	**(M) (O)**

Individual light indication may be inoperative provided:

a. Each associated exit is verified closed and latched before each departure.
b. Associated flight lock is verified to operate normally.

MAINTENANCE (M) PROCEDURES

For airplanes with PSEU P/N 285A1600-102 or later installed:

Deactivate the affected door indication, including the input to the master caution system and the aural takeoff configuration warning system, and verify the associated flight lock is operating normally (AMM 52-00-00/901).

1. To deactivate faulty left aft overwing door indication switch(es), do these steps (AMM 52-00-00/901):
   A. Query maintenance messages to identify failed overwing door switch(es).
   B. Open and collar the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT
   C. For Left Aft Overwing Door Switch A, S1104, remove wire W5160-0007-20 at the PSEU connector D10986 pin 8. Cap and stow the disconnected wire.
   D. For Left Aft Overwing Door Switch A, S1104, jumper the PSEU connector D10986 pin 8 to ground.
   E. For Left Aft Overwing Door Switch B, S1105, remove wire W5160-0022-20 at the PSEU connector D10988 pin 21. Cap and stow the disconnected wire.
   F. For Left Aft Overwing Door Switch B, S1105, jumper the PSEU connector D10988 pin 21 to ground.
   G. Close the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT
   H. Visually inspect the left aft overwing door and confirm that it is closed and latched.
   I. Confirm that the Left Aft Overwing door warning and MASTER CAUTION lights are off after the affected door is closed and latched.
2. To deactivate faulty right aft overwing door indication switch(es), do these steps (AMM 52-00-00/901):
   A. Query maintenance messages to identify failed overwing door switch(es).
   B. Open and collar the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT
   C. For Right Aft Overwing Door Switch A, S1106, remove wire W5160-0019-20 at the PSEU connector D10986 pin 61. Cap and stow the disconnected wire.
   D. For Right Aft Overwing Door Switch A, S1106, jumper the PSEU connector D10986 pin 61 to ground.
   E. For Right Aft Overwing Door Switch B, S1107, remove wire W5160-0017-20 at the PSEU connector D10988 pin 1. Cap and stow the disconnected wire.

MAINTENANCE (M) PROCEDURES 52-71-01-04 PSEU P/N 285A1600-102 or later installed (Cont'd)

F. For Right Aft Overwing Door Switch B, S1107, jumper the PSEU connector D10988 pin 1 to ground.
G. Close the following P6-3 panel circuit breaker:
   1) PSEU PRI
   2) PSEU ALT
H. Visually inspect the right aft overwing door and confirm that it is closed and latched.
I. Confirm that the Right Aft Overwing door warning and MASTER CAUTION lights are off after the affected door is closed and latched.

3. To deactivate faulty left forward overwing door indication switch(es), do these steps (AMM 52-00-00/901):
   A. Query maintenance messages to identify failed overwing door switch(es).
   B. Open and collar the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT
   C. For Left Forward Overwing Door Switch A, S1108, remove wire W5160-0509-20 at the PSEU connector D10986 pin 52. Cap and stow the disconnected wire.
   D. For Left Forward Overwing Door Switch A, S1108, jumper the PSEU connector D10986 pin 52 to ground.
   E. For Left Forward Overwing Door Switch B, S1109, remove wire W5160-0533-20 at the PSEU connector D10988 pin 35. Cap and stow the disconnected wire.
   F. For Left Forward Overwing Door Switch B, S1109, jumper the PSEU connector D10988 pin 35 to ground.
   G. Close the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT
   H. Visually inspect the left forward overwing door and confirm that it is closed and latched.
   I. Confirm that the Left Forward Overwing door warning and MASTER CAUTION lights are off after the affected door is closed and latched.
4. To deactivate faulty right forward overwing door indication switch(es), do these steps (AMM 52-00-00/901):
   A. Query maintenance messages to identify failed overwing door switch(es).
   B. Open and collar the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT
   C. For Right Forward Overwing Door Switch A, S1110, remove wire W5160-0539-20 at the PSEU connector D10986 pin 15. Cap and stow the disconnected wire.
   D. For Right Forward Overwing Door Switch A, S1110, jumper the PSEU connector D10986 pin 15 to ground.
   E. For Right Forward Overwing Door Switch B, S1111, remove wire W5160-0535-20 at the PSEU connector D10988 pin 30. Cap and stow the disconnected wire.
   F. For Right Forward Overwing Door Switch B, S1111, jumper the PSEU connector D10988 pin 30 to ground.
   G. Close the following P6-3 panel circuit breakers:
      1) PSEU PRI
      2) PSEU ALT

MAINTENANCE (M) PROCEDURES 52-71-01-04 PSEU P/N 285A1600-102 or later installed (Cont'd)

H. Visually inspect the right forward overwing door and confirm that it is closed and latched.

I. Confirm that the Right Forward Overwing door warning and MASTER CAUTION lights are off after the affected door is closed and latched.

5. Verify the associated flight lock is operating normally (AMM 52-00-00/901).
   A. Check PSEU BITE and confirm overwing door flight lock faults are not displayed.
6. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Verify the associated door is closed and latched before each departure.

1. Confirm the following:
   A. Exit exterior is flush with the fuselage skin.
   B. The vent panel is closed and faired with the door skin.
   C. The interior door liner is faired with the cabin wall.
   D. The inside handle is rotated to the latched position.
   E. The handle cover is installed.

52-71-02 Door Programming Pin Fault

Interval	Installed	Required	Procedure
C	1	0	

NOTE: If necessary, inhibit the MAINT light for the associated active status message (AMM 31-65-00/201).

INTENTIONALLY
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71-11-01 Fan Cowl Latch Indication Systems (upon incorporation of Boeing Service Bulletin 737-71-1894 or production equivalent)

INTENTIONALLY
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71-11-01 **Fan Cowl Latch Indication Systems (upon incorporation of Boeing Service Bulletin 737-71-1894 or production equivalent)**

Interval	Installed	Required	Procedure
D	2	0	**(M)**

May be inoperative provided the associated fan cowl latch indication system blade(s) is removed.

MAINTENANCE (M) PROCEDURES

Remove the inoperative fan cowl latch indication system blade(s) (AMM 71-00-00/901).

1. Remove the latch indication system forward fairing.
2. Remove the blade assembly and reinstall the bolt for the blade assembly.
3. Reinstall the forward fairing if it is available and not damaged.

   NOTE: The airplane may be operated with the forward fairing missing.

   A. If the forward fairing is not reinstalled, apply speed tape to cover the fan cowl surface area where the forward fairing is normally located, ensuring that the speed tape does not cover the latch or two drain holes near the forward fairing location.
4. Remove the aft outer fairing if it is damaged.

   NOTE 1: The airplane may be operated with the aft fairing missing.

   NOTE 2: Do not remove the inner aft fairing which is attached to the fan cowl skin with adhesive.

   A. Apply speed tape over the inner portion of the aft fairing, ensuring that the speed tape does not cover the latch or two drain holes near the aft fairing location.
5. If necessary due to latch indication system being jammed with the handle in the deployed position, remove the handle.

OPERATIONS NOTE

Apply performance adjustments as appropriate per CDL item 71-11-01 for missing fan cowl latch indication system blades and fairings.

INTENTIONALLY
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73-11-01 Engine Fuel Strainer Sensors

73-21-01 Electronic Engine Control (EEC) Normal Mode – (*DELETED)*

73-30-01 Fuel Control ENG VALVE CLOSED Indicating System

73-31-01 Fuel Flow Indications

73-34-01 Fuel Filter Bypass Indication Systems

73-34-01-01 Engine Main Fuel/Oil Heat Exchanger Sensors

73-34-01-02 Engine Fuel Filter Sensors

INTENTIONALLY
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73-11-01 Engine Fuel Strainer Sensors

Interval	Installed	Required	Procedure
C	2	1	

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

INTENTIONALLY
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73-21-01 Electronic Engine Control (EEC) Normal Mode

Interval	Installed	Required	Procedure

Deleted, Revision 3.

INTENTIONALLY
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73-30-01 Fuel Control ENG VALVE CLOSED Indicating System

Interval	Installed	Required	Procedure
C	2	0	**(M)**

May be inoperative provided associated valve is verified to operate normally.

MAINTENANCE (M) PROCEDURES

Verify associated fuel shutoff valve operates normally (AMM 73-00-00/901).

1. Open and collar the P6-3 Panel ENGINE FUEL ENGINE (1/2) HPSOV IND circuit breaker
2. Start engine and allow it to stabilize.
3. Open the associated engine P6-3 Panel SPAR VALVE ENG circuit breaker.
4. Position engine start lever to CUTOFF.
5. Verify that the engine immediately decelerates.
6. Close the P6-3 Panel SPAR VALVE ENG circuit breaker.

INTENTIONALLY
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73-31-01 Fuel Flow Indications

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided flight deck fuel quantity indicating systems operate normally.

OPERATIONS NOTE

Fuel alerting for associated engine will be inoperative.

INTENTIONALLY
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73-34-01 Fuel Filter Bypass Indication Systems

73-34-01-01 Engine Main Fuel/Oil Heat Exchanger Sensors

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated engine fuel filter sensor operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

73-34-01 Fuel Filter Bypass Indication Systems

73-34-01-02 Engine Fuel Filter Sensors

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated engine fuel/oil heat exchanger sensor operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

74-00-01 Ignition

74-00-01-01 Left Ignition Systems

74-00-01-01A ER Operations Allowed

74-00-01-01B Non ER Operations

74-00-01-02 Right Ignition Systems

74-00-01-02A ER Operations Allowed

74-00-01-02B Non ER Operations

INTENTIONALLY
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74-00-01 **Ignition Systems**

74-00-01-01 **Left Ignition Systems**

74-00-01-01A **ER Operations Allowed**

Interval	Installed	Required	Procedure
B	2	1	**(O)**

May be inoperative provided:

a. Ignition select switch remains in BOTH position.

b. Right ignition systems operate normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Set Ignition Select switch to IGN R for starting the associated engine.
2. After engine start, set Ignition Select switch to BOTH.

74-00-01 **Ignition Systems**

74-00-01-01 **Left Ignition Systems**

74-00-01-01B **Non ER Operations**

Interval	Installed	Required	Procedure
C	2	0	**(O)**

Except for ER operations, may be inoperative provided:

a. Ignition select switch remains in BOTH position.

b. Associated engine right ignition system operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

NOTE: ER operations are not allowed.

1. Set Ignition Select switch to IGN R for starting the associated engine.
2. After engine start, set Ignition Select switch to BOTH.

74-00-01 Ignition Systems

74-00-01-02 Right Ignition Systems

74-00-01-02A ER Operations Allowed

Interval	Installed	Required	Procedure
B	2	1	**(M) (O)**

May be inoperative provided:

a. Associated engine left ignition system is connected to standby power.
b. Left ignition systems operate normally.
c. Ignition select switch remains in BOTH position.

MAINTENANCE (M) PROCEDURES

Connect the associated engine left ignition system to 115V AC standby power (AMM 74-00-00/901).

1. For engine 1, open and collar the following circuit breakers:
   A. ENGINE 1 IGNITION LEFT on the P18-2 panel
   B. ENGINE 1 IGNITION PWR CONT on the P6-5 panel
2. For engine 2, open and collar the following circuit breakers:
   A. ENGINE 2 IGNITION LEFT on the P6-2 panel
   B. ENGINE 2 IGNITION PWR CONT on the P6-5 panel
3. If desired, verify the left ignition systems operate normally by performing Test No. 18 - EEC Channel A Ignition System Test, and Test No. 19 – EEC Channel B Ignition System Test (AMM 71-00-00/501).
   NOTE: It is permitted to find faults with the right ignition system provided the left ignition system passes Test No. 18 and Test No. 19.
4. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Set the Ignition Select switch to IGN L for starting the associated engine.
2. After engine start, set the Ignition Select switch to BOTH.

74-00-01 **Ignition Systems**

74-00-01-02 **Right Ignition Systems**

74-00-01-02B **Non ER Operations**

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

Except for ER operations, may be inoperative provided:

a. Associated engine left ignition system is connected to standby power.
b. Associated engine left ignition system operates normally.
c. Ignition select switch remains in BOTH position.

MAINTENANCE (M) PROCEDURES

Connect the associated engine left ignition system to 115V AC standby power (AMM 74-00-00/901).

1. For engine 1, open and collar the following circuit breakers:
   A. ENGINE 1 IGNITION LEFT on the P18-2 panel
   B. ENGINE 1 IGNITION PWR CONT on the P6-5 panel
2. For engine 2, open and collar the following circuit breakers:
   A. ENGINE 2 IGNITION LEFT on the P6-2 panel
   B. ENGINE 2 IGNITION PWR CONT on the P6-5 panel
3. If desired, verify the left ignition systems operate normally by performing Test No. 18 - EEC Channel A Ignition System Test, and Test No. 19 – EEC Channel B Ignition System Test (AMM 71-00-00/501).
   NOTE: It is permitted to find faults with the right ignition system provided the left ignition system passes Test No. 18 and Test No. 19.
4. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

NOTE: ER operations are not allowed.

1. Set the Ignition Select switch to IGN L for starting the associated engine.
2. After engine start, set the ignition select switch to BOTH.

75-23-01 EEC Cooling Systems *(If Installed)*

75-23-02 Modulated Turbine Cooling (MTC) Systems *(If Installed)*

75-24-01 Turbine Case Cooling Air Flow Systems

75-24-01-01 High Pressure Turbine Active Clearance Control (HPTACC) Valves

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75-23-01 **EEC Cooling Systems** (If Installed)

Interval	Installed	Required	Procedure
D	2	0	**(M)**

May be inoperative provided associated fans are deactivated off.

MAINTENANCE (M) PROCEDURES

Deactivate the associated fans off (AMM 75-00-00/901).

1. For engine 1, open and collar the P6-5 panel ENGINE 1 EEC FAN BLOWERS circuit breaker.
2. For engine 2, open and collar the P6-5 panel ENGINE 2 EEC FAN BLOWERS circuit breaker.
3. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

75-23-02 **Modulated Turbine Cooling (MTC) Systems** (If Installed)

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. At least one MTC valve for the inoperative MTC system is locked open.
b. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Lock one of the MTC valves (left or right) in the open position for the inoperative MTC system (AMM 75-00-00/901).

NOTE: For engines with the simplex MTC system (upon incorporation of CFM Service Bulletin 72-00-0124 or production equivalent with a single MTC valve), the left MTC valve has been removed, and it is necessary to lock the right MTC valve in the open position for the inoperative MTC system.

1. Gain access to the inoperative MTC system left or right MTC valve.
2. Lock the (one) MTC valve in the open position.
3. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

Increase flight planning fuel by either incrementing the current fuel flow correction factor of 1.0% or flight planning fuel by 1.1% for each MTC system inoperative.

75-24-01 **Turbine Case Cooling Air Flow Systems**

75-24-01-01 **High Pressure Turbine Active Clearance Control (HPTACC) Valves**

Interval	Installed	Required	Procedure
C	2	0	**(M) (O)**

May be inoperative provided:

a. Inoperative HPTACC valve is locked in the closed position.
b. Both EECs operate in the normal mode.
c. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Lock the inoperative HPTACC valve in the closed position (AMM 75-00-00/901).

1. Retract and deactivate the leading edge slats.
2. Deactivate the associated engine thrust reverser for ground maintenance.
3. Gain access to the HPTACC valve.
4. Lock the inoperative HPTACC valve in the closed position.
5. Activate the associated engine thrust reverser after ground maintenance.
6. Activate the leading edge slats.
7. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

1. Performance limited weights are reduced by the following amount for one or both HPTACC Valve inoperative.

Takeoff	Enroute Climb	Approach & Landing
11,430 kg	16,057 kg	6,985 kg

NOTE: Enroute climb penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed in the table below - Enroute Diversion Speed Effects.

Diversion Speed	Factor
LRC	1.5
280 KIAS	3.9
290 KIAS	4.5
310 KIAS	5.4
320 KIAS	6.4
330 KIAS	7.5

OPERATIONS (O) PROCEDURES 75-24-01-01 (Cont'd)

2. Increase flight planning fuel by either incrementing the current flow correction factor of 1.6% or flight planning fuel by 1.8% for each inoperative valve.
3. Increase engine-out ETOPS critical fuel reserves by an additional 1.3% on top of the flight planning fuel penalty noted above. The all-engine ETOPS critical fuel reserves would be covered by the flight planning fuel penalty noted above.

77-21-01 Abnormal Start Indication Systems

77-31-01 Engine Vibration Indications

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77-21-01 Abnormal Start Indication Systems

Interval	Installed	Required	Procedure
C	2	0	

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77-31-01 Engine Vibration Indications

Interval	Installed	Required	Procedure
C	2	1	

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

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78-31-01 Thrust Reverser Systems

78-34-01 Reverse Thrust Lever Interlocks

78-34-01A Interlock Released

78-34-01B Interlock Not Released

78-36-01 REVERSER LIMITED Lights (Aft Overhead Panel)

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78-31-01 Thrust Reverser Systems

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

May be inoperative provided:

a. Inoperative reverser is secured in the forward thrust position.
b. Appropriate performance adjustments are applied.

MAINTENANCE (M) PROCEDURES

Deactivate and secure the inoperative thrust reverser in the forward thrust position (AMM 78-00-00/901).

NOTE: Thrust reverser sync locks may be inoperative provided the inoperative sync lock is verified to be in the locked position.

1. Deactivate the inoperative thrust reverser in the forward thrust position.
   NOTE: Thrust reverser deactivation can result in the illumination of the MASTER CAUTION, ENG annunciation, and REVERSER LIMITED indication light associated with the deactivated thrust reverser during engine start or when performing a Master Caution recall. The REVERSER LIMITED indication light associated with the remaining operational thrust reverser must not be illuminated
2. Placard the REVERSER LIMITED indication light associated with the deactivated thrust reverser.
3. Prevent movement of the reverse thrust handle by an appropriate means (lockwire the thrust reverser handle to the associated forward thrust lever).
4. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

OPERATIONS (O) PROCEDURES

NOTE: Thrust reverser deactivation can result in the illumination of the MASTER CAUTION, ENG annunciation, and REVERSER LIMITED indication light associated with the deactivated thrust reverser during engine start or when performing a Master Caution recall.

1. For a wet runway, apply the performance adjustments for one thrust reverser inoperative.
2. Prior to landing, set the GROUND PROXIMITY RUNWAY INHIBIT switch to INHIBIT (if the overrun warning system and GROUND PROXIMITY RUNWAY INHIBIT switch are installed).

NOTE: With Inoperative Thrust Reverser:

1. Use of one engine reverse thrust is left to the discretion of the Captain.
2. On WET runways the use of autobrake is recommended instead of one engine reverse thrust.
3. For Take-off on WET runways, refer to Onboard Performance Tool (OPT).

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78-34-01 **Reverse Thrust Lever Interlocks**

78-34-01A **Interlock Released**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative released.

78-34-01 Reverse Thrust Lever Interlocks

78-34-01B Interlock Not Released

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative not released provided appropriate performance adjustments are applied.

OPERATIONS (O) PROCEDURES

1. For a wet runway, apply the performance adjustments for one thrust reverser inoperative.
2. Prior to landing, set the GROUND PROXIMITY RUNWAY INHIBIT switch to INHIBIT (if the overrun warning system and GROUND PROXIMITY RUNWAY INHIBIT switch are installed).

NOTE: With Inoperative Thrust Reverser:

1. Use of one engine reverse thrust is left to the discretion of the Captain.
2. On WET runways the use of autobrake is recommended instead of one engine reverse thrust.
3. For Take-off on WET runways, refer to Onboard Performance Tool (OTP).

78-36-01 REVERSER LIMITED Lights (Aft Overhead Panel)

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided associated reverser is considered inoperative.

NOTE: Airplane must also be dispatched using MEL item 78-31-01.

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79-21-01 Engine Oil Debris Monitoring System (ODMS)

79-21-01-01 Engine Oil Debris Monitoring System (ODMS) Sensor

79-31-01 Oil Quantity Indication Systems

79-35-01 Oil Filter Bypass Warning Systems

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79-21-01 Engine Oil Debris Monitoring System (ODMS)

Interval	Installed	Required	Procedure
C	2	1	**(M)**

May be inoperative provided:

a. Malfunction is verified to be in the associated engine oil debris monitoring system.
b. Associated engine oil filter bypass warning system operates normally.

MAINTENANCE (M) PROCEDURES

Verify the malfunction is in the associated engine oil debris monitoring system (ODMS).

1. Accomplish inspections of the engine oil system for contaminants (AMM 79-00-00/901). If the engine oil system inspections reveal no contaminants, then the malfunction is in the ODMS.
2. If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

79-21-01 Engine Oil Debris Monitoring System (ODMS)

79-21-01-01 Engine Oil Debris Monitoring System (ODMS) Sensor

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided associated engine oil filter bypass warning system operates normally.

NOTE: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

79-31-01 Oil Quantity Indication Systems

Interval	Installed	Required	Procedure
B	2	1	**(M)**

Except for ER operations, may be inoperative provided:

a. Oil tank is filled to maximum recommended capacity at each refueling.
b. There is no evidence of above normal oil consumption or leakage.

MAINTENANCE (M) PROCEDURES

At each refueling, service affected oil system and verify that there is no evidence of oil leakage or above normal consumption (AMM 79-00-00/901).

1. Within 30 minutes of engine shutdown, replenish engine oil if necessary.
2. Check airplane flight log and maintenance records to make sure oil consumption is within limits.

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79-35-01 Oil Filter Bypass Warning Systems

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided associated engine ODMS, including ODMS sensor, operates normally.

NOTE: An inoperative oil filter bypass warning system is indicated by status message ENG 1 OIL FILTER SENSOR or ENG 2 OIL FILTER SENSOR for the associated engine.

NOTE 1: If status message ENG 1 OIL DEBRIS SENSOR and/or ENG 1 OIL DEBRIS are displayed together with ENG 1 OIL FILTER SENSOR, the associated engine ODMS is not operating normally and dispatch is not allowed.

NOTE 2: If status messages ENG 2 OIL DEBRIS SENSOR and/or ENG 2 OIL DEBRIS are displayed together with ENG 2 OIL FILTER SENSOR, the associated engine ODMS is not operating normally and dispatch is not allowed.

NOTE 3: Illumination of the engine OIL FILTER BYPASS light during the previous flight indicates the oil filter is collecting contamination and dispatch is not allowed. Accomplish inspection per the Fault Isolation Manual and Aircraft Maintenance Manual.

NOTE 4: If necessary, inhibit the MAINT light for the associated active status message(s) (AMM 31-65-00/201).

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80-11-01 Starter Valve Open Alerts

80-11-02 Engine Start Switch Ground Hold/Cutout Systems

80-11-03 Engine Start Valves

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80-11-01 Starter Valve Open Alerts

Interval	Installed	Required	Procedure
C	2	1	**(O)**

May be inoperative provided:

a. Associated engine start valve operates normally.
b. Flight crew verify associated start valve is closed after engine start.

OPERATIONS (O) PROCEDURES

Verify the associated engine start valve is closed after engine start.

1. Use normal engine start procedures except for the following:
   A. Prior to setting ENGINE START switch to GRD, note the duct pressure.
   B. Without automatic ignition:
      As ENGINE START switch releases to OFF, verify that duct pressure returns to the pre-start value or higher.
   C. With automatic ignition:
      As ENGINE START switch releases to AUTO, verify that duct pressure returns to the pre-start value or higher.
2. If duct pressure is below the prestart value, shutdown the associated engine.
   A. Set ISOLATION VALVE switch to CLOSE.
   B. Set affected side PACK switch to OFF.
   C. Set affected engine BLEED air switch to OFF.
   D. For engine 1, set APU BLEED air switch to OFF.
   E. For engine 2, disconnect ground air source.
   F. Set affected engine start lever to CUTOFF.

80-11-02 Engine Start Switch Ground Hold/Cutout Systems

Interval	Installed	Required	Procedure
C	2	0	**(O)**

May be inoperative provided alternate start procedures are established and used.

OPERATIONS (O) PROCEDURES

1. Set ENGINE START switch to GRD and hold if required.
2. Without automatic ignition:
   Set ENGINE START switch to OFF at 63% N2.
3. With automatic ignition:
   Set ENGINE START switch to AUTO at 63% N2.

80-11-03 Engine Start Valves

Interval	Installed	Required	Procedure
C	2	1	**(M) (O)**

Except for ER operations, may be inoperative provided:

a. Associated START VALVE OPEN alert operates normally.
b. Manual override start procedures are used.

MAINTENANCE (M) PROCEDURES

Use alternate procedures for engine start (AMM 80-00-00/901).

1. Establish communications with the flight crew.
2. Manually override inoperative start valve as instructed by flight crew.

OPERATIONS (O) PROCEDURES

NOTE 1: If an inflight engine start is necessary, a crossbleed start will not be available on the associated engine. Ensure that airspeed is sufficient for a windmill start.

NOTE 2: Ground Engine Start: Prior to engine start, the Captain should advise the ground crew that in the event of an aborted engine start, it may be necessary to motor the engine, so ground crew should close the start valve ONLY when instructed by the Captain.

To start the selected engine, perform steps for option 1 if using APU bleed air, or **option 2 if using ground air source or engine crossbleed start:**

1. If bleed air for engine start is supplied by APU, use normal engine start procedures with the following modifications:
   A. Start the selected engine.
   B. When setting selected ENGINE START switch to GRD, instruct ground crew to open the start valve.
   C. Inform ground crew when N2 is rotating.
   D. Instruct ground crew to close the start valve when N2 is 63%.
   E. Verify that the ENGINE START switch moves to OFF.
   F. For engine inflight starts of selected engine, use windmill starts.
2. If bleed air for engine start is supplied by a ground air source or using the engine crossbleed start procedure, accomplish pre-start motoring (steps A through F below) before using normal engine start procedures with the following modifications:
   A. Set the PACK switch to AUTO for the selected engine.
   B. Set the selected ENGINE START switch to GRD and instruct ground crew to open start valve.
   C. Inform ground crew when N2 is rotating.

   CAUTION: IF ENGINE N2 INCREASES OVER 25% DURING ENGINE MOTORING STEP, INSTRUCT THE GROUND CREW TO CLOSE THE START VALVE IMMEDIATELY, AND SET THE ENGINE START SWITCH TO OFF.

OPERATIONS (O) PROCEDURES (Cont'd)

D. Motor the engine with the start valve open and associated PACK switch in AUTO for 4 minutes.
E. After 4 minutes of motoring, instruct the ground crew to close the start valve.
F. Set the ENGINE START switch to OFF.
G. Wait 5 minutes to allow the engine starter to cool.

NOTE: If the engine is not started within 15 minutes of completing the engine motoring, repeat the steps above to do the engine motoring again.

H. Set both PACK switches to OFF per normal engine start procedure.
I. Start the selected engine.
J. When setting selected ENGINE START switch to GRD, instruct ground crew to open the start valve.
K. Inform ground crew when N2 is rotating.
L. Instruct ground crew to close the start valve when N2 is 63%.
M. Verify that the ENGINE START switch moves to OFF.
N. For engine inflight starts of selected engine, use windmill starts.

Caribbean Airlines

B737-8 NONESSENTIAL EQUIPMENT AND FURNISHINGS (NEF) PROGRAM

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Caribbean Airlines
B737 Nonessential Equipment and Furnishings Program

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Approved by TTCAA	
Signature:	Date: 01-Dec-2023

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Caribbean Airlines
B737 Nonessential Equipment and Furnishings Program

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Caribbean Airlines
B737 Nonessential Equipment and Furnishings Program

INTRODUCTION

Caribbean Airlines has developed a Nonessential Equipment and Furnishings policy and procedures program based in part through the use of the following documentation:

- FAA Advisory Circular AC 20-168,
- MMEL Global Change GC-138 (PL-116),
- NEF Universal List (attachment to PL-116), and
- TTCAR No. 7

Caribbean Airlines operates its aircraft under TTCARs operating rules.

Caribbean Airlines must submit their NEF deferral program to the TTCAA.

NONESSENTIAL EQUIPMENT AND FURNISHINGS (NEF)

NEF are those items installed on the aircraft as part of the original type certification, supplemental type certificate, or other form of alteration that have no effect on the safe operation of flight and would not be required by the applicable certification rules or operational rules. They are those items that, if damaged, inoperative, or missing, have no effect on the aircraft's ability to be operated safely under all operational conditions. These nonessential items may be installed in areas including, but not limited to:

- Cargo Areas
- Crew Rest Areas
- Flight Deck Area
- Galley Area
- Lavatories
- Passenger Compartment
- Service Areas

NEF PROGRAM

The **Caribbean Airlines** NEF Program is as follows:

A. A NEF List has been developed and NEF Items are tracked through the use of the aircraft's MEL Deferred Defect Control Sheet.

B. The NEF List includes the following procedures for each NEF item:
   - Maintenance (M) Procedure
   - Operation (O) Procedure
   - Placarding (P) Procedure

C. The NEF Item evaluation process will include the following items:
   - Is the item required for the operational rules in which the aircraft is operated?
   - Does it create the potential for fire/smoke or other hazardous conditions?
   - Could it have an adverse effect on other required systems or components?
   - Does its condition potentially affect the safety of crew, passengers, or service personnel?
   - Could it have a negative impact on emergency or abnormal procedures?
   - Does it create additional workload for the crew at critical times of flight or flight preparation?
   - Crewmembers may need to evaluate the deferred NEF on a flight-by-flight basis.

   NOTE: THE ABOVE EVALUATION PROCESS MUST BE ACCOMPLISHED FOR THE DAMAGED, INOPERATIVE, OR MISSING ITEMS AT ITS FACE VALUE, AND ALSO FOR THE UNDERLYING CAUSE OF THE DISCREPANCY.

D. Repair and/or replacement of items listed in the NEF List are required within 120 calendar days from the date of discovery.

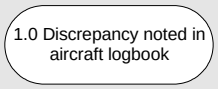
E. The NEF List and program description will be kept in the aircraft specific Minimum Equipment List (MEL) Binder.

F. The aircraft's MEL has incorporated the requirements of MMEL Global Change GC-138 (PL116) in ATA chapter 25.

G. If a discrepancy is discovered that is not covered by the aircraft's Configuration Deviation List (CDL), MEL or NEF List. The Flight Crew, with the assistance of the Maintenance Department, may perform the NEF Item Process to determine if the discrepancy can be added to the NEF List. The discrepancy must meet the intent of the NEF Item Process or it will require the issue to be resolved before further flight.

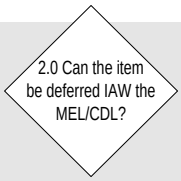
NEF DECISION TREE DISCUSSION

1.0 Discrepancy noted in aircraft logbook. The inoperative, damaged or missing item must be identified and documented in the aircraft logbook (or other approved location) by:



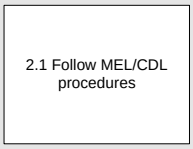
1. Flight Crew; or
2. Company maintenance personnel; or
3. Personnel authorized and approved to perform such functions as outlined in the maintenance program.

2.0 Can the item be deferred in accordance with (IAW) the MEL, CDL, or the MEL?



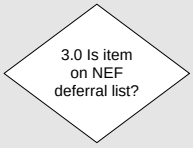
1. If the inoperative, damaged, or missing item is listed in the MEL or CDL, then the deferral procedures for that item must be followed. If the item is a subcomponent of a primary system identified in the MEL/CDL, where no previous relief was authorized, the subcomponent **may not be deferred** in accordance with the NEF procedures outlined in Chapter 25 of the MEL.

2.1 Follow MEL/CDL procedures.



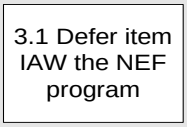
1. If the item is identified in another part of the MEL/CDL, then the procedures approved for the deferral of such item must be followed.

3.0 Is item on the NEF deferral list?



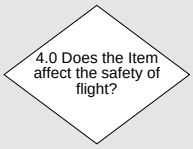
1. Is the item on the NEF list? If yes, then follow the NEF deferral procedures in step 3.1. (Items that are not previously on the NEF list should proceed to step 4.0.)

3.1 Defer item IAW the NEF deferral program.



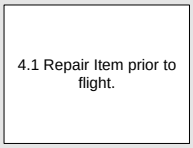
1. If the item is identified in the NEF deferral list, then the procedures approved for the deferral of such item shall be followed.

4.0 Does the item affect the safety of flight?



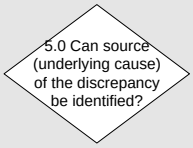
1. Is it obvious from a maintenance or operational perspective that the item, in and of itself, could have an adverse effect on the safe conduct of flight? If there is an obvious safety-of-flight issue, then the inoperative, damaged, or missing item may not be deferred and step 4.1 shall be followed.

4.1 Repair item prior to flight.



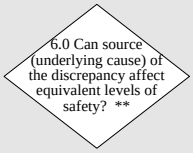
1. The item may not be deferred and must be repaired prior to flight.

5.0 Can source (underlying cause) of the discrepancy be identified? (If applicable)



1. Can the source of the discrepancy be identified? This step may or may not apply to the item under consideration. If the source can be identified, then proceed to step 6.0, otherwise proceed to step 4.1.

6.0 Can source (underlying cause) of discrepancy affect equivalent levels of safety?

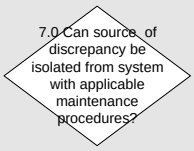


1. If the source (underlying cause) of the discrepancy affect equivalent levels of safety, then it must be determined if it can be isolated from all other systems so as to alleviate any safety concern.

**** Note:** In making this determination, very close coordination between the Flight Crew, maintenance and operations personnel may be required.

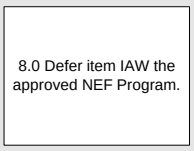
2. If, after review, the source of the discrepancy could be considered a safety-of-flight concern, the item must be repaired prior to flight (step 4.1). If the source of the discrepancy is not a safety-of-flight concern then defer the item in accordance with the approved NEF procedures in step 8.0. If it cannot be determined, or is uncertain, that the source of the discrepancy is a safety-of-flight concern then proceed to 7.0.

7.0 Can source (underlying cause) of discrepancy be isolated from the system with applicable maintenance procedures?



1. If applicable, the source (underlying cause) of the discrepancy must be isolated from all other systems so as to alleviate the safety-of-flight concern.
2. If the item cannot be safely isolated then the item must be repaired prior to flight (step 4.1).
3. If isolated, the isolation of the source must pass the entire test identified in the evaluative process (steps 4.0-7.0) for the item.
4. If source can be isolated then proceed to step 8.0.

8.0 Defer Item IAW the approved NEF program.



1. Defer the item if, after completing the previous 8 steps, the item can be deferred IAW the NEF program.

Note: Before an item can be deferred as an NEF item, the NEF program evaluation process for determining shall be followed if an item can be considered a NEF. Although NEF items are not safety-of-flight items, they have not been evaluated through the normal AEG review process and therefore require the concurrence of the Flight Crew, Maintenance, and Operational personnel, if applicable. NEF items are **not** deferred under the authority of an airframe and powerplant certificate but rather the item is deferred under the NEF program.

The evaluation process should determine items such as:

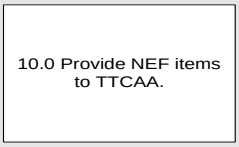
- **a.** Is the item required for the operational rules in which the aircraft is operated?
- **b.** Does it create the potential for fire/smoke or other hazardous conditions?
- **c.** Could it have an adverse effect on other required systems or components?
- **d.** Does its condition potentially affect the safety of passengers, crew, or service personnel?
- **e.** Could it have a negative impact on emergency or abnormal procedures?
- **f.** Does it create additional workload for the crew at critical times of flight or flight preparation?
- **g.** Crewmembers may need to evaluate the deferred NEF on a flight-by-flight basis.

Note: The above evaluation process must be accomplished for the inoperative, damaged, or missing items at its face value, and also for the underlying cause of the discrepancy.

9.0 Update NEF deferral list as required.

1. **Caribbean Airlines** will continually add items to the NEF list as they see fit.

10.0 Provide NEF Items to TTCAA.



1. **Caribbean Airlines** will provide the items placed in the NEF program to the TTCAA for review. This review is not to be conceived as a requirement to approve the NEF items. It is merely a means to provide oversight to ensure the program is effective.

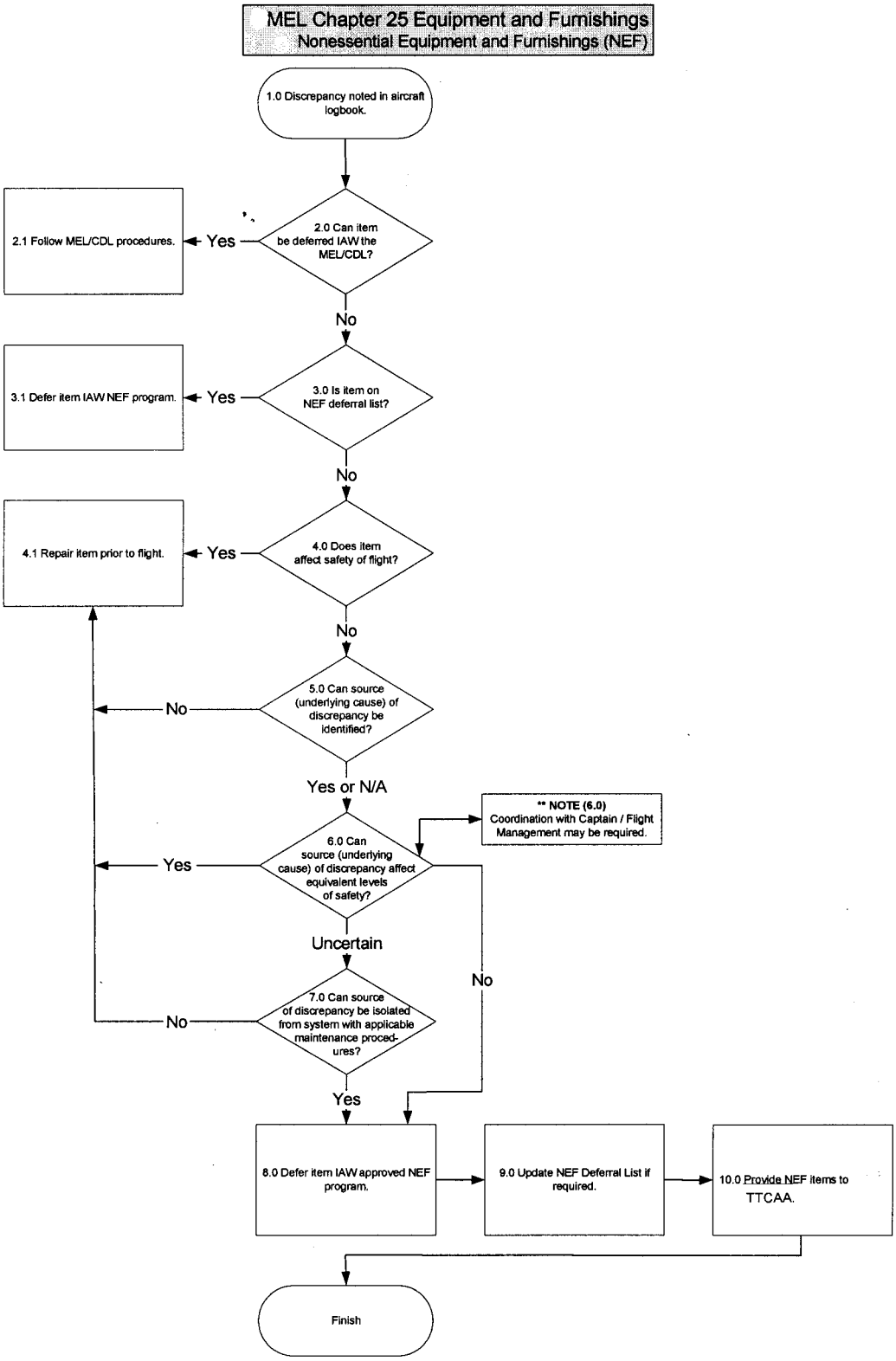


Figure 1: NEF Decision Tree

NEF LIST

The following items may be deferred in accordance with the NEF program. The repair interval is Category D (120 days).

COCKPIT

1. Sunshades: May be inoperative or missing
   (M) Procedure – None Required
   (O) Procedure – None Required
   (P) Procedure – Display an Inoperative Placard in a prominent position to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Cup Holders: May be inoperative or missing
   (M) Procedure – None Required
   (O) Procedure – None Required
   (P) Procedure – Place Inoperative Placard on or next to Cup Holder and note it on the MEL Deferred Defect Control Sheet.

3. Cushions: May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.
   (M) Procedure – None Required
   (O) Procedure – None Required
   (P) Procedure – Display an Inoperative Placard on or near the cushions and note it on the MEL Deferred Defect Control Sheet.

4. Vent knobs: May be inoperative or missing as long as the air flow through the vent is acceptable to the Flight Crew.
   (M) Procedure – None Required
   (O) Procedure – None Required
   (P) Procedure – Display an Inoperative Placard on or near the vent knob and note it on the MEL Deferred Defect Control Sheet.

5. Gaspers: May be inoperative as long as the air flow through the gasper is acceptable to the Flight Crew.
   (M) Procedure – None Required
   (O) Procedure – None Required
   (P) Procedure – Display an Inoperative Placard on or near the gasper and note it on the MEL Deferred Defect Control Sheet.

6. Upholstery: May be worn, torn, or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard on or near the upholstery and note it on the MEL DEFERRED DEFECT CONTROL SHEET.

7. Decorative Trim / Trim Strips: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position near the trim to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

8. Smoke Goggle Stowage Bags: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard on the affected bag(s) and note it on the MEL Deferred Defect Control Sheet.

9. Side Panel Map Cases: may be inoperative or missing as long as the crew has an alternate location for map storage.

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the map case and note it on the MEL Deferred Defect Control Sheet.

10. Foot Rests: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the foot rest to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

11. Yoke Clips: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard on or near the clip and note it on the MEL Deferred Defect Control Sheet.

12. Hand Microphone Holders / Headset Holders: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard on the holder and note it on the MEL Deferred Defect Control Sheet.

13. Access Compartment Latches: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the latch(es) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

14. Ashtrays: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the ashtray(s) and note it on the MEL Deferred Defect Control Sheet.

15. Coat and / or Hat Hooks: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the hook(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

16. Compartment Doors: May be inoperative or missing

    (M) Procedure – Maintenance will secure the door closed. Flight Crew may perform task if Maintenance is not available.

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected door(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

17. Document Holders: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the holder(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

18. Spring Clips: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the clip(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

19. Seat Belt Tidy Clips: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

20. Spare Bulb Kit and Contents: May be inoperative or missing

    (M Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the kit location to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

21. Windshield / Side Window Heat Contact Covers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the contact cover(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

22. Vanity Door Mirror: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the mirror to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

23. Carpet: May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

24. Logbook Holder: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the logbook holder to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

25. Observer’s Storage Compartment: May be inoperative or missing

    (M) Procedure – Maintenance will secure the compartment closed. Flight Crew may perform task if Maintenance is not available.

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the compartment to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

26. Electrical Outlets and Covers: The outlets may be inoperative. The covers may be damaged or missing.

    (M) Procedure – If required, Maintenance will electrically isolate the affected outlet by pulling and collaring the appropriate circuit breaker. Flight Crew may perform task if Maintenance is not available. If the cover is damaged or missing, no maintenance action is required.

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the outlet to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

27. Spare Headsets: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

28. Trim: seat, sidewall, overhead, etc.: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected trim to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

29. Note Pads and Holders: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the holder to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

30. Pens / Pencils and Holders: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the holder to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

31. Side Window Pull Out Shades: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard on the affected shade retainer(s) and note it on MEL Deferred Defect Control Sheet.

32. Outlet Unit: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the outlet to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

GALLEY

1. Coffee Makers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected coffee maker(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Coffee / Tea / Brewing Pots: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected pot(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

3. Convection Oven and Controls: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected oven(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

4. Hot Cup and Controls: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected hot cup to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

5. Hot Water Dispenser: May be inoperative or missing

   (M Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dispenser to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

6. Ice Container(s): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected container(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

7. Pots, Pans, Utensils: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected storage area to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

8. Catering Containers: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected containers location to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

9. Boiler: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected boiler to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

10. Waste Containers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected containers to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

11. Work Lights: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lights to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

12. Galley Heating and Controls: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected controls to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

13. Galleys Sinks and Drains: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected sink/drain to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

14. Work Tables: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected table(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

15. Coat Hangar Clips: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected clips to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

16. Paper Clips: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected clips to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

17. Mirrors: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected mirror(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

18. Sink Drain and Handle: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected drain handle to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

19. Electrical Outlets / Outlet Units / Power Supplies including Covers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected unit to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

20. Water System: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position near the water system controls to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

21. Water Heater: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected water heater to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

22. Gaspers: May be inoperative as long as the air flow through the gasper is acceptable to the Flight Crew.

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected gasper(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

23. Refrigerator / Wine Chiller: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected equipment(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet

CABIN

1. Miscellaneous Cabin Forms: May be missing or needs replenishing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on the cabin log to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Appearance Items: May be Worn, Soiled, Frayed, Torn, Damaged. Loose, Missing (must not present hazard to pax/crew or impede emergency egress)

a. Cabin Interior Trim

b. Carpet/Floor Coverings

c. Curtains / Tiebacks

d. Wall Coverings (including sidewall panels; excluding sidewall return air grilles)

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected items to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

3. Main Door Acoustic Cover: May be inoperative or missing

(M) Procedure – Maintenance will secure door in open position. Flight Crew may perform task if Maintenance is not available.

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

4. Cabin Lighting / Signs (only items not covered by MEL ATA 33): May be inoperative or missing

   a. Reading Lights

   b. Table lights

   c. Indirect/Overhead Light

   d. Floor Illumination Lights

   e. Light Lens / Covers (including emergency light): Loose, Damaged but light functions properly

   f. Lavatory Occupied Lights

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

5. Safety Demo Equipment: May be inoperative or missing

   a. Demo Tape / CD (notify flight attendant, physical demo may be required)

   b. Demo Equipment: Safety Belt, Life Vest, O2 Mask (consider availability of safety video)

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected equipment storage location to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

6. Spare Life Vests: Missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected storage area for the spare life vests to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

7. Magazine Rack / Restraint: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the magazine rack to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

8. Bulkhead Video Screens: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected screen(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

9. Stereo Equipment: May be inoperative or missing

   (M) Procedure – None Required

   (O Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected equipment to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

10. Gaspers: May be inoperative as long as the air flow through the gasper is acceptable to the Flight Crew.

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard on the affected gasper(s) and note it on the MEL Deferred Defect Control Sheet.

11. Seat Back Monitor / Seat Display Unit: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected Monitor / Unit to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

12. Entertainment Controllers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected controller location to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

13. Ash Trays: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected ash trays to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

14. Headphones / Remote Jack Units (RJU); USB Outlet Units & Outlet Units: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected unit(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

15. Decorative Trim / Trim Strips: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected trim to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

16. Cushions / Upholstery / Carpets: may be worn, torn, frayed or stained as long as they are determined to be serviceable by the Flight Crew, cabin crew or maintenance.

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

17. Inflight Entertainment System: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected system to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

18. Acoustic Doors / Curtains: May be inoperative or missing

    (M) Procedure – Maintenance will secure doors or curtains in open position. Flight Crew may perform task if Maintenance is not available.

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected door/curtain to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

19. Electrical Outlets and Covers: The outlets may be inoperative. The covers may be damaged or missing.

    (M) Procedure – If required, Maintenance will electrically isolate the affected outlet by pulling and collaring the appropriate circuit breaker. Flight Crew may perform task if Maintenance is not available. No maintenance action required for the covers.

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected outlet to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

20. Cabin Mirrors: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected mirror(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

21. Seat Track Covers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected seat track to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

22. Cabin Windows: May be

    a. Crazed, Moisture Between Panes

    b. Interior Panel: Damaged, Needs Cleaning

    c. Shades: Damaged, Inoperative

    d. Window Trim: Loose, Damaged, Missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected window(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

23. Fire Extinguisher Seals (safety pin intact and fire extinguisher otherwise fully serviceable: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected seal(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

24. Cabin / Galley Handset Cradle (handset fully operational): May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected cradle to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

25. Ventilation Grill: Dirty

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected grill to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

26. Therapeutic Oxygen Fittings: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected fitting to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

27. Cabin Divider Panels: May be Operable but Difficult to Position Closed

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected panel(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

28. Attendant Work Table: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected table to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

29. Attendant Storage Box: May be inoperative or missing

(M) Procedure – None Required

(O Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected box to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

30. Seat Belt Extensions: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected seat belt extension storage area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

31. Boarding Music: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected control area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

32. Cabin Speakers: Static but Operational

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected speaker(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

33. Seatback / Bulkhead / Sidewall Literature Pockets: May be inoperative or missing

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected pocket to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

34. Aircraft Removable Equipment: May be inoperative or missing

a. Dirty Linen Storage Containers

b. Waste Containers

(M) Procedure – None Required

(O) Procedure – None Required

(P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected container location to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

LAVATORIES, CLOSETS AND VESTIBULE

FORWARD/AFT LAVATORIES

1. Door Spring: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected spring to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Sanitizer Dispenser: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dispenser to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Seal Strips on Lavatory Door: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected door to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

4. Privacy Stop on Lavatory Door: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected stop to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

5. Vanity Mirror: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected mirror to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

6. Toilet Paper / Roller: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected roller to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

7. Vanity Lights: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected light(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

8. Lights: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lights to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

9. Mirror Lights: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lights to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

10. Diaper Changing Table: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected table to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

11. Sink Drain Screen: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected sink to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

12. Assist Handles: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected handle(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

13. Toilet Seat Lid: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lid to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

14. Toilet Seat: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected seat location to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

    NOTE: DISPATCH SHOULD BE IN ACCORDANCE WITH B737-8 MEL ITEM 38-30-01B.

15. Toilet: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected toilet to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

    NOTE: DISPATCH SHOULD BE IN ACCORDANCE WITH B737-8 MEL ITEM 38-30-01B.

16. Soap Holder/Soap Dispenser: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected holder/dispenser to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

17. Faucet Aerator: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected faucet to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

18. Hot and Cold Faucet Indicators: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected faucet(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

19. Cup Holder: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected cup holder to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

20. Coat Hook: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected hook to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

21. Deodorizer Holder: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected holder to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

22. Inside Door Ashtrays: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected ashtray(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

23. Miscellaneous Placards: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected placards to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

24. Paper Towels / Holder: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected towel holder to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

25. Interior Wall Trim / Coverings: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected trim to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

26. Air Grill Covers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected grill to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

27. Tissue Dispenser: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dispenser to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

28. Coat Rods / Hangers: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected coat rod to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

29. Towel Rods / Towels: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected towel rod to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

30. Water System/Water Heater: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected water system/heater to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

31. Waste Container: May be inoperative or missing

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected waste container to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

32. Upholstery: May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

    (M) Procedure – None Required

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected area to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

33. Electrical Outlets and Covers – The outlets may be inoperative. Covers may damaged or missing

    (M) Procedure – If required, Maintenance will electrically isolate the affected outlet by pulling and collaring the appropriate circuit breaker. Flight Crew may perform task if Maintenance is not available. If the cover is damaged or missing, no maintenance action is required.

    (O) Procedure – None Required

    (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected outlet(s) to be seen by the Flight Crew and note it on the MEL Deferred Defect Control Sheet.

FORWARD/AFT CLOSETS

1. Closet Doors/Curtains: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected door/curtain to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Manual Storage Compartments: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected compartment to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

3. Coat Rods: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected coat rod to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

4. Coat Hangers: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected coat hanger to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

5. Coat Hangers: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected coat hanger to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

6. Placards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected placard location to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

7. Decorative Trim / Trim Strips: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected trim area to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

8. Lights: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected light(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

9. Carpet: May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected area to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

VESTIBULE

1. Crew Name Plaque: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected plaque location to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Decorative Trim / Trim Strips: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected trim to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

3. Radio Rack Decorative Closeouts: May be inoperative, damaged or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected rack cover to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

4. Carpet / Floor Liner: May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Make note of defect on MEL Deferred Defect Control Sheet.

5. Overhead Lights (excluding Emergency Lights): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected light(s) to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

SERVICE BAYS

EXTERNAL POWER CONNECTIONS

1. Hold open rods or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switch to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

POTABLE WATER SERVICE BAY

1. Hold open rods or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

2. Compartment lighting lenses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lenses to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

3. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switches to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

4. Dust cover caps for service ports: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

5. Water quantity indication: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected gauge/indicator to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

6. Dust cover cap chains or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

TOILET SERVICE

1. Hold open rods or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Compartment lighting lenses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lenses to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switches to be seen by Flight Crew and note it on MEL Deferred Defect Control Sheet.

4. Dust cover caps for service ports: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

5. Dust cover cap chains or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

HYDRAULIC SERVICE BAY

1. Hold open rods or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Compartment lighting lenses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lenses to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switches to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

4. Dust cover caps for service ports: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

5. Dust cover cap chains or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

6. Manual Service Hoses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected hose connection area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

GROUND AIR SERVICE BAY

1. Hold open rods or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Compartment lighting lenses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lenses to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switches to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

4. Dust covers and lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected dust cover area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

AFT FUSELAGE/TAIL ACCESS

1. Compartment lighting lenses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lenses to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switches to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Hold open rods: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

FUEL SERVICE BAY

1. Ground Service Control Panel: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected panel to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Hold open rods or lanyards: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected access door to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Compartment lighting lenses: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected lenses to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

4. Compartment lighting switches: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected switches to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

PASSENGER SEATS

PASSENGER VIDEO SYSTEM

1. Seat Display Unit (In-arm / Wall Mounted / Console Mounted): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected unit(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Video Monitor(s) (wall- or overhead-mounted units): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected monitor to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. IFE Video Monitor Shroud(s): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected monitor to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

4. In-flight Entertainment System (IFE): May be inoperative or missing

   (M) Procedure – Determine if the Prerecorded Passenger Announcement System is operational. If not, apply MEL for the Prerecorded Passenger Announcement System MEL item 23-31-02.

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected unit to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

PASSENGER AUDIO ENTERTAINMENT

1. Remote Jack Units (RJU): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected unit(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. In-flight Entertainment System: May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected system control panel to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Speaker(s): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected speaker to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

4. Passenger Control Unit / Handset (Including Cradle): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected unit(s) to be seen by Flight Crew and note it on the MELME Deferred Defect Control Sheet.

PASSENGER CALL

1. Passenger Seat Call Light(s): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected call light to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Passenger Seat Call Button(s): May be inoperative or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected seat call button to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

LIGHTING

1. Passenger Reading Light(s): May be inoperative or missing

   (M) Procedure – None Required

   (O Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected light(s) to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

MISCELLANEOUS

1. Passenger Seat Tray Table Assembly (& Latch): May be inoperative or missing

   (M) Procedure – Maintenance will secure the tray table in stowed position. Flight Crew may perform task if Maintenance is not available.

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected tray table to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

2. Passenger Seat Headrest(s): May be inoperative or missing

   (M Procedure – Maintenance will secure the headset in stowed position. Flight Crew may perform task if Maintenance is not available.

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected headrest to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

3. Passenger Seat Foot Rest(s): May be inoperative or missing

   (M) Procedure – Maintenance will secure the foot rest in stowed position. Flight Crew may perform task if Maintenance is not available.

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the Affected Foot Rest To Be Seen By Flight Crew And Note It on the MEL Deferred Defect Control Sheet.

4. Passenger Seat Trim: May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

5. Passenger Seat Cushion(s): May be worn, torn, frayed or stained as long as the item is otherwise serviceable as determined by the Flight Crew.

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Display an Inoperative Placard in a prominent position on or near the affected area to be seen by Flight Crew and note it on the MEL Deferred Defect Control Sheet.

6. Airline Magazine and advertisement in the cabin: May be missing, worn, torn or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Note it on the MEL Deferred Defect Control Sheet.

7. Log Book covers: May be worn, torn or missing

   (M) Procedure – None Required

   (O) Procedure – None Required

   (P) Procedure – Note it on the MEL Deferred Defect Control Sheet.

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